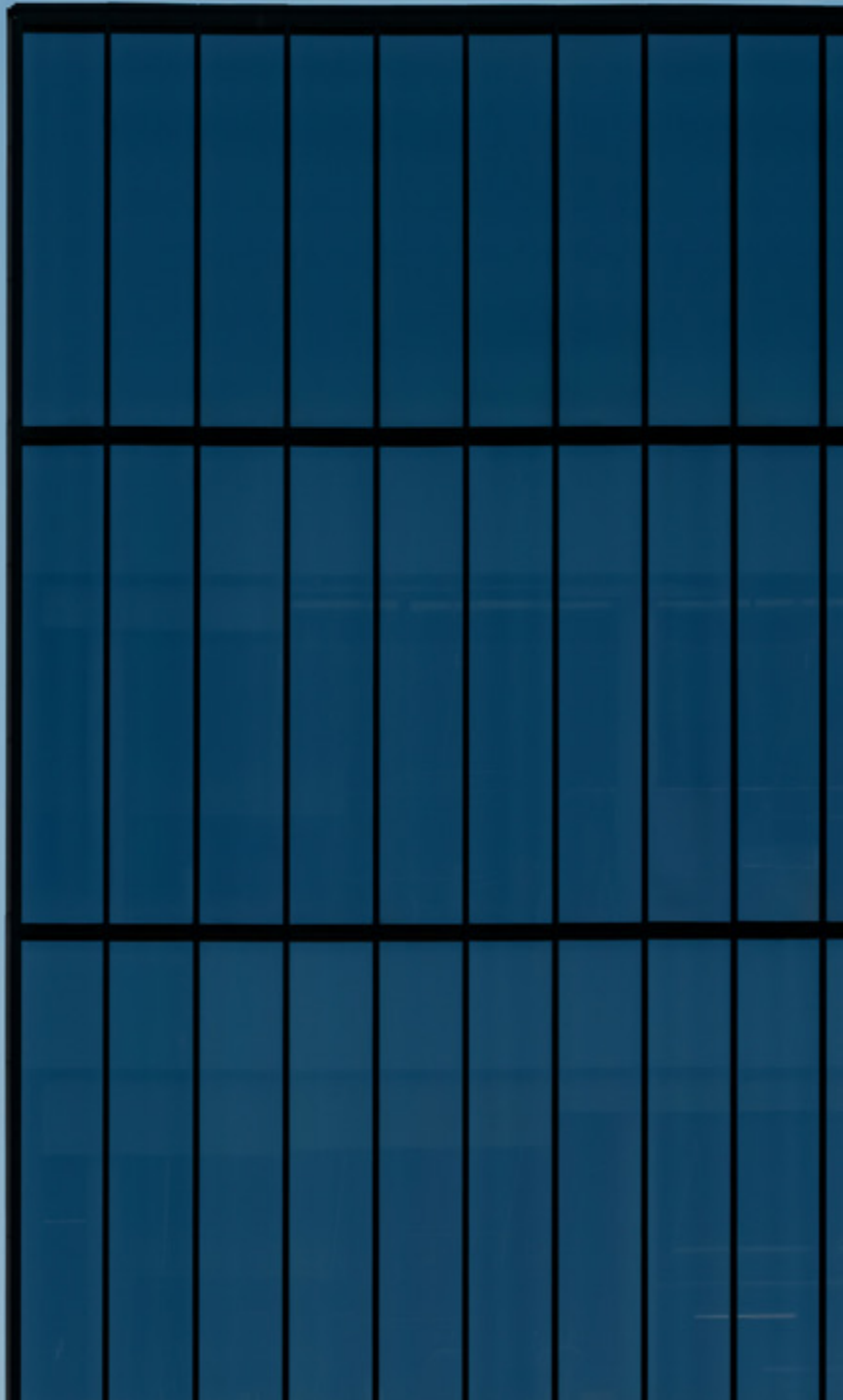


Allianz Global Insurance Report 2026: The future of insurance in a fragmenting world

28 May 2026



Content

Page 3-4

Executive Summary

Page 5-11

Looking back: From boom to normalization

Page 12-15

The changing risk landscape

Page 16-22

Looking ahead: Insurance remains a growth industry

Page 23-25

The next frontier: Climate risk, affordability and the future of insurability

Page 26-27

Appendix

Executive Summary



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- **The insurance industry is cooling from exceptional growth, not slowing into weakness.** Global premiums rose by +7.1% to EUR6.9trn in 2025, adding EUR456bn to the global premium pool. Although growth moderated from the exceptional +9.4% recorded in 2024, it remained comfortably above the ten-year CAGR of +5.6%, confirming that the industry's growth drivers remain firmly intact. Life insurance remained the largest segment (EUR2,861bn; +EUR185bn in 2025), followed by P&C (EUR2,320bn; +EUR85bn) and health (EUR1,688bn; +EUR185bn).
- **Asia remains the structural center of growth even as advanced markets outperform cyclically.** Western Europe grew nearly twice as fast as its 2015–2025 average, while emerging market growth softened as China slowed to +7.4%, more than 3pps below its long-term average. Excluding China and Japan, however, Asia remained the industry's fastest-growing major region, with premium growth accelerating to +10.9% versus a long-term average of +4.9%.
- **Much of the industry's recent expansion reflects higher prices rather than materially broader coverage.** Global insurance penetration rose only modestly to 7.2% of GDP in 2025 and remains below levels seen a decade ago. The disconnect is most visible in P&C, where penetration has stagnated around 2.5% of GDP despite strong premium growth. Trends across business lines are diverging: health-insurance penetration rose from 1.4% of GDP in 2015 to 1.8% in 2025, while life insurance at 3.0% remains well below the levels above 4% seen two decades ago.
- **The P&C boom is normalizing while life and health remain structurally strong.** Global P&C growth slowed to +3.8% in 2025 as pricing cycles matured and claims inflation stabilized. North America remained dominant, accounting for 52% of global P&C premiums, while Asia continued to exhibit the world's largest protection gap, with penetration of just 1.3% compared with 4.3% in North America. Life insurance grew by a still robust +6.9%, though the US annuity boom is fading. Health insurance remained the industry's strongest growth story, expanding by +12.3%, the fastest pace since 2014, driven by rising medical costs and growing demand for private healthcare protection.
- **Despite Asia's rise, global insurance remains overwhelmingly dominated by the US.** North America increased its share of global premiums from 42.5% to 46.4% over the past decade, meaning that almost every second euro written globally now originates from the region. China has emerged as a clear number two, increasing its market share from 7.5% to 10.9%. Yet, at EUR746bn in premiums, it remains less than a quarter of the size of the North America market at EUR3,191bn. Western Europe, by contrast, will continue to lose relative weight across most business lines.

- Geopolitics and fragmentation are becoming central forces shaping the insurance industry.** The Iran war is acting as a major external supply shock, disrupting energy markets, trade flows and supply chains. In our central scenario, global GDP growth is expected to slow to +2.6% in 2026, while Eurozone growth will fall to just +0.8%. If the conflict is not resolved during the summer, we expect additional upward pressure on inflation and a materially worse global growth outlook. More broadly, geopolitical fragmentation is creating a structurally more complex operating environment, challenging assumptions around global integration, capital mobility and cross-border risk diversification. Insurers will need to adapt by building more regionally resilient operating models, integrating geopolitical analysis more directly into underwriting and capital allocation and developing products tailored to emerging risks such as cyber escalation. While fragmentation raises costs and operational complexity, it also increases demand for protection and resilience, reinforcing the strategic relevance of insurance in a more uncertain global environment.
- Protection remains a structurally growing need in an increasingly uncertain and ageing world.** The global insurance market is expected to expand by +5.3% annually over the next decade, driven by rising demand for protection, demographic change, higher interest rates and growing health and retirement needs. Health insurance is projected to remain the fastest-growing segment at +6.7% p.a., followed by life insurance at +4.9% and P&C at +4.7%. Asia, particularly China and India, will remain the main growth engine, adding 5pps to their global market share, while North America is expected to retain its dominant global market position at roughly 46%. Europe, by contrast, is likely to continue ceding global market share over the coming decade (-4pps).
- Asia will generate most of the industry's future growth, with India emerging as the standout opportunity.** The global premium pool is expected to expand by EUR5,260bn by 2036, with more than half of additional premiums originating in Asia. China will remain the region's dominant market, but India is expected to emerge as the fastest-growing major insurance market. Despite already ranking among the world's ten largest, India remains significantly underinsured, with penetration of just 3.8% of GDP and per capita spending of roughly EUR85. Rising incomes, demographic change and recent reforms are expected to drive annual premium growth of around +10.7% over the next decade.
- Climate change is turning insurance affordability into a structural challenge for the industry.** Insured NatCat losses are rising by +5–7% annually in real terms, while higher premiums are increasingly colliding with weakening household purchasing power. The result is widening protection gaps, pushing lower-income households out of coverage. As affordability deteriorates, more climate losses are shifting onto already strained public balance sheets. Preserving long-term insurability will require moving beyond repricing toward stronger adaptation incentives, resilience-focused underwriting, targeted affordability mechanisms and deeper public-private risk-sharing frameworks, supported by large-scale investment in climate adaptation and resilient infrastructure.



Looking back: From boom to normalization

The global insurance industry returned to a more sustainable growth trajectory in 2025, with total gross written premiums (GWP) rising by +7.1% to EUR6.9trn. The expansion added EUR456bn to the global premium pool over the course of 2025. The life segment remained the largest (EUR2,861bn; +EUR185bn in 2025), followed by P&C (EUR2,320bn; +EUR85bn) and health (EUR1,688bn; +EUR185bn). However, growth decelerated from the exceptional +9.4% recorded in 2024, returning the industry toward a more sustainable trajectory after two years of super-charged growth. Crucially, even at this more moderate pace, the 2025 rate remained comfortably above the ten-year CAGR of +5.6%, confirming that the structural growth drivers that have fueled the global insurance market expansion since the mid-2010s remain in place. Over the full decade from 2015 to 2025, total GWP expanded from EUR3.97trn to EUR6.87trn – a cumulative addition of more than EUR2.9trn, driven by rising insurance penetration, expanding middle classes in emerging economies and the continued broadening of coverage across all three major segments.

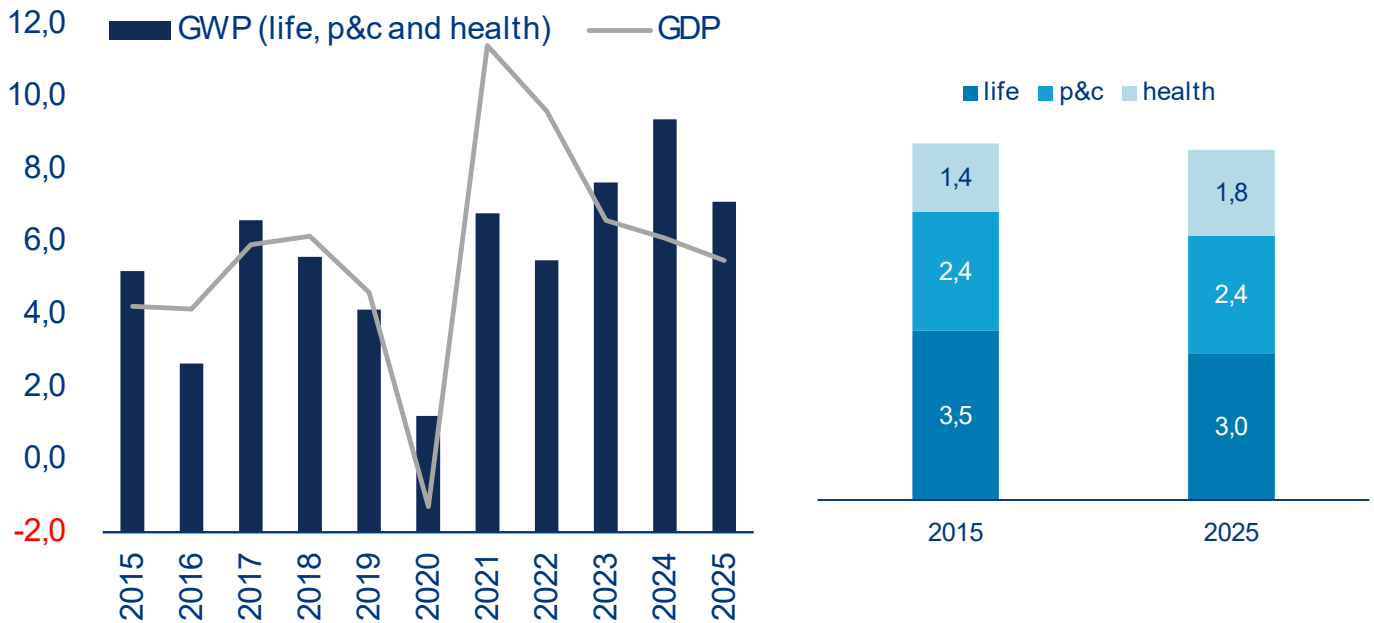
Much of the industry's recent premium growth reflects higher prices rather than a comparable deepening of insurance protection. With consumer price inflation running at approximately 2–3% across major markets in 2025, part of the headline growth merely reflects inflationary pass-through rather than additional coverage written. Insurance penetration, measured

as premiums relative to GDP, increased only modestly to 7.2% in 2025 from 7.0% in 2024 and remains below the 7.3% recorded a decade earlier. This divergence between nominal premium growth and real insurance deepening has become a defining feature of the post-2022 insurance cycle. While the industry has expanded substantially in monetary terms, the share of economic activity devoted to insurance has yet to recover to its long-run level. The disconnect is particularly visible in P&C, where higher claims costs have driven pricing increases without materially expanding insurance penetration.

The trend in global insurance penetration is increasingly uneven across business lines, with P&C broadly stable, health steadily gaining relevance and life insurance still recovering from the ultra-low-rate era. P&C penetration has remained broadly stable at around 2.5% of GDP, while health insurance has steadily gained importance, rising from 1.4% of GDP in 2015 to 1.8% in 2025 as demand for private healthcare protection increased. Life insurance, by contrast, continues to recover only gradually from the prolonged period of ultra-low interest rates. Although penetration has edged back up since 2022 to 3.0% of GDP in 2025, it is still below the 3.5% a decade earlier as the segment remains well below historical norms (Figure 1). The longer-term decline is even more striking: two decades ago, global life insurance penetration exceeded 4% of GDP.

Figure 1: Premium growth remains ahead of GDP

Global gross written premiums* and nominal GDP growth* (y/y, in %) and global gross written premiums* as % of GDP by segments



*The conversion into EUR is based on 2025 exchange rates.

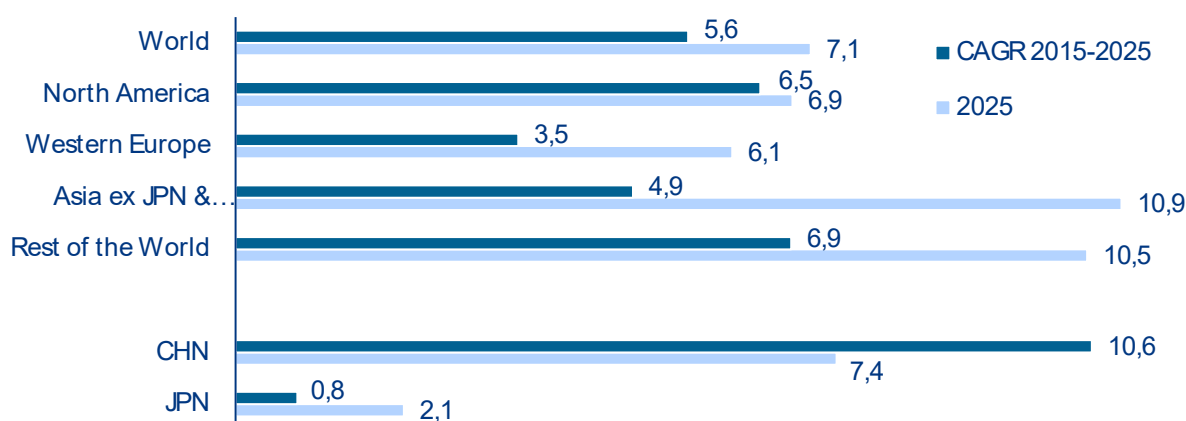
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

Growth remains strong globally, but the center of momentum is shifting as advanced markets outperform cyclically while Asia retains its structural growth lead. While all major regions grew faster than their respective long-term averages in 2025, advanced markets were the main drivers of this acceleration, with Western Europe (+6.1%) expanding nearly twice as fast as its 2015-2025 average. Emerging market growth, by contrast, softened relative to the past decade, largely reflecting slower expansion in China, which accounts for almost 60% of the emerging market aggregate. Chinese premium growth slowed to +7.4% in 2025, more than 3pps below its long-term average. While this is

to a large extent a function of past excellence rather than a weak current performance, nominal growth has also been held back by annual inflation registering just above 0%. Meanwhile Asia¹ (when excluding China and Japan) continues to significantly outperform Europe and North America, with premium growth reaching +10.9% in 2025 compared with a long-term average of +4.9%, reinforcing its position as the industry's key structural growth market.

¹ Throughout the report, Asia always refers to the region excluding China and Japan. If both countries are included, we use the term wider Asia.

Figure 2: 2025 growth outpaces long-term average
Gross written premium growth*, 2025 and CAGR 2015-2025 by region in %



*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The global P&C market is entering a phase of normalization after several years of exceptional pricing-driven growth. The segment grew by +3.8% in 2025 to reach EUR2,320bn globally, a notably softer performance than both the overall market and the segment's own ten-year CAGR of +5.6%, as well as last year's growth of +8.5%. The moderation reflects a maturing pricing cycle. After several years of sharp rate increases across motor, property and specialty lines, pricing has started to normalize in some regions, while claims costs, although still elevated, are no longer accelerating at the pace seen in 2022 and 2023. North America² remains the undisputed center of global P&C activity, accounting for 52.0% of world P&C premiums, with EUR1,205bn written in 2025. However, regional growth slowed to +2.2% from +9.7% in the previous year, reflecting both a more mature rate cycle and a -6.3% decline in US accident & health premiums³ that weighed on overall market growth (US: +2.2%); excluding A&H, however, the US P&C market is estimated to have grown by 4.1%⁴(Figure 3).

Western Europe's P&C market posted solid growth in 2025, although the exceptional post-inflation growth surge is now beginning to normalize. Premiums grew by +5.3% to EUR529bn, a solid expansion but notably below the exceptionally strong growth rates of +8.5% in 2023 and +6.8% in 2024 as inflation pressures continued to ease. Performance across the region was increasingly uneven. Southern European markets remained particularly dynamic, with several recording

high single-digit growth above +8%, while Scandinavia, the UK, Belgium and Switzerland expanded more modestly at around +1% to +4%. Among the largest Eurozone markets, growth remained comparatively robust, led by Germany at +7.7%, France at +7.0%, Spain at +6.5% and Italy at +6.4%. Wider Asia P&C premiums grew by +4.1% to EUR391bn in 2025, broadly in line with the previous year. Yet despite a population more than six times larger than Western Europe's, the region's P&C market remains significantly smaller. The gap is equally visible in insurance penetration and spending levels. P&C penetration stands at just 1.3% in wider Asia and 1.1% in China, compared with 2.5% in Western Europe and 4.3% in North America. Per capita P&C spending reached only EUR98 across Asia (China: EUR135), compared with EUR1,232 in Western Europe and EUR3,111 in North America. These disparities underline the scale of structural underinsurance and the region's long-term catch up potential.

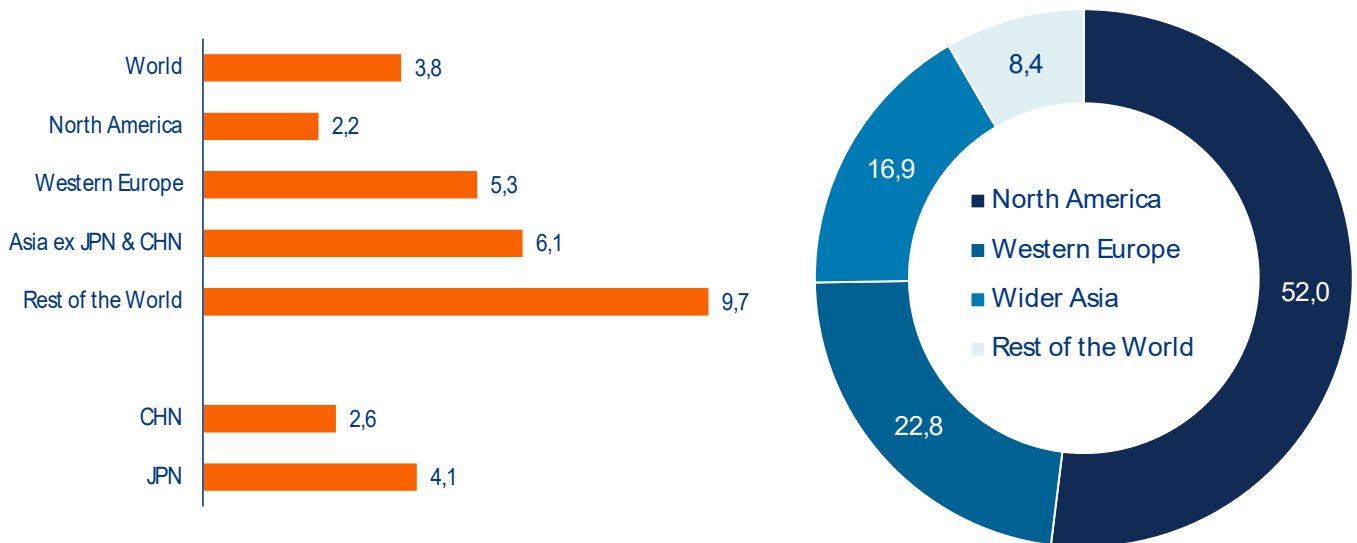
China is by far the largest market in the wider region, accounting for almost every second euro written. At +2.6% China registered one of the weakest performances in recent years outside the Covid-19 period and well below the country's average growth rate of +6.6% over the past decade. The slowdown largely reflects China's relatively muted nominal growth dynamics. All other insurance markets (rest of the world) recorded growth of +9.7% in the P&C segment, driven by strong (largely inflationary) increases of above +10% in Eastern Europe and Latin America.

² 96% of P&C premiums in North America are written in the US; the rest is attributable to Canada.

³ In this report, Accident & Health (A&H) premiums written by Life/A&H entities are included in the US P&C market figures, consistent with our general approach of classifying accident lines as part of the P&C market.

⁴ This estimate is based on growth in net premiums written as direct written premium figures had not yet been published at the time of writing.

Figure 3: Europe and Wider Asia power ahead
P&C gross written premium growth* and market share by region, 2025 in %



*The conversion into EUR is based on 2025 exchange rates.

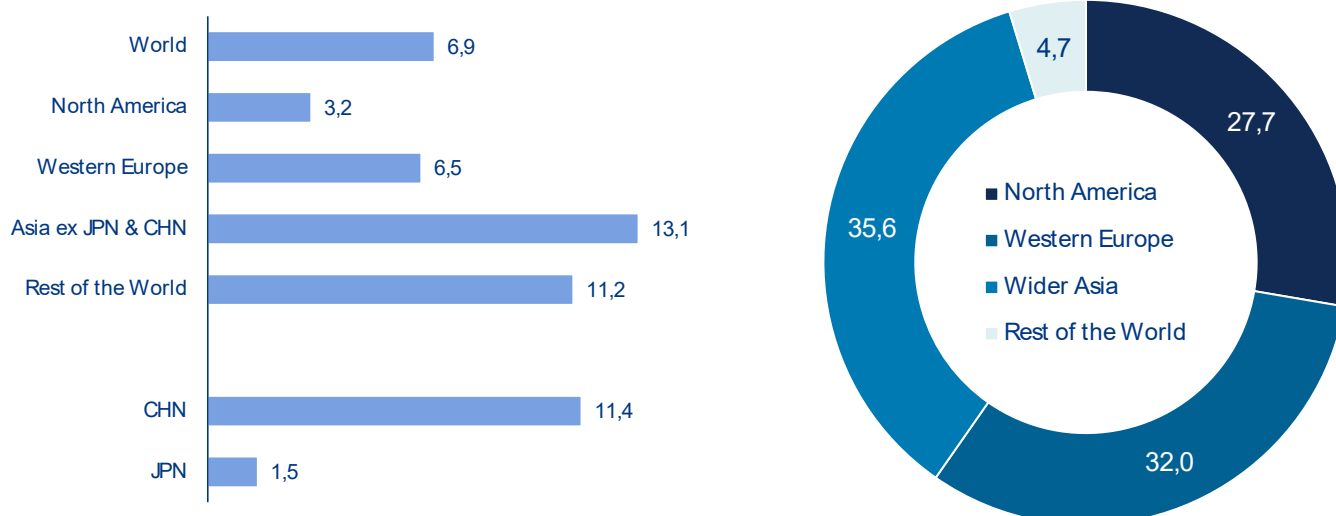
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The life insurance boom is cooling in North America, while Asia is reasserting itself as the industry's main growth engine. Global life insurance premiums grew by +6.9% in 2025, a clear moderation from the exceptionally strong +11.3% recorded in 2024 but still a robust performance by historical standards. The slowdown was driven primarily by North America, where the annuity boom triggered by households locking in higher interest rates has begun to lose momentum. Growth leadership has now shifted back toward Asia (Figure 4).

Higher interest rates continue to support life insurance globally, but regional momentum is increasingly diverging. In Western Europe, life premium growth moderated to +6.5% in 2025 from +9.9% in the previous year, although higher rates continued to support savings products. Performance remained uneven across markets. France, the Netherlands, Finland and Portugal posted another year of double-digit growth, while Switzerland continued to lag. Germany returned to positive territory with growth of +5.3%. Wider Asia recorded healthy +9.9% growth in 2025, with the majority of markets posting strong double-digit gains.

As a result, Asia as a whole remains the largest life insurance market, well ahead of North America and Europe (Figure 4), reflecting both demographic pressures and weaker public pension systems. This is also reflected in insurance penetration, which stands at 3.3% in Asia, driven by the popularity of life products in advanced Asian markets such as South Korea, Taiwan and Singapore. But China at 2.5% and India at 2.8% already approach North American levels of 2.8%, although still below Western Europe's 4.3%. These relatively high Asian penetration rates reflect greater reliance on private retirement savings in the absence of comprehensive public pension systems. By contrast, North America's comparatively low penetration partly reflects the widespread use of occupational retirement schemes such as 401(k) plans.

Figure 4: Asia pulls further ahead
Life gross written premium growth* and market share by region, 2025 in %



*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

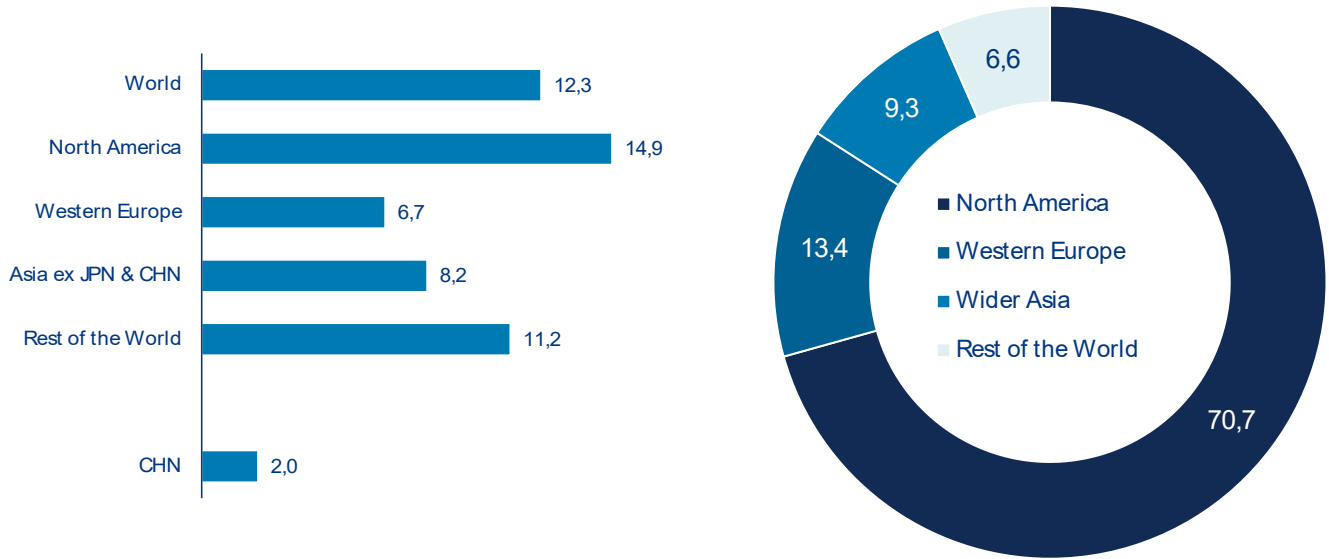
Health insurance⁵ remains the industry's fastest-growing segment, driven by rising medical costs and increasing demand for private protection. The US continues to dominate the global health market, accounting for more than 70% of worldwide health premiums (Figure 5). In many other markets, private health insurance is still a niche segment, albeit a very dynamic one: global premium growth averaged +8.5% p.a. over the last decade with particularly strong rates in recent years as Covid-19 boosted demand for additional health protection. In 2025, global health premiums increased by +12.3%, the strongest expansion since 2014, driven largely by North America where medical inflation pushed growth to +14.9%.

Despite a recent normalization in parts of Asia, the region's low health-insurance penetration continues to point to substantial long-term growth potential. Health-insurance growth in wider Asia slowed to +3.4% in 2025, reflecting a sharp deceleration in China, where growth dropped to just +2%. However, many markets

including Indonesia, Singapore, Hong Kong, Thailand and India continued to record double-digit growth, with several accelerating further in 2025. This strong momentum partially reflects the still low penetration in the region, which is below 1% in all markets except Taiwan. Even more than in life insurance, demand is mainly driven by the status of the social security system, i.e. the coverage and quality of the public healthcare system. This explains why even in some developed markets, such as the Scandinavian markets, private health insurance is almost non-existent.

⁵ In general, data visibility of health-insurance premiums is still rather low. Often, health premiums are included in other segments as health coverage is seen as part of other products and not separately reported; in Japan, for instance, health is treated as "third sector products" within life.

Figure 5: No slowdown in sight
Health gross written premium growth* and market share by region, 2025 in %



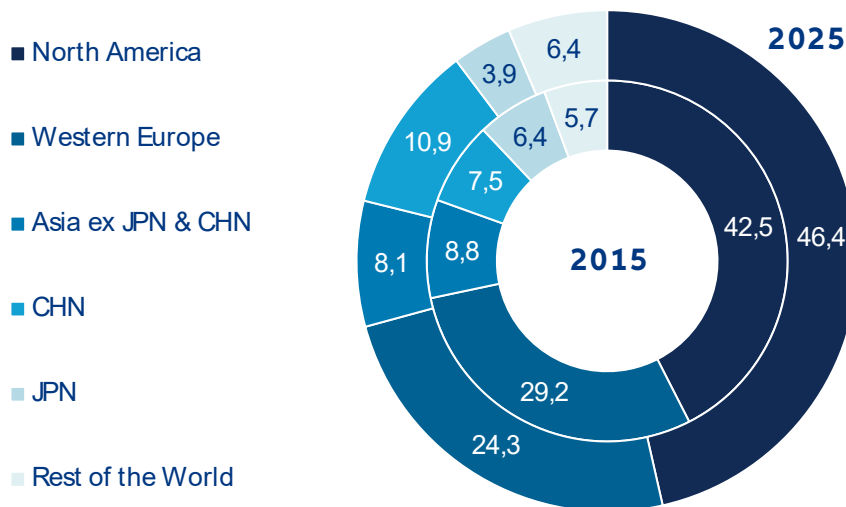
*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The global insurance industry remains overwhelmingly dominated by the US despite the rise of China and broader Asia. Unlike many other industries where traditional markets are losing importance to new emerging markets, the global insurance industry is still dominated by the US. In fact, over the past decade, the North American insurance market has increased its share of global premiums from 42.5% to 46.4%, meaning that almost every second euro written globally now originates from the region. On the other hand, other „old“ markets such as Western Europe (-4.9pps) and

Japan (-2.5pps) behaved more or less as expected, losing market share mainly to China, which boosted its global share from 7.5% to 10.9%. However, the gap with the North America market is still huge: while the Chinese market is already the clear number two in terms of premium income (EUR746bn in 2025), it is dwarfed by the North America market, which generates more than four times as much premium income (EUR3,191bn). Given this huge discrepancy, it does not look like the US dominance in global insurance will end any time soon (Figure 6).

Figure 6: US dominance remains unchanged
Total gross written premiums*, 2015 and 2025 by region in %



*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

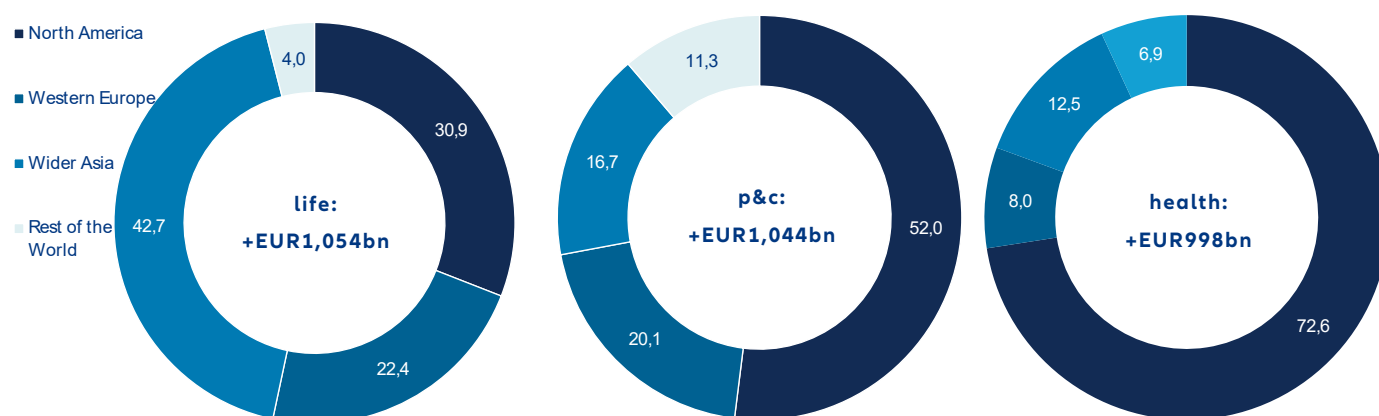
The past decade has reshaped the global insurance market unevenly across regions and business lines.

North America accounted for 52% of additional global P&C premiums and 71% of additional health premiums over the past ten years, underlining the region's overwhelming importance in both segments. In life insurance, however, the picture is more balanced. Despite the recent US annuity boom, only around one-third of additional life premiums originated in North America, while wider Asia captured by far the largest share of global growth. Western Europe, by contrast, continues to lose relative weight across all major business lines, falling behind not only the US market but increasingly also wider Asia.

Another notable shift is the growing convergence in absolute premium growth across segments.

Although life insurance remains substantially larger in size at EUR2,861bn compared with EUR2,320bn in P&C and EUR1,688bn in health, the absolute increase in premium volumes has recently been broadly similar. While life insurance growth was constrained for years by ultra-low interest rates, health insurance has benefited strongly from the post-Covid increase in risk awareness and demand for private health protection (Figure 7).

Figure 7: Western Europe continues to lose ground
Share of absolute premium growth* by region, 2015 - 2025 in %



*The conversion into EUR is based on 2024 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.



The changing risk landscape

Economic and capital market outlook

The global economy has entered a period of heightened uncertainty as the Iran war acts as a significant external supply shock, disrupting energy markets, raising transport costs and fragmenting supply chains. The result is renewed inflationary pressure alongside weaker growth at a time when the global outlook was already fragile. The scale and persistence of the economic impact remain highly uncertain and will depend critically on how the conflict evolves.

In our central scenario, in which a US-Iran agreement struck during the summer months leads to a normalization of oil and gas trade before the end of the year, we forecast global GDP growth of 2.6% in 2026, revised down by 0.5pp from pre-conflict expectations. The impact across major economies is uneven. In the US, growth is expected to remain

relatively more robust at +2.0% as substantial investment in artificial-intelligence infrastructure and stronger private consumption driven by tax cuts largely offset the energy shock. China would also prove comparatively resilient at +4.7%, supported by a more diversified energy mix and solid external demand. Europe, however, would bear the heaviest burden. Eurozone growth would nearly halve to +0.7% as the region's greater dependence on energy imports adds to longstanding structural weaknesses that continue to constrain economic momentum.

Inflation is set to rise on both sides of the Atlantic in 2026, interrupting the gradual normalization seen over the past year. In the US, the energy shock will push inflation back toward levels last seen during the post-pandemic surge. CPI inflation is forecast to rise to 3.6% in 2026, peaking at around 4.0% around mid-year

before gradually easing as energy prices stabilize and weaker activity dampens underlying price pressures. The Fed is expected to raise rates by 25bps in 2026, largely looking through the energy-driven spike as the risk of inflation expectations becoming de-anchored appears contained. After all, unlike in 2022, labor markets are softer and monetary conditions are no longer highly accommodative, limiting the scope for broad-based second-round effects. In the Eurozone, inflation dynamics would be similar in direction but more persistent in profile. Headline CPI is forecast at 3.1% for 2026 as the pass-through from higher wholesale gas prices into household utility bills keeps inflation elevated for longer than in the US. This leaves the ECB in an uncomfortable position: inflation is rising above target just as growth weakens sharply. We forecast the ECB to hike rates by 50bps in 2026, with a focus on keeping inflation expectations in check.

Government bond markets are undergoing a sharp repricing as the energy shock pushes inflation expectations higher while simultaneously weakening the growth outlook. Long-term sovereign yields are rising as investors demand greater compensation for inflation risk and increasingly question the sustainability of large fiscal deficits in a weaker growth environment. On balance, we expect 10-year Treasury yields to reach 4.6% by year-end, with persistent fiscal deficits of around 7% of GDP adding further upward pressure on US borrowing costs. In Europe, 10-year Bund yields are forecast at 3.0%, as weaker growth and safe-haven demand from investors diversifying away from US assets partially offset rising inflation expectations. In the periphery, however, Italian and French borrowing costs

are increasing more sharply than Germany's as markets grow more concerned about the capacity of heavily indebted sovereigns to absorb energy-related fiscal pressures without compromising fiscal consolidation commitments – a dynamic likely to intensify as France approaches its 2027 election cycle.

Risks to the outlook remain clearly skewed to the downside. A prolonged closure of the Strait of Hormuz into fall would produce a materially more adverse outcome. In this downside scenario, the Eurozone economy would fall into technical recession, with annual growth in 2026 slowing to just +0.2%, while US growth would weaken sharply to +1.5% as equity market corrections feed through to consumer confidence and spending. Inflation would rise to 4.1% in the Eurozone and 4.2% in the US, forcing central banks into a more aggressive tightening response despite deteriorating growth conditions – a policy dilemma with few good options. Beyond the direct energy channel, a prolonged conflict would amplify second-round effects through weaker consumer confidence, rising insolvencies, tighter financial conditions and growing fiscal strain across energy-importing economies. At a time of weakening international policy coordination and increasingly constrained public finances, policymakers may have less capacity to cushion additional shocks. In such an environment, non-linear dynamics could emerge quickly, turning an initial energy shock into a broader and more persistent global downturn.

How geopolitical fragmentation impacts insurance

The insurance industry was built for a world of open capital flows, stable rules and predictable cross-border commerce, and that world is receding. A series of geopolitical shocks, from the US-China trade conflict and the Covid-19 pandemic to the Russia-Ukraine war and the US-Iran conflict, has accelerated the fragmentation of trade, finance and supply chains, pushing the global economy toward a more fractured geoeconomic order. For insurers, this is not simply a challenging macroeconomic backdrop but a structural shift in the foundations of the global insurance business model.

The question is no longer whether fragmentation will reshape global insurance, but how far and how fast. The assumptions that supported global operating models for decades are weakening simultaneously: capital is becoming less mobile, regulatory regimes are diverging and borders are increasingly functioning as strategic barriers rather than manageable operational frictions. For insurers, this directly challenges the foundations of global risk pooling. Insurance depends on the ability to diversify exposures across geographies and move capital and reinsurance capacity freely across borders. Fragmentation weakens both. As cross-border activity becomes more constrained, volatility in claims, currencies and investment returns increases, while regional pools struggle to replicate the efficiency and resilience of global ones.

Regional pools are structurally less efficient because fragmentation increases correlation across exposures while reducing the fungibility of capital and reinsurance. Political shocks increasingly affect currencies, asset prices, trade flows and claims simultaneously, weakening diversification benefits precisely when they are most needed. At the same time, sanctions risk, capital controls and regulatory divergence increase the likelihood that capital becomes trapped within jurisdictions during periods of stress, forcing insurers to hold more capital against the same underlying risks. This undermines one of the key advantages of global insurance groups: efficient cross-border capital allocation and diversification. As trapped capital and duplicated solvency buffers rise, returns on

capital are likely to come under structural pressure. The challenge is therefore no longer preserving the hyper-globalized insurance model of the past, but maintaining sufficient diversification and risk-sharing capacity in a more regionalized world.

Regulatory fragmentation is reinforcing these pressures. The decades of supervisory convergence that produced Solvency II, IAIS standards and equivalence regimes are giving way to more modular and bloc-specific systems. Cross-border reinsurance recoveries, sanctions compliance and contract enforceability can no longer be treated as stable assumptions. Meanwhile, the weakening of multilateral cooperation is amplifying the very risks insurers must absorb. Climate change, cyber threats, pandemics and financial instability all require international coordination to contain; as geopolitical rivalry undermines that cooperation, these risks become larger, more interconnected and harder to price.

The investment consequences are equally significant. Insurers collectively manage trillions in long-duration assets and fragmentation permanently raises the cost and complexity of deploying that capital globally. When the movement of capital increasingly requires political approval rather than responding to market logic, global integration loses efficiency by design. Currency and convertibility risks rise alongside political tensions, sovereign risk becomes harder to hedge and friendshoring creates exposures that are often less liquid and operationally more complex.

The effects are unevenly distributed across business lines. Political-risk insurance is experiencing a structural increase in demand as multinational companies move from treating geopolitical exposure as episodic to embedding it permanently within risk-management frameworks. In trade credit, tariffs are squeezing margins and raising insolvency risk, particularly among smaller companies exposed to disrupted supply chains. Marine and aviation insurers must navigate rerouted shipping lanes and persistent geopolitical volatility; the closure of major maritime chokepoints has shown how quickly coverage availability and pricing can deteriorate when political

risk crystallizes into physical disruption. State-sponsored cyber threats are escalating, with attribution challenges straining the boundaries of standard policy language. In D&O, rising regulatory investigations and extra-territorial enforcement actions are increasing claims exposure for internationally active firms. Retail insurance faces a more indirect but persistent transmission through slower growth, higher inflation and weaker household purchasing power.

At the same time, fragmentation is also creating new insurable risk categories that reward analytical capability and underwriting specialization. The push for energy independence is increasing demand for specialized engineering insurance, while industrial policy initiatives including the CHIPS Act, European sovereignty investment programs and rising defense spending are generating substantial new property and infrastructure exposures. Geopolitics is on balance a net headwind for the sector's long-term growth prospects, but rising investment in defense, energy security, cyber resilience and strategic infrastructure is creating offsetting opportunities for insurance, risk transfer and long-duration capital deployment.

The fact that the insurance sector has remained broadly stable through successive geopolitical shocks so far is no guarantee of future success. Capital positions have remained healthy, combined ratios have held and capacity has not materially decreased. This is testament to the resilience of the insurance industry. However, each shock raises the structural baseline, which is not fully reversed during recovery periods. This means that the industry has to reprice for a genuinely higher level of structural risk. This makes strategically adapting to a fragmenting world a matter of current urgency.

The industry's strategic response must operate on several levels simultaneously.

- **First, operating models must become less centralized and more regionally resilient.** In a more fragmented world, resilience and adaptability matter more than pure scale efficiency. Capital must be increasingly pre-positioned within jurisdictions rather than assumed freely transferable, and entities must

be capable of operating with greater autonomy during periods of geopolitical disruption. Scenario planning must also replace narrow point forecasts. Tail risks such as severe bloc bifurcation or financial decoupling are no longer remote enough to ignore in strategic planning, even if the most likely trajectory remains one of escalating but contained protectionism.

- **Second, portfolio and underwriting decisions must incorporate explicit geopolitical assessment as a systematic input rather than a qualitative overlay.**

Insurers should classify markets along a clear spectrum from core to optional to potential exit candidates, combining underwriting analysis with regular geopolitical, sanctions and affordability assessments rather than making decisions reactively under pressure. Stress testing must extend beyond financial variables to include sanctions regimes, reinsurance enforceability and diversification failure under fragmentation scenarios. The industry cannot assume that mechanisms designed for a globalized world will continue to function under sustained geopolitical strain.

- **Third, product strategy must evolve to address the new exposures fragmentation is creating rather than relying primarily on exclusions and silent coverage.** Recent conflicts have exposed significant gaps in affirmative cover for war risk, political violence, trade disruption, cyber escalation and supply-chain interruption. Insurers that develop explicit and well-priced solutions for these exposures will benefit from a structural increase in corporate demand. Those that rely primarily on exclusions risk marginalization from some of the fastest-growing segments of the commercial market.

Fragmentation will reshape how the global insurance industry operates. A more regionalized world implies higher capital requirements, weaker diversification benefits and greater operational complexity. Yet the same forces that reduce efficiency are also increasing demand for protection, resilience and specialized risk transfer. Insurance may become structurally more complex and capital-intensive, but also strategically more important.



Looking ahead: Insurance remains a growth industry

Insurance will remain a structural growth industry over the next decade, although growth will become more regionally uneven. We expect the global insurance market to expand by +5.3% annually over the next ten years, broadly in line with last year's long-term outlook. Beneath this stable headline picture, however, regional dynamics are shifting meaningfully.

North America is likely to remain the industry's dominant profit and growth pool. Compared to last year, our outlook for the region has become more optimistic across all three business lines, particularly in P&C. Three forces are driving this shift: the expected AI investment boom and associated productivity gains, continued increases in household spending on protection – especially in health insurance – and elevated interest rates supporting life and annuity products.

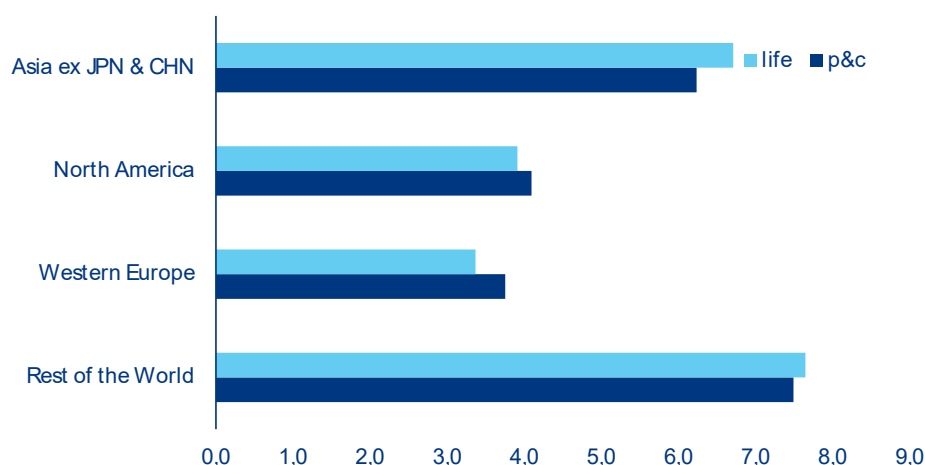
Europe faces a structurally weaker outlook. Our view on Western Europe has become more cautious compared with last year. The region is likely to continue struggling with the fallout from higher energy costs triggered by the escalation in the Middle East (see Chapter 3 for more details), slower than expected progress on strategic sovereignty and both weaker AI investment and slower AI adoption than the US. As a result, we expect European P&C growth to lag the US over the next decade at +3.8% versus +4.1%. Although

rising natural-catastrophe losses should continue to support premium growth, we now expect only a limited increase in penetration rates, rising by just +0.1pp to 2.6% by 2036.

Asia will remain the industry's principal long-term growth engine despite rising geopolitical headwinds. We expect P&C growth in Asia to slow modestly relative to previous forecasts as trade tensions and supply-chain restructuring weigh on economic momentum. Even so, projected annual growth of +6.2% in wider Asia and +6.3% in China remains comfortably above expected growth rates in Europe and North America. Overall, we expect global P&C premiums to expand by +4.7% annually through 2036 as demand for protection continues to rise worldwide.

Life and health insurance will remain the industry's key structural growth segments. We expect life insurance to grow at a CAGR of +4.9% over the next decade, benefiting from elevated interest rates and rising retirement needs. Asia and China should remain the principal growth engines as demographic ageing accelerates and pension systems continue to deepen. Health insurance is expected to remain the industry's fastest growing segment at +6.7% annually, supported by rising healthcare costs and still substantial unmet demand, particularly across Asia (Figure 8).

Figure 8: Solid growth
Gross written premium* growth, CAGR 2026 -2036 by region and segment in %



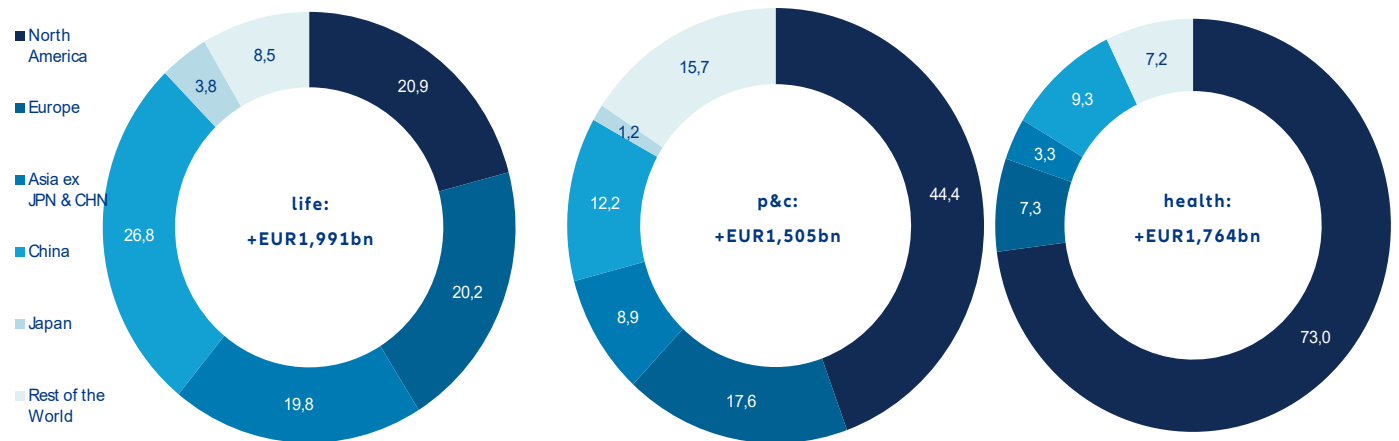
*The conversion into EUR is based on 2025 exchange rates.

Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research..

Asia will generate most of the industry's additional premium growth. The global premium pool is expected to expand by EUR5,260bn by 2036 (Figure 9), with life insurance contributing the largest share at EUR1,991bn. More than half of all additional premiums will originate in Asia including China, exceeding the combined contribution of North America and Europe. While China will remain the region's dominant market in absolute terms with growth of +7.6% annually, India is expected to emerge as the fastest-growing major market at +10.5% annually, positioning it to challenge Japan as Asia's second-largest life insurance market by 2036.

Despite Asia's rise, the US insurance market will remain unrivaled in scale. Although North America's share of incremental P&C growth is expected to decline relative to the previous decade, the US market will continue to dominate globally. By 2036, US P&C premiums are projected to exceed EUR1,800bn, almost five times larger than China's market and more than twice the size of Europe's. In health insurance, US dominance is even more pronounced, with more than two-thirds of expected global premium growth projected to originate from the US market.

Figure 9: Life remains on top
Share of additional gross written premiums* by 2036, by region in %

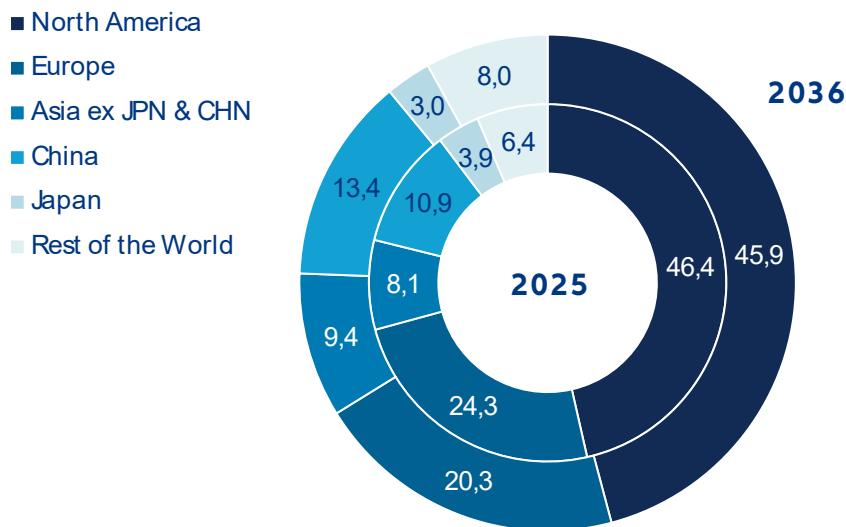


*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

The global insurance map will continue shifting eastward, albeit gradually. North America is expected to retain a global market share of roughly 46% through 2036, surrendering only marginal ground over the next decade (-0.5pp). India and China, by contrast, are expected to continue gaining relevance, together adding almost 4pps of global market share. Western Europe will

continue to lose relative weight. A glimmer of hope for the Old Continent: while it lost 5.3pps of market share in the last decade, it may lose „only“ 4pps in the next decade (Figure 10).

Figure 10: North America, a slightly smaller giant
Total gross written premiums*, 2025 and 2036 by region in %



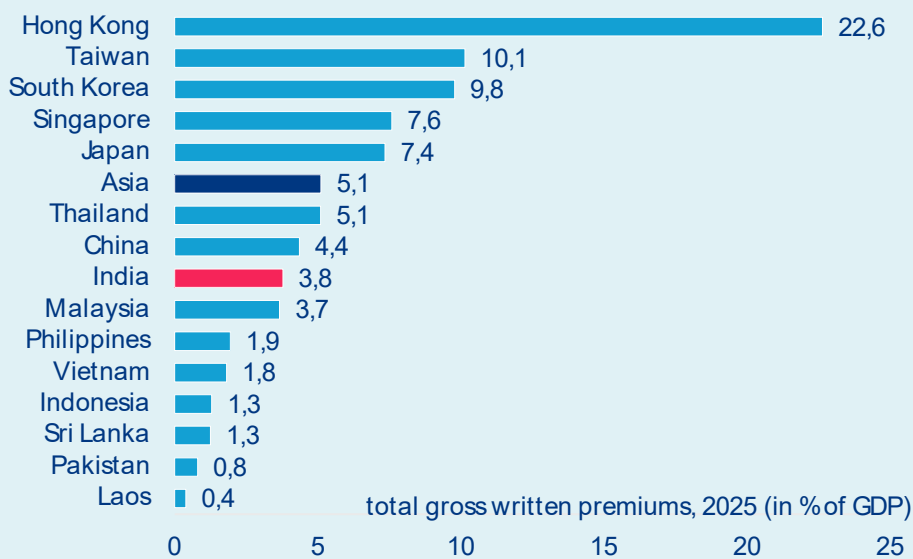
*The conversion into EUR is based on 2025 exchange rates.
Sources: National financial supervisory authorities, insurance associations and statistical offices, Axco, LSEG Datastream, Allianz Research.

Indian insurance market deep dive – turning scale into depth

India ranks among the world's ten largest insurance markets. But by almost every measure of depth, it remains significantly underdeveloped. Despite double-digit growth rates and total gross written premium income of EUR124bn in 2025, insurance penetration and density still indicate substantial pent-up demand. In 2025, total gross written premium income corresponded

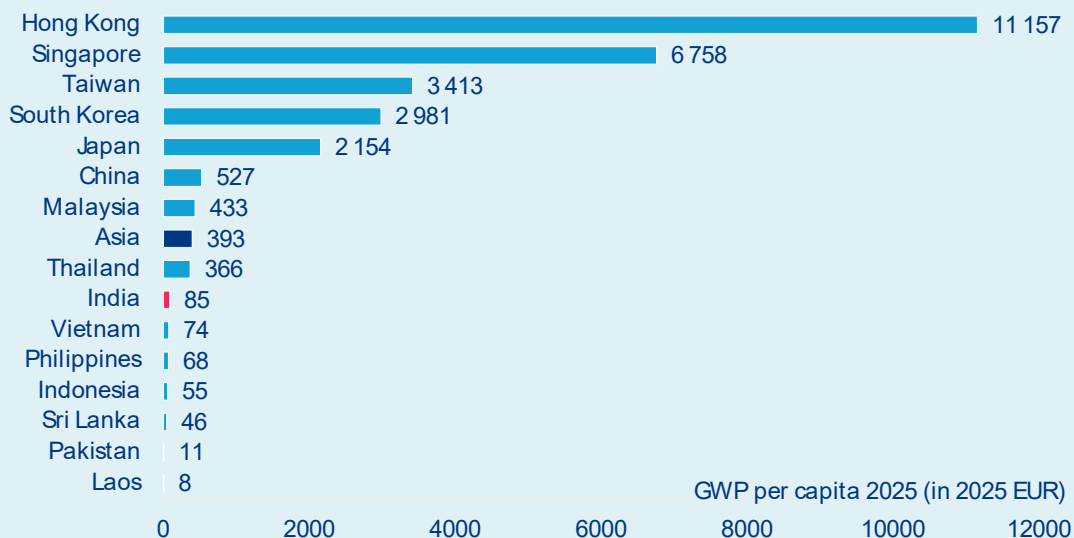
to 3.8% of GDP compared to 4.4% in China or 5.1% in Thailand. And while per-capita spending for insurance amounted to the equivalent of around EUR527 in China and EUR366 in Thailand, it was merely EUR85 in India (Figure 11 and 12).

Figure 11: Insurance penetration, selected countries (gross written premiums in percent of GDP)



Source: National financial supervisory authorities, insurance associations and statistical offices..

Figure 12: Insurance density, selected countries (gross written premiums per capita, in 2025 EUR)



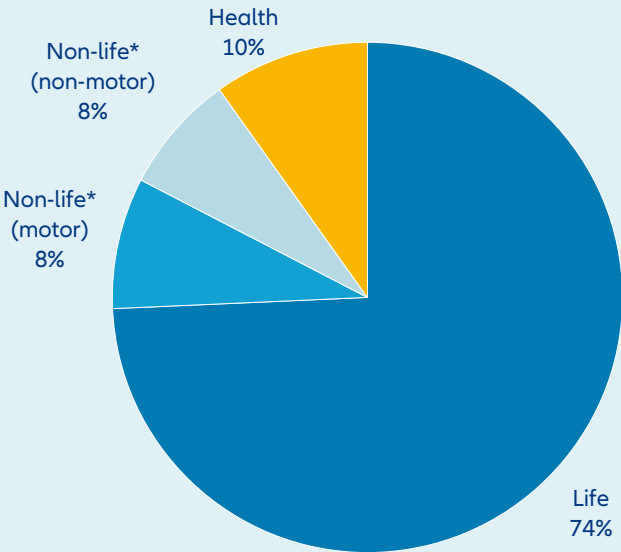
Source: National financial supervisory authorities, insurance associations and statistical offices.

Life insurance dominates the market, but the non-life and health segments reveal where the structural gaps and the growth potential are most acute. India’s insurance market is dominated by the life insurance segment, which accounted for 74% of total gross written premium income in 2024. Non-life insurance premiums, including personal accident and travel insurance, accounted for 16% and health insurance accounted for 10% of total gross written premiums (see Figure 13).

Mandatory motor coverage is the backbone of non-life insurance yet, half of all vehicles on Indian roads remain uninsured, illustrating the gap between legal requirements and market reality. As in most countries, motor insurance was the most important business line in the non-life segment (excluding health), with a share of 52% in fiscal year 2024–2025. Third-party liability business made up more than half of the motor segment

as such coverage is mandatory for all vehicles driven on public roads. Nevertheless, it is estimated that over 50% of vehicles remain uninsured⁶. Given the importance of the agricultural sector for the Indian economy, crop insurance accounted for 16% of general premium income. However, there were significant differences in the composition of premium income across individual states, largely reflecting their respective economic structures (see Figure 14).

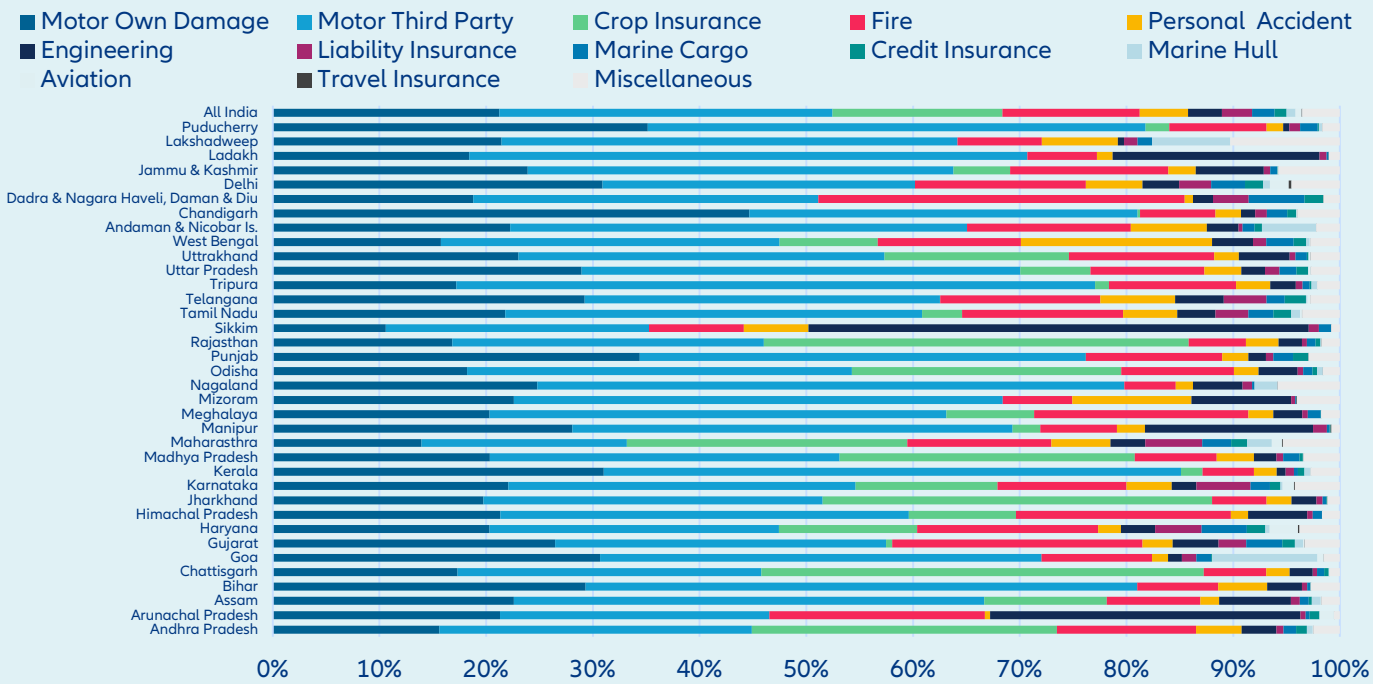
Figure 13: Gross written premiums by business line (in percent)



*including personal accident and travel insurance premiums
Source: IRDAI, Allianz Research.

⁶ See Axco (2025): Insurance market reports (non-life). India (as of March 2026),

Figure 14: Non-life insurance premium by line of business and state (in percent)

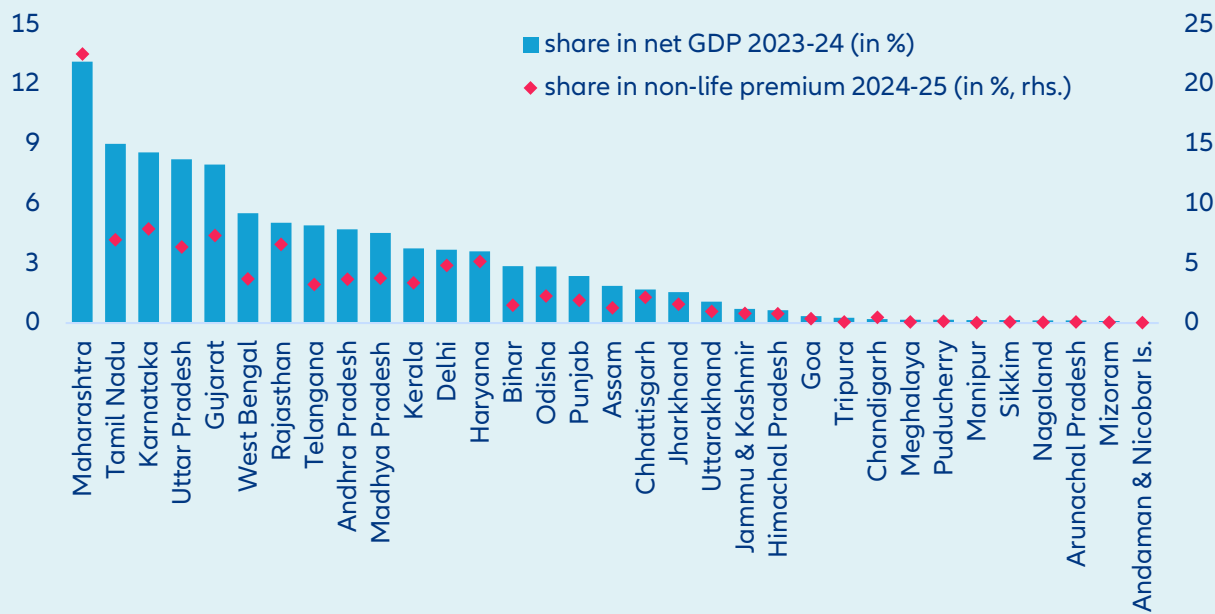


Source: IRDAI

Premium income is as unevenly distributed as economic activity itself, and product design that ignores this geography will systematically miss the largest underserved populations. Five states – Gujarat, Karnataka, Maharashtra, Rajasthan and Uttar Pradesh – accounted for 48% of general premium income, with Maharashtra alone accounting for 20% (see Figure 15). In health insurance, Karnataka, Maharashtra and Tamil Nadu accounted for 49.2% of premium income in

2023–2024, with 22.5% generated in Maharashtra. Despite signs of economic convergence, marked regional differences persist. In 2023–2024, per-capita net state domestic product ranged between EUR570 in Bihar, representing around a third of the national average of EUR1,790, to EUR5,568 in Sikkim. These differences need to be taken into account when it comes to the design and distribution of insurance products.

Figure 15: Non-life premium income by state



Source: IRDAI, Reserve Bank of India

The most powerful long-term growth driver for India's insurance market, alongside rising household wealth, is the still incomplete and uneven reach of the country's social protection system. Market expansion is therefore expected to be driven primarily by pent-up demand and the growing need for supplementary private risk protection, as public social security remains at a relatively early stage of development. According to the latest available ILO data, only 64.3% of India's population was covered by at least one branch of the social security system at all. Effective pension coverage is even more limited: the ILO estimates that only 45.8% of the working-age population was covered by the pension system, compared with close to universal coverage in most industrialized countries. Without targeted government initiatives such as Atal Pension Yojana and the e-Shram portal aimed at informal workers, as well as the supplementary state-level pension schemes, this share would be significantly lower. This is particularly relevant given that only around 12% of India's workforce is employed in the formal sector and therefore benefits from standard pension coverage. Against the backdrop of rising life expectancy, these structural gaps point to a growing need for supplementary, capital-funded private pension solutions to secure retirement income. A similar dynamic is evident in healthcare. Despite the successful rollout of Pradhan Mantri Jan Arogya Yojana (PM-JAY), out-of-pocket spending still accounts for 43.9% of total health expenditure - high by international standards. This underscores both the need for further reforms to broaden public healthcare coverage and the resulting demand for supplementary private health insurance. In this context, medical inflation in India, which is among the highest in Asia, poses a significant financial challenge for households and health insurance providers alike.

Regulatory reform and an ambitious universal coverage target are creating both the policy tailwind and the competitive catalyst for the next growth phase. Recent insurance market reforms, notably the opening up to foreign insurance participation, will increase the efficiency of the insurance sector and together with the IRDAI's „Insurance for All by 2047“ program are expected to spur double-digit insurance market growth of an estimated average 10.5% in life insurance, 10.2% in non-life and 12.5% in health insurance per annum until 2036.

India's insurance sector will expand not just in scale but in structural sophistication, shifting from a peripheral to a systemically important pillar of the financial system. Rising incomes and a growing middle class will lift demand for protection and savings products, driving higher penetration and density. Demand is expected to shift from basic, often mandatory covers such as motor and simple life towards more complex products including health, pensions and wealth-linked insurance. Insurers will increasingly act as institutional investors, supporting capital market development and long-term financing, particularly for infrastructure. Higher incomes and improving financial literacy would also help reduce underinsurance and encourage more proactive risk management by households and firms. Together with demographic trends, notably ageing, this would further boost demand for private health and retirement solutions. India's still-developing social security system implies a greater reliance on private provision compared to other markets such as China. Overall, sustained growth is likely to transform the sector into a more mature and systemically important pillar of the financial system.



The next frontier:

Climate risk, affordability and the future of Insurability

Affordability is emerging as a key structural challenge facing the insurance industry. In some markets and lines of business, the cost of adequate coverage is beginning to outpace households' capacity to pay, driven by claims severity, inflation and broader economic pressures. Where this gap becomes persistent, it translates into structural underinsurance, concentrating uninsured risk where financial resilience is weakest. Climate risk is where this dynamic is already most visible and most advanced.

Global insured NatCat losses are rising by 5–7% annually in real terms, underlining that climate risk is now a structural underwriting issue rather than a cyclical volatility event. Climate change is an undisputed driver, but observed loss trends are also shaped by socioeconomic factors, including exposure growth in hazard-prone areas and inflation rather than by climate effects alone⁸. Rising premiums are increasingly colliding with falling household purchasing power. Climate shocks alongside other socioeconomic pressures such as demographic change, wars and geopolitical instability are driving both insurance costs and greater financial stress. This dynamic is already

visible in household insurance markets: In the US, homeowners' insurance premiums rose by +38% between 2019 and 2024, roughly twice the rate of inflation. The same climate shocks driving up premiums also erode the capacity to pay: droughts reduce crop yields and push up food prices, while heat stress lowers labor productivity and wages. Global food prices could be around 13% above baseline by 2050 in a +2°C warming scenario, with Asia-Pacific facing increases of up to 27%, while the most exposed European economies could see cumulative GDP losses of 5–7% over five years⁹. Households are therefore being squeezed from both directions: rising insurance costs and weakening income resilience.

Lower-income households face a self-reinforcing vulnerability trap as higher premiums and weaker incomes push them out of insurance coverage. The poorest and most climate-exposed households are hit hardest, creating a cycle in which rising risks deepen both poverty and underinsurance. Evidence from developing countries shows that the poorest 40% experience climate-driven income losses 70% larger than the average¹⁰, while a 1°C increase in temperature raises poverty rates by 0.9 to 2.3pps¹¹. Similar patterns are emerging in developed

⁷ Swiss RE (Sigma 1/2025)

⁸ The primary driver to date has been exposure growth (urbanization, asset accumulation and rising construction costs) which accounts for over 80% of the long-term increase in weather-related insured losses. Climate change effects have so far been a smaller but growing contributor, and Swiss Re cautions that for certain perils and regions, such as wildfire in North America and severe convective storms in Europe, hazard intensification is already accelerating losses well beyond what exposure alone can explain.

⁹ Feeding a warming world: Securing food and economic stability in a changing climate | Allianz

¹⁰ Climate change through a poverty lens | Nature Climate Change

¹¹ World Bank Document

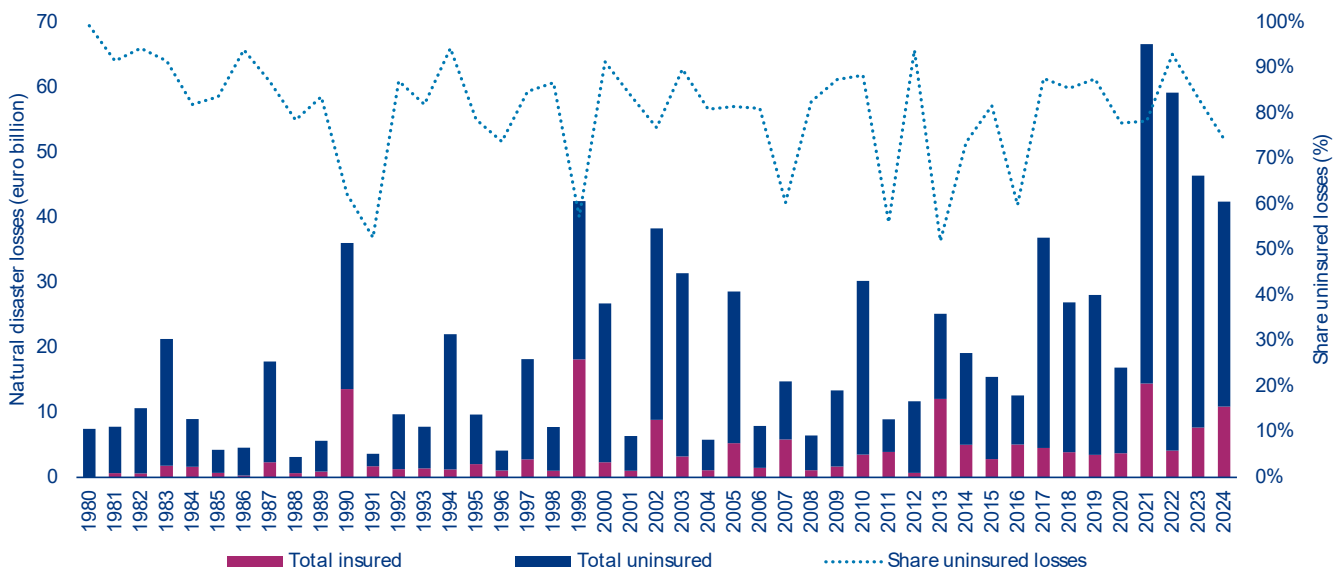
economies, where climate change also drives chronic rather than temporary poverty, making exclusion from insurance more persistent. As premiums rise, disaster insurance risks becoming a product accessible only for wealthier households. Evidence from Florida shows that without public investment in risk reduction and affordability mechanisms, both traditional and parametric insurance products become inaccessible for low- and middle-income households¹².

As private insurance affordability declines, climate losses increasingly shift onto public balance sheets that are themselves being weakened by climate shocks.

In the EU, uninsured direct disaster losses averaged EUR59bn annually between 2021 and 2023 (Figure 16), highlighting how a large share of climate losses continues to sit outside formal insurance coverage and is ultimately absorbed by households, firms and governments¹³. This gap is significantly demand-driven: EIOPA research highlights¹⁴ that low risk awareness, expectations of government

compensation and lack of product clarity mean many households and businesses simply do not purchase available NatCat coverage, a challenge distinct from the supply-side retreat seen in parts of the US market. This is also visible in the rapid expansion of state-backed insurance schemes. Heat stress alone could worsen European fiscal balances by around 0.5% of GDP annually. France faces additional fiscal pressure of 2.2% of GDP, while Italy and Spain face 1.9% and 1.7% respectively. The governments most exposed to climate risk are often those with the least fiscal space to subsidize insurance or invest in adaptation, creating a dangerous feedback loop of rising losses, weaker coverage and declining fiscal resilience. In the US, California's FAIR Plan increased insured exposure from USD220bn in September 2022 to USD750bn by early 2026¹⁵, and following the Los Angeles wildfires required a USD1bn emergency levy on insurers to maintain solvency. Florida's Citizens scheme peaked at USD618bn of insured exposure in 2023, showing how quickly public contingent liabilities can grow when private insurers retreat¹⁶.

Figure 16: Development of NatCat losses in Europe and share of uninsured losses



Source: CATDAT

¹² Insurance and climate risks: Policy lessons from three bounding scenarios | PNAS

¹³ Climate and nature risks fuel multi-billion insurance protection gaps | WWF

¹⁴ Climate insurance protection gaps: a demand-side challenge - European Insurance and Occupational Pensions Authority

¹⁵ California FAIR Plan Association (March 2026)

¹⁶ Citizens Property Insurance Corporation (Policies in Force)

Europe is increasingly moving toward mandatory and state-backed NatCat systems, but these solutions must avoid creating long-term moral hazard. Public-private NatCat systems are becoming a structural feature to adapt to extreme weather events. France already operates a mandatory state-backed NatCat reinsurance scheme funded by a surcharge on all property insurance policies, and after repeated climate-related losses that charge was nearly doubled last year to 20%. Since 2025, all businesses in Italy have been required to carry NatCat insurance, with the government covering up to half of the exposure through its state reinsurer. Germany is about to introduce a mandatory scheme, and the German Insurance Association has proposed a new private-sector pool for high-risk residential properties backed by a government stop-loss mechanism for catastrophe losses above EUR30bn. At the EU level, supervisors are exploring a Europe-wide risk-sharing system combining a premium-financed pool with a public loan-funded mechanism. These schemes can preserve insurability, but if public support becomes too strong, pricing signals could weaken and incentives for resilience investment, relocation and land-use discipline could be reduced. Equally, even well-designed public-private schemes face structural limits if investment in climate-change adaptation, from enforced flood defenses and resilient infrastructure to smart agriculture and water management systems, does not scale fast enough to reduce costs of the rising loss trends. Without combined strategies, the fiscal and insurance capacity underpinning the public-private NatCat schemes will be progressively eroded.

The deeper issue is that climate risk is exposing the limits of pricing alone as a long-term solution to maintaining insurability. Insurance pricing continues to rely primarily on claims experience, probabilistic NatCat models and increasingly granular hazard maps that capture location-specific risk differentials. These tools remain broadly effective for pricing incremental climate risks and preserving underwriting discipline. However, annual repricing becomes increasingly difficult when physical risks and claims severity rise faster than households' ability to absorb higher

premiums. For insurers, the challenge is no longer simply to reprice risk, but to preserve insurability in economies where affordability is structurally weakening.

Preserving long-term insurability requires stronger adaptation incentives, efficiency gains, targeted affordability measures and public-private solutions. This requires moving beyond pure risk transfer toward stronger risk reduction and resilience incentives. Insurance will increasingly need to incentivize adaptation through pricing, deductibles and prevention management, while efficiency gains become essential to offset structurally rising claims severity driven by technological complexity, specialized repair services and legal costs. Automation, AI and digital claims management already show significant potential to reduce operational costs without compromising coverage quality. Critically, however, affordability cannot be preserved through artificially suppressed premiums. Underpricing weakens incentives for risk reduction and ultimately accelerates rather than closes protection gaps. Where lower-income households are being priced out of coverage, targeted support mechanisms such as means-tested vouchers or public reinsurance are likely to prove more effective and transparent than below-risk pricing. In areas where physical risks have shifted permanently, managed relocation may ultimately be more sustainable than maintaining coverage at unsustainably low prices. More broadly, protection gaps cannot be solved by private insurance alone. Public-private partnerships, sovereign backstops and targeted affordability mechanisms will become increasingly necessary where risks become privately uninsurable, but must always be combined with credible adaptation investment at scale. Without enforced building codes, flood defenses and resilient infrastructure, public-private schemes absorb rising losses without addressing their source, progressively eroding the fiscal and insurance capacity that underpins them.

Appendix

Appendix A

Insurance markets

2025²⁾ KPIs

	Total premium income (in EUR bn) ¹⁾			Penetration (as % of GDP)			Density (per capita, in EUR) ¹⁾		
	p&c	life	health	p&c	life	health	p&c	life	health
Argentina	11,8	1,5	0,03	2,4	0,3	0,01	258	33	0,6
Australia	40,5	14,0	18,1	2,5	0,9	1,1	1 500	517	670,6
Austria	15,6	5,2	3,4	3,0	1,0	0,7	1 710	568	376,8
Bahrain	<i>0,4</i>	<i>0,1</i>	<i>0,2</i>	<i>1,1</i>	<i>0,2</i>	<i>0,5</i>	<i>260</i>	<i>41</i>	<i>134,6</i>
Belgium	<i>13,3</i>	<i>17,8</i>	<i>2,5</i>	<i>2,1</i>	<i>2,8</i>	<i>0,4</i>	<i>1 127</i>	<i>1 516</i>	<i>214,7</i>
Brazil	24,0	10,5	53,6	1,2	0,5	2,7	113	49	251,6
Bulgaria	<i>2,2</i>	<i>0,3</i>	<i>0,2</i>	<i>3,7</i>	<i>0,5</i>	<i>0,3</i>	<i>326</i>	<i>48</i>	<i>28,6</i>
Canada	<i>49,9</i>	<i>66,6</i>	<i>25,4</i>	<i>2,5</i>	<i>3,3</i>	<i>1,3</i>	<i>1 244</i>	<i>1 659</i>	<i>633,0</i>
Chile	4,9	9,1	1,1	1,5	2,9	0,4	247	460	57,6
China	190,9	433,2	121,5	1,1	2,5	0,7	135	306	85,8
Colombia	5,8	6,8	1,1	1,4	1,6	0,3	109	127	21,0
Croatia	1,6	0,3	0,2	1,7	0,4	0,2	411	87	39,4
Czech Republic	7,3	2,1	0,6	2,1	0,6	0,2	684	196	60,2
Denmark	<i>12,8</i>	<i>36,2</i>	<i>0,5</i>	<i>3,1</i>	<i>8,8</i>	<i>0,1</i>	<i>2 134</i>	<i>6 034</i>	<i>88,7</i>
Egypt	<i>0,9</i>	<i>1,1</i>	<i>0,3</i>	<i>0,3</i>	<i>0,3</i>	<i>0,1</i>	<i>8</i>	<i>9</i>	<i>2,6</i>
Finland	4,8	6,0	0,8	1,7	2,2	0,3	850	1 075	141,3
France	<i>98,2</i>	<i>192,1</i>	<i>48,4</i>	<i>3,3</i>	<i>6,4</i>	<i>1,6</i>	<i>1 474</i>	<i>2 882</i>	<i>726,7</i>
Germany	103,6	95,0	54,1	2,3	2,1	1,2	1 232	1 130	643,9
Greece	2,7	2,8	0,6	1,1	1,1	0,2	267	283	55,6
Hong Kong	<i>3,3</i>	<i>77,0</i>	<i>2,3</i>	<i>0,9</i>	<i>21,1</i>	<i>0,6</i>	<i>441</i>	<i>10 411</i>	<i>305,0</i>
Hungary	3,0	2,1	0,1	1,3	0,9	0,1	312	214	13,4
India	19,7	<i>92,1</i>	12,4	0,6	<i>2,8</i>	0,4	13	<i>63</i>	8,5
Indonesia	6,2	8,2	1,4	0,5	0,7	0,1	22	29	5,0
Ireland	<i>4,6</i>	<i>24,1</i>	<i>3,9</i>	<i>0,7</i>	<i>3,8</i>	<i>0,6</i>	<i>874</i>	<i>4 539</i>	<i>741,0</i>
Italy	<i>48,0</i>	<i>122,6</i>	5,0	<i>2,1</i>	<i>5,4</i>	0,2	<i>812</i>	<i>2 073</i>	84,0
Japan	58,5	206,7	n/a	1,6	5,7	n/a	475	1 679	n/a
Kazakhstan	1,2	1,8	0,1	0,4	0,7	0,1	55	86	6,6
Kenya	<i>0,9</i>	<i>1,4</i>	<i>0,6</i>	<i>0,8</i>	<i>1,2</i>	<i>0,5</i>	<i>16</i>	<i>25</i>	<i>10,9</i>
Laos	<i>0,1</i>	<i>0,01</i>	n/a	<i>0,3</i>	<i>0,1</i>	n/a	<i>6</i>	<i>1</i>	n/a
Malaysia	<i>4,7</i>	10,6	<i>0,3</i>	<i>1,1</i>	2,5	<i>0,1</i>	<i>130</i>	296	<i>7,4</i>
Mexico	18,1	22,0	8,7	1,1	1,3	0,5	137	167	65,6
Morocco	2,8	2,7	0,5	1,7	1,7	0,3	72	72	12,6
Netherlands	<i>16,7</i>	<i>15,7</i>	66,5	<i>1,4</i>	<i>1,3</i>	5,6	912	855	3 624,3
Nigeria	<i>0,9</i>	<i>0,4</i>	n/a	<i>0,4</i>	<i>0,2</i>	n/a	<i>4</i>	<i>2</i>	n/a
New Zealand	<i>5,5</i>	<i>1,6</i>	n/a	<i>2,5</i>	<i>0,7</i>	n/a	<i>1 039</i>	<i>303</i>	n/a
Norway	10,7	15,2	0,5	2,3	3,3	0,1	1 910	2 708	87,5
Pakistan	<i>1,5</i>	<i>1,2</i>	n/a	<i>0,4</i>	<i>0,4</i>	n/a	<i>6</i>	<i>5</i>	n/a
Peru	<i>2,1</i>	3,5	<i>0,4</i>	<i>0,7</i>	1,2	<i>0,1</i>	<i>61</i>	103	<i>12,5</i>
Philippines	2,1	5,6	<i>0,2</i>	0,5	1,4	<i>0,1</i>	18	48	<i>1,8</i>
Poland	<i>14,5</i>	<i>3,6</i>	1,7	<i>1,6</i>	<i>0,4</i>	0,2	<i>381</i>	<i>95</i>	43,8
Portugal	6,2	8,1	1,8	2,0	2,7	0,6	600	783	171,1
Romania	3,5	0,8	<i>0,2</i>	0,9	0,2	<i>0,1</i>	183	44	<i>11,7</i>
Saudi Arabia	<i>5,8</i>	<i>2,8</i>	<i>10,7</i>	<i>0,5</i>	<i>0,3</i>	<i>1,0</i>	<i>169</i>	<i>81</i>	<i>308,4</i>
Singapore	3,7	33,5	2,5	0,7	6,4	0,5	625	5 705	428,3
Slovakia	1,4	0,6	0,2	1,0	0,5	0,1	253	115	36,1
South Africa	11,8	40,2	n/a	3,0	10,2	n/a	183	622	n/a
South Korea	79,8	74,2	n/a	5,1	4,7	n/a	1 545	1 437	n/a
Spain	36,5	35,9	13,5	2,2	2,1	0,8	763	750	280,9
Sri Lanka	0,4	0,6	0,1	0,4	0,7	0,1	16	27	2,7
Sweden	10,8	39,4	n/a	1,8	6,5	n/a	1 017	3 697	n/a
Switzerland	22,1	24,3	14,4	2,4	2,6	1,5	2 462	2 710	1 609,0
Taiwan	9,8	56,1	13,0	1,3	7,2	1,7	424	2 428	560,3
Thailand	8,0	14,0	4,2	1,6	2,7	0,8	112	196	58,2
Türkiye	16,5	3,5	4,2	1,3	0,3	0,3	188	40	47,9
United Arab Emirates	7,3	1,9	8,2	1,5	0,4	1,7	642	170	726,3
United Kingdom	<i>122,0</i>	<i>275,3</i>	<i>10,4</i>	<i>3,5</i>	<i>7,9</i>	<i>0,3</i>	<i>1 754</i>	<i>3 958</i>	<i>150,1</i>
United States	<i>1155,5</i>	725,6	1167,6	<i>4,4</i>	2,8	4,5	<i>3 327</i>	2 089	3 362,1
Vietnam	<i>2,7</i>	<i>4,9</i>	n/a	<i>0,6</i>	<i>1,2</i>	n/a	<i>26</i>	<i>48</i>	n/a

¹⁾ 2025 exchange rates.

²⁾ Estimates are shown in italics.

Appendix B**Insurance markets****Long-term development**

	CAGR 2015-2025 (in %)			CAGR 2026-2036 (in %)			Total premium income 2036 (in EUR bn) ¹⁾		
	p&c	life	health	p&c	life	health	p&c	life	health
Argentina	61,0	54,4	61,0	17,5	18,8	22,8	69,8	10,1	0,3
Australia	6,5	-8,1	4,3	4,7	3,1	4,2	67,1	19,5	28,5
Austria	3,8	-2,3	5,6	3,1	1,2	5,0	21,8	5,9	5,9
Bahrain	1,2	-6,3	6,7	3,4	3,7	5,6	0,6	0,1	0,4
Belgium	3,2	0,9	5,2	3,0	4,0	6,1	18,4	27,4	4,8
Brazil	7,1	11,6	9,2	5,5	11,4	5,8	43,0	34,5	99,9
Bulgaria	10,5	7,2	16,9	7,2	7,4	15,4	4,7	0,7	0,9
Canada	4,1	5,3	4,8	3,7	4,1	4,7	74,8	104,0	42,1
Chile	8,2	9,3	11,1	6,4	7,3	6,3	9,7	19,9	2,2
China	6,6	11,3	18,2	6,3	7,6	8,1	374,3	966,7	285,6
Colombia	9,1	12,4	14,6	6,7	8,0	11,5	11,9	15,8	3,7
Croatia	7,0	-0,4	13,7	4,4	2,6	12,2	2,5	0,4	0,5
Czech Republic	6,6	-2,7	8,9	4,7	2,4	8,6	12,0	2,7	1,6
Denmark	2,8	6,2	7,2	3,0	5,6	8,1	17,8	66,0	1,3
Egypt	20,4	20,2	30,5	11,4	9,8	20,9	3,0	3,0	2,5
Finland	1,8	0,2	9,0	3,3	4,5	9,8	6,8	9,8	2,2
France	4,5	3,7	2,8	4,1	3,5	3,2	153,5	280,8	68,7
Germany	4,6	0,5	3,7	4,3	2,5	3,9	165,3	124,3	82,7
Greece	2,3	4,5	14,4	5,4	3,6	6,8	4,7	4,2	1,1
Hong Kong	2,3	8,2	6,8	4,1	5,2	5,9	5,1	135,1	4,2
Hungary	10,7	5,1	16,1	6,8	4,5	12,6	6,2	3,3	0,5
India	10,9	10,4	18,3	10,2	10,5	12,5	57,3	277,0	45,2
Indonesia	8,0	3,7	13,1	7,3	7,9	12,9	13,3	18,8	5,4
Ireland	5,1	8,0	4,9	3,5	3,4	4,9	6,8	34,8	6,7
Italy	2,9	0,5	8,3	3,6	2,8	7,6	70,6	167,0	11,1
Japan	1,9	0,5	n/a	2,5	2,9	n/a	77,2	282,9	n/a
Kazakhstan	12,0	32,8	12,7	7,1	11,8	12,1	2,4	6,1	0,5
Kenya	5,5	12,9	12,7	7,8	12,1	12,5	2,1	5,0	2,3
Laos	11,3	35,0	n/a	10,4	15,0	n/a	0,2	0,0	n/a
Malaysia	3,3	6,1	2,1	5,6	5,2	6,7	8,5	18,6	0,5
Mexico	9,2	9,8	12,2	8,2	8,5	9,7	43,0	53,8	24,0
Morocco	5,8	10,8	5,2	5,9	7,1	6,8	5,2	5,8	1,0
Netherlands	2,5	0,0	4,4	3,2	2,7	4,3	23,8	21,1	105,7
Nigeria	21,6	21,7	n/a	15,7	20,0	n/a	4,7	3,3	n/a
New Zealand	7,1	3,9	n/a	3,6	3,6	n/a	8,0	2,3	n/a
Norway	7,1	6,2	12,8	5,0	4,4	8,7	18,3	24,4	1,2
Pakistan	27,7	11,5	n/a	13,8	13,8	n/a	6,4	5,2	n/a
Peru	5,0	10,7	8,3	5,2	8,0	5,2	3,7	8,3	0,8
Philippines	7,6	8,7	n/a	8,5	10,0	n/a	5,1	16,0	n/a
Poland	6,9	-3,8	24,1	4,2	3,6	8,7	22,9	5,3	4,2
Portugal	6,1	-2,2	11,2	4,0	3,9	8,1	9,7	12,5	4,2
Romania	9,6	11,1	27,8	3,2	3,3	21,6	4,9	1,2	1,9
Saudi Arabia	5,8	26,8	10,5	6,2	11,7	8,2	11,3	9,4	25,3
Singapore	4,2	7,5	23,9	5,6	5,5	8,4	6,7	60,2	6,1
Slovakia	3,6	-5,4	n/a	4,0	3,5	10,5	2,1	0,9	0,6
South Africa	7,0	5,0	n/a	5,6	5,6	n/a	21,6	73,0	n/a
South Korea	5,4	0,8	n/a	4,9	4,4	n/a	135,6	119,0	n/a
Spain	4,1	3,4	5,8	3,4	3,6	6,0	52,9	52,8	25,6
Sri Lanka	8,8	15,9	10,6	10,6	12,8	8,3	1,1	2,4	0,2
Sweden	4,8	5,4	n/a	3,6	4,1	n/a	16,1	61,2	n/a
Switzerland	2,0	-3,3	3,1	2,9	1,6	2,4	30,4	28,9	18,8
Taiwan	5,9	-1,3	4,1	4,8	4,3	4,8	16,4	89,2	21,6
Thailand	2,5	0,5	9,7	4,5	3,6	8,6	12,9	20,8	10,4
Türkiye	40,5	43,8	48,1	11,8	14,4	15,3	56,3	15,6	20,0
United Arab Emirates	7,7	-0,4	11,2	4,7	5,3	6,3	12,0	3,4	16,1
United Kingdom	7,8	4,5	4,8	3,4	3,4	3,3	176,1	397,2	14,9
United States	5,7	4,9	9,0	4,1	3,9	6,9	1 799,6	1 103,9	2 437,4
Vietnam	10,6	16,4	n/a	9,5	7,0	n/a	7,3	10,2	n/a

¹⁾ 2025 exchange rates.



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Too hot to grow

The economic costs of extreme heat

28 May 2026

Content

Page 3-4

Executive Summary

Page 5-12

The economic transmission of heat stress

Page 13-16

The macroeconomic implications of heat stress

Page 17-19

The fiscal burden

Page 20-23

From exposure to resilience: the three-pillar adaptation plan

Page 24-26

Appendix

Executive Summary



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- **Extreme heat is emerging as a structural economic risk, with Europe highly exposed.**

Heat stress events have multiplied sevenfold since the 1980s while the average death toll per event has risen fivefold. That share partly reflects measurement: vital registration and excess-mortality surveillance are far more developed in Europe than in much of Africa and South Asia, where heat deaths go largely uncounted. But genuine structural vulnerability is also at work – ageing populations, dense urban building stock designed to retain warmth and severely underdeveloped cooling infrastructure, with air-conditioning penetration averaging 19% across Europe against roughly 90% in the US.

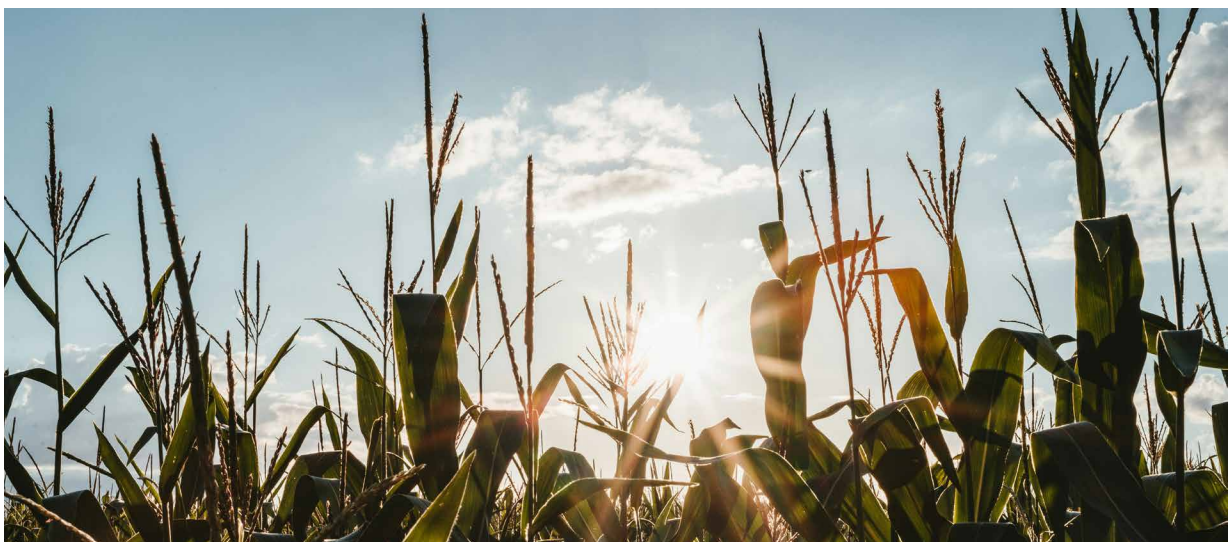
- **The economic transmission of heat stress is non-linear, with a critical threshold around 30°C beyond which productivity losses intensify sharply.** Below this level, warming reduces heating costs and is associated with modest productivity gains. Above this level, the relationship reverses and both channels worsen with each additional degree. The dominant effect operates through labor: output per hour declines by approximately USD1.3 (constant PPP, ~3% of mean hourly output in our 2014-2024 sample) for every degree across the 30-35°C range. Wage adjustments follow productivity with a lag, so the short-run cost falls disproportionately on firm profitability before gradually transmitting to household income and consumption. A second, smaller channel runs through energy: consumption rises by around 1.2% per degree, raising firms' input costs at exactly the temperatures where labor productivity is falling.

- **To gauge the macroeconomic stakes, we construct a stress scenario in which the five hottest years observed in each country between 2014 and 2024 are replayed in ascending order over 2026–2030 – the fifth-hottest year in 2026, the fourth in 2027 and so on, culminating in the country's hottest year on record in 2030.** Under this trajectory, cumulative implied GDP losses (2026 – 2030) could reach 5–7% for the most exposed economies: USD240bn for France, USD354bn for Japan, USD147bn for Italy, USD131bn for Germany and USD120bn for Spain. More consequentially for long-run growth, in such a scenario the decline in fixed capital formation systematically exceeds consumption losses, reaching 8% on average across affected countries: As heat compresses expected returns on capital, investment falls, reducing future productive capacity in a self-reinforcing drag. Moreover, stagflationary dynamics should be expected, with rising prices alongside rising unemployment, placing monetary authorities in a binding trade-off that is especially acute in the Eurozone, where a single policy rate must serve economies with sharply diverging climate exposures.

- **The fiscal consequences fall most heavily on the economies least able to absorb them.** The loss of economic output due to heat reduces tax revenues: Estimated annual losses would reach 1.8% in France, 1.3% in Italy and Spain and 0.7% in Germany – partly because progressive tax systems mean revenues tend to fall faster than output itself, amplifying the fiscal drag beyond the headline GDP loss. Simultaneously inflation-indexed transfers, healthcare costs and emergency infrastructure repair raises public

expenditure. Fiscal balances deteriorate by around 0.5% of GDP annually on average. Italy and Spain risk breaching the Maastricht deficit ceiling (again) once heat-related pressures are incorporated. France, already carrying a projected deficit of –4.9% of GDP, faces additional heat-related pressure of 2.2%.

- **Insured losses remain a small fraction of total damages, reflecting a structural mismatch between what heat destroys and what conventional insurance was designed to cover.** In 2022, total climatological losses in Europe reached EUR46bn while the insured share rose only marginally. Most heat damage accumulates through excess mortality, lost working hours, healthcare-system pressure and infrastructure stress – channels that indemnity contracts are not designed to handle. This makes extreme heat harder to insure than other climate risks because losses are widespread and are often indirect – like lower productivity or health impacts – making them difficult to measure and price. Closing the protection gap is therefore as much a product-design challenge as a capacity one, and the insurance toolkit is already evolving in response: parametric instruments that pay out on objective temperature or duration thresholds, public–private risk-sharing arrangements for systemic exposures and dedicated public backstops where private capacity cannot reasonably reach.
- **Heat-adaptation policy in Europe is built mainly to compensate for losses rather than prevent them.** Following the multi-actor frame of IPCC, closing the gap in Europe requires coordinated action on four fronts: labor regulation, buildings, public finance and households. A workable occupational regime needs binding temperature thresholds, automatic work restrictions when those thresholds are crossed, paid compensation for lost hours and coverage that reaches fixed-term, seasonal and platform workers. No major European economy has all four, and the gap is concentrated on the last: protections were designed around standard contracts and leave the workers most exposed to heat largely outside the regime. Prevention itself is also underused, with shifted hours, partial mechanization and indoor cooling still rare. For buildings, four pieces have to fit together: overheating standards in new construction, mandatory passive cooling in renovation, cooling access for vulnerable households as a social entitlement and grid-adequacy planning that accounts for coincident summer cooling demand and the thermal derating of generators. The revised EU Energy Performance of Buildings Directive delivers the first; the gap is the other three, where indoor temperatures, mortality and peak electricity demand are actually held down. On fiscal architecture, every major European economy has a national adaptation strategy but almost none has translated it into a multi-year budget envelope, so the response defaults to ad-hoc emergency packages – and each episode quietly consumes the fiscal space that ex-ante adaptation would have used to lower the next. The missing layer is households. EU households hold almost EUR40trn in financial assets, including a very large stock of deposits, while much of Europe’s housing stock remains poorly adapted to hotter summers. Mobilizing even a small, well-targeted share – through incentives for retrofit, passive cooling and affordable parametric cover – could close part of the gap that public budgets alone cannot reach. But this is not a private-finance solution: the households most exposed are often not those with the greatest liquid savings, so public guarantees, subsidies and distributional safeguards are what turn household wealth into resilience rather than into inequality.



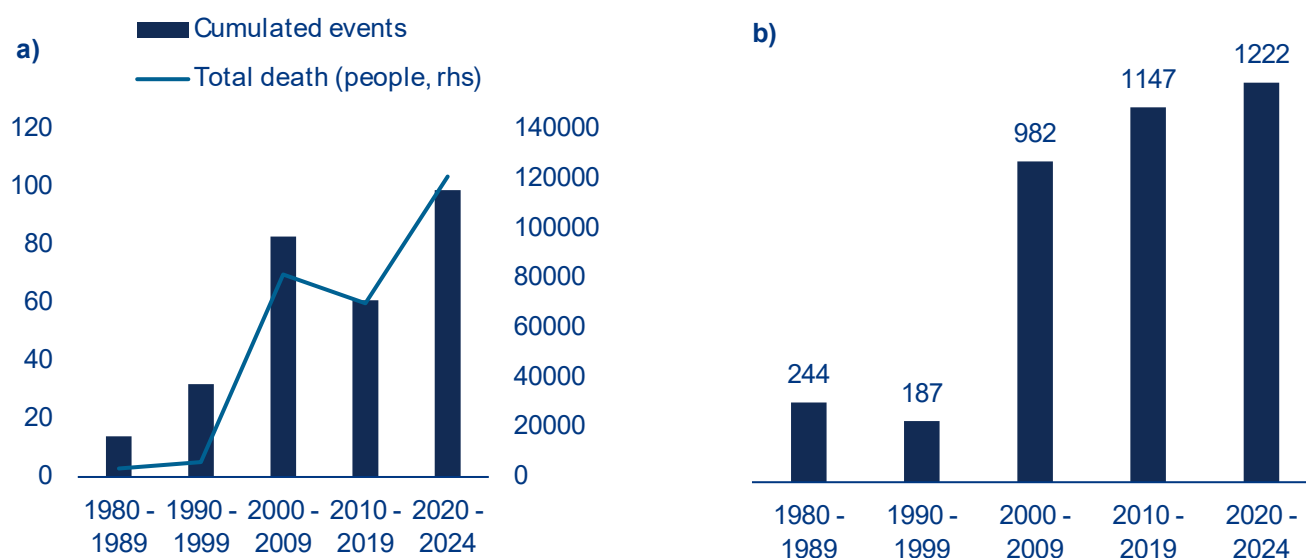
The economic transmission of heat stress

From one decade to another, the data are recording more frequent and more lethal heat stress events.

Between 1980 and 1989, only 14 heat-wave events¹ were catalogued globally; the figure more than doubled in the 1990s, jumped to 83 in the 2000s and has already reached 99 in the truncated 2020–2024 period, meaning five years have nearly matched what entire decades produced two cycles earlier (Figure 1a). Mortality has followed an even steeper trajectory: from around 3,400

deaths in the 1980s to more than 121,000 in the first half of the 2020s alone. The most analytically significant metric is the average death toll per event, which has risen fivefold from 244 in the 1980s to 1,222 in 2020–2024. This decoupling between event frequency and per-event lethality indicates that the intensity and duration of individual episodes are increasing faster than societies can adapt (Figure 1b).

Figure 1: Increasing exposure to heat stress: a) Evolution of number of events and total death (1980 – 2024); b) Average number of death per heat stress event (1980 – 2024)



Sources: EM-DAT, Allianz Research

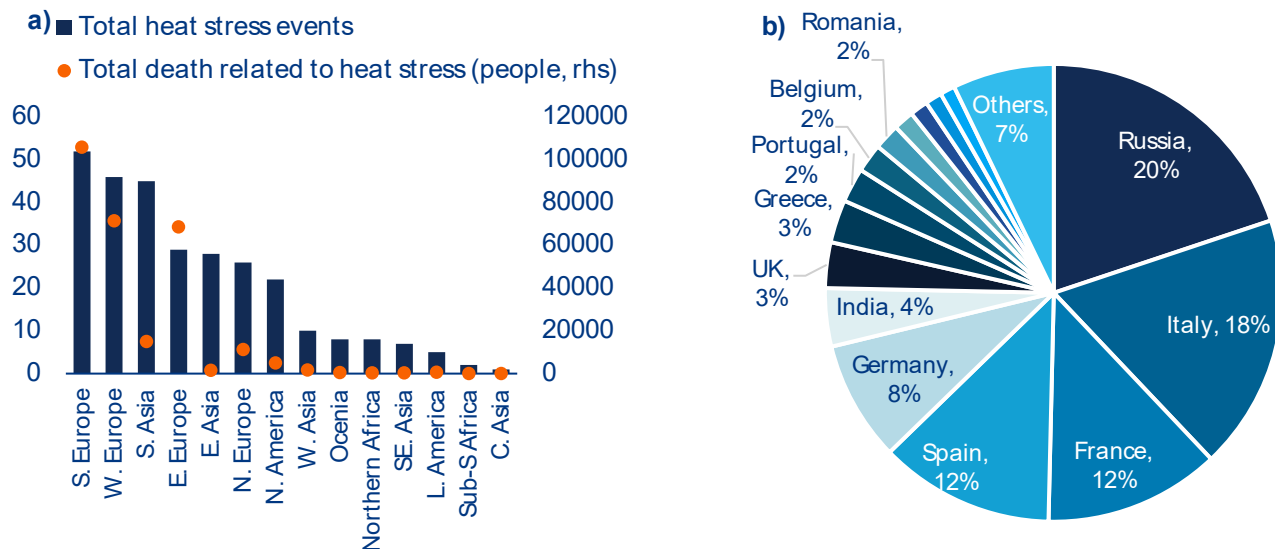
¹ EM-DAT defines a heat wave as a period of abnormally hot and/or humid weather lasting typically two or more days, with location-specific thresholds. Only events with ≥ 10 deaths, ≥ 100 people affected, a declared emergency, or an international assistance call are catalogued.

The cross-sectional distribution of heat stress events and mortality adds a second layer of insight, showing that exposure and impacts do not fully align geographically (Figure 2a). At the regional level, Southern and Western Europe concentrate the highest number of recorded events (52 and 46 respectively), closely followed by South Asia (45). Yet, this apparent convergence in frequency masks a profound divergence in outcomes. Southern Europe alone accounts for more than 105,000 deaths, while Western and Eastern Europe contribute a further 71,000 and 68,500 respectively. By contrast, South Asia, despite a comparable number of events, records fewer than 15,000 deaths over the same period. This asymmetry becomes even more pronounced when examined at the country level. A limited set of European countries dominates the global mortality profile: Russia (20%), Italy (18%), France (12%), Spain (12%) and Germany (8%) together account for 68% of total deaths. The concentration is too strong to be explained by hazard occurrence alone. Recent valuations suggest that heat-related mortality in European cities already entails welfare losses of EUR192–314 per adult per year, on par with air pollution, with avoided mortality alone worth more than

EUR150bn annually.² This is only one channel of impact. On the productivity side, heatwaves cost 0.3–0.5% of European GDP in anomalously hot years such as 2003, 2010, 2015 and 2018, and more than 1% in the most exposed southern regions. Losses are projected to rise fivefold by 2060 in the absence of further mitigation or adaptation.³

The pattern in Figure 2b needs to be read carefully: EM-DAT counts reflect reporting practices as much as biophysical exposure. In hot regions such as the Gulf, India and interior Australia, populations have adapted over time through acclimatization, building design and work practices, so many high-temperature episodes do not register as disaster events, even though heat continues to kill outdoor workers, the poor and the elderly. Europe sits at the opposite end, combining ageing populations with impaired thermoregulation, dense historic urban form, uneven air-conditioning access and stronger vital registration. The European mortality concentration therefore reflects exposure meeting structural vulnerability and more complete death recording, not exceptional hazard frequency.

Figure 2: Global distribution of heat stress events: a) Number of events and total death across regions (1980 – 2024); b) Heat stress related death by country (1980 – 2024)



Sources: EM-DAT, Allianz Research

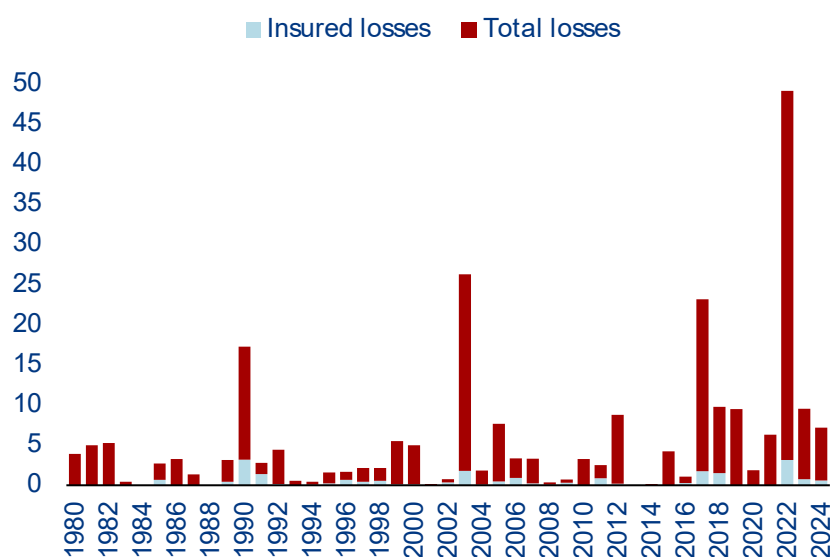
² [Economic valuation of temperature-related mortality attributed to urban heat islands in European cities | Nature Communications](#)

³ [Current and projected regional economic impacts of heatwaves in Europe | Nature Communications](#)

Insured and uninsured losses tell a parallel story. Total losses from climatological hazards in Europe have risen across 1980–2024, with sharp peaks in 2003 and 2022, while the insured share has stayed limited (Figure 3). In 2022, total losses reached around EUR46bn, of which insurance covered only a small fraction. The gap is not simply a matter of market lag; it also reflects what heat actually damages. Compared with storms or floods, extreme heat less often shows up as direct damage to insured assets. Most of its cost appears through excess mortality, lost working hours, healthcare-system pressure and the disruption that follows when grids, railways and water systems are pushed past their design limits. The more conventionally insurable parts, such as direct property damage, heat-amplified wildfire losses, some crop and equipment failures etc., make up only part of the total. EEA data for 1980–2022⁴ show that only around 10% of economic losses from heatwaves, droughts, wildfires and other climatological events in Europe were insured, against more than a third for storms, hail and wind. These figures cover direct asset damage only and exclude mortality, healthcare costs and lost productivity.

Heat is also hard to insure through standard indemnity contracts. Heatwaves can be defined meteorologically, but turning a temperature episode into a compensable loss is diffuse, lagged and hard to attribute. Losses tend to be highly correlated across large regional portfolios, which limits diversification. And the largest impacts – deaths, morbidity, lost output, public-service stress – are not the kind of damage private indemnity contracts were written for. Heat therefore sits at the boundary of conventional insurability: some asset-based losses can be transferred, but the largest welfare and productivity costs are hard to transfer at scale. Closing the gap will require new instruments, such as parametric cover, public–private risk-sharing and dedicated public backstops, without which much of the cost will keep falling on households, firms and public budgets.

Figure 3: Climatological (including heat waves) insured and uninsured losses in Europe (1980 – 2024)



Sources: EEA, Allianz Research

⁴ Economic losses from weather- and climate-related extremes in Europe | Indicators | European Environment Agency (EEA)

Heat translates into measurable macroeconomic cost through two channels: additional energy demand and lower labor productivity.⁵ Both relationships are sharply nonlinear, with a critical threshold around 30°C beyond which damages intensify. The remainder of this section estimates the magnitude of each channel using panel regressions covering 35–49 countries over three decades, and quantifies what those estimates imply when each country experiences its recent peak-heat conditions relative to a 1991–2010 climatological baseline. The output of these two channels feeds directly into the global macroeconomic scenario presented in Section 2.

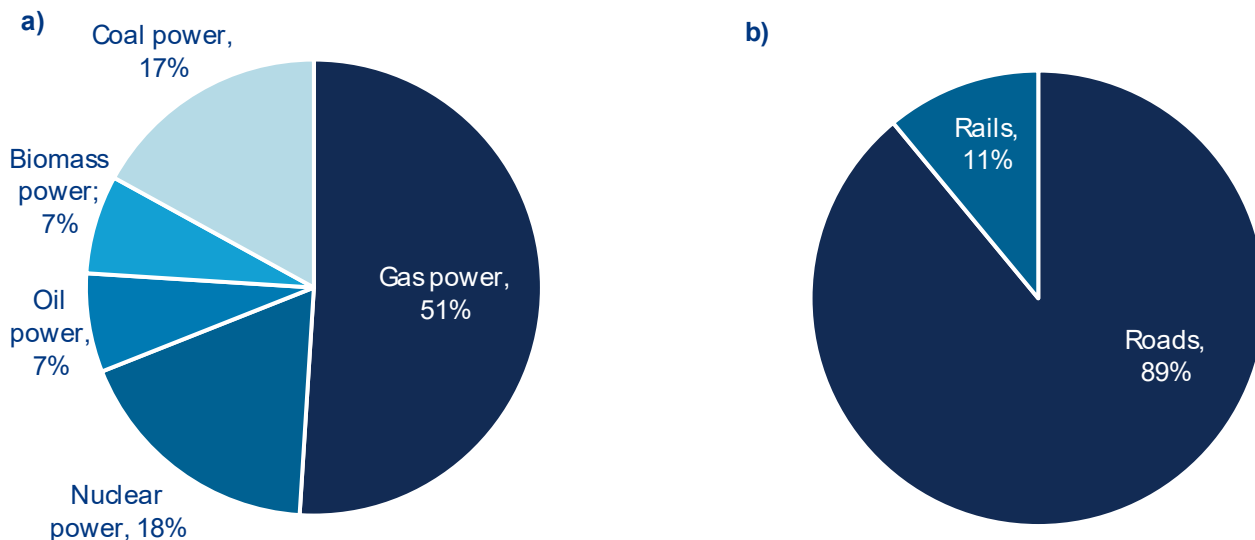
Energy demand

The relationship between temperature and energy demand is inherently non-linear, and the structure of Europe’s power mix amplifies the risk on both demand and supply sides. At moderate temperatures, milder conditions reduce space-heating needs and consumption falls; above a threshold, cooling dominates and consumption rises sharply, producing a convex U-shaped response (Auffhammer and Mansur, 2014)⁶. The supply side magnifies the demand-side pressure. A large share of European generation capacity exposed to heat stress relies on thermoelectric technologies – gas (51%), nuclear (18%) and coal (17%) – all of which

depend on water availability and cooling efficiency. Elevated air and river temperatures directly constrain these systems, forcing output reductions precisely when cooling-driven electricity demand surges (Figure 4a).

Recent episodes illustrate the systemic nature of the resulting grid stress, and analogous pressures affect transport infrastructure. During the August 2020 California heatwave, electricity demand peaked near 47,000 MW while generation capacity was simultaneously constrained, producing the first rolling blackouts in nearly two decades. During the 2019 French heatwave, nuclear output was reduced due to cooling constraints, tightening supply and triggering sharp electricity-price spikes. Heat also degrades transmission networks, lowering the carrying capacity of generators, transformers and power lines, while even non-thermal renewables, in particular solar PV, experience efficiency losses at high temperatures. Transport infrastructure faces analogous physical pressures: road networks account for nearly 89% of heat-exposed transport assets and rail for 11% (Figure 4b), with asphalt softening and rutting under sustained heat and rail tracks prone to thermal expansion, illustrated by the UK in July 2022, when temperatures first exceeded 40°C and rail operators imposed strict speed limits or suspended services entirely.

Figure 4: Exposure of critical European infrastructure to heat stress : a) Energy infrastructure; b) Transport infrastructure



Sources: Forzieri et al. (2018)⁷, Allianz Research

⁵ What to watch | July 01, 2025 | Allianz

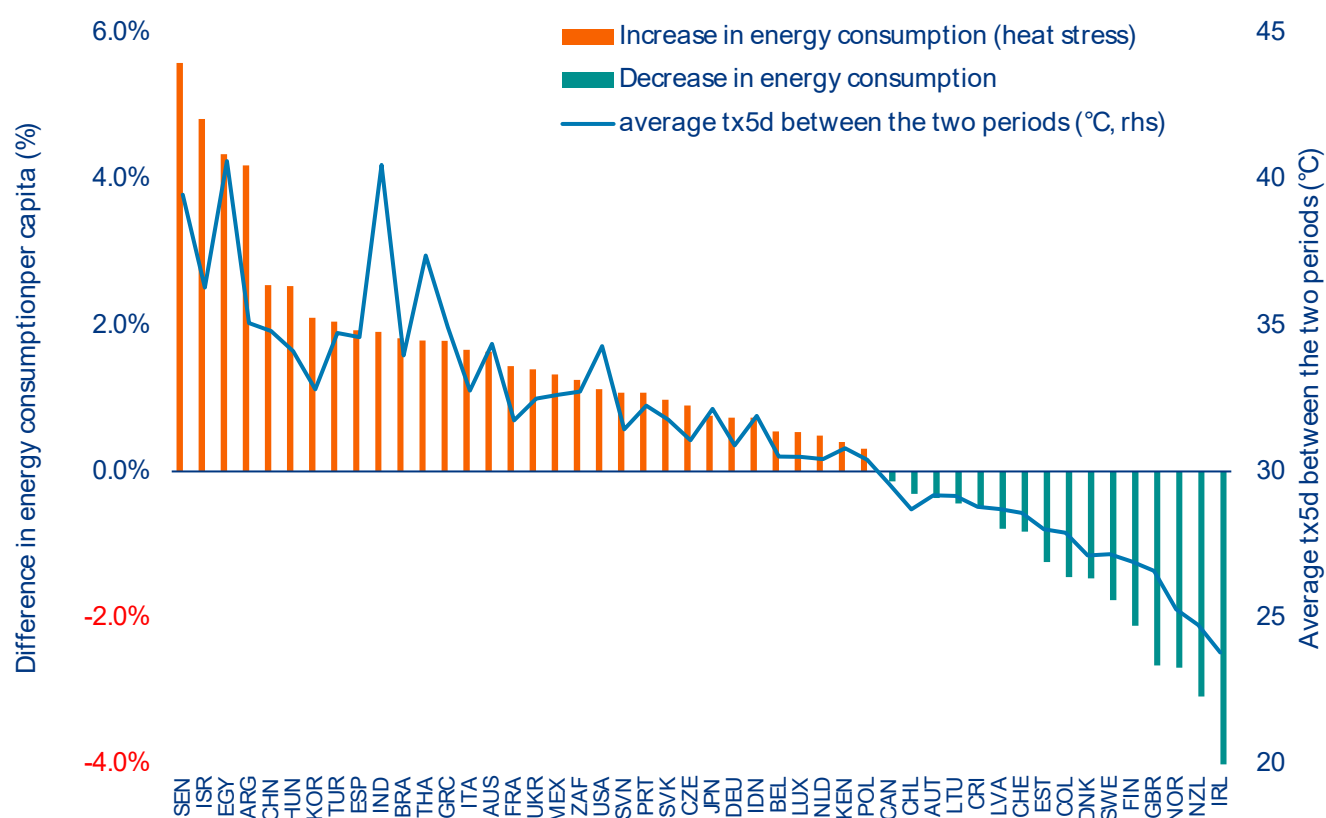
⁶ Measuring climatic impacts on energy consumption: A review of the empirical literature - ScienceDirect

⁷ Escalating impacts of climate extremes on critical infrastructures in Europe - ScienceDirect

Our empirical results confirm the non-linear structure and identify a threshold close to 30°C (Appendix 1). Using the Tx5d indicator, which is the mean daily maximum temperature over the hottest five consecutive days of the year, on an unbalanced panel of 49 countries over 1991–2023, the estimated marginal effect of heat stress on per capita energy consumption is sign-switching and temperature-dependent. At 25°C, a one-unit increase in Tx5d is associated with an approximately 0.6% decline in per capita energy consumption. At 35°C, the relationship reverses: the same marginal increase implies a 1.2% rise. The twofold differential illustrates how disproportionately energy-system stress intensifies once thermal thresholds are crossed.

The country-level counterfactual reveals a sharply asymmetric pattern, with heat already operating as a structural driver of energy demand across much of the G20 (Figure 5). Comparing per capita energy consumption during each country's most intense post-2014 heat episode (maximum Tx5d) with its 1991–2010 baseline average, countries already operating at high temperature levels (Senegal, Egypt, China, South Korea, Spain and Brazil) register increases above 4–5%, pushed further into the convex segment of the response. Almost all G20 economies in the tropics see rising energy demand across sectors, while only Europe and Oceania post aggregate reductions, driven by lower residential heating needs in cooler latitudes. The cooling gains are structurally fragile and likely to erode as temperatures continue to rise.

Figure 5: Per capita energy demand under peak heat stress (maximum Tx5d after 2014) vs. the 1991–2010 average baseline



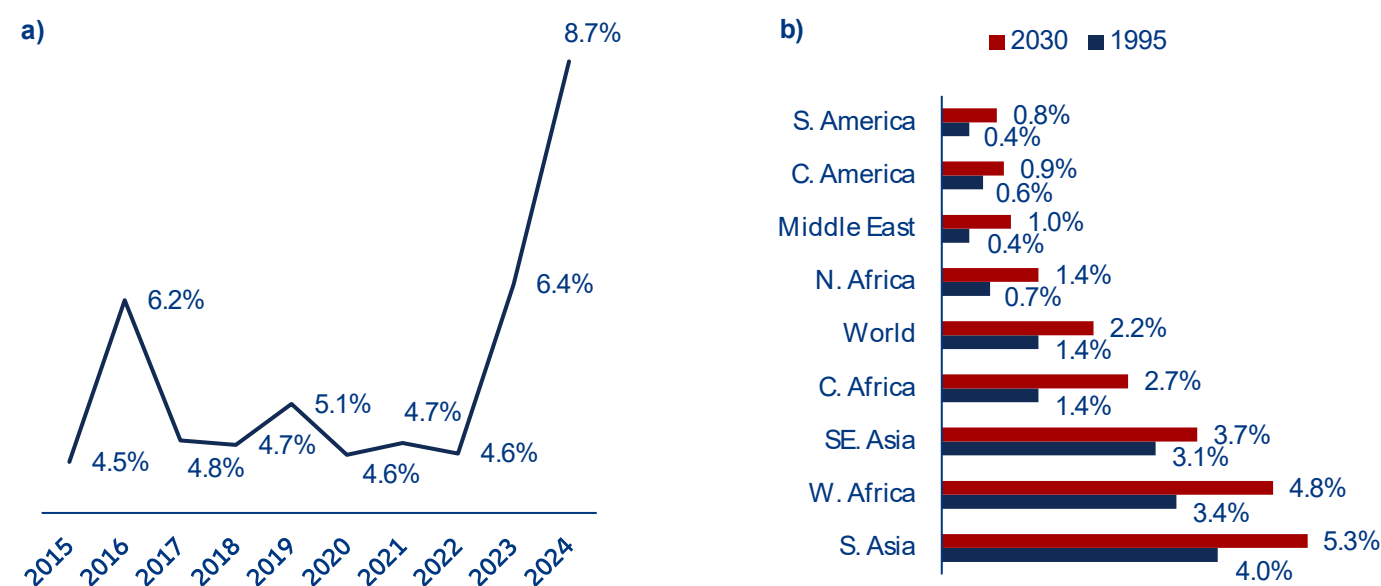
Sources: China National Customs, Allianz Research

Labor productivity and wages

Heat stress reduces labor output through physiological, cognitive and sleep-related channels that extend well beyond outdoor work (Figure 6). At the worker level, sustained exposure above the thermoneutral zone induces cardiovascular strain, dehydration and cognitive impairment, with field studies consistently documenting productivity losses across sectors (Ebi et al., 2021)⁸. The share of global working hours lost to heat stress is projected to rise from 1.4% in 1995 to 2.2% by 2030, with much higher impacts in already-warm regions – 5.3% in South Asia, 4.8% in West Africa – and a near doubling even in traditionally less exposed regions

(Figure 6b). Indoor workers are not insulated: night-time temperatures, which are rising faster than daytime averages in many regions, increased total sleep time lost by around 6% over 2020–2024 relative to the 1986–2005 baseline, reaching a record 8.7% in 2024, with up to 12 additional hours of sleep loss per person per year in the most affected locations (Figure 6a). Degraded sleep duration and quality has measurable next-day consequences for cognition, cardiovascular health and decision-making, transmitting heat impacts to indoor as well as outdoor labor.

Figure 6: Heat stress and productivity : a) Loss of sleeping time for the period 2015 – 2024, compared to a baseline 1986 – 2005 (%); b) Decline in working hours (%)



Sources: The Lancet Countdown (2025), ILO (2019), Allianz Research

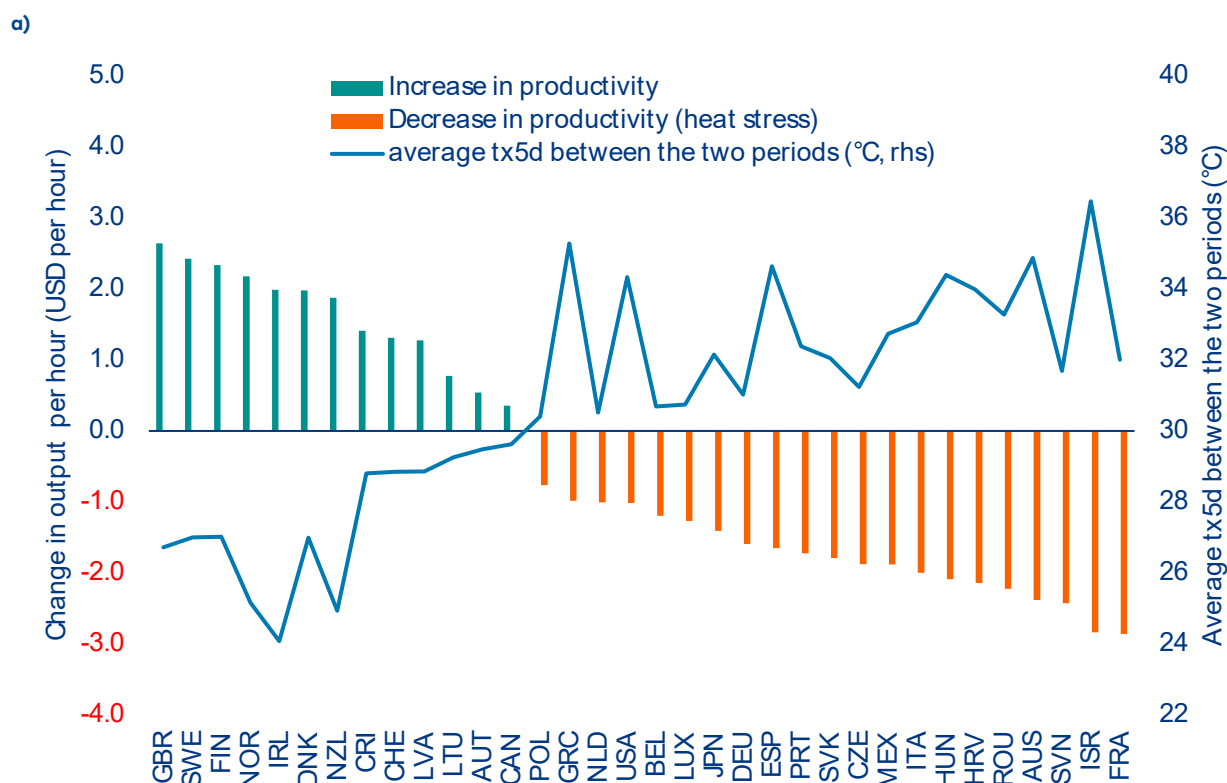
Our regressions on output per hour reveal a clear threshold effect at 30°C, consistent with the empirical consensus (Appendix 2). Estimated on a panel of 35 countries over 1997–2023, the coefficient on Tx5d below 30°C is positive and marginally significant (+0.59), consistent with mild productivity gains at moderate temperatures in temperate economies. Above 30°C, the interaction coefficient turns negative and statistically significant (–1.30), implying that each additional degree of heat beyond the threshold reduces output per hour by

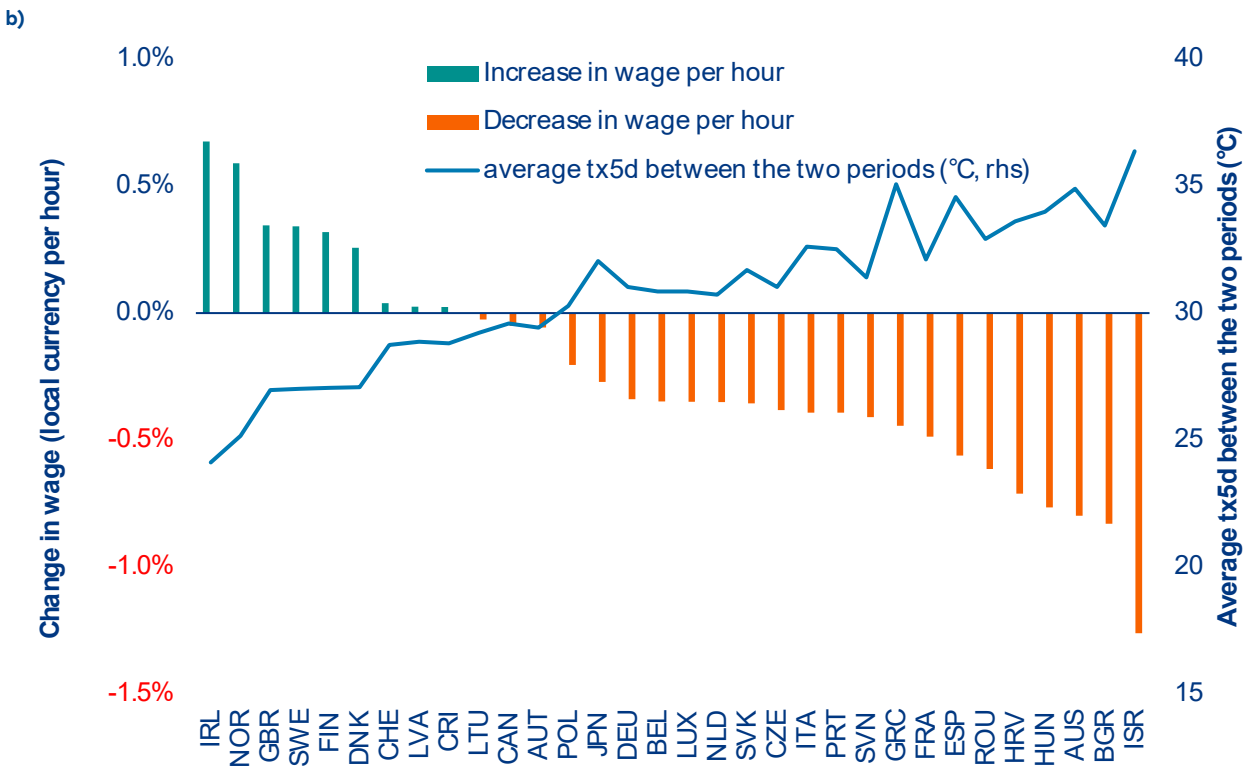
approximately USD1.30 (constant PPP, representing 3% of average output per hour in our sample between 2014 and 2024). Applied to the country-level counterfactual, where each country experiencing its maximum Tx5d observed over 2014–2024, against its 1991–2010 average, France, Slovenia, Australia and Italy emerge as the most affected, with output per hour declining by up to USD 1–3 in the most exposed cases. The UK, Finland, Ireland, New Zealand and Canada, all below the 30°C threshold, exhibit modest productivity gains (Figure 7a).

The wage response materializes with a lag, reflecting institutional rigidities in labor markets (Appendix 3). Using a one-year lag specification on a panel of 33 countries over 1995–2023, the relationship follows an inverted U-shape: below 30°C, higher heat stress in year t is associated with a modest acceleration of wage growth in year $t+1$ (approximately +0.2% at 25°C); above 30°C, the relationship reverses (approximately –0.2% at 35°C). The lag is consistent with the combination of collective-bargaining frictions, nominal wage rigidities and minimum-wage floors, all of which delay the transmission of productivity losses into labor compensation. The counterfactual maps the same threshold logic onto observed peak heat: Israel, Bulgaria, Australia, Hungary, Romania and Spain register wage declines of approximately 0.7–1.3%, while Ireland, Norway, the UK, Sweden and Finland post wage gains of 0.3–0.7% (Figure 7b).

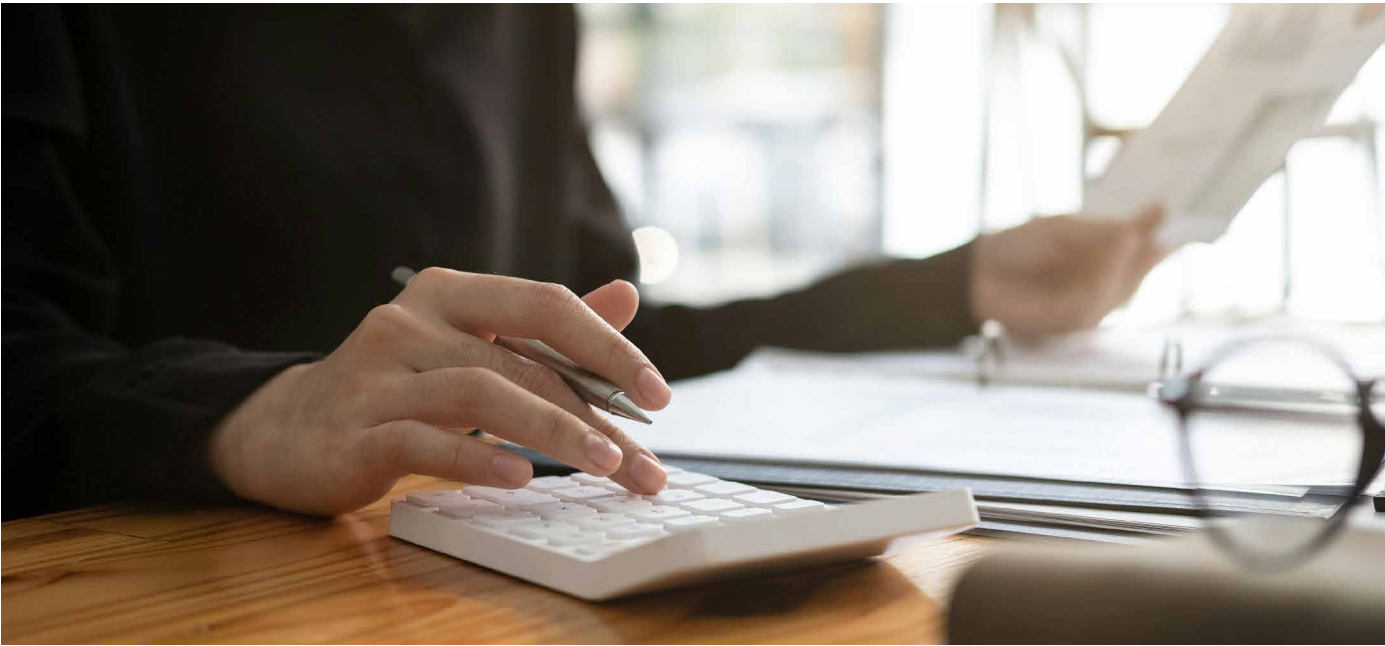
The distributional implication is structurally important and motivates the macroeconomic scenario that follows. In the short run, the cost of heat stress falls disproportionately on firm profitability as wage rigidities prevent compensation from adjusting in line with productivity. Over time, however, that wedge closes: real wages decline, household purchasing power erodes and consumption weakens, precisely in the economies already most exposed to rising temperatures. The two channels estimated in this section, additional energy demand and lower labor productivity, transmitted partially and with a lag into wages, are the inputs to the global macroeconomic simulation in the next section, which quantifies the general-equilibrium consequences in terms of GDP, prices, employment and fiscal balances.

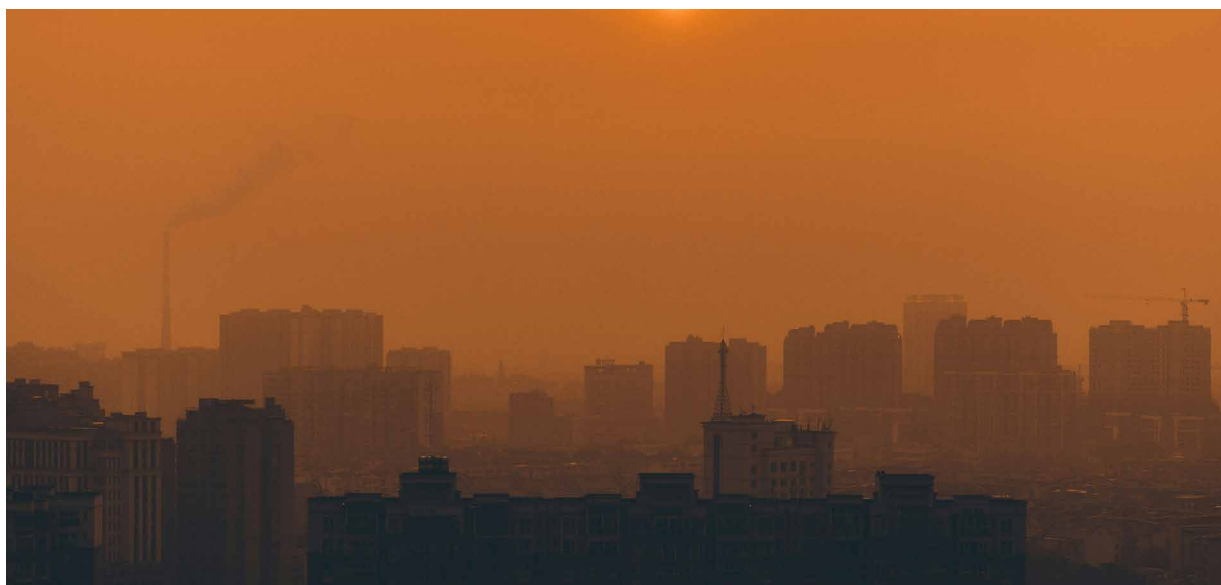
Figure 7: The change in labor productivity under peak heat stress (maximum Tx5d post-2014) relative to the 1991–2010 baseline, measured as (a) output per hour in USD and (b) wage per hour (%).





Source: Allianz Research





The macroeconomic implications of heat stress

To quantify the macroeconomic cost of these effects, we translate the empirical estimates into a structural scenario using the Oxford Economics Global Economic Model. The counterfactual exercises described above provide country-specific measures of the productivity and energy demand impact of heat stress, but they remain partial-equilibrium estimates, as they capture the direct effect of temperature on each variable in isolation, without accounting for the general-equilibrium feedbacks that propagate through the broader economy. As the recent climate-macroeconomics literature has increasingly recognized, reduced-form empirical estimates of climate damages, while essential for identification, cannot capture the full macroeconomic cost because they omit the general-equilibrium feedbacks, through trade, prices, investment and fiscal balances, that amplify the initial shock.⁹

To capture these interdependencies, we impose three simultaneous shocks on the model, each calibrated directly from our regression results. The first is a shock to potential output¹⁰, reflecting the estimated decline in output per hour worked under extreme heat conditions. Entering the productivity loss through potential output ensures that the shock operates on the

supply side of the economy, reducing the economy's productive capacity. The second shock targets wages and salaries per employee¹¹, capturing the lagged and partial transmission of productivity losses to labor compensation that our analysis identifies. The exogenous wage shock therefore captures the residual labor market adjustment that our regressions document beyond the model's endogenous response. The third shock targets energy demand¹², reflecting the additional per capita energy consumption induced by heat stress. This channel translates higher cooling needs into increased household and firm expenditure, higher energy imports for net-importing economies and upward pressure on consumer prices.

To assess the macroeconomic cost of heat stress, we construct a counterfactual scenario and compare it against a baseline projection. The baseline simulates economic outcomes over the period 2026–2030 under climatological normal conditions, defined as the average Tx5d per country observed over the 1991–2010 reference period. This baseline represents the economic trajectory each country would follow if heat stress remained at its historical norm. Against this baseline, we impose a stress scenario in which each country experiences,

⁹ [Climate Change through the Lens of Macroeconomic Modeling | NBER](#)

¹⁰ YHAT in Oxford Economics

¹¹ PEWFP in Oxford economics

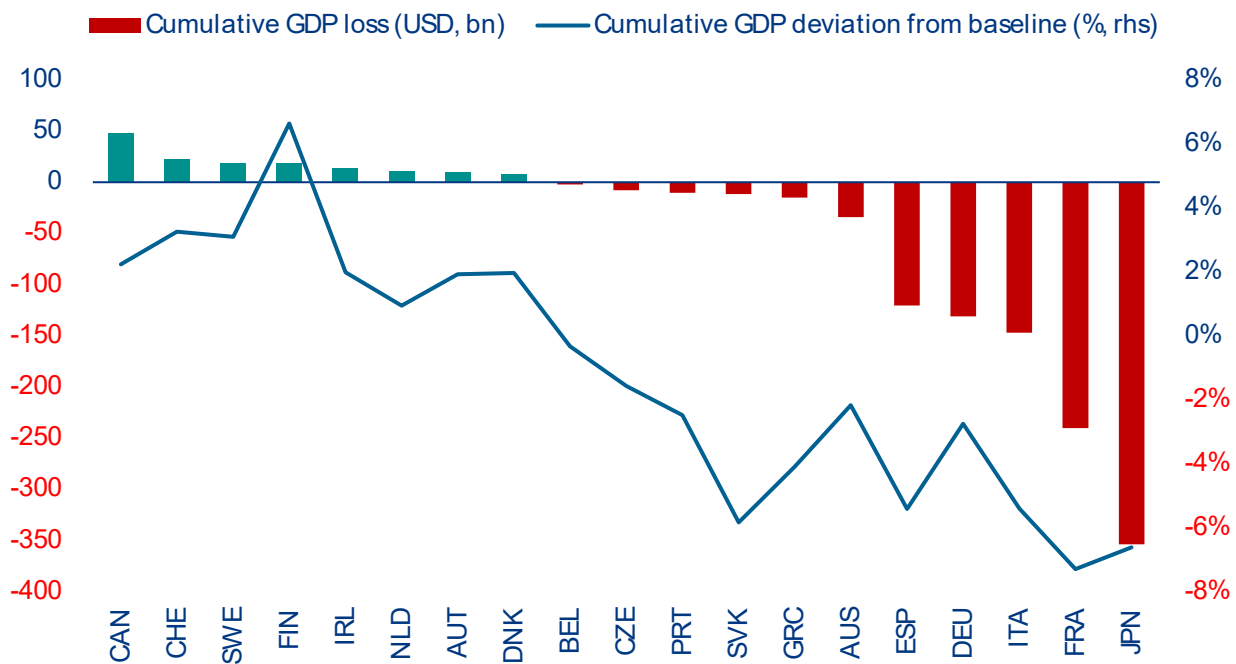
¹² DENERGY in Oxford economics

consecutively from 2026 to 2030, five years with the most important heat waves recorded during the 2014–2024 period, ranked by Tx5d. This is not a climate scenario based analysis but a historically grounded exercise: it asks what the macroeconomic consequences would be if the extreme heat conditions already observed in the recent past were to recur over a sustained five-year window.

The results reveal a clear divide across economies (Figure 8). Countries where average Tx5d remains below the 30°C threshold under the stress scenario – Canada, Switzerland, Sweden and Ireland – experience cumulative GDP gains of 1% to 3% relative to the historical climate baseline, leading to absolute gains

ranging from USD10bn to USD47bn.¹³ These gains reflect the net balance of reduced heating costs and marginally improved working conditions, benefits that are structurally fragile and likely to erode as temperatures continue to rise. Beyond this group, the transition into net losses is sharp. France, Japan, Slovakia, Spain and Italy register the highest cumulative GDP losses of 5% to 7% below baseline, reflecting the direct productivity drag that our shocks impose. The largest absolute losses, however, are concentrated in the biggest economies in the sample: Spain (USD120bn), Germany (USD131bn), Italy (USD147bn), France (USD240bn) and Japan (USD354bn).

Figure 8: Cumulative GDP loss from sustained extreme heat stress (2026–2030) relative to historical climate baseline



Sources: Oxford economics, Allianz Research

¹³ Finland is an outlier, with a potential cumulative GDP gain of 7% under sustained extreme heat stress

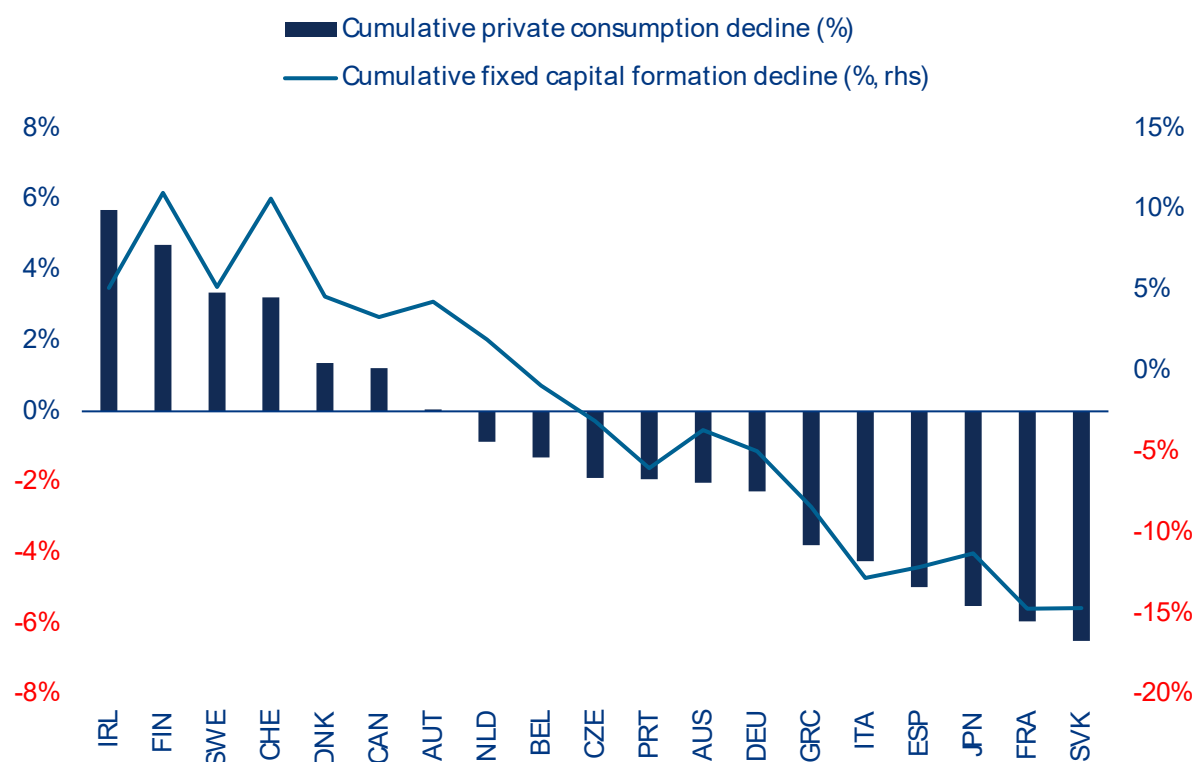
Decomposing the GDP impact into its demand-side components reveals that the macroeconomic cost of heat stress operates through two reinforcing channels: weaker household consumption and lower investment leading to a long-run decline in fixed capital formation (Figure 9).

In countries where moderate warming produces net economic benefits – Ireland, Finland, Sweden, Switzerland and Denmark – both private consumption and fixed capital formation rise relative to baseline, with investment gains generally exceeding consumption gains. The pattern inverts sharply for heat-exposed economies, and here the most striking feature of the results is the systematic amplification through investment. In virtually every country where the heat stress scenario produces economic losses, the decline in fixed capital formation exceeds the decline in private consumption, often by a wide margin. Portugal sees a consumption decline of 1.9% but an investment decline of 6.0%. Italy registers consumption losses of 4.2% against an investment collapse of 12.8%. France and the Slovak Republic, the two most affected economies, experience investment declines of 14.7% alongside consumption losses of 5.9% and 6.5% respectively. This wedge between the consumption and investment

response reflects a powerful amplification mechanism: as heat stress reduces potential output and compresses firm margins, the expected return on new capital falls, discouraging investment. Lower investment in turn reduces future productive capacity, reinforcing the initial productivity shock and creating a self-reinforcing drag on the economy's growth trajectory. This asymmetry between the investment and consumption response is consistent with Casey et al. (2022)¹⁴, who show that standard climate-economy models systematically understate investment damages because climate change disproportionately affects capital-intensive sectors exposed to outdoor heat.

The consumption response, while smaller in percentage terms, is nonetheless consequential. It captures the combined effect of lower real wages, which our wage shock imposes directly, and higher energy expenditure, which erodes household purchasing power. In the most affected economies, cumulative consumption declines of 4% to 6.5% compared to the baseline, representing a material reduction in living standards over a five-year horizon, with implications for domestic demand, retail activity and ultimately tax revenues.

Figure 9: Impact of sustained extreme heat stress on private consumption and fixed capital formation, 2026–2030 (cumulative deviation from baseline)



Sources: Oxford economics, Allianz Research

¹⁴ [Understanding climate damages: Consumption versus investment - ScienceDirect](#)

The joint behavior of consumer prices (CPI) and the unemployment rate under the stress scenario reveals the stagflationary nature of the heat stress shock, a finding with direct implications for monetary and fiscal policy. In cooler economies – Finland, Ireland, Denmark, Austria and Switzerland – both indicators improve relative to the baseline: prices fall by 0.5% to 2.5% cumulatively, while unemployment declines by 0.14 to 0.88pp. In these countries, the modest productivity gains associated with below-threshold heat stress expand supply, easing price pressures and strengthening labor demand simultaneously.

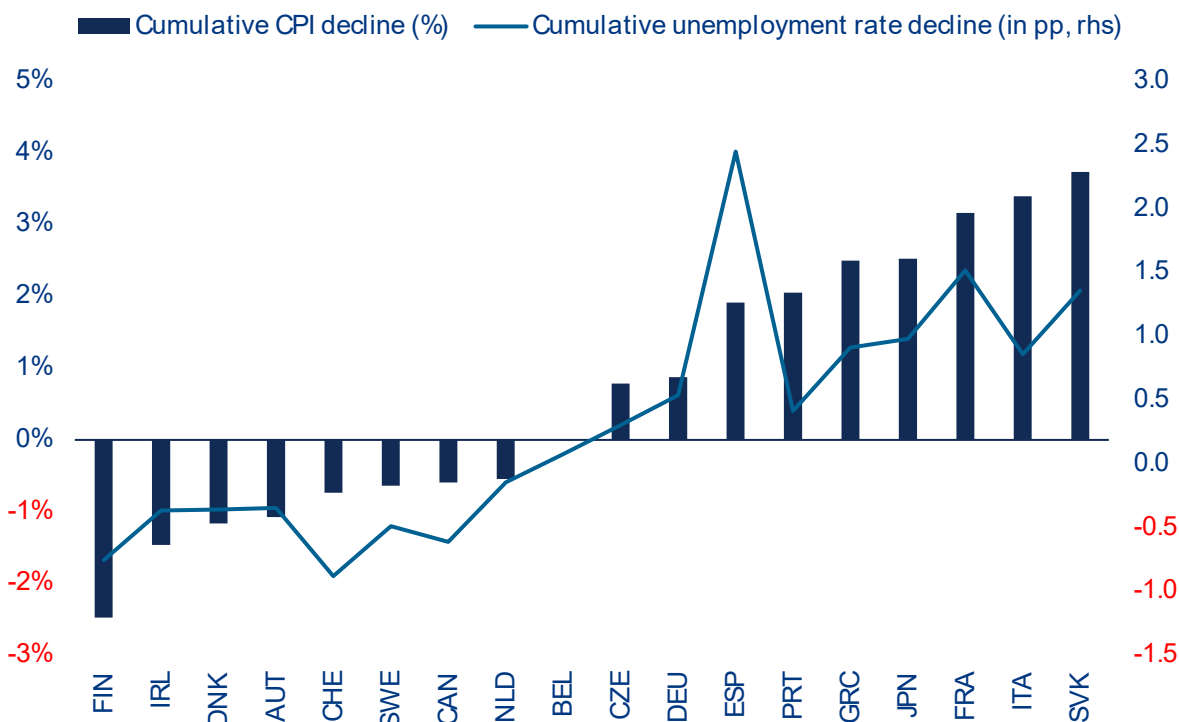
The picture reverses as heat exposure rises. For hotter countries in our samples – the Czech republic, Germany, Spain, Portugal, Greece, Japan, France, Italy and Slovakia – both prices and unemployment move in the same adverse direction, which is a defining characteristic of a negative supply-side disturbance. Spain registers a cumulative CPI increase of 1.9% alongside a 2.45pp rise in unemployment. France, Italy and Slovakia experience price increases of 3.2% to 3.7% combined with unemployment increases of 0.86pp to 1.51pp. This co-movement of rising inflation and rising unemployment is the hallmark of stagflation, and it stands in sharp contrast to the pattern produced by a standard demand

shock, where the two variables move in opposite directions along the Phillips curve. In the New Keynesian framework, Blanchard and Galí (2007)¹⁵ show that such a trade-off arises when real imperfections, such as wage rigidities, prevent the economy from adjusting smoothly to supply-side shocks. The heat stress shock in our simulation operates through precisely this type of supply-side channel: reduced potential output pushes up unit costs and prices, while simultaneously contracting wages and salaries.

The result of such heat stress scenario is a policy environment in which conventional stabilization tools face a binding tradeoff: tightening monetary policy to contain inflation would deepen the employment losses, while loosening policy to support the labor market would accommodate further price increases.

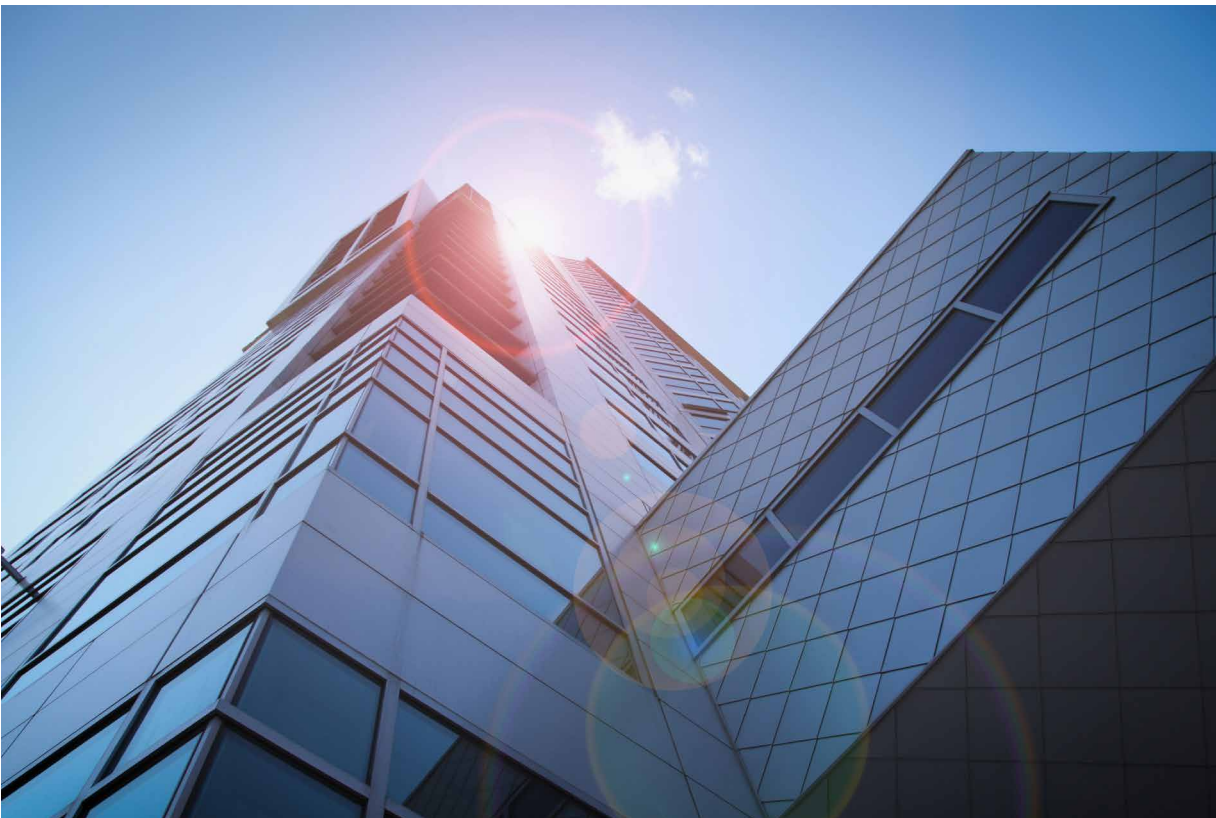
This dilemma is especially acute in the Eurozone context, where a single monetary policy must simultaneously address economies experiencing positive supply effects from moderate warming, such as Finland and Austria, and economies facing stagflationary pressures from extreme heat, such as Germany, France, Italy and Spain, a form of climate-induced macroeconomic divergence that existing policy frameworks are not designed to manage.

Figure 10: Consumer prices and labor market response to sustained extreme heat stress, 2026–2030 (cumulative deviation from baseline)



Sources: World Bank, Microsoft, Allianz Research

¹⁵ Real Wage Rigidities and the New Keynesian Model - BLANCHARD - 2007 - Journal of Money, Credit and Banking - Wiley Online Library

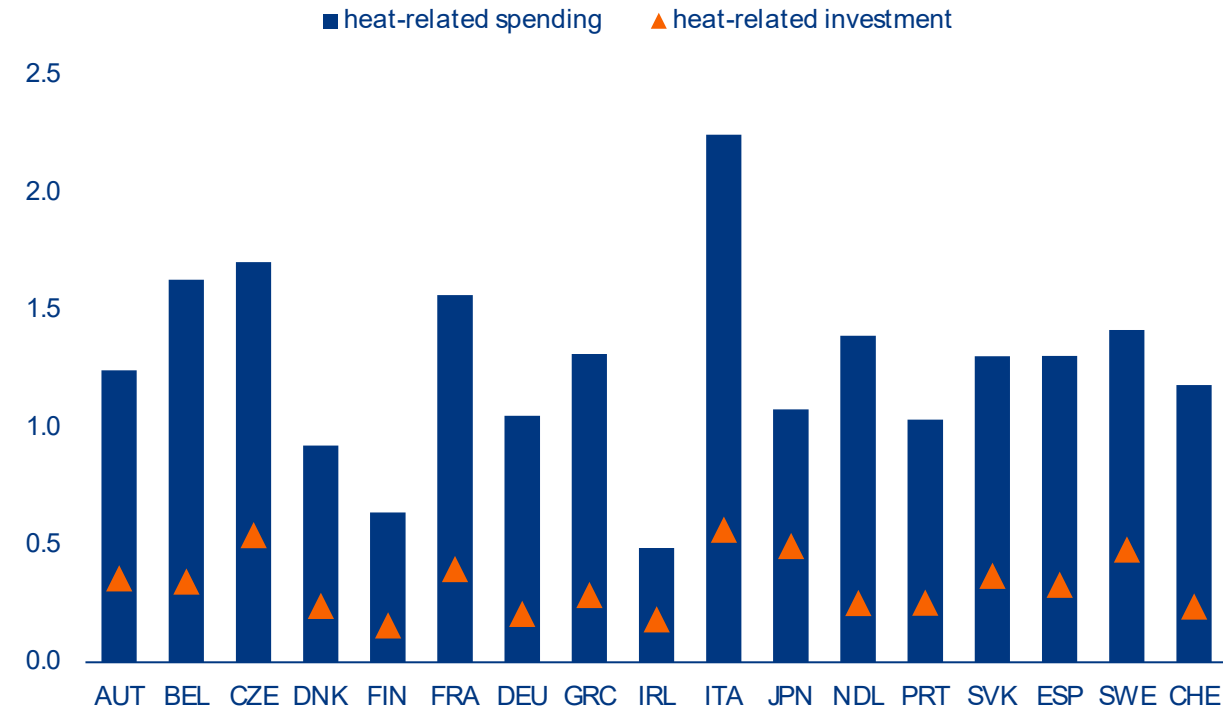


The fiscal burden

Even though it already receives about 50% of climate-related investments, extreme heat puts a strain on public finances by cutting revenues and raising adaption costs. Heat reduces GDP and productivity, which lowers tax collection, while also disrupting supply chains and energy systems, which further weakens revenues. Estimated annual revenue losses are -1.8% in France, -1.3% in Italy and Spain, -1.0% in Greece and -0.7% in Germany – higher than estimated GDP losses because progressive tax systems mean revenues tend to fall faster than output itself, amplifying the fiscal drag beyond the headline GDP loss. At the same time, public spending rises. Governments face higher inflation-linked costs (e.g. wages, pensions and social benefits) as well as heat-specific expenditures (e.g. healthcare, energy subsidies, disaster response, infrastructure repair and

social protection). Demand for emergency investments in cooling infrastructure also increases. In Europe, heat-related spending accounts for around 50% of climate-related public expenditure (2.1% of GDP), but this figure varies widely: 40% in Austria, 51% in Germany, 58% in France, 63% in Spain and 65% in Italy. Flooding accounts for about 35%, with the remainder going to wildfires, drought and other areas like biodiversity. As a percentage of GDP, heat-related spending ranges from 2.2% in Italy to 1.6% in France, 1.3% in Spain, 1.1% in Germany and 0.5% in Ireland (Figure 11). Adaptation to heatwaves accounts for a significant proportion of more than a quarter of heat-related budgets, primarily for cooling and urban greening measures. However, overall investment in heat resilience is still lacking to provide for a heat proof society.¹⁶

¹⁶ From invisible to investible: An investment taxonomy for climate adaptation | Allianz

Figure 11: Heat-related spending and investment, in % of GDP 2023

Sources: OECD, Allianz Research

On average, the fiscal balances of European countries could deteriorate by around 0.5% of GDP annually due to heat stress, creating a structural tension between the fiscal damage caused by heat and the investment required to adapt to it. A rapidly growing

literature has established that climatic risks impose economically significant costs on public finances¹⁷, operating through both the erosion of tax revenues as climate damages reduce economic output and the increase in public expenditure as governments face rising costs for existing commitments and new demands for adaptation. Our results confirm both channels operating simultaneously. On the revenue side, the productivity and wage shocks documented in our simulations translate directly into lower tax receipts: economies producing less and paying lower real wages generate less income tax, social security contributions and value-added tax revenue. On the expenditure side, the stagflationary character of the heat shock through rising prices combined with contracting output, inflates the cost of government programs indexed to prices or wages. The share of public spending subject to such

indexation varies considerably across our sample, from approximately 20% in Ireland to 58% in Finland, creating heterogeneous fiscal exposure to the same inflationary impulse.

We identify three groups of countries based on the interaction between heat-related fiscal pressure and existing budgetary constraints. The first group comprises countries where this tension is most acute: heat-related fiscal pressure is significant and public finances are already strained exceeding the Maastricht deficit criterion of –3%. France is the most exposed case. Heat stress could worsen an already high expected fiscal deficit of –4.9% (average 2025-2030) by an additional 2.2% of GDP. At this level, fiscal space for discretionary adaptation spending is severely compromised. Germany, with an estimated deficit of –3.6% and additional heat-related pressure of 0.9%, faces a tighter version of the same constraint. Italy's fiscal deficit of –3.1% could worsen by an additional 1.9% of GDP while Belgium, where 0.2% of additional pressure meet a deficit of –4.0%, completes this group

¹⁷ [Climate Change Impacts on Public Finances Around the World | Annual Reviews](#)

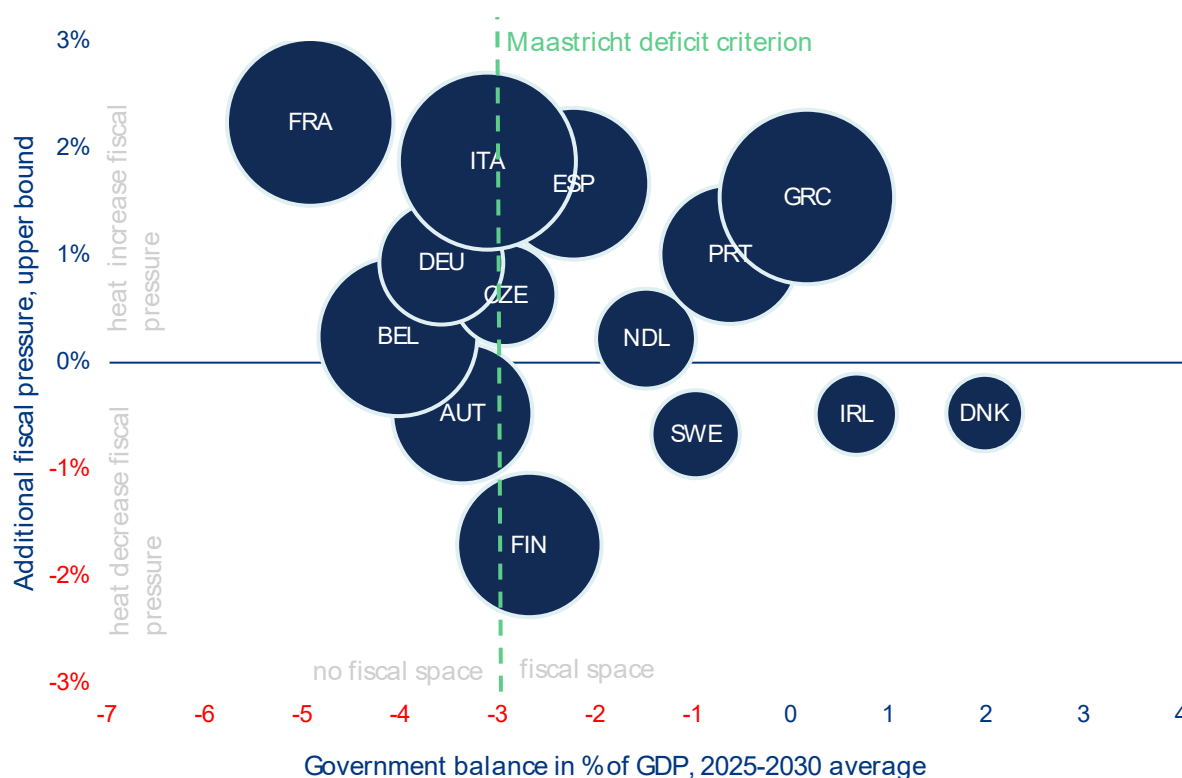
in the upper quadrant (Figure 12). For these economies, the fiscal damage from heat is actively eroding the budgetary room from which the adaptation response must be financed. Considering in addition the projected debt burden of a country also makes clear that the heat-related fiscal pressure on countries like France (121% of GDP on average between 2025 and 2030, bubble size), Italy (136% of GDP) or Belgium (109% of GDP) is more worrying than for countries like Germany with a projected debt burden of about 66%.

The second group includes countries where the fiscal deficit still provides some room but where heat-related pressures from lower growth, higher inflation and adaptation investment needs are large enough to threaten the Maastricht threshold. Italy and Spain, with estimated deficits of –2.8% and –2.4% respectively, could clearly breach the –3% criterion once heat-related pressures of 1.9% and 1.7% of GDP are factored in. The Czech Republic faces a similar dynamic, with 0.6% of heat pressure meeting a deficit hovering just above

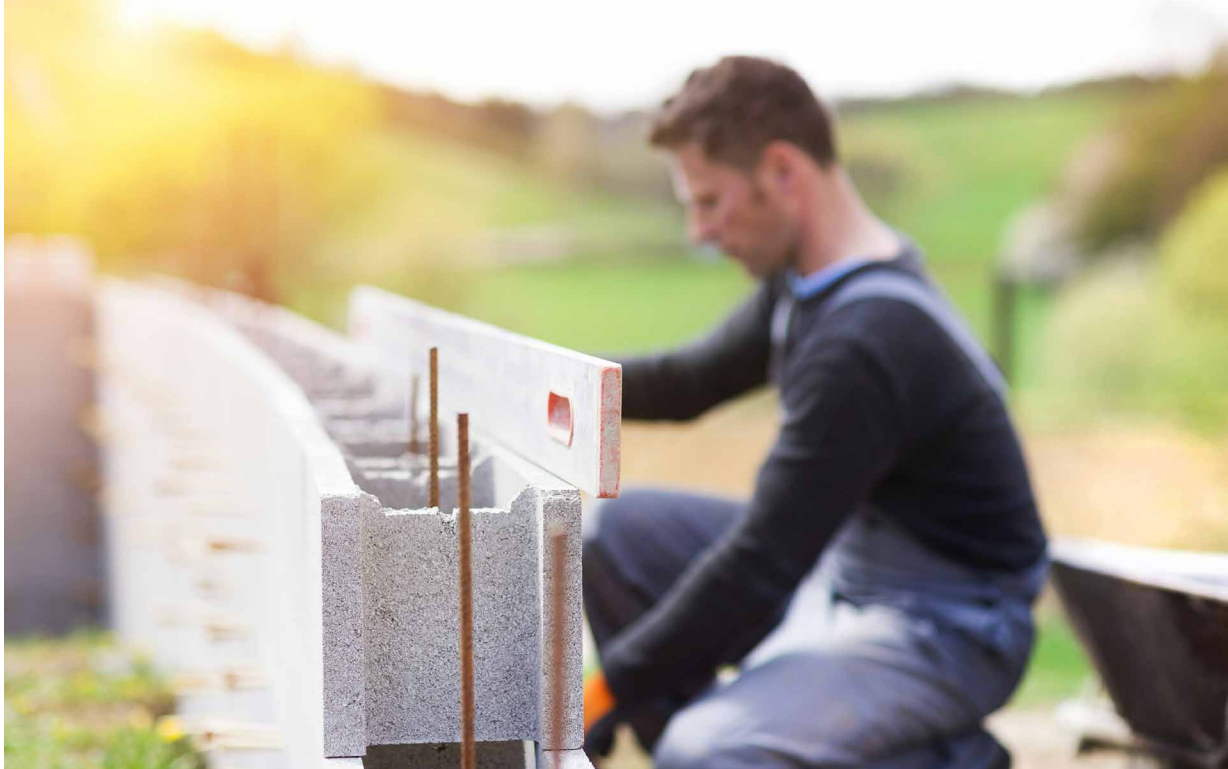
–3%. Other countries in this group, such as Greece (1.5% of heat pressure), Portugal (1.0%) and the Netherlands (0.2%), have more fiscal room and are thus less immediately constrained, but the trajectory might point toward increasing pressure.

The third group, which contains cooler regions such as the Nordic countries, may experience an improvement in their fiscal positions as moderate warming boosts economic activity through reduced heating costs and marginally improved working conditions. For these countries, the fiscal space created by a temporarily favorable climate position should be used for preventive adaptation investment, including updating building codes, establishing heat-health action plans, introducing workplace heat protocols, before the temperature distribution shifts further. Preventive action taken now, when fiscal conditions permit, yields substantially higher returns than remedial action taken later under fiscal constraint.

Figure 12: Average fiscal pressure due to heat in % of GDP (annually between 2026 and 2030) and government balance in % of GDP (2025-2030 average)



Sources: OECD, COFAS, Oxford economics, Allianz Research. Note: Bubble size indicates the projected debt burden by country 2025-2030 on average. These figures take into account the average change in inflation due to heat between 2026 and 2030, the share of indexed spending by country and heat-specific expenditure as a share of total climate-related public expenditure



From exposure to resilience: the three-pillar adaptation plan

Pillar 1. Protecting workers and productivity

The labor-productivity exposure shows that above 30°C, each additional degree of Tx5d reduces output per worker-hour by approximately USD1.30. Globally, without a tailored adaptation strategy, working hours lost to heat stress are projected to rise from 1.4% in 1995 to 2.2% by 2030 (ILO 2019). Four regulatory mechanisms shape how much of this will be mitigated: quantitative temperature thresholds, automatic work-restriction or suspension triggers, compensation for hours lost and contractual extension to fixed-term, seasonal, self-employed and platform workers.

No major European economy currently pulls all four, and the open cells cluster on the dimensions most relevant to outdoor and non-standard workers (Table 1). Spain comes closest, combining binding indoor thresholds (Royal Decree 486/1997) with up to four days of paid adverse-weather leave (Royal Decree-Law 8/2024), though the indoor focus and standard-contract design leave outdoor and non-standard workers partially exposed. France introduced procedural employer obligations from July 2025 but stopped short of quantitative thresholds or compensation. Germany's ASR A3.5 sets graduated thresholds at 26°C, 30°C and 35°C with a recognized compliance route under workplace-safety law, but does not by itself trigger automatic work restrictions. Finland regulates exposure time above 28°C through binding work-rest limits. The UK and Ireland rely on general workplace-safety duties.

Table 1: Coverage matrix — occupational heat-stress regulation across selected economies, scored on four mechanism dimensions.

Country / Economy	Quantitative threshold	Automatic work restriction / suspension trigger	Compensation for lost time	Extension to non-standard workers
Spain	● indoor	● adverse-weather trigger	● RDL 8/2024	● standard contract bias
France	○	● procedural only	○	○
Germany	● ASR A3.5 graduated	○ compliance route only	○	○
Finland	● 28°C / 33°C	● work-rest limits	○	○
Italy	○	● protocols	● wage support for heat stoppages	○
UK & Ireland	○	○	○	○
Slovenia	● indoor reported	○	○	○
Australia	○ no national heat-specific mechanism	○	○	○

Legend: ● fully covered, ● partial or conditional, ○ not covered

Source: Allianz Research

Read in this format, the gap is not that most countries are silent, it is that no country closes all four cells, and the open cells concentrate where exposure is highest.

Of the four economies most affected in the output-per-worker-hour counterfactual (France, Italy, Slovenia and Australia) none appears to combine comprehensive binding heat thresholds, automatic work-restriction rules and broad compensation coverage. France has recently strengthened procedural obligations; Italy provides some wage-support channels during extreme heat, recognized for temperatures above 35°C and, in some cases, below it depending on perceived temperature and work conditions; Australia relies mainly on general work-health-and-safety duties rather than a national

heat threshold and Slovenia's coverage should be treated separately depending on whether the analysis focuses on indoor thresholds or outdoor exposure. Several EU countries have no explicit heat-specific workplace provision. The most consistent open cell across all groups is contractual: compensation provisions, where they exist, are built around standard employment relationships; they may cover fixed-term and seasonal employees in principle, but access is weaker in practice, while self-employed and platform workers are often excluded or only indirectly covered. The cost of adaptation therefore falls disproportionately on the workers least able to bear it.

Pillar 2. Cooling buildings and stabilizing the grid

The exposure is two-sided and must be addressed simultaneously: poorly adapted buildings on the demand side, where indoor temperatures rise and cooling demand and heat mortality follow, and thermoelectric generation on the supply side, which derates under heat stress precisely when cooling demand peaks. Four regulatory levers shape how much of this is mitigated: overheating thresholds in building codes, mandatory passive cooling in renovation, cooling access as social policy and summer adequacy planning that models coincident cooling demand and supply derating. No major European economy currently pulls all four, and the gaps cluster on renovation, social-policy access and adequacy planning (Table 2). The

revised EPBD (Directive 2024/1275), with a transposition deadline of 29 May 2026, brings overheating and passive cooling into the energy-performance framework but is silent on cooling access and grid resilience. At the national level, France's RE2020 is the most advanced overheating regime, integrating a degree-hour indicator calibrated on a 2003-heatwave reference; England's Approved Document O and the Netherlands' TOjuli indicator play comparable new-build roles. Spain and Germany sit a tier below – Spain capping cooling energy demand without a dedicated overheating definition and Germany addressing summer thermal protection through DIN 4108-2 rather than a dedicated regime.

Table 2: Coverage matrix — building heat resilience and summer peak-demand management across selected economies, scored on four mechanism dimensions.

Country / Economy	Quantitative overheating threshold (new build)	Mandatory passive cooling in renovation	Cooling access as social policy	Heat-adjusted summer adequacy & distribution planning
EU EPBD 2024/1275 (from May 2026)	● EPC reflection, passive cooling in min. standards	● depends on transposition	○ out of scope	○ out of scope
France	● RE2020 DH indicator	● new-build focus	○	● RTE summer adequacy assessment
Germany	● DIN 4108-2 (indirect)	○	○	○ winter-oriented
Spain	● cooling-energy cap (CTE)	● partial	○	○ no documented mechanism
United Kingdom (England)	● Approved Doc O	○ new-build only	○	○ winter-peak focus
Italy	○ no dedicated national overheating indicator	○	○	○
Netherlands	● TOjuli indicator	● partial	○	○
Legend: ● fully covered, ● partial or conditional, ○ not covered				

Source: Allianz Research

Read in this format, the gap is twofold: cooling access remains far less institutionalized than winter heating support, and even the strongest national codes apply mainly to new build rather than to the existing stock where most exposure sits. The forthcoming EU floor in May 2026 will raise the minimum on overheating and passive cooling but does not address cooling access or grid resilience, and its bite depends entirely on transposition. The most consistent open cell is cooling access: no major European economy has developed a systematic summer counterpart to the winter heating-support architecture, and the gap falls on the groups that drive the European mortality concentration – older people, low-income households and tenants in the least-insulated dwellings. A workable national plan therefore rests on closing the four open cells together: binding overheating thresholds extended from new build to deep renovation; mandatory passive cooling triggered by a transparent renovation threshold; cooling access as a coordinated package – passive retrofits and efficient-appliance subsidies first, targeted bill support and medically prioritized access second, paired with a regulated right-to-install subject to structural and heritage constraints and heat-adjusted summer adequacy planning that explicitly models coincident cooling demand and supply derating, paired with residential demand response.

Pillar 3. Income protection and adaptation finance

Pillar 3 addresses the income-transmission channel of heat risk: when productivity losses become wage losses, the burden falls unevenly across workers, households and public budgets. The wage compression documented in previous sections lands first on outdoor workers, fixed-term, seasonal and platform workers and lower-income households. Existing stabilizers, such as unemployment insurance, short-time work, indexed transfers and energy-poverty support, absorb part of the shock, but with a structural twist: the same indexation and triggering that protect household purchasing power also raise public expenditure or compress tax receipts, lifting the fiscal cost of each heat episode.

This is the fiscal-stabilizer paradox at the heart of Pillar 3. The more effectively a welfare state cushions households from heat-driven income and price shocks, the more it shifts the climate burden onto public budgets. If that burden is met through ex-post compensation rather

than ex-ante adaptation, it consumes the fiscal space needed to lower future exposure. The paradox has two legs: a household-income leg, where existing social protection determines who absorbs the shock, and a fiscal-architecture leg, where public finance determines whether the resulting cost goes into prevention or compensation.

Only Spain (Royal Decree-Law 8/2024 paid adverse-weather leave, including heatwave situations) and partially Italy (CIG/CIGS wage supplementation activated for extreme heat) have operational heat-specific income-protection mechanisms. The remaining major economies – France, Germany, the UK, the Netherlands and most other member states – have general schemes that could in principle absorb heat-driven income loss but require ad-hoc activation each time, with no automatic heat trigger in legislation. Parametric risk-transfer for residual income volatility remains essentially absent at the national level in Europe; the European Commission's Climate Resilience Dialogue (2024) classifies it as an evolving tool requiring further analysis.¹⁸ National adaptation strategies are now in place across all major European economies but remain strategic documents rather than multi-year budget envelopes. At EU level, several frameworks enable national action without amounting to a European income-protection or adaptation-budget regime.

To reduce the gap, income protection, residual risk-transfer, budget planning and investment appraisal have to be redesigned as one fiscal architecture, working on both sides of the problem. Existing social-protection instruments should carry explicit heat triggers conditional on workplace adaptation by employers. Parametric instruments calibrated to occupational heat thresholds can complement social insurance for the residual income volatility it does not absorb, with prioritized access for outdoor and non-standard workers and a public-backstop component for affordability. Multi-year adaptation budget envelopes, set out within medium-term expenditure frameworks, should replace reliance on ex-post emergency packages, paired with climate-risk screening across public investment management and coordinated oversight bringing together the health, housing, environment, energy, transport and labor ministries.

¹⁸ Climate Resilience Dialogue - Final Report - Zurich Climate Resilience Alliance

Appendix

A1 – The impact of heat stress on energy demand

To empirically assess the relationship between heat stress and energy demand, we construct a panel regression framework that captures how rising heat stress translates into higher energy consumption – primarily through the increased use of cooling technologies such as air conditioning and refrigeration. As climate change drives more frequent and intense heat extremes, this mechanism is expected to become an increasingly important driver of energy demand, with significant implications for grid stability, energy security and the pace of the low-carbon transition.

Our analysis draws on two data sources. Energy consumption data are sourced from the IEA's World Energy Balances 2025, expressed in per capita megajoules (MJ) to allow for cross-country comparability. To capture heat stress, we use the Tx5d indicator from the ERA5 reanalysis dataset, defined as the mean daily maximum temperature over the hottest five consecutive days of a given year. This metric is particularly well-suited to our analysis, as it reflects the kind of sustained heat episodes most likely to trigger a behavioral and technological response in energy use, rather than isolated daily peaks.

The empirical model is estimated on an unbalanced panel covering 49 countries over the period 1991–2023. It provides both broad geographic coverage and sufficient temporal depth to identify meaningful climate-energy relationships. We employ a two-way fixed effects specification, controlling for both country-level heterogeneity and common time trends, with robust standard errors clustered at the regional level to account for potential spatial correlation in residuals. The regression model reads

$$\ln(\text{energy}) = \beta_1 \text{tx5d} + \beta_2 \text{tx5d}^2 + C(\text{ISO3}) + C(\text{Year}); \text{Eq. 1}$$

The relationship between heat stress and energy consumption is nonlinear, following a convex U-shaped curve (Table A1), confirming previous literature results.¹⁹ At moderate temperature levels, energy demand declines as milder conditions reduce the need for space heating and other thermal energy services. This downward trend reverses once temperatures exceed approximately 30°C, a commonly used threshold for heat stress²⁰, at which point cooling demand begins to dominate, driven primarily by the widespread use of air conditioning.

This non-linearity is reflected in the regression results, which show that the marginal effect of heat stress on energy consumption is both temperature-dependent and sign-switching. At 25°C, the effect is negative: a one-unit increase in the Tx5d indicator is associated with an approximately 0.6% decrease in per capita energy consumption. At 35°C, the relationship reverses: the same marginal increase in heat stress is associated with a 1.2% rise in energy consumption. This twofold difference in magnitude illustrates how the energy implications of heat stress intensify disproportionately at higher temperature levels. A dynamic with important consequences for energy planning and infrastructure resilience in a warming climate.

¹⁹ [Electricity Consumption and Temperature: Evidence from Satellite Data, WP/21/22, February 2021](#)

²⁰ [Keeping cool in a hotter world is using more energy, making efficiency more important than ever – Analysis - IEA](#)

Table A1: Regression results: The impact of heat stress on energy consumption

Variable	Coefficient	p-value
Intercept	10.3390	0.000
tx5d	-0.0514	0.005
tx5d ²	0.0009	0.008

Source: Allianz Research

A2 – The impact of heat stress on productivity

The adverse effects of heat stress on labor productivity are well established in the academic literature. Growing evidence from both micro and macro studies shows that exposure to extreme temperatures affects health, labor supply, and labor productivity in the short run.²¹ At the physiological level, heat stress affects workers through physiological and behavioral responses, in turn reducing labor supply, productivity and overall labor capacity, with the relationship largely non-linear, declining sharply beyond peak temperature thresholds.²² Despite considerable heterogeneity across work sectors and study designs, nearly all field studies consistently find a reduction in productivity due to occupational heat exposure.²³

Building on this evidence, we adopt the same empirical framework used for our energy consumption analysis to derive an econometric estimate of heat stress impacts on productivity. Heat stress is measured using the Tx5d index, as defined previously. For productivity, we draw on two complementary indicators from the OECD database²⁴, output (gross value added) per hour worked, expressed in USD, and wages per hour worked, which together capture both the economic output and the compensation dimension of labor performance. The models reads as follow

$$GVA_{\text{per hour}} = \text{tx5d} + \text{tx5d}_{\geq 30} + C(\text{ISO3}) + C(\text{Year}); \text{Eq. 2}$$

$$\Delta \ln(\text{wage}_{\text{per hour}}) = \text{tx5d}_{(t-1)} + \text{tx5d}_{(t-1)}^2 + C(\text{ISO3}) + C(\text{Year}); \text{Eq. 3}$$

Equations 2 and 3 depart from the baseline specification of Equation 1 in two important ways, each motivated by the distinct channels through which heat stress affects labor productivity. Equation 2 models gross value added per hour as a function of both the Tx5d index and an interaction term, Tx5d when heat stress exceeds 30°C, to account for the non-linear intensification of heat impacts at higher temperature levels. Equation 3, which models the log change in wages per hour, follows the same structure as the energy consumption model (Equation 1), but introduces a one-year lag specification for both the linear and quadratic Tx5d terms. This lagged structure reflects the hypothesis that wage adjustments to heat stress occur with a delay, as labor market responses, such as renegotiated contracts or sectoral reallocation, tend to materialize more slowly than changes in physical output.²⁵

²¹ Reflections—Temperature Stress and the Direct Impact of Climate Change: A Review of an Emerging Literature | Review of Environmental Economics and Policy: Vol 10, No 2

²² Heat stress and the labour force | Nature Reviews Earth & Environment

²³ Frontiers | Occupational heat stress, heat-related effects and the related social and economic loss: a scoping literature review

²⁴ OECD Data Explorer • Productivity database

²⁵ Economic impact of labor productivity losses induced by heat stress: an agent-based macroeconomic approach | Climatic Change | Springer Nature Link

The output per hour regression (Eq. 2) is estimated on a panel of 35 countries over the period 1997–2023 (Table A2).

The results reveal a clear threshold effect consistent with the non-linear relationship hypothesized in our framework. Below 30°C, the coefficient on Tx5d is positive and marginally significant at the 10% level (0.59), suggesting that moderate heat is associated with a slight increase in output per hour – potentially reflecting seasonal productivity gains in temperate economies. Above 30°C, however, the picture changes markedly: the coefficient on the interaction term is negative and statistically significant at the 5% level (–1.30), indicating that each additional degree beyond this threshold is associated with a reduction in output per hour of approximately 1.30 USD. These findings underscore the non-linearity of heat’s impact on labor productivity, where the results point to 30°C as a critical inflection point beyond which heat stress begins to impose measurable economic costs on labor output.

Table A2: Regression results: The impact of heat stress on output per hour

Variable	Coefficient	p-value
Intercept	13.7401	0.142
tx5d	0.5900	0.072
tx5d _{≥30}	-1.3039	0.042

Source: Allianz Research

The wage regression (Eq. 3) is estimated on a panel of 33 countries over the period 1995–2023 (Table A3). The results are broadly consistent with those obtained for output per hour, with a turning point again emerging around 30°C. The use of a one-year lag structure reflects the hypothesis – supported by the data – that labor compensation adjusts to heat stress with a delay, as wage-setting mechanisms respond more slowly to productivity shocks than output itself.

The relationship between heat stress and wage growth follows an inverted U-shape. In the range below 30°C, higher heat stress in year t is associated with a modest acceleration in wage growth in year $t+1$, consistent with the productivity gains observed at moderate temperatures in the output per hour specification. Above 30°C, this relationship reverses: additional heat stress dampens wage growth the following year, reflecting the transmission of heat-induced productivity losses into labor compensation over time.

The marginal effect of heat stress on wage growth is both temperature-dependent and sign-switching. At 25°C, the effect is positive, with a one-unit increase in Tx5d associated with an approximate 0.2% increase in wage growth. At 35°C, the relationship reverses, with the same marginal increase in heat stress associated with a 0.2% decline in wage growth. While the magnitude of the effect may appear modest, its consistency with the output per hour results and its statistical significance lend confidence to the finding that heat stress above 30°C exerts a meaningful drag on labor compensation, with implications for household income, consumer demand and the broader macroeconomic costs of climate change.

Table A3: Regression results: The impact of heat stress on per hour wage growth

Variable	Coefficient	p-value
Intercept	-0.0756	0.364
tx5d _(t-1)	0.0120	0.027
tx5d _(t-1) ²	-0.0002	0.013

Source: Allianz Research

A photograph showing a variety of hands of different skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text 'Our team' is overlaid in the center, with 'Our' in white and 'team' in orange.

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How AI is rewiring global trade

Concentrating Power, Dependencies, and Supply Chains

21 May 2026

Content

Page 3-4

Executive Summary

Page 5-11

An AI boom built in Asia

Page 12-20

The growth in trade in services and data flows is accelerating with the diffusion of AI

Page 21-24

The world is flat: AI reconfigures the geography of trade and dependencies

Page 25-29

The race for AI dominance is increasingly also being fought through industrial policy

Executive Summary



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AI and trade are no longer separate policy domains. AI growth depends on globalized supply chains for semiconductors, computing infrastructure and digital services while trade is increasingly shaped by who controls AI infrastructure, data flows and cloud capacity.



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- **Trade openness is a structural precondition for AI-driven productivity gains.** Open economies benefit disproportionately from cheaper inputs, faster innovation diffusion and AI adoption spillovers. Trade openness accounts for 23% of the variation in AI adoption across countries with highly open economies, such as Singapore, UAE and Ireland, leading in diffusion. However, while AI can significantly boost growth, its benefits are unlikely to be evenly distributed.



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- **Export volumes of AI-enabling goods have surged from USD1trn in 2014 to USD3.8trn in 2025 (+280%), accounting for 15% of global trade and far outpacing the 40% growth in goods trade overall.** Asia dominates the supply side, accounting for 65% of global AI-related exports and seven of the top ten exporters, led by China (18% of AI-related exports), Taiwan (12%) and Hong Kong (11%). The composition remains concentrated in intermediate inputs (76%) and equipment (23%), reflecting deep dependence on semiconductors and data-center infrastructure. On the demand side, the US has tripled its AI-related imports since 2023, underpinned by 5,427 operational data centers, 45% of the global total. Europe's import growth of just +40% underscores a widening infrastructure gap.



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- **Services imbalances could scale with AI.** ICT services trade reached USD900bn in 2024 (11% of global services trade), with Ireland alone exporting USD173bn, exceeding both the rest of the EU (USD142bn) and the US (USD108bn), and reflecting its role as a billing hub for US multinationals. However, it masks structural dependence, with the EU excluding Ireland running a USD45bn ICT services deficit (mainly with Ireland), highlighting its reliance on US digital ecosystems. And things could get worse. Hypothetically, if the AI services subscription penetration rate of US providers such as ChatGPT Plus and Claude Pro increases from 3% to 50% in a high adoption scenario in the Eurozone, annual payments to US providers could reach EUR34bn (currently EUR2.7bn), equivalent to 20% of the current EU–US goods services deficit. AI thus acts as a scalable, recurring channel that exacerbates structural EU–US digital imbalances and has the potential to lower the US overall trade deficit.



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Beyond the headline data, three structural dynamics are reshaping the global AI trade order:

1. Supply chain concentration as single points of failure

The AI supply chain is concentrating rapidly at the technological frontier. Taiwan, South Korea, the US and the Netherlands dominate the production of advanced chips, high-bandwidth memory and semiconductor equipment, creating critical single points of failure with no near-term redundancy. Unlike broader goods trade, which has partially reoriented along geopolitical lines, AI-related goods remain largely unfragmented across geopolitical blocs, with semiconductors and key components still flowing globally except at the technological frontier (notably advanced US export controls) due to deep interdependencies, challenging the narrative of technological decoupling.

2. Infrastructure as geopolitical power

Control over data centers, cloud infrastructure and computing capacity is emerging as a primary source of geopolitical leverage, distinct from traditional goods trade dynamics. Europe's strategic exposure is acute. With less than 10GW of operational data-center capacity, four times below the US (60GW), and a development pipeline six times smaller, the gap is not expected to narrow. US hyperscalers already control 35% of European computing capacity and account for nearly half of the upcoming pipeline, consolidating a 70% cloud market share. Structural constraints, fragmented regulation, complex permitting processes, grid connection delays, no domestic hyperscaler and limited VC or state-backed funding reinforce this dependency. Finally, reliance on Asian hardware inputs complements the US dependence that will increase without sufficient EU investment in the area. Under that backdrop, Europe is permanently under the threat of a US "kill switch" on cloud data. In the meantime, this effective expansion of this key infrastructure is dependent on inputs sourced in Asia, notably in China (14% import ratio for AI datacenter components) and Taiwan for semiconductor, and as a result exposed to current supply-chain disruption risks along new geopolitical tensions in Middle East. Regaining control and strengthening sovereignty on AI good production but also service delivery is critical for the region to avoid suffering a twin external dependence.

3. Industrial policy: from subsidies to protectionism

Global tariffs on AI-enabling goods have fallen from 5.6% in 2015 to 2.8% in 2025, well below the 7.8% average for manufacturing overall, but non-tariff measures have surged, driven primarily by the US and China, reflecting a strategic shift from financial support toward technology protectionism. Over 3,600 AI subsidy measures targeting critical materials, semiconductors, GPUs, computing equipment and optical fibers are now in force globally, with China having nearly doubled its trade coverage over five years, followed by South Korea, Malaysia, the US and Japan. The overlap of national and multilateral governance regimes is concentrating production, infrastructure and modelling capabilities in the hands of a small number of economies. Regulatory fragmentation is becoming a binding constraint on AI services trade, with ICT trade restrictiveness already explaining 5% of the cross-country variation in AI services flows. The EU and US are advancing incompatible regulatory frameworks, creating compliance friction that reinforces existing structural imbalances.



An AI boom built in Asia

AI fundamentally depends on global trade in goods and services, with a complex, internationally fragmented value chain spanning raw materials, semiconductors, computing hardware, cloud infrastructure, data and final applications (AI-enabling goods). Trade enables access to critical inputs, such as advanced chips, specialized equipment and energy resources, while facilitating the cross-border exchange of intermediate goods, services and knowledge needed to build and scale AI systems. However, the strong geographic concentration of key stages of the AI supply chain coupled with heightening frictions tightening the exchange channel (tariffs, shutdown of the Strait of Hormuz) raises concerns about unequal access and the risk of a widening technological divide.

Over the past decade, global trade in AI-related goods has doubled, far outpacing the growth of overall goods trade and of non-AI-related goods in particular. Global AI-related goods trade has expanded rapidly over the past decade, doubling from USD1.9trn in 2014 to USD3.8trn in 2025. Meanwhile, overall global trade in goods increased more moderately, rising by +40% to USD26trn in 2025. This growth was

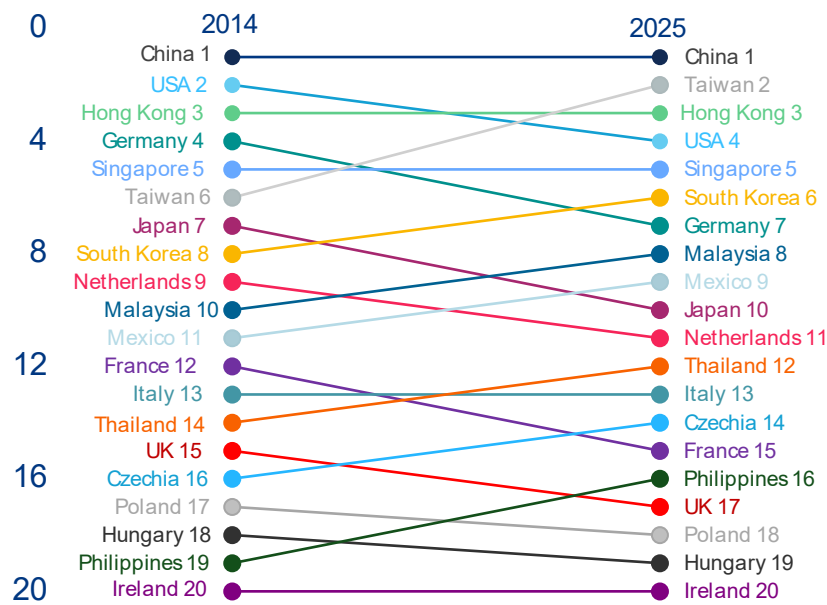
increasingly supported by AI-related sectors, with trade in non-AI goods rising by only a third. This expansion has seen three distinct acceleration phases. The first acceleration occurred in 2017–2018 (+15% y/y and +11% y/y respectively), driven by fundamental developments in need for AI-related goods and the introduction of the transformative architecture, which underpins most large language models. A second surge followed in 2021 (+30% y/y), driven by the release of early large-scale models, including GPT-3 (2020) and DALL-E (2021) and paving the way for the emergence of generative AI in 2022, which needed more confounding technical infrastructure for model innovation and computation. However, momentum weakened in 2022 amid a challenging macroeconomic environment. Higher interest rates and elevated inflation combined with the introduction of US export controls on advanced AI chips and semiconductors, led to a marked deceleration in AI-related goods trade growth (+4% y/y in 2022), followed by a contraction in 2023 (-7% y/y), with export values falling below their 2021 peak. More recently, advances in multimodal and agentic AI systems have reignited investment momentum. As AI became increasingly available and embedded in the economic system, AI-related goods exports reached new heights in 2025

(+22% y/y). Early 2026 data, covering January and February, show that the boom continues, especially in countries such as Taiwan (+62% y/y) and South Korea (+105% y/y).

Asia has firmly established itself as the center of global trade in AI-enabling goods. The region accounts for 65% of global exports and dominates the entire value chain. The region is at the heart of an increasingly concentrated landscape, with seven of the top ten exporters representing 80% of global trade. China leads with a 18% share, followed by Taiwan (12%) and Hong Kong (11%). Singapore (7%), South Korea (6%), Malaysia (4%) and Japan (3%) also reinforce the region’s strength. The US (8%), Germany (5%) and Mexico (4%) are the only non-Asian economies in this group, highlighting the strong presence of AI trade in Asia. This dominance has deepened over time as competitive dynamics have shifted in the region’s favor, causing

advanced economies to lose ground. Despite a slight decline in its share (-3pps), China remains the leading exporter, while Taiwan’s rise from 6th place in 2014 to 2nd place in 2025 (+7pps) marks the most significant shift. This was supported by a +53% year-on-year export surge in 2025 (Figure 1). Hong Kong has also gained prominence, further reinforcing Asia’s lead. At the same time, emerging players are gaining traction: Mexico recorded the fastest growth in 2025 (+62% year on year), while Malaysia, Thailand (+49% year on year) and the Philippines are expanding as alternative hubs. Together, they accounted for 11% of global trade (+3pps) and 18% of additional trade in 2025. By contrast, most advanced economies, including those in Europe, have lost ground. The US, for example, slipped from second to fourth place, reflecting a relative decline in manufacturing-based AI trade despite continued strength in high-value innovation.

Figure 1: Top 20 AI-enabling goods exporters ranking (2014 vs 2025)

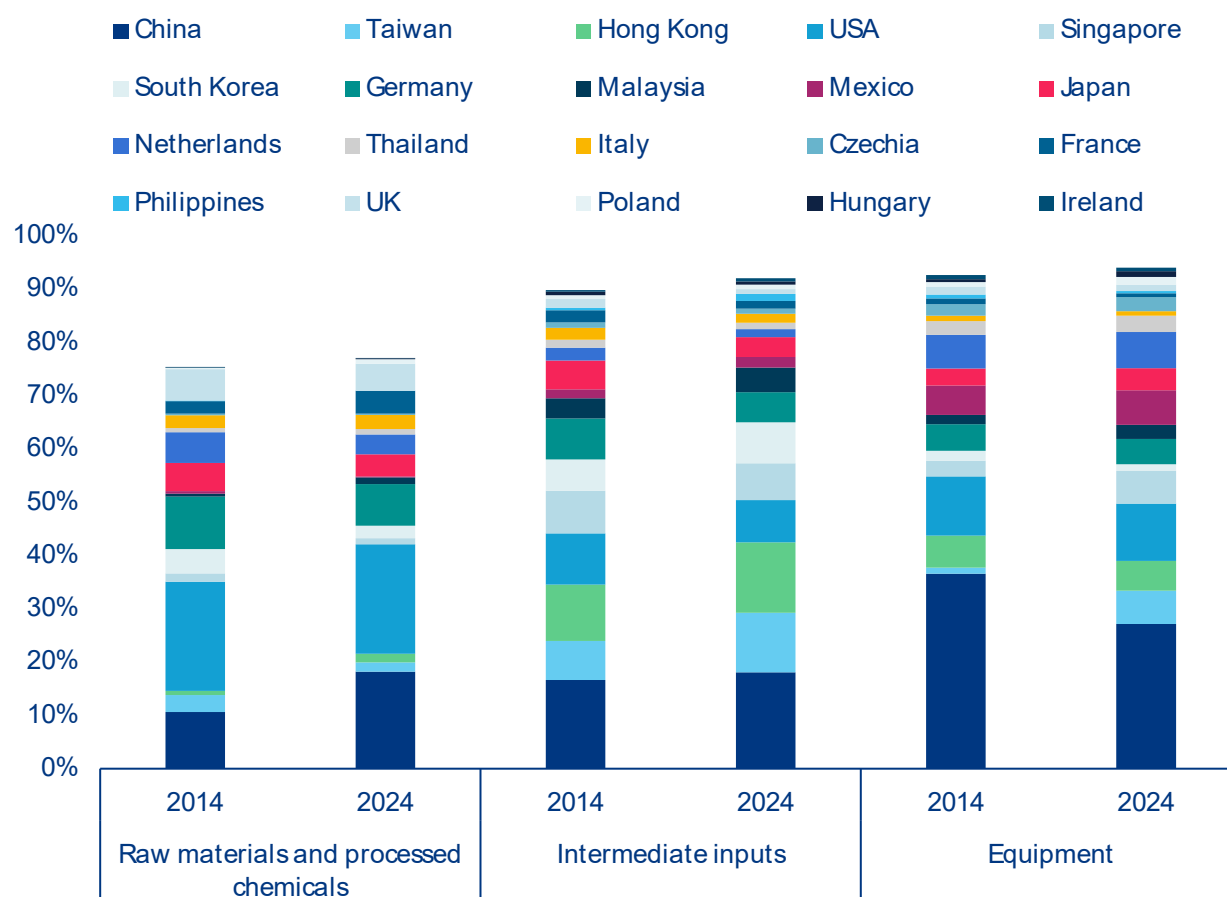


Sources: UN Comtrade, National Customs, Allianz Research

Although AI-enabling goods trade is already highly concentrated globally, the value chain reveals even greater specialization. Leading exporters cluster at distinct stages, particularly in raw materials, processed chemicals and equipment. AI-enabling goods can be categorized into three groups: raw materials and processed chemicals, intermediate inputs and equipment. Raw materials and processed chemicals constitute critical substances that are fundamental to the fabrication of AI equipment. At the aggregate level, this category represents the most diversified segment of the AI value chain. The top 20 exporters in this category account for three-quarters of global exports. This indicates significant concentration of the primary inputs for AI development, and this concentration only strengthens further upstream in the value chain, with the top 20 exporters representing over 90% of exports in the intermediate inputs (processed or partially manufactured goods) and equipment categories (specialized tools and machine components essential for AI development). China and the US have a particularly pronounced footprint at the extremes of the value chain.

Together, they account for 39% of global exports of both raw materials and processed chemicals, as well as equipment (Figure 2). For intermediate inputs, the level of concentration remains high in absolute terms, but it is comparatively more evenly distributed, with China, Hong Kong and Taiwan collectively representing 43% of global exports, and each country holding a similar share. Notably, shifts in countries' global export shares of intermediate inputs have been linked to changes in the overall ranking. Taiwan and South Korea have gained +4pps and +2pps, respectively, reaching 11% and 8% of the global export share. Similarly, Hong Kong's share has increased by +3pps to 13% over the past decade. However, this has not allowed it to become the second-largest exporter, as Taiwan has made a notable entry at the high end of the value chain. It has gained +5pps to reach a 6% export share in equipment; it is the only country, alongside Singapore, to have increased its export share in this category. This has come at the expense of China, which has lost -9pps and stands at a 27% export share.

Figure 2: Decomposition of the AI value chain (% share of global trade for each category 2014 vs 2024)



Sources: UN Comtrade, National Customs, Allianz Research.

An ultra-segmented AI supply chain, with Europe holding a modest role. The global AI ecosystem is sharply segmented geographically. Upstream activities – chip design, semiconductor equipment, data-center infrastructure and telecoms – are concentrated in Western economies, with the US dominating GPU/CPU architecture and Europe and Japan leading in lithography and wafer-production machinery. Downstream, Asia takes over almost entirely: China holds a critical position in rare earth processing, battery manufacturing and electronic components; South Korea and Taiwan anchor the foundry landscape, the former specializing in memory chips and the latter as the undisputed leader in advanced logic manufacturing. Malaysia and Thailand have developed strong packaging and assembly capabilities, with Thailand increasingly focused on data-center hardware. Critical raw materials are similarly dispersed – nickel from Indonesia and Australia, cobalt from the DRC, lithium from Chile, helium from Qatar, silicon from China. Europe's position across this architecture is, overall, low to moderate, leaving the region with limited strategic leverage and meaningful exposure to supply-chain disruptions at any of its key upstream or downstream nodes.

The trade of goods related to AI is dominated by intermediate inputs and equipment, reflecting a surging demand for high-performance AI infrastructure. Although raw materials and processed chemicals account for only 2% of the trade in AI-enabling goods, exports are dominated by intermediate inputs (76% of total AI-related exports) and equipment (23%). This masks significant country-level specialization and exposure to supply risks. The UK stands out in the upstream sector, with a 7% share of raw materials (+5pps above the global average). At the intermediate stage, the strongest concentration is in Asia: South Korea and the Philippines rely heavily on this segment (95% and 91% respectively), as do Hong Kong and Taiwan, all of which are well above the global average of 75%. European exporters such as Italy (84%) and France (83%) exhibit comparable, albeit less pronounced, patterns. At the downstream stage, equipment exports are more prevalent in the Netherlands (57%), Mexico (49%), the Czech Republic (45%) and Thailand (43%). Meanwhile, China and the US have a more balanced approach, with each country having a 68% share of intermediate inputs and equipment shares of 31% and 28%, respectively.

Box: The Middle East crisis could send semiconductor prices surging

Over the past two years, two-thirds of the expansion in Asian semiconductor exports were due to price increases and one-third due to higher volumes. Risks of energy shortages from the Middle East crisis could send prices even higher, given the already tight and extremely concentrated supply. Looking at the recent results of the biggest foundry worldwide located in Taiwan (70% market share in 2025) as a proxy for the advanced-node segment of the supply chain, wafer shipments grew +28% year-on-year in Q1 2026, a robust rate by historical standards. However, revenue in USD terms expanded by +40.6% over the same period, leaving a gap of approximately 12pps that cannot be attributed to volume alone. The explanation lies primarily in a structural shift in product mix rather than in nominal price increases on comparable goods. Taiwan's production capacity base has migrated progressively toward 3nm and 5nm process technologies, which carry substantially higher per-wafer prices than the legacy nodes they are displacing in the revenue composition; advanced technologies now account for 74% of total wafer revenue despite representing a modest share of physical units shipped. However, a look at China's trade statistics suggest that inflationary effects are rather broad-based as even for the less advanced chips used for industrial purpose – above 28nm – we observe a diverging trend between export volumes (downward) and value (upward) in Q1 (Figure 3, right). Chinese data show that strong volume on AI products is masking rising disruption risks in manufacturing production (over -10% contraction of global shipments of smartphone and computer estimated this year, Figure 3, left), as well as for some electric and mechanical devices (i.e., transistors, printed circuits, HVAC, batteries) that are critical for building up the data-center fleet.

Figure 3: China exports value, total & high-tech goods (left) and China exports value & volume, semiconductor in USD, 12-month trailing yoy% (right)



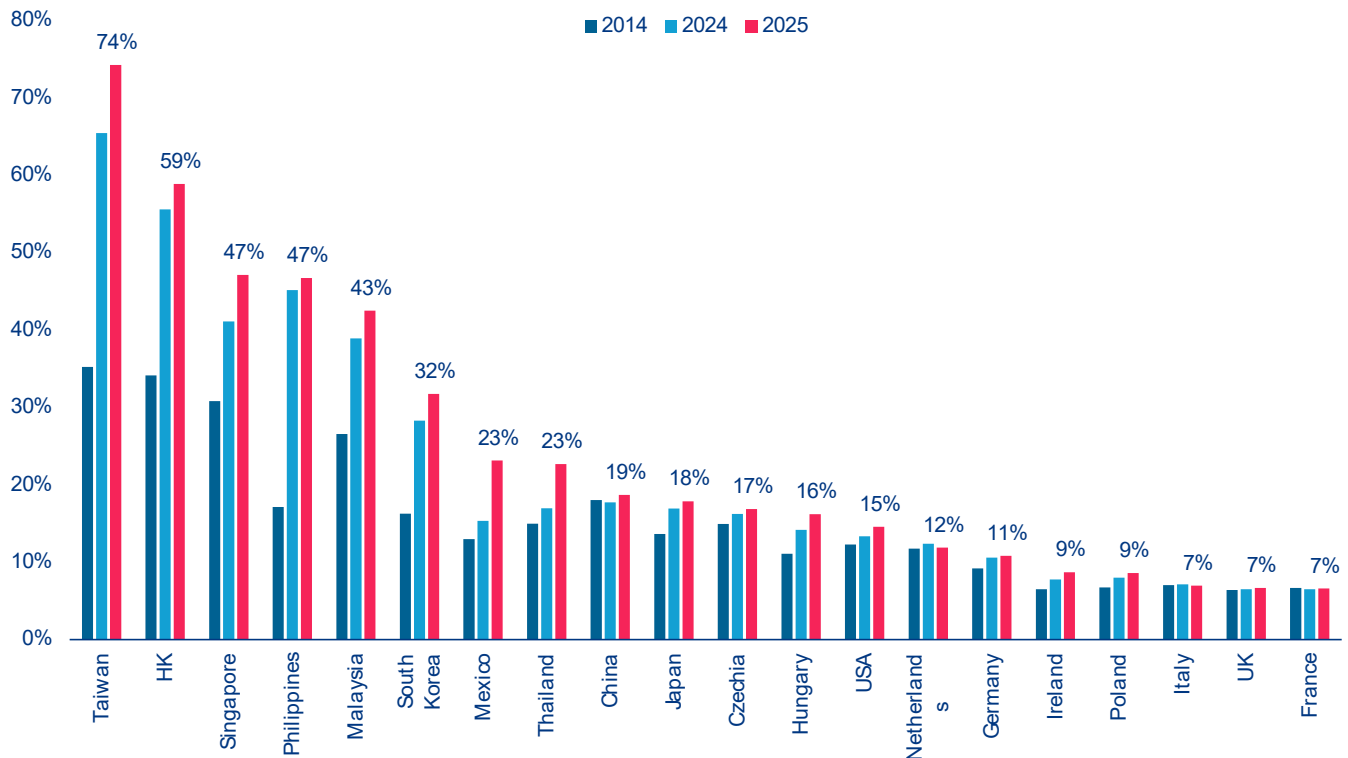
Sources: China National Customs, Allianz Research

The sustained increase in semiconductor pricing is best understood as a demand-led phenomenon, with supply-side constraints serving as an amplifying mechanism. The commercial deployment of large language models from late 2022 onwards ramped up the compute requirements of hyperscale cloud operators. Since then, the transition from AI model training to large-scale inference has extended and broadened the requirement for advanced silicon across a wider range of end-users. Naturally, this is squeezing supply: Leading-edge foundry capacity requires three to five years and capital commitments in the tens of billions of dollars to come online. The imbalance between demand urgency and supply inelasticity has given foundries meaningful pricing power, with Taiwanese semiconductor producers' gross margins expanding from 58.8% to 66.2% within a single year. Supply-side cost pressures related to critical materials, notably gallium, germanium, tungsten and antimony, all subject to successive rounds of Chinese export controls since 2023, are another increasingly consequential factor, with affected material costs rising in the range of 30-50%, though they are more of a medium-term geopolitical risk for now.

The fact that the majority of AI-enabling goods exports are driven by a small group of countries also makes their trade model highly vulnerable to potential shocks.

Taiwan and Hong Kong are the most exposed to an AI bubble burst, with 74% and 59% of their exports being related to AI goods respectively (Figure 4). They are followed by Singapore and the Philippines (both 47%), Malaysia (43%) and South Korea (32%). In contrast, trade in AI-enabling goods accounts for 15% of US exports, indicating that American exports are relatively well diversified. Similarly, Mexico and Thailand (both 23%), China (19%), and Japan (18%) have a moderate level of export concentration in AI-enabling goods.



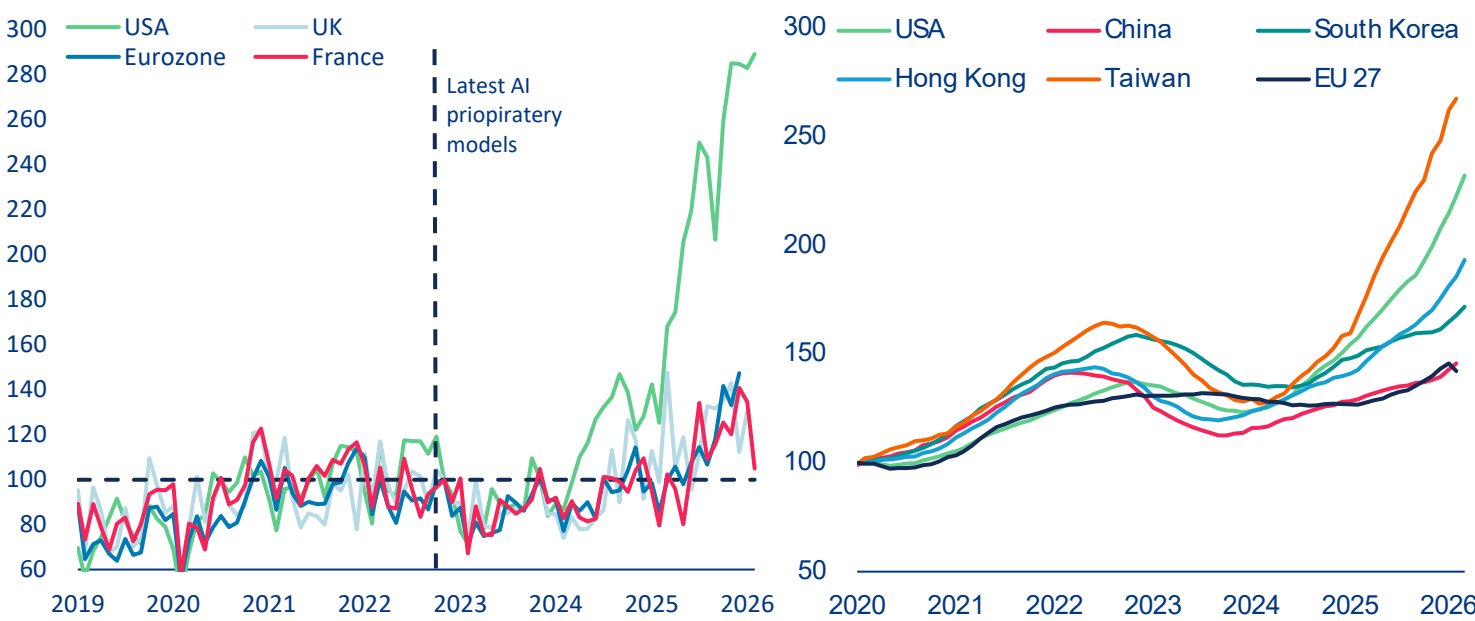
Figure 4: Share of AI-related goods (% of countries' total exports)

Sources: UN Comtrade, National Customs, Allianz Research

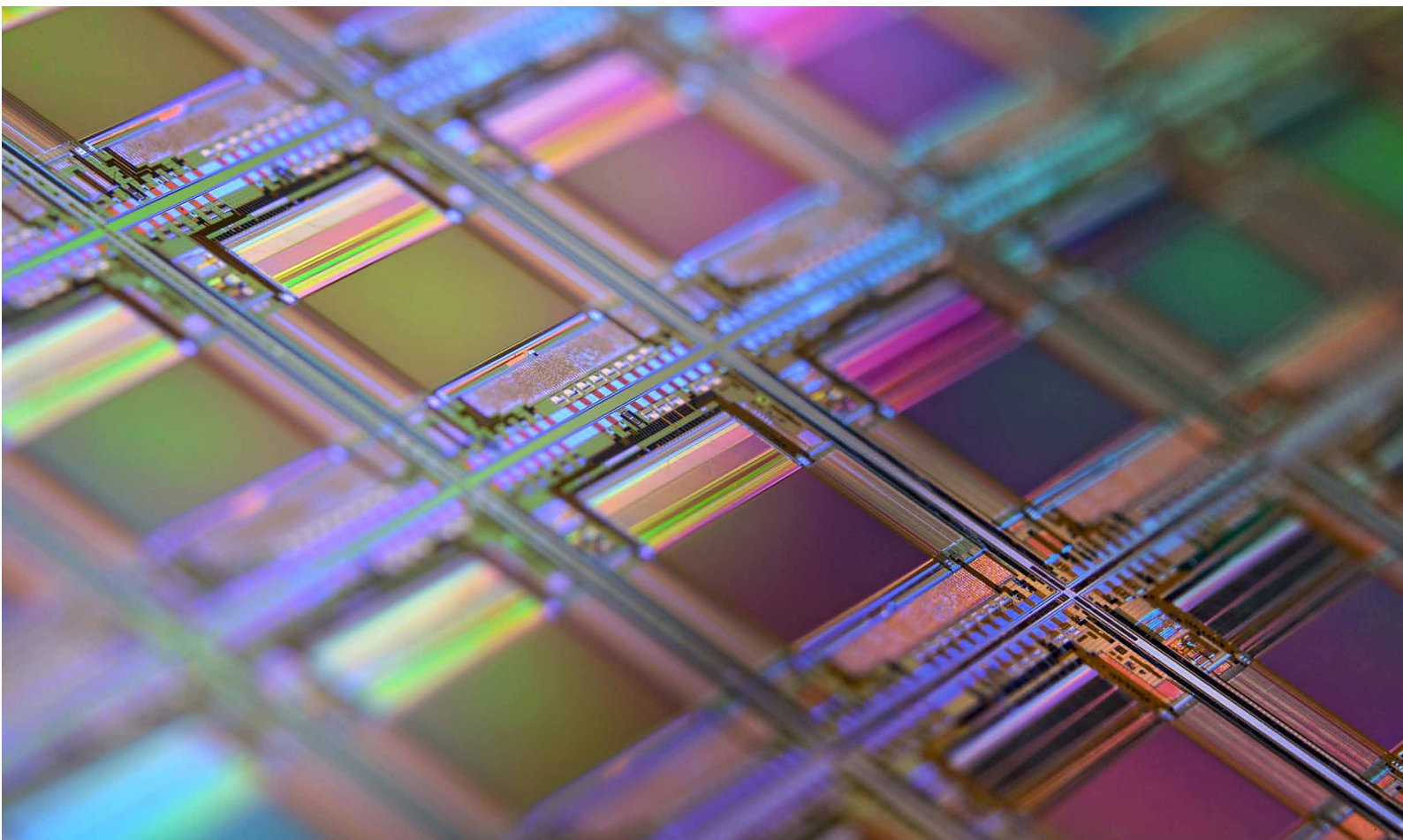
However, AI-related import demand is highly concentrated in US and partially Asia. Since 2023, Taiwan and the US have recorded the largest increases in AI-related imports, reflecting their pivotal positions in the AI supply chain: Taiwan through the import of capital equipment for semiconductor manufacturing and the US through demand linked to its dominance in AI services and data-center infrastructure. Since the mass-market deployment of large language models since 2023, the US has tripled its imports of advanced AI-related products (Figure 5, left), reflecting massive domestic investment in AI and continued reliance on foreign semiconductor supply. This surge has been reinforced by the rapid expansion of US data-center infrastructure, with 5,427 facilities accounting for 45% of global capacity in 2025 and 10 times the count of any other country. By comparison, Europe's AI-related imports have grown by only around 40% since 2023 (Figure 5, right), reflecting weaker investment in AI infrastructure, limited data-center capacity (Germany 2nd with 529, UK 3rd with 523, France

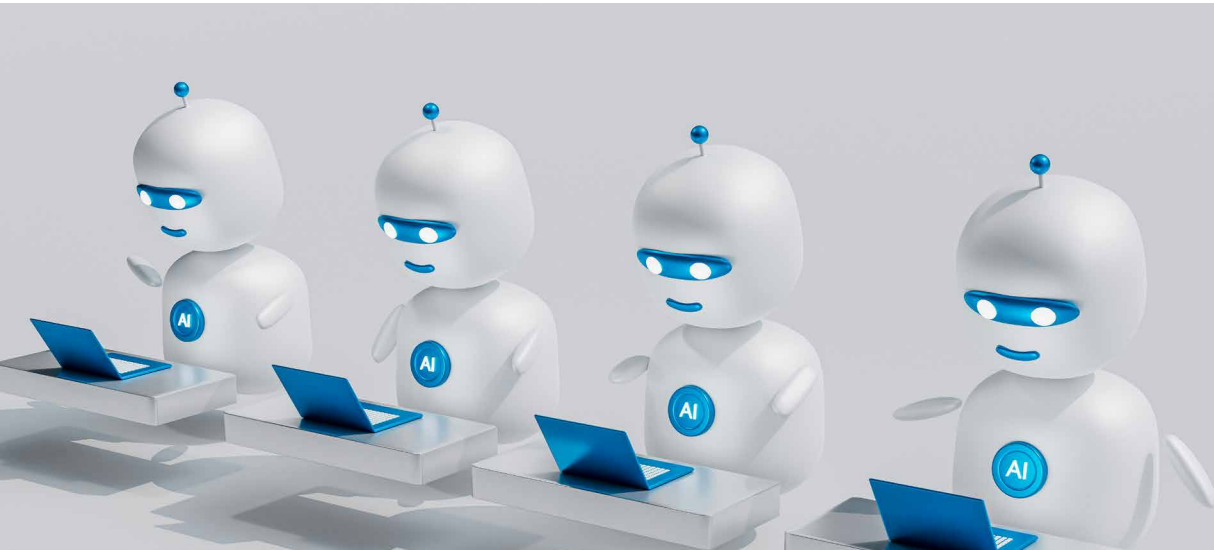
6th with 322 data-centers) and restricted access to advanced chips compared to US hyperscalers. Growth slowed further in early 2026. In Asia, beyond Taiwan, the strongest import growth has been seen in South Korea, reflecting its dual role in semiconductor production and consumers of advanced AI components, and Hong Kong, highlighting its important roles in semiconductor production and technology flows into China due to US export controls. Overall, the AI semiconductor market remains highly concentrated geographically: the US being the dominant end market for the most advanced generation of AI semiconductors, the EU and China as the other two large but structurally distinct demand pools. This creates significant exposure to shifts in demand, trade policy and regulation across a small number of economies: for AI hardware manufacturers any deterioration in US demand, tightening of trade and tariff policy, or regulatory disruption in Europe or China would have outsized consequences for the suppliers currently running at capacity to serve them.

Figure 5: US imports growth from selected AI products have skyrocketed, index (left) and Imports of overall AI-related products, in USDbn (right)



Sources: Trade Map, Allianz Research





The growth in trade in services and data flows is accelerating with the diffusion of AI

Data flows and AI-related services trade are the backbone of transforming economic models into the future. It differs structurally from the trade of goods in several key respects. Firstly, it is deeply integrated into broader digital and intra-firm flows, with much of its value being captured through cloud services, software licensing, data infrastructure and model access, which are frequently bundled together rather than being traded as individual services. This helps to explain the outsized role of hubs such as Ireland, which function less as traditional exporters and more as invoicing and coordination centers for global technology firms, particularly US multinationals. Consequently, bilateral imbalances, such as those between the EU excluding Ireland ("EU26") and Ireland, understate the true extent of dependence on US-based AI ecosystems. Indeed, the EU26 entity runs a large ICT services trade deficit of USD45bn, mostly explained by the deficit with Ireland, at close to USD55bn. India and the UK are far behind in terms of ICT exports to the EU and the US comes fourth, exporting less USD5bn directly to the EU26. Secondly, AI services act as the operating system for the trade of AI goods: advanced chips, servers and hardware only generate value when combined with cross-border

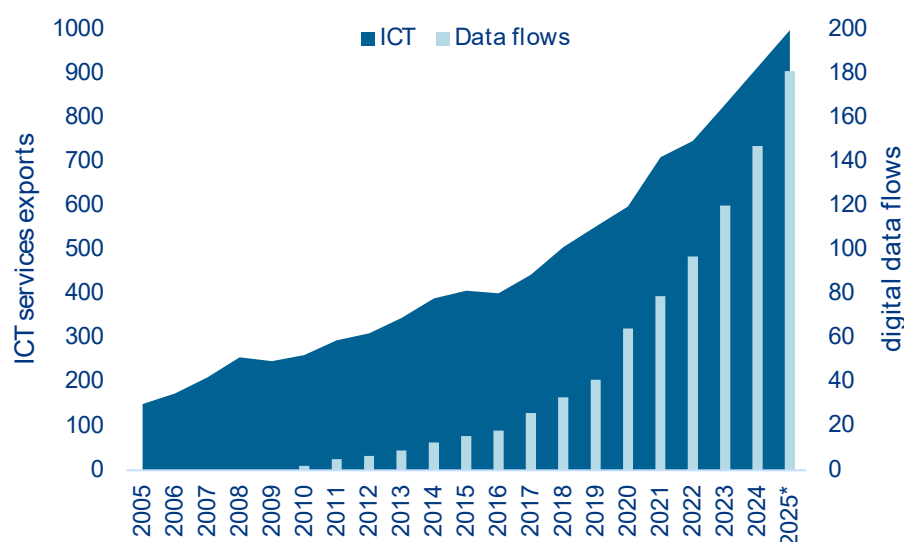
flows of computing power, software and continuous model updates. This makes services a prerequisite rather than an adjunct to the trade of goods. Thirdly, the high concentration of trade reflects platform-based business models rather than pure comparative advantage, with a small number of economies capturing disproportionate value through intellectual property, data and digital infrastructure. Finally, the trade of AI services is increasingly shaped by regulatory and data governance regimes, which influence where services are hosted, billed and exported from. This reinforces fragmentation across jurisdictions. Together, these features imply that not only is AI-related services trade concentrated, it is also foundational, determining how value is distributed across the global AI economy.

An essential for the development and distribution of AI-systems are cross-border data flows. As AI systems rely on large, diverse and continuously updated datasets that only generate value when combined with software, hardware and expertise, the flow of these data internationally is essential for AI development and adoption. As data is non-rivalrous, it can be reused by multiple users and applications, creating strong

economies of scale and scope. Free data flows enable firms to combine information across markets, build more accurate and generalizable models and accelerate innovation. By contrast, restrictions limit scale, raise costs, reduce interoperability and slow diffusion. The growing importance of this issue is reflected in broader trends: data-driven services such as computing, telecommunications and finance now account for almost half of the global trade in services. ICT services have evolved rapidly over time making up 0.53% of nominal global GDP in 2015 and have increased by 56% to 0.85%

in 2025, while cross-border data volumes have grown more than tenfold from 15.5 zettabytes in 2015 to 181 zettabytes in 2025 (Figure 6). Together, these dynamics demonstrate that seamless cross-border data flows are a key driver of AI performance and innovation, as well as being a fundamental pillar of the global digital economy and the rise in AI-related services. And although data is generated and flows globally, it is often processed, stored and monetized in a small number of digital hubs with advanced infrastructure, which concentrates value geographically.

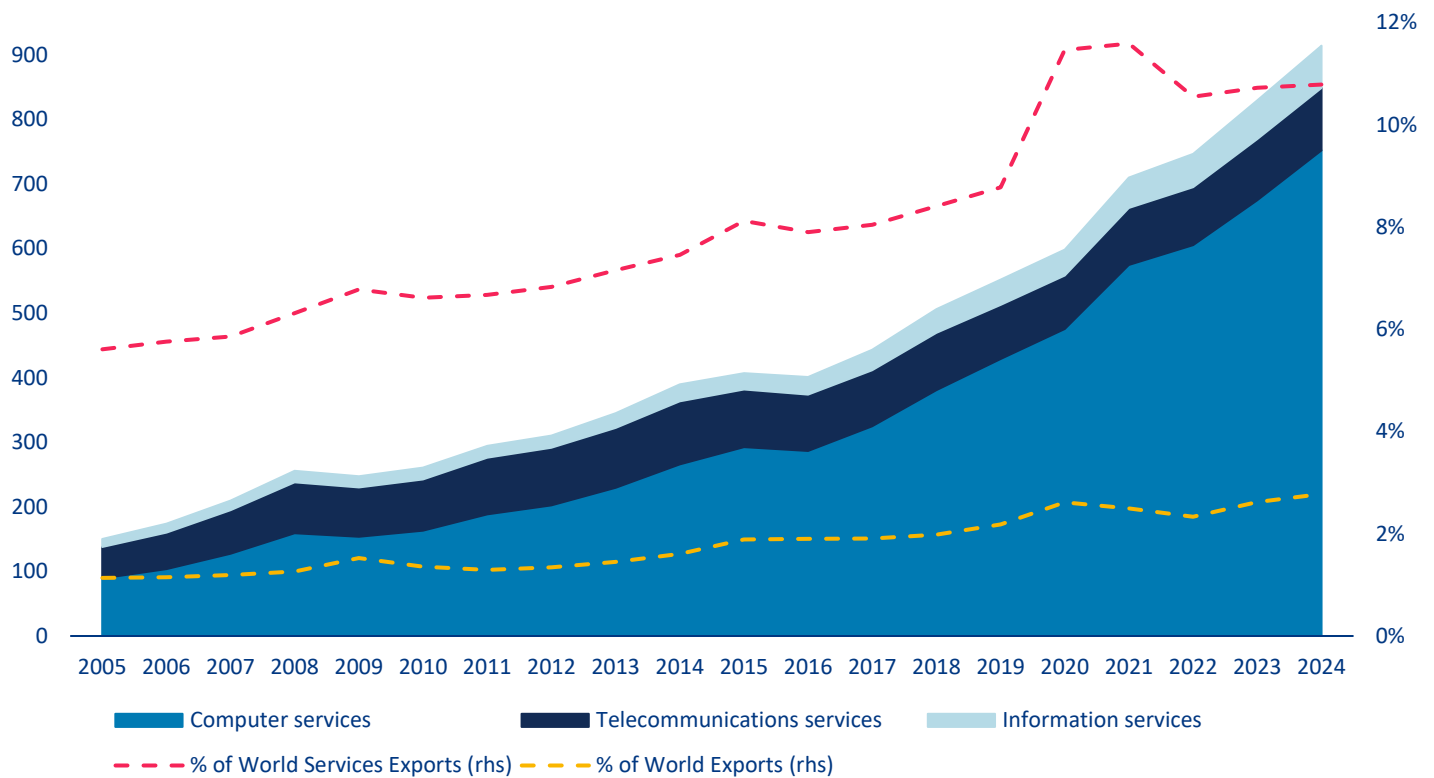
Figure 6: Cross-border global trade in ICT services (in USDbn) and digital data flows (in zettabytes)



Sources: OECD BATIS, Statista, Allianz Research. Notes: 2025* is an interpolated approximation for ICT services as no data is yet available.

The value of international trade in ICT services is approaching USD1trn, growing more rapidly than overall services (11% of global services trade, up from 8% a decade ago). This growth has been driven by rapid expansion since 2016–17, with the sector being highly concentrated in a few hubs. The sector, which is dominated by computer services such as software, consulting, infrastructure and support, reached approximately USD900bn in 2024 (around 3% of the total value of goods and services traded globally, and 11% of the value of services) (Figure 7). Ireland stands out as a key hub for US tech multinationals providing digital services to European core markets. It exported ICT services worth USD173bn in 2024, accounting for

an exceptionally high 60% of its total exports. This is largely due to intra-firm flows linked to datacenters and licensing revenues. Consequently, Ireland has become the world's largest ICT services exporter, surpassing the EU (excluding Ireland) with USD142bn and the US with USD108bn of ICT exports to the rest of the world, respectively. Within the EU (for which we count both extra and intra-EU trade), major ICT exporters include Germany, the Netherlands and France. Meanwhile, India remains a major player, while China lags behind in providing digital services to the world.

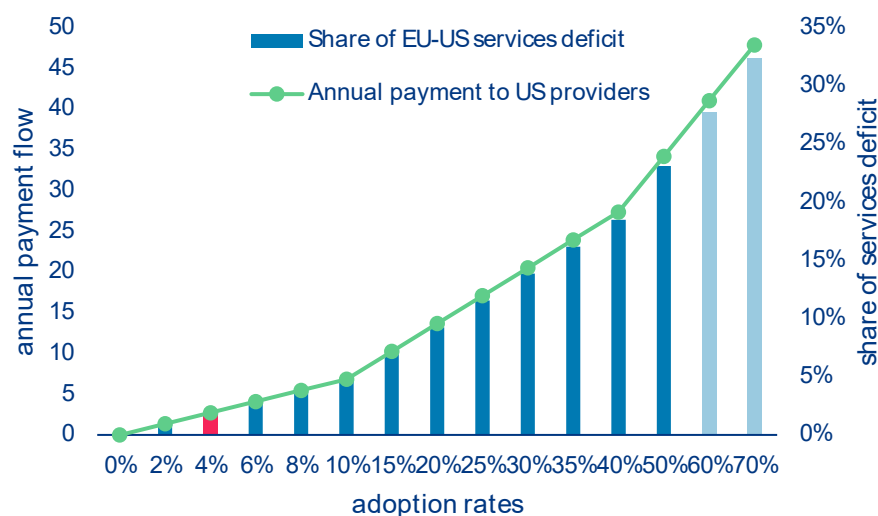
Figure 7: AI-related ICT services trade breakdown (in USD bn)

Sources: OECD, WTO, Allianz Research

AI diffusion will be the next big driver in EU-US services imbalances due to EU dependence on AI model provision. Adopting AI in the Eurozone could significantly increase the transatlantic services imbalance through recurring subscription payments to US AI providers. Assuming an adult population of around 285mn in the Eurozone and a Claude Pro or ChatGPT Plus-style price of EUR20 per month (EUR240 per year, the current standard pricing for individual users), even modest adoption would result in significant cross-border services imports. With an estimated current penetration rate of around 4%, this equates to approximately EUR2.7bn per year in payments to

US providers; already a significant portion of digital services imports, albeit still modest compared to the EUR148bn EU-US services deficit in 2024. Increasing adoption to a 10% penetration rate would raise outflows to approximately EUR6.8bn, and to around EUR17.1bn at a 25% penetration rate (Figure 8). At this point, AI subscriptions alone would be comparable to a tenth of the entire current services deficit. Under a high-adoption scenario in which 50% of adults subscribe, annual payments would total approximately EUR34bn, implying an additional impact of over one-fifth of the current EU-US services deficit, representing a significant increase in US services exports driven purely by AI diffusion.

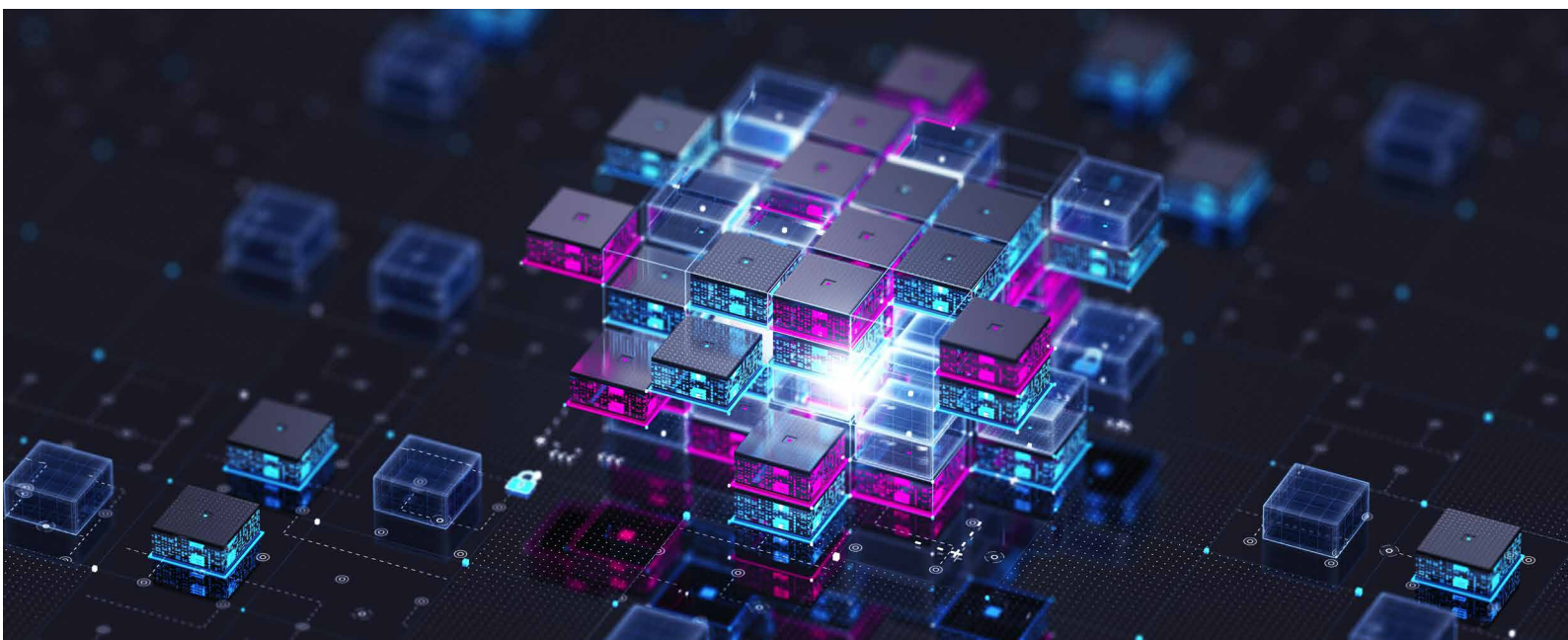
Figure 8: Back-of-the-envelope calculation of annual payment flow from the EU to the US (in EURbn) and share of EU-US services trade deficit (in %) along AI penetration rates in the adult population



Source: Allianz Research. Notes : Assumes an adult European population of 285mn along hypothetical AI adoption rates, a subscription fee of EUR20/month (current standard pricing of ChatGPT Plus or Claude Pro for individual users) and a EU-US services trade deficit of EUR148bn.

More broadly, AI acts as a scalable, recurring channel for importing services to the US economy via software subscriptions, cloud computing, APIs, IP licensing and digital advertising. These flows are intangible, highly scalable and concentrated in US hyperscalers. This matters because the EU's apparent digital surplus is already distorted by profit shifting and US tech revenues booked via Ireland. Stripping this out suggests a structural digital services deficit exceeding EUR100bn. AI accelerates this dynamic further as, unlike infrastructure-heavy digital services, subscriptions

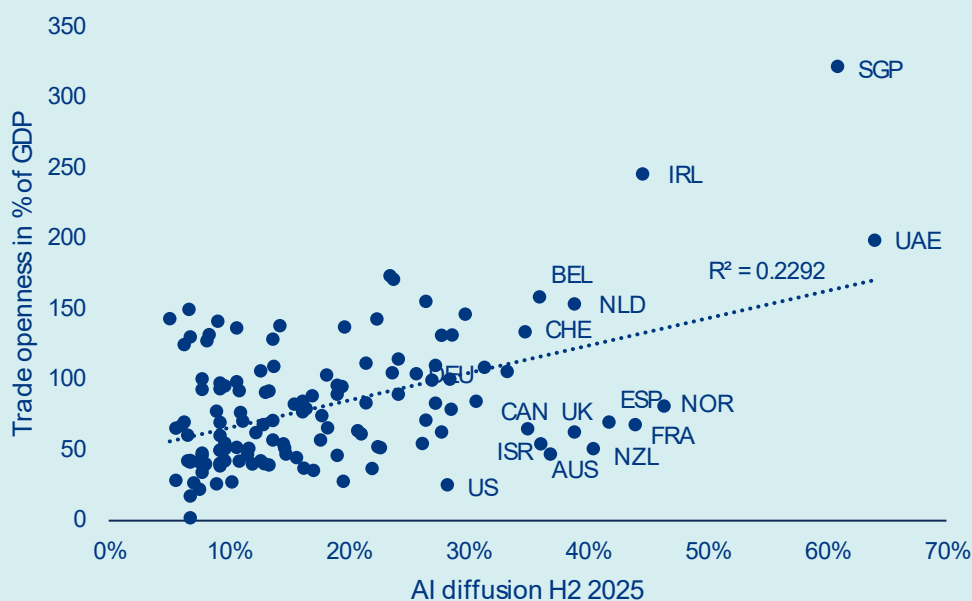
represent near-pure cross-border revenue transfers with limited domestic value creation. Furthermore, the impact is likely underestimated in the simple subscription model: enterprise AI usage (APIs, Copilot integrations and SaaS embedding) already exceeds consumer subscriptions, and productivity effects could also reduce Europe's own services exports. Therefore, unless offset by strong domestic AI platforms, European digital champions or localization of AI infrastructure and billing, AI diffusion risks becoming a persistent structural driver of widening EU-US services imbalances.



Box: Europe struggles with strategic dependencies

Despite its openness to trade, Europe faces uneven AI adoption, restricting its ability to compete with leading rivals. Trade openness explains 23% of the variation in AI diffusion across countries (Figure 9), with highly open economies such as Singapore, the United Arab Emirates and Ireland also recording the highest diffusion rates. But Europe contradicts this trend. Though its economies are highly open and integrated into global value chains, AI adoption remains uneven due to strict data privacy and regulation, limiting the continent's ability to benefit from AI-driven productivity gains. Overall, while AI can significantly boost productivity and growth, its benefits are unlikely to be evenly distributed, potentially increasing inequality between capital and labour, high- and low-skill workers, and larger versus smaller firms.

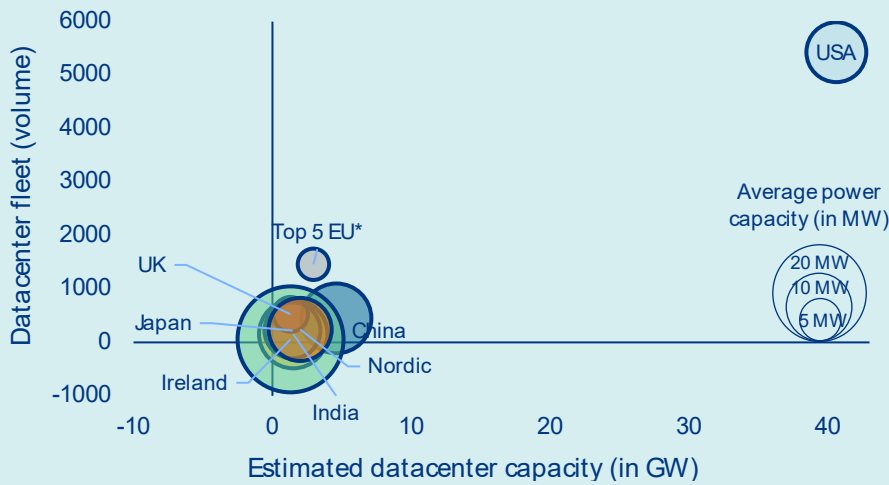
Figure 9: Trade openness (total trade as % of GDP) versus AI diffusion in H2 2025 (in %)



Sources: World Bank, Microsoft, Allianz Research

When it comes to data centers, the US is racing far ahead. With an estimated 40+ GW of installed capacity today (Figure 10), projected to surpass 100 GW by 2030, the US already has the world's largest computing infrastructure. This is underpinned by over USD600bn in capital expenditure committed by US hyperscalers alone this year. By contrast, Europe remains structurally underequipped. Its data-center capacity is heavily concentrated in five core markets (FLAP-D: Frankfurt, London, Amsterdam, Paris and Dublin), accounting for around 60% of the total European capacity. Meanwhile, emerging hubs such as the Nordic countries are only just beginning to attract significant interest, thanks to competitive energy costs and an increasing proportion of renewable energy sources. The challenge for Europe is not merely one of scale, but also of capital and political architecture. Unlike the US, where private hyperscalers flood the market with investment, or China, where the state substitutes for private capital, Europe suffers from a dual deficit: insufficient private investment and an absence of coordinated public policy. The fragmentation of Europe's regulatory landscape prevents the emergence of a clear, common framework dedicated to AI infrastructure, leaving European ambitions dangerously reliant on external resources.

Figure 10: Data-center fleet & estimated capacity (in GW) per region

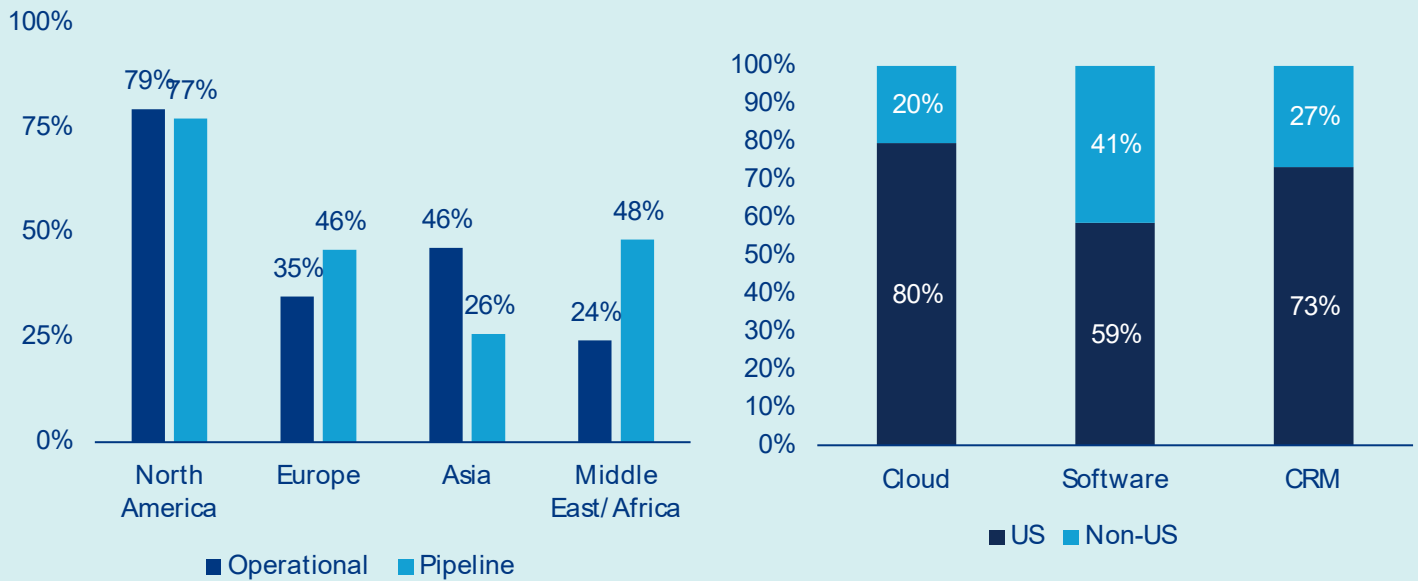


Sources: Cloudscene, Cushman & Wakefield (Datacenter H2 2025 update), Allianz Research. Notes: *Country group including Germany, France, Italy, Spain & Netherlands.

To avoid falling into a technology dependency trap, regulatory and capital constraints need to be cleared. Building data-center infrastructure in Europe is a formidable undertaking, constrained by a unique combination of structural, regulatory and financial hurdles that are unparalleled in the US or China. Limited land availability in urban areas, lengthy and complex permitting processes and increasingly stringent environmental regulations – particularly regarding water consumption and carbon footprint – can extend development timelines significantly, often stretching from planning to commissioning to over four years. Perhaps the most acute bottleneck is power connectivity: grid connection queues in key European markets have grown dramatically, with some operators reporting wait times of five years or more for adequate power access. This is a direct consequence of ageing grid infrastructure and competing demand from industrial and residential users. Financing conditions add another layer of complexity, as the capital-intensive and long-term nature of data-center projects does not align with the risk appetite of many European institutional investors. This is due to the absence of the deep and liquid private credit markets that underpin US hyperscaler investment.

The consequences of this investment gap are already evident in the ownership structure of infrastructure. As of 2025, US hyperscalers accounted for roughly 40% of total data-center operational capacity across Europe, but almost half of the pipeline (Figure 11, left) as large US tech firms are taking profits from weak private capital market in Europe and low competition from local players to consolidate their presence within a key market for them. The digital services layer tells an even starker story: US firms currently capture 80% of European cloud revenue, 59% of enterprise software revenue and 73% of CRM expenditure, leaving non-US players, including European vendors, to compete for the remaining margins (Figure 11, right). This dominance is no accident. US hyperscalers' substantial financial resources enable them to absorb permitting delays, pre-fund grid upgrades and negotiate directly with national governments on energy and land access, structural advantages that no European operator can currently match on an equivalent scale. Their ample cash reserves effectively enable them to fill the void left by the lack of traction in private and public investment on the continent, as they steadily expand their physical and commercial footprint with each passing planning cycle.

Figure 11: Weight of US hyperscalers in datacenter capacity (current & future) across the globe (left) and market share of IT services and software applications revenue in Europe per firm nationality (right)



Sources: CBRE, Synergy Research, European Software & Cyber dependencies report (European Commission, Dec. 2025), Allianz Research. Notes: Data as of Q4 2025.

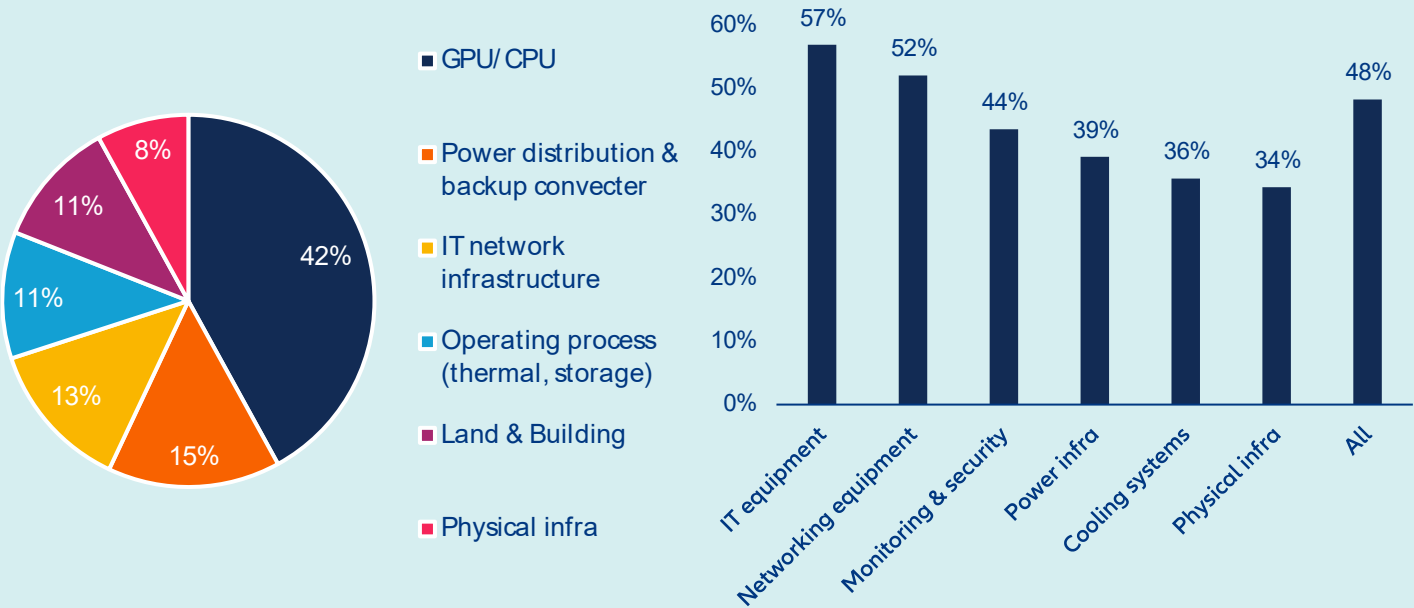
Initiatives such as Mistral AI and sovereign computing capacity projects in France and Sweden are promising, albeit modest, counterweights. Without decisive, coordinated action from Europe, including the mobilization of sovereign funds and development banks, as well as the establishment of a harmonized permitting framework, the continent risks becoming dependent on foreign infrastructure providers, losing not just market share, but also strategic autonomy over the digital backbone of its economy.

Strong industrial-linked specialization and regulatory constraints undermine European ambitions to build up a domestic semiconductor ecosystem and take part to AI race. Historically, European industrial strategy oriented domestic semiconductor production toward application-specific and industrial-grade chips, well suited to its automotive and manufacturing base but leaving the continent without meaningful capacity in the advanced nodes where AI demand is now concentrated. In the meantime, the relatively small weight of the ICT industry in Europe (less of 1% of GVA) but also the absence of a large domestic hyperscaler with ample cash reserves like in the US meant there was never sufficient demand to justify leading-edge foundry investment at scale, a gap that policy has failed to compensate for. Regulatory complexity and slow subsidy disbursement have further deterred the large infrastructure commitments that catching up would require. The cancellation last year of Intel's plan to build a foundry in Magdeburg is a good example. Underlying both is a structural weakness in private capital markets: With only 6% of global AI funding flowing to European firms, the continent lacks.

Finally, Europe's data-center ambitions depend on fragile, Asia-heavy supply chains. Building AI-dedicated data-centers is, first and foremost, a hardware problem, and Europe currently owns very little of it. GPUs and CPUs alone account for around 42% of the total capital cost of a 1GW AI data center (Figure 12, left), and this expenditure largely flows outside of Europe: non-EU import ratios exceed 57% for IT equipment and 52% for networking gear (Figure 12, right). Five Asian economies (Taiwan, China, South Korea, Malaysia and Vietnam) collectively supply around two-thirds of this expenditure. Taiwan's near-monopoly on sub-3nm logic chips and South Korea's dominance in memory represent structural dependencies, not cyclical ones, and cannot be meaningfully substituted in the short term. The next tier of components – power distribution, IT network infrastructure and cooling systems, each representing 10–15%

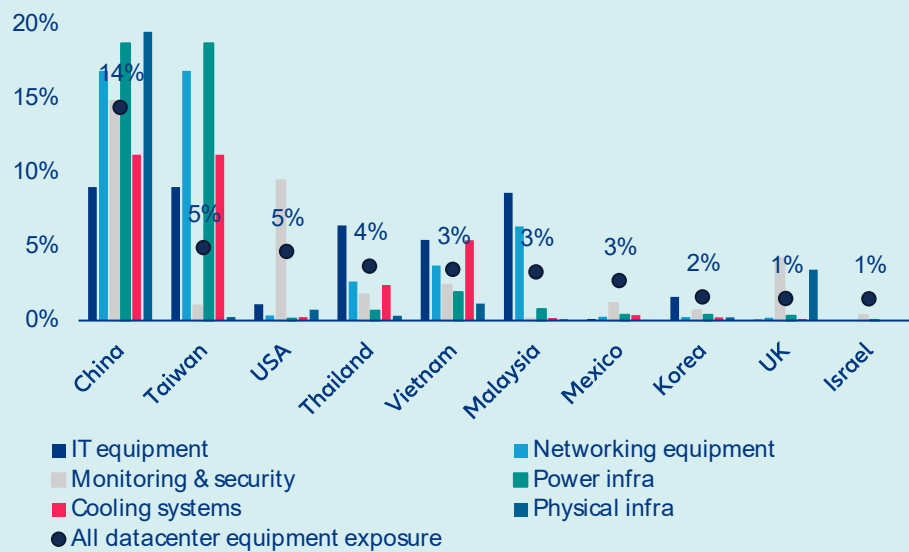
of datacenter value – shows only modestly better numbers: monitoring and security sits at 44% non-EU exposure, power infrastructure at 39%, and cooling systems at 36%. Even physical infrastructure, the least technology-intensive segment, runs at 34% external reliance. At the country level, China stands out with a total exposure share of 14% across all datacenter equipment categories (Figure 13), nearly three times the weight of both the US and Taiwan, reflecting its dominance in advanced electronics and physical and cooling inputs. Vietnam primarily supplies computers and electronic transmission devices, while Turkey is a notable partner for cooling systems. Despite its influence on the software and services stack through its hyperscalers, the US has a more modest hardware footprint concentrated in IT and networking. Overall, Europe has little domestic manufacturing depth in advanced electronics and limited battery and power storage capacity. Its supply chain is also acutely exposed to two live geopolitical risks: renewed trade tensions between Washington and Beijing (currently on pause but far from resolved) and any energy disruption in Asia stemming from Middle East instability. For a continent with sovereign AI ambitions, this is a precarious foundation.

Figure 12: Estimated cost breakdown of an AI-dedicated 1GW capacity data center* (left) and share of non-EU imports ratio for critical materials used in AI data-center infrastructure construction, 2025 (right)



Sources: Bernstein, Eurostat, Allianz Research.** GPU share based on Nvidia GB200 cost.

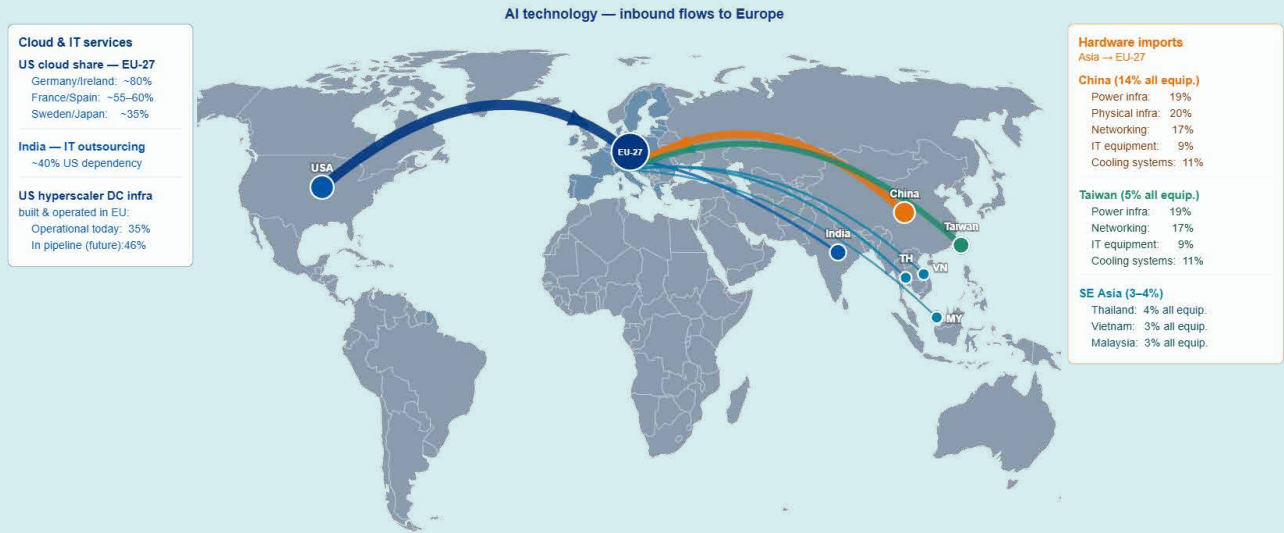
Figure 13: Top 10 non-EU suppliers of critical materials for AI data-center infrastructure construction



Sources: Eurostat, Allianz Research

This structural dependency has direct macroeconomic consequences. Europe’s limited control over AI infrastructure and its reliance on imported hardware and foreign hyperscalers imply that a growing share of AI-related value added will be captured abroad, dampening domestic productivity multipliers and GDP gains from AI adoption. At the same time, higher data-center build costs, prolonged permitting cycles and constrained grid access raise the effective cost of AI deployment in Europe, slowing diffusion into the broader economy. This combination risks a dual drag: weaker productivity growth relative to the US and higher unit costs for digital and industrial transformation. Over time, it could also widen the transatlantic gap in investment intensity, reinforce Europe’s structural current account outflows in digital services and increase vulnerability to external price-setting in critical digital inputs.

Figure 14: Top 10 non-EU suppliers of critical materials for AI data-center infrastructure construction



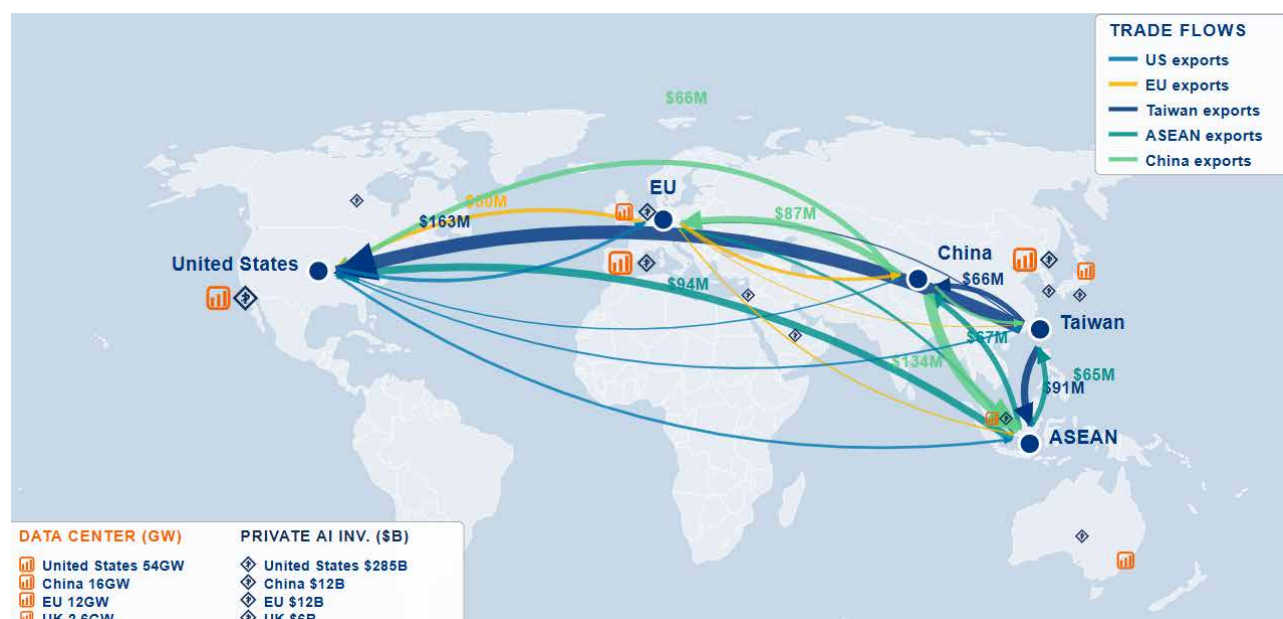
Sources: Allianz Research

The world is flat: AI reconfigures the geography of trade and dependencies

The AI supply chain is becoming more geographically distributed in assembly and packaging, but more concentrated and fragile at the technological frontier, with diversification creating the appearance of resilience while reinforcing dependency on a small number of critical nodes. The most critical parts of the AI supply chain (advanced chips, high-bandwidth memory, and the equipment to produce them) remain highly concentrated in a small set of countries, especially Taiwan, South Korea, and the Netherlands, and this

concentration has intensified since 2022 due to export controls and the AI buildout. This creates systemic single points of potential failure in the global AI system. Partial diversification took place only in the middle layers of the supply chain. Server assembly, packaging and networking are shifting geographically under the influence of US policy and export controls, with gains for Vietnam, Mexico, Japan, Taiwan and parts of Eastern Europe. But this friendshoring is still limited in scale and far from replacing established hubs.

Figure 15: AI supply chain concentrated in several key geographies

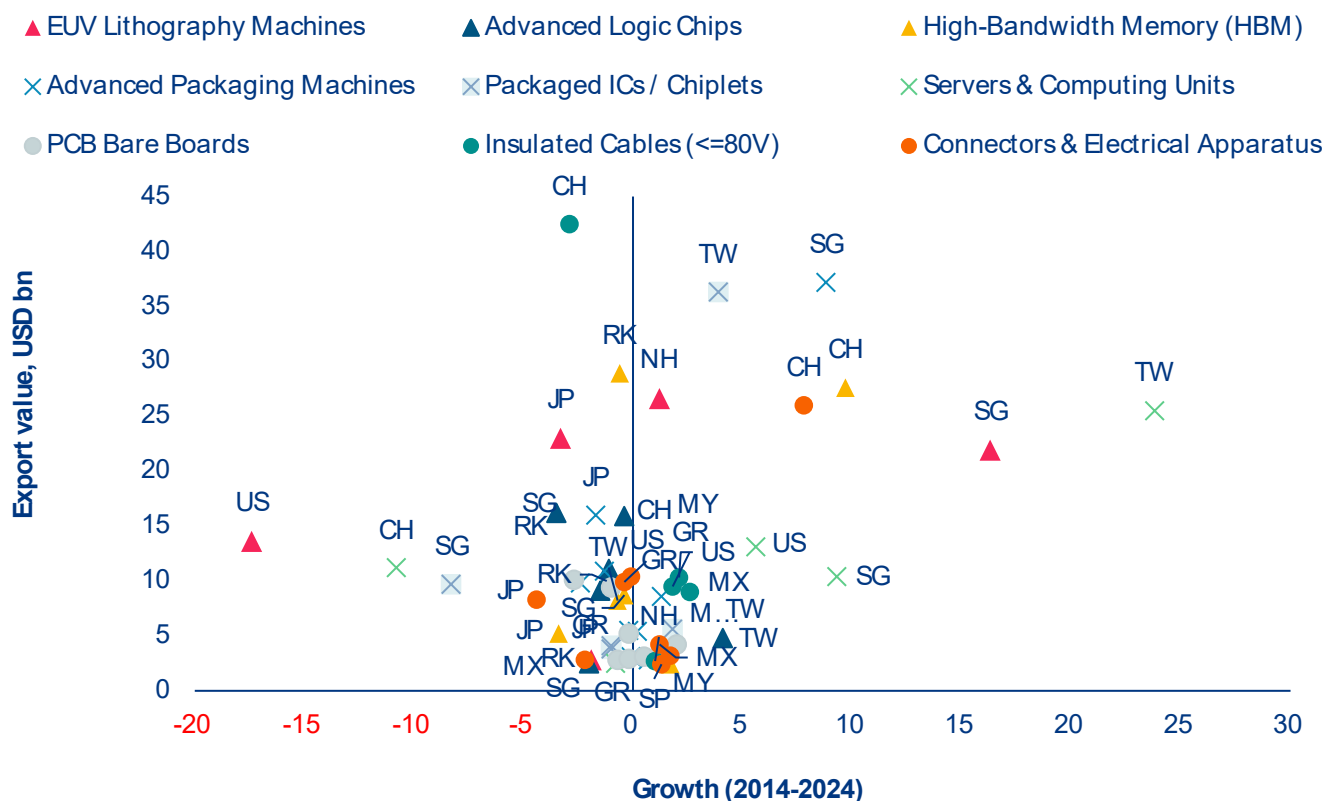


Sources: Stanford HAI 2026, Synergy Research, Trade Map, Allianz Research

Despite friendshoring efforts, diversification away from China remains limited. A decade of diversification investment has produced incremental shifts in specific sub-segments but no geography has emerged as a credible large-scale alternative. Vietnam and Mexico have attracted investment and attention as potential alternative bases for electronics assembly, and there are signs of incremental share gains in specific sub-segments, but the scale of Chinese capacity in these

categories means that meaningful diversification remains a multi-year undertaking for most supply chains. China continues to dominate low-value, high-volume components that are essential but underappreciated (Figure 16). Diversification away from China in these segments is slow, meaning supply chains remain exposed to geopolitical shocks despite rhetoric of decoupling. The risk here is overexposure to Chinese production in case of a new geopolitical shock or policy change in Beijing.

Figure 16: Concentration towards Asia in the higher value added parts of the AI supply chain, while China further integrates low-value, high-volume



Sources: UNComtrade, Allianz Research. Note: Shapes of the markers correspond to different sections of AI-supply chain: triangle = higher value, cross = middle value, circle = lower value.

Hidden upstream vulnerabilities emerge with the current conflict in the Middle East. Shortages of key inputs for semiconductor production, notably helium, would impact the most leading-edge fabs, which could translate to losses above USD10bn a month. Qatar accounts for a third of global production of helium, an upstream input of the AI supply chain that is also almost entirely shipped through Gulf terminals. While the inflationary cost impact is relatively small on semiconductor production compared to the rest of the supply chain, the production shortage would be much more impactful, with an asymmetric impact. Semiconductor producers in South Korea are most at risk, given their exposure of imports from Qatar, and early estimates suggest 4 to 6 months of reserves. However, everything hinges on gas storage in non-producing countries. The US holds the largest known reserves and has historically been the dominant producer, but its federal helium reserve has been

progressively wound down under legislation mandating its disposal. Japan, one of the world's largest helium consumers given its electronics and semiconductor manufacturing base, holds no domestic reserves and is almost entirely import-dependent. New countries could increasingly offer their helium reserves to the market, including Russia, Australia and Algeria, as well as Tanzania, which holds what may be one of the world's largest undeveloped helium deposits in the Rift Valley, though it remains years away from commercial production.

US trade policy has reinforced rather than fundamentally reshaped semiconductor supply chains. China now accounts for just 0.7% of US semiconductor imports in tariffed categories, subject to a 25% tariff, while production (or rerouting) shifted toward Taiwan, Vietnam, Malaysia, Thailand and Mexico. Despite headline tariff threats of up to 46% on Vietnam and 36% on Thailand, semiconductors remain largely exempt from new trade measures, with

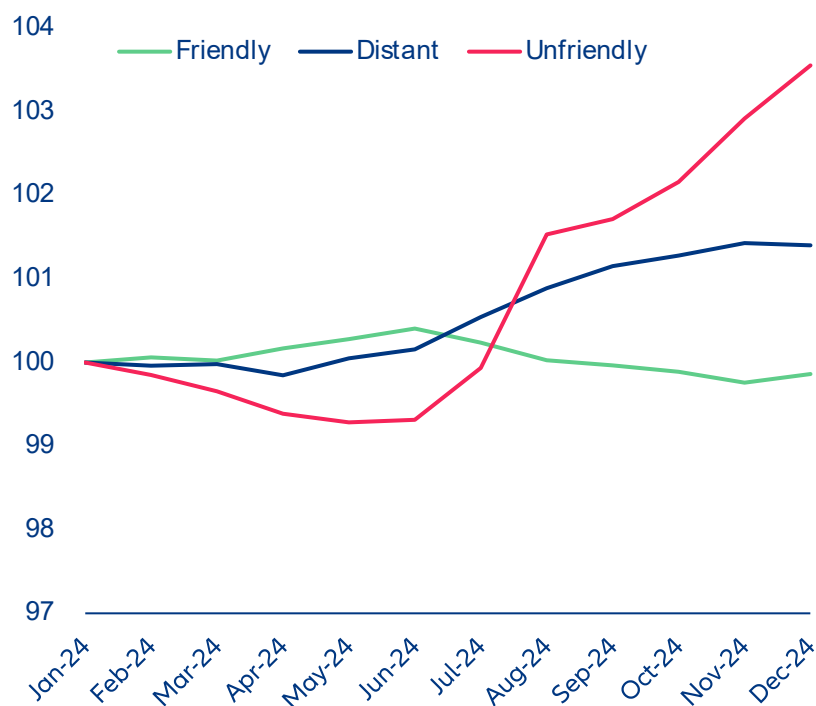
effective tariff rates near zero for most major suppliers and around 1% for Malaysia. As a result, the post-2018 semiconductor supply chain continues to operate under near free-trade conditions. The more ambitious reshoring effort has been the CHIPS Act which redirected USD52bn toward domestic manufacturing and research. Its investment allocation has been substantial, but the production results remain limited and represent only a small fraction of Taiwan's output.

Geopolitics is changing where semiconductors are manufactured rather than with whom they are traded.

The commercial logic of the supply chain remains largely intact: Taiwan, South Korea and the broader Southeast Asian production ecosystem continue to sell to every major economy without any need of political alignment. This creates a form of structural interdependence that cuts across geopolitical blocs in ways that policy narratives tend to obscure: China remains deeply integrated into the commercial semiconductor market for everything below the frontier of advanced logic and memory, and continues to source the vast majority of its chip requirements from

the same Taiwanese and Korean suppliers that serve US hyperscalers. The meaningful exceptions are narrow but strategically significant, US export controls have effectively excluded China from sub-7 nanometre logic, high-bandwidth memory for AI applications, and the lithography equipment required to produce either domestically, creating a two-tier market in which the frontier is controlled and the mainstream is not. This commercial logic is increasingly visible in the trade data. For most goods, trade has fragmented along political lines, rising between geopolitically „friendly“ countries and declining between „unfriendly“ ones, and until recently AI-related goods followed the same pattern. Since 2024, however, this trend has inflected, with trade in AI-related goods between „unfriendly“ countries stabilising and subsequently rising, even as broad goods trade between the same countries has continued to fragment (Figure 17). This divergence suggests that, outside the narrow frontier segment subject to export controls, commercial interdependence in the AI supply chain is proving more resilient than political alignment.

Figure 17: Growth of AI-related goods between unfriendly countries experiences the largest increase during 2024

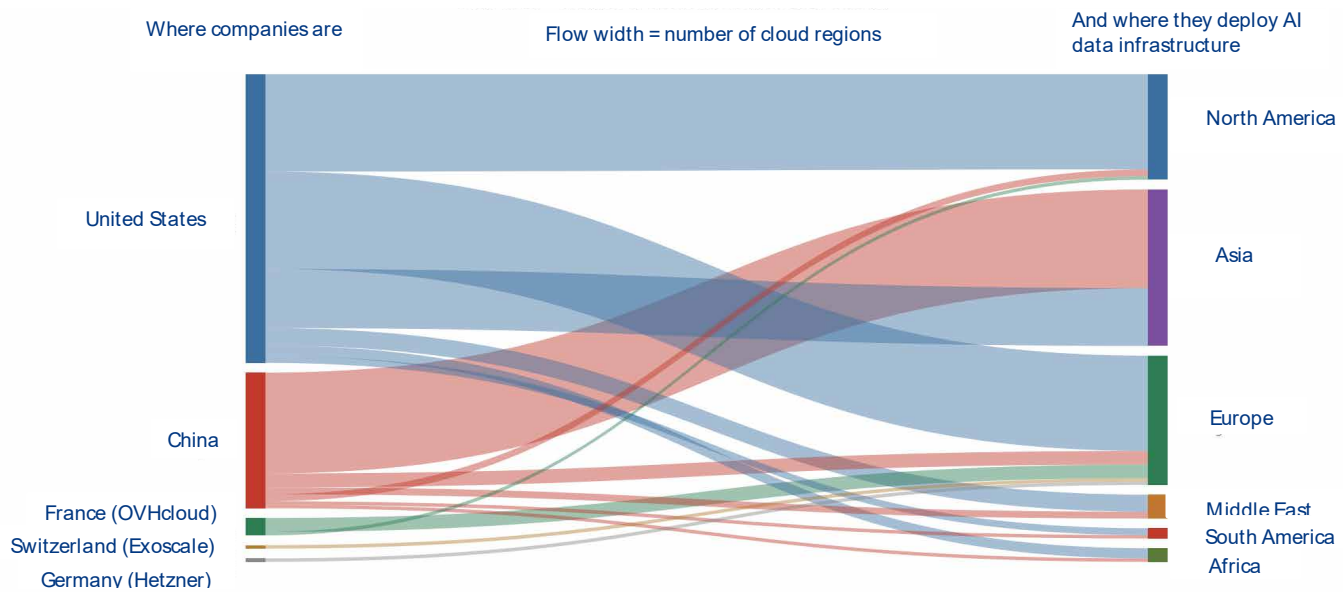


Sources: UN Comtrade, Allianz Research

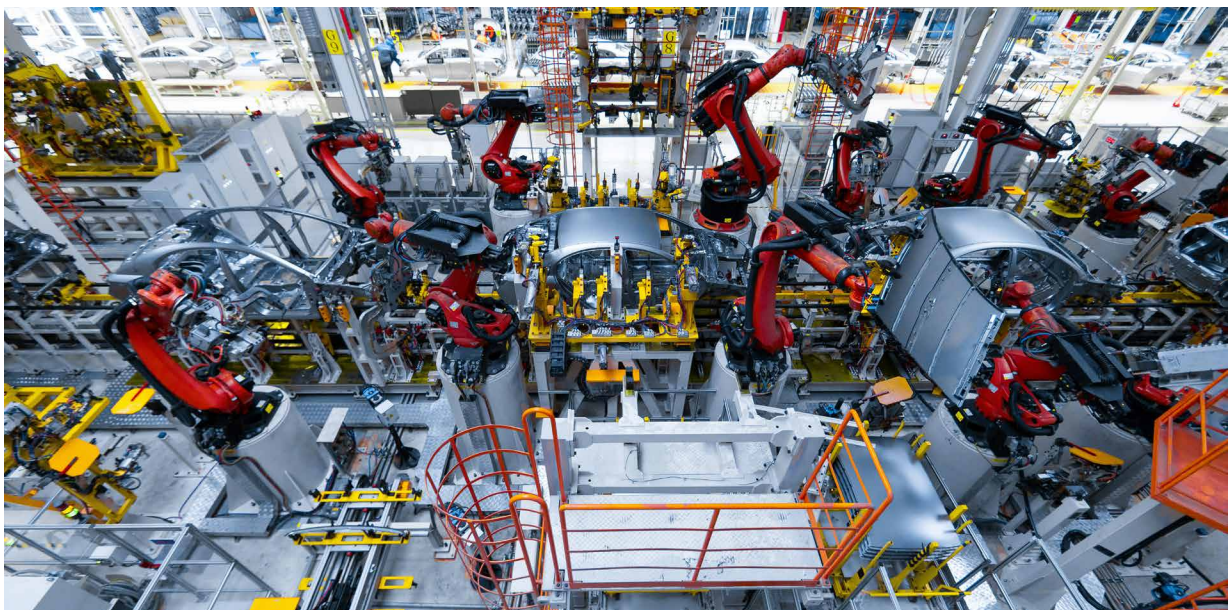
AI-related production and infrastructure provision will remain concentrated. Geopolitics is reshaping semiconductor investment globally, but despite massive subsidy programs across the US, Europe, China, Japan and India, the likely outcome is not full supply security but a more geographically distributed industry that remains highly dependent on a small number of leading-edge producers for the most critical AI and defense technologies. The geographic concentration of large-scale AI-capable data centers is becoming a new source of geopolitical power, as control over compute infrastructure increasingly determines which economies

can develop, export and benefit from AI services. The US dominates the global AI infrastructure landscape in the Western hemisphere, while China dominates Asia (Figure 18). Large parts of Africa and Latin America remain largely excluded from the compute capacity and so far depend half-half on the US and China with no dominance decided yet. This growing concentration risks reinforcing technological dependence, widening digital divides and shifting economic and strategic influence toward the countries that control cloud infrastructure, data flows and AI platforms.

Figure 18: Distribution of headquarters of AI companies and AI data infrastructure deployment



Sources: WTO (2025), Lehdonvirta, Wu and Hawkins (2024), Allianz Research



The race for AI dominance is increasingly also being fought through industrial policy

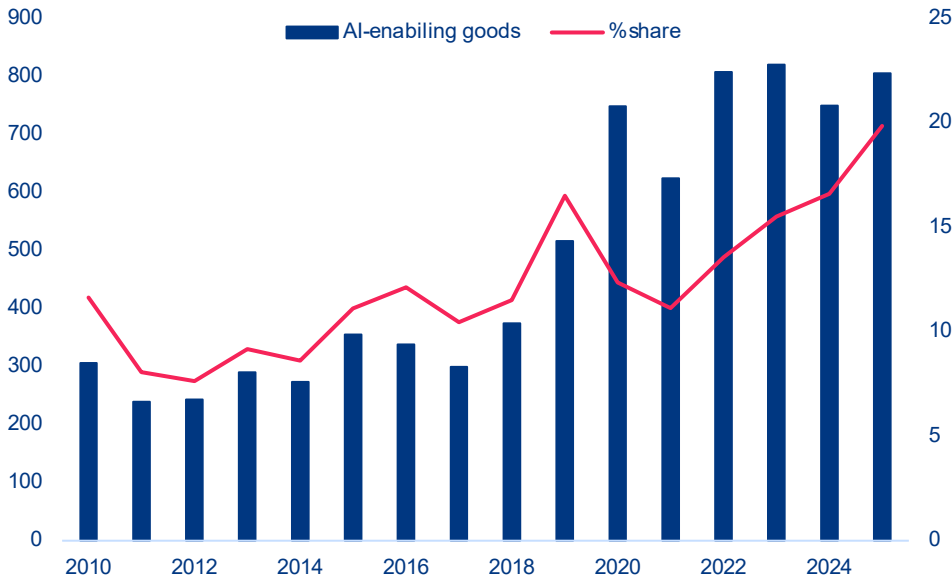
Tariffs on AI-enabling products remain low. Industrial policy and trade rules are becoming central to competition in the field of AI, influencing access to critical minerals, semiconductors, cloud infrastructure, IT services and cross-border data flows, all of which are essential for the development and deployment of AI. Open and predictable trade regimes support innovation, productivity and the integration of AI into global value chains. However, these tools are increasingly being weaponized in an increasingly geopolitical world. Despite rising tensions, tariffs on AI-enabling products remain relatively low. Worldwide, most-favored-nation (MFN) applied tariffs on AI-related goods fell from 5.6% in 2015 to 2.8% in 2025, compared with 7.8% on manufacturing goods overall. The US maintains an average tariff of 1.4%, the EU of 2.0%, the UK of 1.3%, and EFTA countries effectively zero, while China applies the highest average tariff in the group at 3.5%. Overall, AI-related goods continue to be traded under relatively open conditions, reflecting the strong interdependence of global AI supply chains and the strategic importance of maintaining access to computing infrastructure, digital technologies and knowledge flows.

Non-tariff measures (NTMs) on AI-enabling goods account for a fifth of all harmful NTMs in force.

Over the last decade, NTMs rose from 355 in 2015 to 805 in 2025 (Figure 19) with their share of the total constantly increasing since the deployment of AI to the general public in 2022. These measures, including export restrictions and financial assistance, increasingly reflect efforts to protect technological leadership and secure strategic industries. China has been the top player in implementing trade restrictions with a peak 2020, while the US has more recently picked up since 2022 overtaking China only in 2025. The US and China implemented the most restrictions in 2025 with 208 new restrictions from the US and 159 from China. At the same time, the US is the most affected nation as well, followed by Germany, Malaysia, Thailand, China, the Netherlands and Japan. While the most common new measure implemented in 2025 were import tariffs, this has shifted clearly from financial assistance in foreign markets which was top of the list since 2011 and peaked in 2020 with 431 new measures, followed by tariffs and trade finance and the occasional export

ban on AI-enabling goods. While such restrictions may slow technology diffusion in the short term, they also carry significant costs, are difficult to enforce and can accelerate innovation and diversification elsewhere, limiting their long-term effectiveness unless backed by strong domestic industrial capabilities and coordinated policy frameworks.

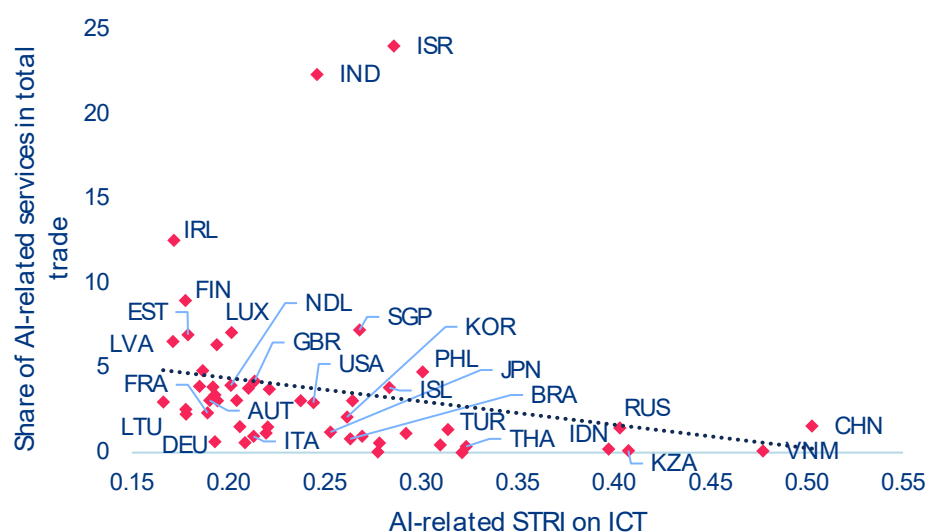
Figure 19: NTMs on AI-enabling goods, number of cumulative measures implemented (left) and percentage share (right)



Sources: GTA, Allianz Research

Trade in AI-enabling services is limited by restrictive regulations. The trade of AI-related services is increasingly being shaped by restrictions on digital services and cross-border data flows. These restrictions are becoming a key determinant of global AI market share. Although AI is making services such as finance, software and telecommunications more tradable, these sectors are still limited by foreign equity restrictions, licensing regulations, localization requirements and barriers to cross-border delivery. Economies with more restrictive ICT and digital trade regimes capture smaller shares of AI-enabled services exports. In fact, ICT trade restrictiveness explains around 5% of a country's share of AI-related services trade (Figure 20). There is significant regulatory fragmentation: the OECD Digital Services Trade Restrictiveness Index (STRI) records 78 economies requiring a local presence out of 129 countries covered

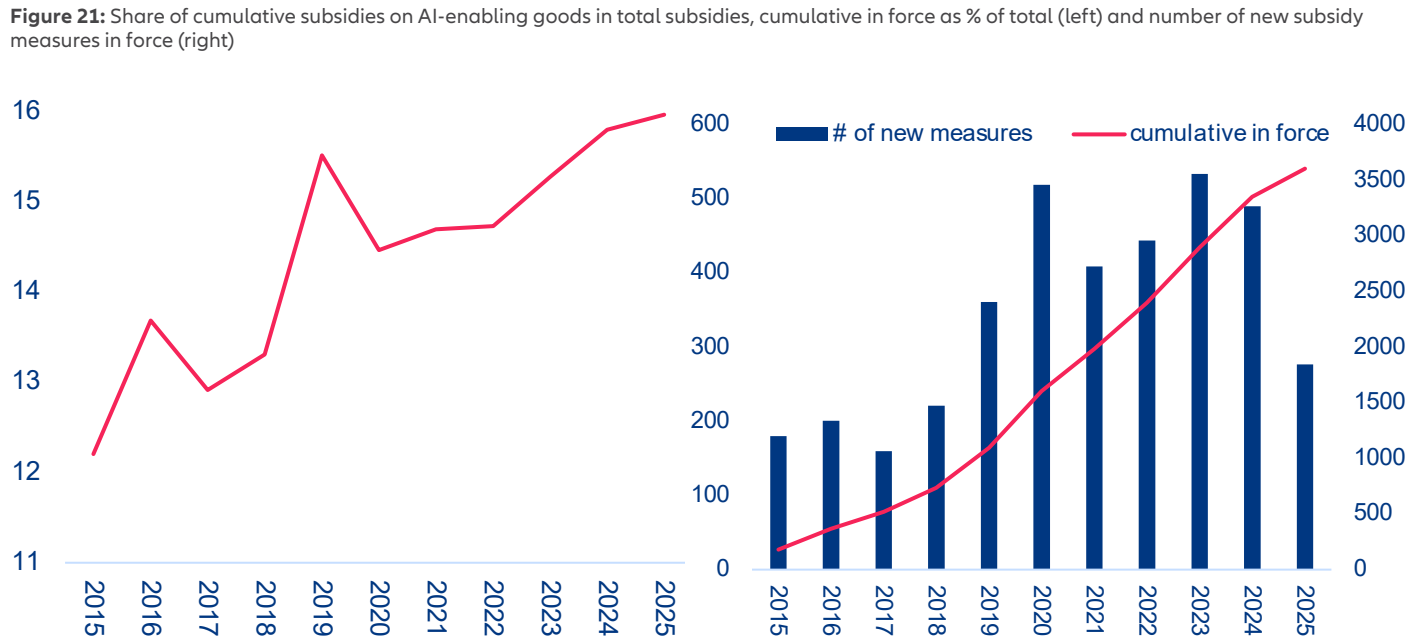
and 46 imposing data localization rules in 2025. The number of countries requiring data to be processed or stored locally nearly doubled in the last decade. These barriers disproportionately affect AI-intensive sectors such as telecommunications, banking, insurance and software, where cross-border supply and commercial presence remain tightly regulated. Furthermore, fragmented data governance amplifies divergence by raising compliance costs, limiting access to cloud infrastructure and global datasets, and weakening participation in AI innovation networks, particularly for smaller economies. While some regulation is necessary for trust and security, openness to digital trade and data flows is now a key factor in achieving AI competitiveness. Economies with restrictive regimes are increasingly failing to capture AI-driven growth.

Figure 20: AI-related services share in total trade (in %) to AI-related services trade restrictiveness index

Sources: OCED TIVA, OECD STRI, Allianz Research. Notes: Sector shares are as reported in OECD TiVA VA exports for 2022. AI-related sectors are ICT services. Services restrictiveness based on 2025 data only on ICT services sectors.

The growing financial support for AI is exacerbating global technological disparities. Higher-income economies are implementing large-scale AI strategies with the backing of fiscal resources, industrial policy, and targeted support for domestic champions. In contrast, lower-income economies are hampered by fiscal and infrastructure constraints, which limit their ability to take comparable action. This is widening the structural gaps in innovation and trade capacity. The scale of AI-related subsidies is also expanding: the proportion of AI-enabling goods covered by subsidies increased from 12% in 2015 to almost 16% in 2025 (Figure 21, left), and the total number of AI-related subsidy measures now exceeds 3,600 worldwide (Figure 21, right). Although the introduction of new measures slowed in 2025, the overall stock continues to rise. China is in the lead by a large margin, with its subsidy coverage of AI-related goods trade increasing from 14.5% in 2015 to 55% in 2020 and continuing to expand nearly twofold since then. It is followed by South Korea, Malaysia, the US and Japan as the top five AI-related subsidy measure players to expand or protect their position in the AI value chain. These measures primarily target critical materials, advanced semiconductors, AI accelerators (GPUs), semiconductor manufacturing equipment, optical fibres and data-centre components. They are funded through grants, tax credits, loans, state-backed equity injections, electricity subsidies, preferential financing and public support for R&D and infrastructure.

Initiatives to develop and regulate AI are expanding rapidly, but they are also creating a fragmented policy landscape. Regulation now extends beyond algorithms to encompass the entire physical and digital infrastructure of AI systems, including data centers, semiconductor supply chains, energy-intensive computing facilities, and cross-border data flows that facilitate AI-driven trade. Governments are no longer merely shaping trade in AI-enabling inputs; they are also actively building industrial ecosystems to secure positions in AI value chains. 23% of countries have set up national AI governance bodies (32 in Europe, 10 in Asia, four in Africa, five in North America, four in Central and South America, and two in Oceania), and 28% have adopted AI regulations, guidelines or standards. These frameworks are increasingly integrating regulation, intellectual property, competition policy, skills development, labor markets and infrastructure investment in order to foster innovation while managing concentration risks. Regional approaches differ significantly. North America has traditionally relied on lighter regulatory frameworks, allowing faster innovation with fewer constraints. China's model is more state-driven, with AI governance closely tied to industrial policy and oversight. Europe occupies a middle ground: while regulation strengthens trust, transparency and adoption, key factors in scaling AI, it also introduces



Sources: GTA, Allianz Research

compliance and security requirements that can slow innovation cycles, even though many processes are becoming increasingly automatable. Overall, however, regulation is not currently a binding constraint in Europe. AI governance is emerging as a global trade regime in its own right, which could further reinforce the existing dominance of a select few. At the multilateral level (Table 1), frameworks such as the OECD AI Principles and the G7 Hiroshima Process promote interoperable standards and trusted data flows in order to reduce friction in the trade of AI services. At the same time, national strategies are becoming increasingly geopolitical. The US ‘Pax Silica’ approach links AI directly to trade security through export controls, investment screening, and domestic incentives for the production of chips and computing power.

Meanwhile, China’s Digital Silk Road initiative integrates AI infrastructure exports, such as datacenters and cloud platforms, into broader trade and connectivity strategies. Middle Eastern economies are leveraging their energy abundance and sovereign wealth to establish themselves as global compute hubs with AI-friendly regulatory environments. Other initiatives, including the UNESCO AI Ethics Framework and the UN Global Digital Compact, seek to establish shared norms for data governance, model oversight and infrastructure efficiency. Together, these overlapping regimes are increasingly determining where AI is produced, how it is governed and how AI-enabled goods and services flow globally, effectively turning AI governance into an emerging trade regime in its own right.

Table 1: Major AI-trade-related international policy initiatives

Initiative	Lead Actors	Core Focus Areas	Trade Relevant Impacts
OECD AI Principles (2023)	OECD countries	Responsible AI, transparency, data governance	Supports interoperable standards for cross border data flows and cloud services; reduces regulatory fragmentation.
G20 AI Principles (2019)	G20 economies	Human centric AI, innovation, digital economy	Encourages harmonized data governance approaches that facilitate AI enabled services trade.
UNESCO AI Ethics (2021)	UNESCO members	Ethical standards, inclusiveness, data rights	Shapes global norms for data handling, influencing access to datasets and model training resources.
G7 Hiroshima Process & AI Code of Conduct (2023)	G7	Frontier model safety, AI supply chain security	Promotes common benchmarks that support trusted AI supply chains; potential trade conditions for high risk AI systems.
Bletchley Declaration (2023)	28 countries incl. US, UK, EU, China	Safety of frontier models	Introduces shared risk management expectations, influencing compliance requirements for cross border AI deployment.
UN General Assembly AI Resolution (2024)	UN member states	Safe, inclusive global AI governance	Calls for capacity building in digital infrastructure; facilitates AI adoption in developing countries.
Council of Europe Framework Convention on AI (2023)	Council of Europe	Human rights, democratic governance of AI	Sets obligations that may become preconditions for AI trade with EU markets.
International Network of AI Safety Institutes (2024)	Multinational	Safety research, model evaluations	Could lead to standardized audits that function as “technical passports” for AI model trade.
UN AI Advisory Body Final Report (2024)	UN	Global governance mechanisms	Recommends infrastructure investment frameworks and compute governance strategies.
UN Global Digital Compact (2024)	UN	Digital cooperation, data flows, public infrastructure	Promotes open digital markets and trusted cross border data exchange.
International Scientific Report on Advanced AI Safety (2024)	Multilateral science groups	Scientific evaluation of AI risks	Provides technical benchmarks and safety thresholds that may condition AI exportability.
US Pax Silicia Strategy	United States	Chip controls, AI supply chain security, cloud regulation	Reshapes global trade in semiconductors, cloud capacity, and advanced models; creates new dependencies.
China’s Digital Silk Road	China	Digital infrastructure, cloud, AI services exports	Embeds AI capabilities into BRI corridors, strengthening China centric digital trade routes.
Middle East AI Hub Strategy	UAE, Saudi Arabia, Qatar	Compute clusters, cloud zones, energy powered AI	Turns the region into a global exporter of compute and AI services; attracts multinational AI firms via regulatory free zones.
OECD Cross Border Data Flows	OECD	Data governance, interoperability, trust frameworks	Provides stock taking of global data flow policies; supports common G7/G20 approaches for enabling AI driven digital trade.
OECD Going Digital: Trade & Data Flows	OECD	Policy landscape for international data flows	Identifies how data regulations affect international trade and AI-enabled services; informs trade negotiators.
OECD aligned research on AI & digital trade rules	OECD affiliates, legal scholars	Next gen rules for AI, digital trade, cross border data	Explores regulatory divergence, national security restrictions, and future international digital trade agreements.

A photograph showing a group of diverse hands of various skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text "Our team" is overlaid in the center, with "Our" in white and "team" in orange.

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Allianz Research | 19 May 2026

Transatlantic spread – from monetary signal to fiscal mirror

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In Summary

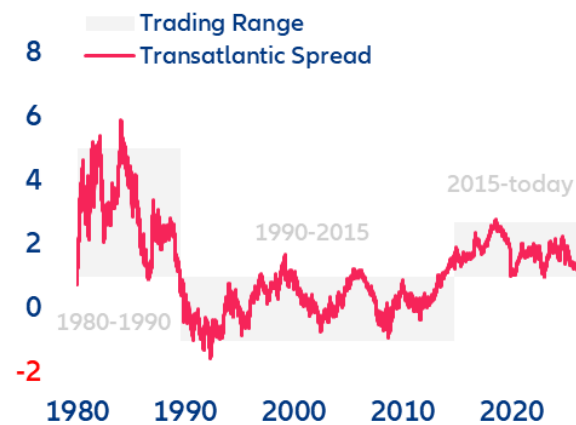
- **Is one of the world's most important pricing signals entering a new era?** Since 2015, the 10-year spread between US Treasury yields and Eurozone German Bunds has ranged between 100bps and 270bps. Since early 2025, pushed first by Germany's fiscal pivot and then by converging rate expectations (hawkish ECB vs Fed easing bias), the 10y US–Bund spread has narrowed by roughly 90bps. If this continues, the transatlantic spread could finally break through the lower trading range that has held for more than a decade.
- **The latest bond sell-off suggests that the convergence trade has run its course: We expect a gradual reversal rather than a new tighter-for-longer regime.** Over the last few weeks, the spread has stopped narrowing and widened to around 140bps. In the short run (next 6–12 months), the move should remain modest (+20bps) and still driven by rate expectations as markets reprice a less hawkish ECB against a Fed that stays on hold longer than consensus expects or even proceeds to hike.
- **In the medium term (3–5 years), the transatlantic drivers will shift decisively from rate expectations to risk premia.** We see the US neutral rate rising by 10–20bps on stronger AI-driven productivity gains while the Eurozone neutral rate drifts 10bps lower, consistent with a historical 20–50bps widening of the real-rate spread. On top of that, the US Treasury term premium looks underpriced relative to Bunds: the US convenience yield has flipped from -30bps to +20bps as Treasury supply erodes the safety premium, while the Bund retains a -40bps convenience yield anchored in collateral scarcity. The ECB's faster QT has already cheapened Bund duration by 35bps versus 10bps for Treasuries and we expect that gap to close. Adding a structural +10bps from the OIS/swap-spread channel, we see the spread widening by ~75bps to around 200bps.
- **Ultimately, the transatlantic spread is evolving from a pure monetary-cycle gauge to a barometer of fiscal credibility and sovereign liquidity.** For investors, this means the transatlantic spread should be monitored not only through the lens of rate differentials, but equally as a mirror of fiscal sustainability and sovereign market liquidity. For duration positioning, this means active positioning in Bunds versus US Treasuries not only in terms of rate expectations but also where the term premium repricing will be most pronounced (longer maturities). In addition, the marginal buyer of Treasuries will increasingly demand a higher term premium, supporting a steeper US curve and a structurally weaker dollar against the euro. For corporate issuers, the EUR–USD funding-cost differential is expected to widen, favoring euro issuance (reverse Yankee issuance) that may support EUR investment-grade bonds over their US counterparts.

The transatlantic spread reaches the lower bound of its long-term trading range

Is transatlantic spread entering a new lower-for-longer era? The transatlantic spread, which refers to the differential between US Treasury yields and Eurozone benchmark yields (German Bunds), is among the most important pricing signals in global financial markets. It influences two of the largest fixed-income markets and moves the EURUSD, the most important currency pair in global trade and financial transactions.

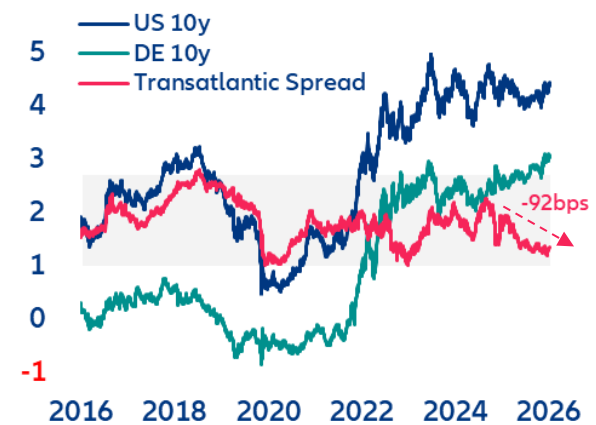
Any sustained shift in the transatlantic spread carries transmission effects across global capital flows, equity valuations, corporate funding and hedging costs. Since the 1980s, transatlantic spread has experienced three distinct phases. From 1980 to 1995, covering the Volcker Era and the post oil-shock disinflation, it was characterized by a high and volatile spread, fluctuating between +500bps and -150bps for the ten-year maturity. The period from 1995 to 2015 marked a phase of moderation and convergence, during which the spread appeared capped at an upper bound of 100bps and rarely fell below -100bps. With the onset of ECB quantitative easing in 2015, which pushed long-term yields in the Eurozone persistently lower, a third phase began, with the ten-year spread ranging between 100bps and 270bps (Figure 1). Since 2025, the transatlantic spread has narrowed by ~90bps, approaching the lower bound of the third phase trading range. This raises the question of whether we are at the start of a new lower-for-longer regime or should expect a near-term reversal of the narrowing trend (Figure 2).

Figure 1: Long-term 10y transatlantic spread
Spread in %



Sources: LSEG Workspace, Allianz Research

Figure 2: Transatlantic spread at lower bound
Yield and spread in %



US and Eurozone yields: from term-premium suppression to repricing rate expectations

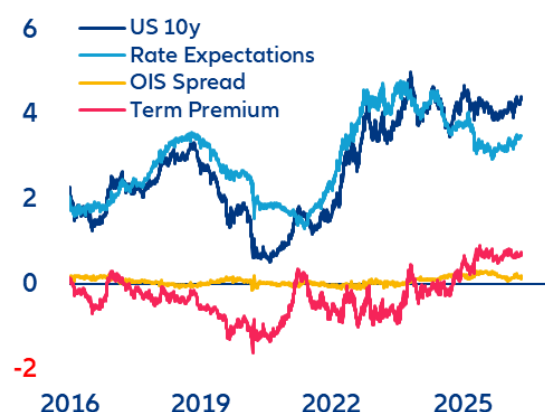
At its core, the transatlantic spread signals divergence in the monetary policy stance as well as in the growth and inflation outlooks. But it also embodies risk premia for the uncertainty around future inflation and neutral rates in addition to premia for fiscal risk and market liquidity. Term-structure models help to refine this view by decomposing observed market yields and rates into three components:

- **Rate expectations** capture growth and inflation expectations as expressed by expected nominal short-term rates.
- **Term premium** reflects the uncertainty around the future path of short-term rates and inflation expectations. It also captures duration supply/demand pressure with fiscal policy (debt issuance) on the supply side and changes in central bank bond purchases and holdings – Quantitative Easing (QE) or Quantitative Tightening (QT) – as a major factor on the demand side.
- **OIS spread**, defined here as the difference between government bond yields and the swap rate, represents the premia for market liquidity. Its level and sign signal whether government bonds are scarce or abundant, and whether they are cheap or expensive to finance and hedge (market liquidity). The OIS spread is an indicator of imbalances in government bond supply and demand. The higher the OIS spread, the more constrained governments are in issuing additional debt without paying a liquidity premium to the market.

The weight of these components varies overtime. If we look at the composition of US and German 10y yield over the last 10 years, we can clearly see these shifts. Pre-Covid, both yields were subject to prolonged suppression of term premia (negative), reflecting QE duration demand, while rate expectations reflected different monetary policy stances: tightening in the US versus stable in the Eurozone. Post-Covid, both yields surged, driven by rapidly rising rate expectations. During this time, the term premium shifted from a dampening to an amplifying factor once central banks reduced duration demand through QT while governments increased supply by fiscal expansion. Since 2025, US rates have held steady as a rising term premium was offset by falling rate expectations, reflecting the Fed's

easing bias amid a late-cycle slowdown. German Bund yields, however, rose steadily. First this was driven by a higher term premium following Germany's fiscal pivot (*Sondervermögen*). In a second phase, from mid-2025 onwards, the yield increase was driven by a shift in rate expectations as markets priced upside risk to euro short-term rates on the back of an emerging economic recovery. Over this period, market liquidity played a minor role for US Treasuries in absolute terms. Structurally, however, OIS spreads switched into positive territory in 2025, marking a shift from the US government being a receiver of a liquidity premium to being a payer. For German Bund yields on the other hand, OIS spreads were a major driver – at times even equal to the term premium. Until 2022, this reflected the relative scarcity of German government debt in the euro sovereign market. Investors paid a substantial premium to hold these ultra-safe assets, contributing to structurally low Bund yields. With the fiscal pivot, this privilege has at least diminished, reflecting the increased supply of German government bonds (Figure 3 and 4).

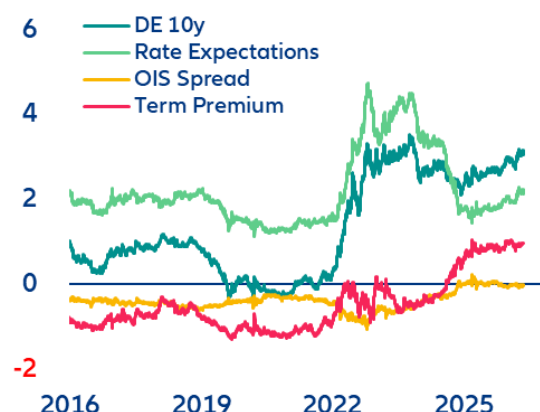
Figure 3: Decomposition of US 10y Treasury yield
Yield and components in %



Notes: Term structure decomposition based on Abrahams et al. (2016)

Sources: LSEG Workspace, Allianz Research

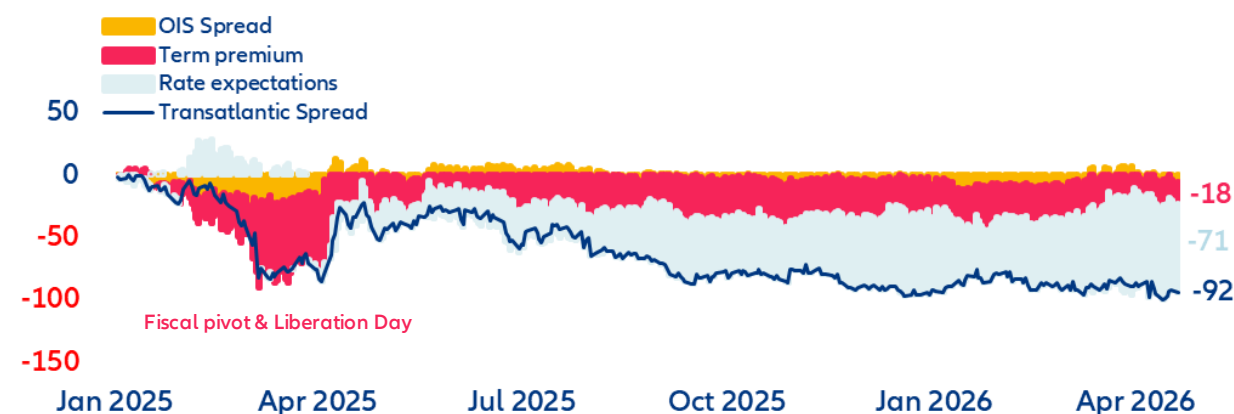
Figure 4: Decomposition of 10y German Bund yield
Yield and components in %



Latest transatlantic spread narrowing driven by rate expectations not risk premia

The latest narrowing of the transatlantic spread (-92bps since 2025) was initially driven by a term-premium repricing after the German fiscal pivot, reinforced by the “Liberation Day” trade shock. After that, however, it increasingly became the product of converging rate expectations (reflecting the easing bias of the Fed versus an ECB on standby) and divergent growth dynamics (slowdown in US vs early recovery in Eurozone). The energy-price shock triggered by the Iran war reinforced this trend but inverted the signs of the rate expectations. Markets now expect the Fed to stay on pause while pricing at least two ECB rate hikes by end -2026 (Figure 5). If this trend continues, the transatlantic spread could finally break through the lower trading range of the last 10 years.

Figure 5: Contributions to the 10y transatlantic spread since 2025
Yield and components in bps



Notes: Term structure decomposition based on Abrahams et al. (2016)

Sources: LSEG Workspace, Allianz Research

However, we believe the current convergence trend is nearing its end and is about to reverse. In the near term, markets are pricing in 25bps of excess ECB rate hikes through year-end, in our view. This should push the transatlantic spread wider again in H2 2026 as long-term inflation expectations remain largely aligned, offering little offset. In the medium term, we see real rates as a major driver for a wider transatlantic spread. The expected real rate differential currently sits at its long-term average but we estimate the US neutral rate will rise by 10-20bps over the next five years, driven by stronger productivity growth. In contrast, we see the Eurozone neutral rate declining by 10bps. Historically, such divergence in neutral rates has been followed by a widening of the expected real rate spread between US and Eurozone of at least 20–50bps

Figure 6: Difference in neutral rates and expected rates
Real yield in %

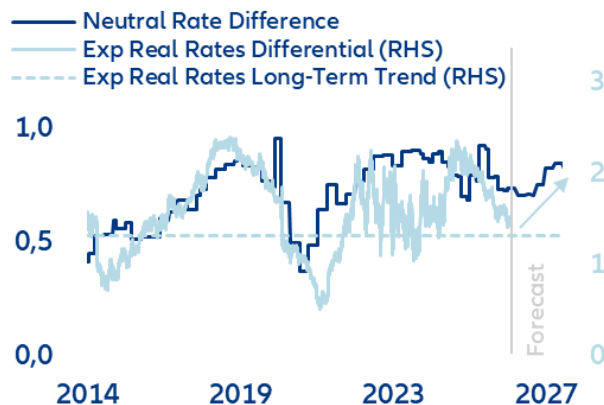
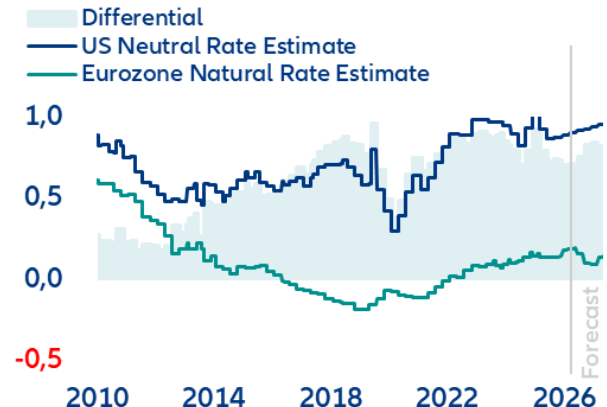


Figure 7: US vs Eurozone neutral rates
Real yield in %



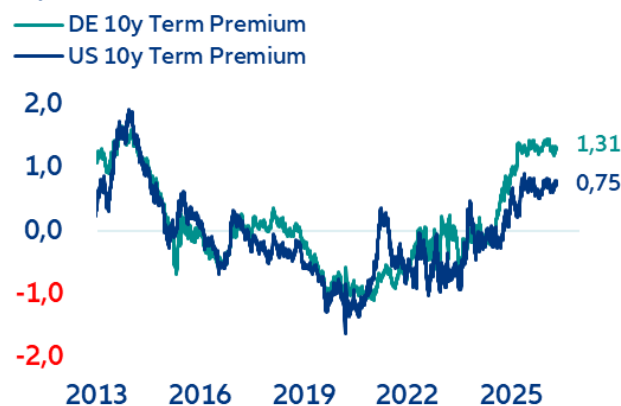
Notes: US neutral rates estimate as median value of Lubik & Matthes (2015), Del Negro et al. (2020), Davis, Fuenzalida, Huetsch, Mills, and Taylor (2024), Ferreira & Shousha (2023), Holston, Laubach and Williams (2017) and Ferreira, Lott and Richards (2025)
Eurozone neutral rate estimate as median of Lubik & Matthes (2015), Brand et al. (2024), Davis, Fuenzalida, Huetsch, Mills, and Taylor (2024), Ferreira & Shousha (2023), Holston, Laubach and Williams (2017) and Ferreira, Lott and Richards (2025)

Sources: LSEG Workspace, Allianz Research

Diverging QT pace drives term premium

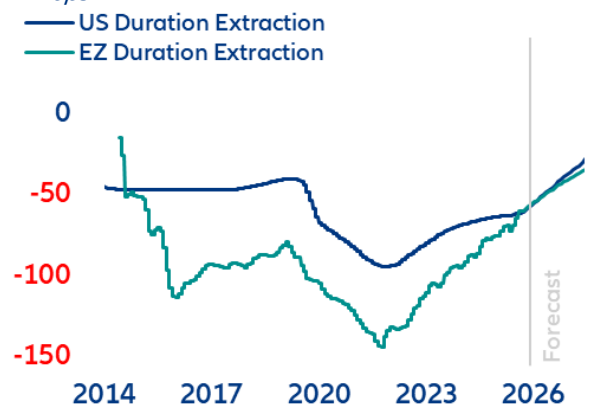
On the term premium side, we also see no further impulse for convergence in US and Eurozone sovereign yields. The term premiums for 10y Bund and 10y US Treasury tracked closely for an extended period. From mid-2024, however, the Bund term premium began rising significantly faster than its US counterpart. Today, the German term premium stands 60bps above the US level (Figure 8). We consider this gap excessive. Notably, half of the divergence is attributable to the ECB's faster pace of quantitative tightening (QT). We estimate that the run-off of ECB holdings has cheapened 10y Bund duration by 35bps, versus only 10bps in the US (Figure 9).

Figure 8: German term premium exceeds US by 60bps
in %



Notes: Term structure decomposition based on Abrahams et al. (2016)
Sources: LSEG Workspace, Allianz Research

Figure 9: QE/QT effect on 10y term premium
in bps



Notes: for US: Li & Wei (2013) ; Eurozone: Eser et al. (2019)

We expect this QT effect to equalize over the next years. In particular, recent Fed T-Bill purchases do not cause a renewed yield-dampening effect. Confined to very short-dated securities, these purchases do not drain duration from the market, therefore the upside pressure on US duration pricing from Fed portfolio run-off remains intact. The elevated German term premium relative to the US also appears stretched given the US's higher risk profile. Strong expansion of Treasury supply (fiscal deficit) has eroded the US convenience yield – the premium investors pay for an asset's combined safety and liquidity properties. US Treasuries still carry a significant liquidity privilege based on the USD-based collateral (repo) ecosystem. But the safety component has deteriorated to a point where the US convenience yield flipped from a dampening effect of -30bps at end-2024 to +20bps most recently. The Bund convenience yield has also declined with rising issuance (fiscal pivot) but remains at -40bps, which is 60bps below US Treasuries (Figure 10). The scale of the US Treasury risk premium is also visible when comparing the term premium of the US 10y Treasury bond with the one implied in the USD 10y swap rate. The two tracked almost identically until “Liberation Day”, after which markets have priced an additional +50bps premium for the sovereign (Figure 11). In our view, this is not yet sufficiently priced relative to the Bund. On this basis, we see clear upward pressure on the US Treasury term premium relative to the Bund.

Figure 10: Convenience yield US vs Germany, 10Y maturity in %

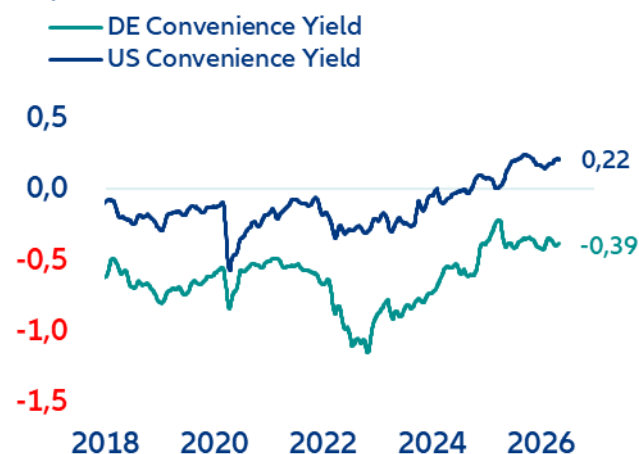


Figure 11: US 10Y Treasury risk premium vs swap in %

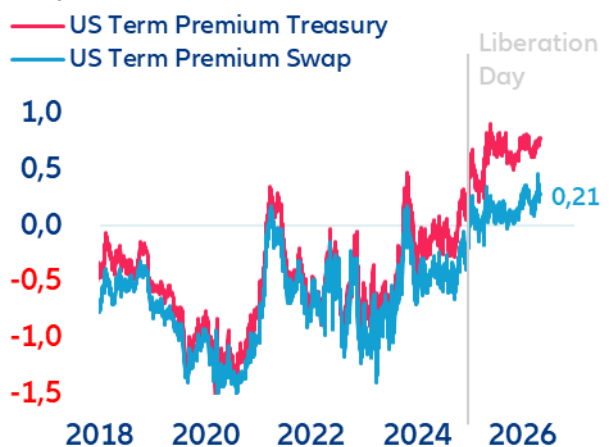


Fig.10 Notes: to illustrate the effect on the total yield, the sign of the convenience yield is inverted. A negative value represents a yield dampening effect, a positive value represents an upside pressure on the total yield. Convenience yield is calculated as the average of synthetic proxies mentioned in Lustig et al. (2024) and the intersovereign convenience yield in Du et al. (2018) Fig 11. Notes: Term structure decomposition based on Abrahams et al. (2016)

Sources: LSEG Workspace, Allianz Research

The transatlantic liquidity divide will drive OIS spreads

The transatlantic OIS spread differential remains structurally biased toward widening in Europe's favor, driven by fundamentally asymmetric liquidity and supply dynamics. In the US, persistently high net Treasury issuance (USD2-3trn/year), structurally constrains dealer balance sheets and weaker foreign demand keeps US swap spreads biased wider. Near-term regulatory relief (Supplementary Leverage Ratio adjustment) could provide a temporary tightening impulse, but the underlying trend remains intact. In the Eurozone, the picture is the mirror image. Despite the step-up in Bund supply (EUR170-180bn additional issuance), superior bank absorption capacity, ample excess reserves and persistent high-quality collateral scarcity in repo markets support a structural tightening bias on euro swap spreads. The specialness¹ of German Bunds is still driven by collateral scarcity rather than credit. This is the repo market signal for of the higher Bund convenience yield. It reinforces this dynamic, as does solid foreign demand on the back of euro strength and a more favorable hedging profile for non-US investors. The net result is a widening US–EU liquidity differential, with euro swap spreads tightening relative to their US counterparts over the medium term. We see potential of around +10bps effect on the transatlantic spread.

¹ Specialness being the premium of a strongly desired collateral which allows cheaper borrowing in repo transactions.

We expect the 10Y spread to widen by approximately +20bps in the short run. In the medium term (next 3 to 5 years) we see a widening of +75bps to around 200bps, reversing the bulk of its compression since 2025 (Table 1). However, we expect a clear shift in the drivers of that move. The transatlantic spread is transitioning from a predominantly rate-expectations-driven signal reflecting monetary policy divergence to one equally shaped by risk premia repricing.

Table 1: Transatlantic spread components

Component	Sub-component	Since 2025	Short-term Outlook	Mid-Term Outlook
Rates Expectations	Real Rates Expectations	-67bps	+15bps	+30bps
	Inflation Expectations	-5bps	+5bps	+10bps
Term Premium	QT Effect	-27bps		
	Risk premium	+12bps		+25bps
OIS Spread	OIS Spread	-5bps		+10bps
TOTAL		-92bps	+20bps	+75bps

Sources: LSEG Workspace, Allianz Research

We find the US Treasury term premium underpriced relative to the Bund, given the scale of US fiscal deficits, eroding convenience yield and tighter liquidity. By contrast, the Eurozone benefits from a supportive convenience yield and more favorable liquidity conditions. This expected change in the composition of the transatlantic spread shows how it is evolving from a pure monetary cycle signal into a barometer of sovereign fiscal credibility. Treasury supply dynamics will increasingly set the floor for US long-end yields as the US is forced to offer elevated real rates to retain foreign capital inflows. Europe, meanwhile, stands to benefit from a relative repricing: even as Bund supply rises, the Eurozone's stronger structural liquidity position and lower fiscal risk premium provide a durable anchor. For investors, this means the transatlantic spread should be monitored not only through the lens of rate differentials, but equally as a gauge of fiscal sustainability and sovereign market liquidity. A cross-curve duration positioning between Bunds and US Treasuries should therefore not only reflect rates expectations but take into account which curve segment will see the most pronounced term premium repricing. A structurally higher term premium in the US should also add steepening pressure on the US curve and act as a weakening force on the USD versus EUR. On credit and issuance, we expect the EUR-USD funding cost differential to widen, favoring euro issuance for global borrowers (reverse Yankee supply). This could have a tightening effect on EUR investment-grade spreads relative to their USD counterparts. In short, the transatlantic spread is moving from a "*Fed versus ECB*" story to a "*Treasury versus Bund*" story, and portfolios built for the former need to be repositioned for the latter.

These assessments are, as always, subject to the disclaimer provided below.

FORWARD-LOOKING STATEMENTS

The statements contained herein may include prospects, statements of future expectations and other forward-looking statements that are based on management's current views and assumptions and involve known and unknown risks and uncertainties. Actual results, performance or events may differ materially from those expressed

or implied in such forward-looking statements.

Such deviations may arise due to, without limitation, (i) changes of the general economic conditions and competitive situation, particularly in the Allianz Group's core business and core markets, (ii) performance of financial markets (particularly market volatility, liquidity and credit events), (iii) frequency and severity of insured loss events, including from natural catastrophes, and the development of loss expenses, (iv) mortality and morbidity levels and trends,

(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,

and (xi) general competitive factors, in each case on a local, regional, national and/or global basis. Many of these factors may be more likely to occur, or more pronounced, as a result of terrorist activities and their consequences.

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Behind the gate: The promises and perils of private markets democratization

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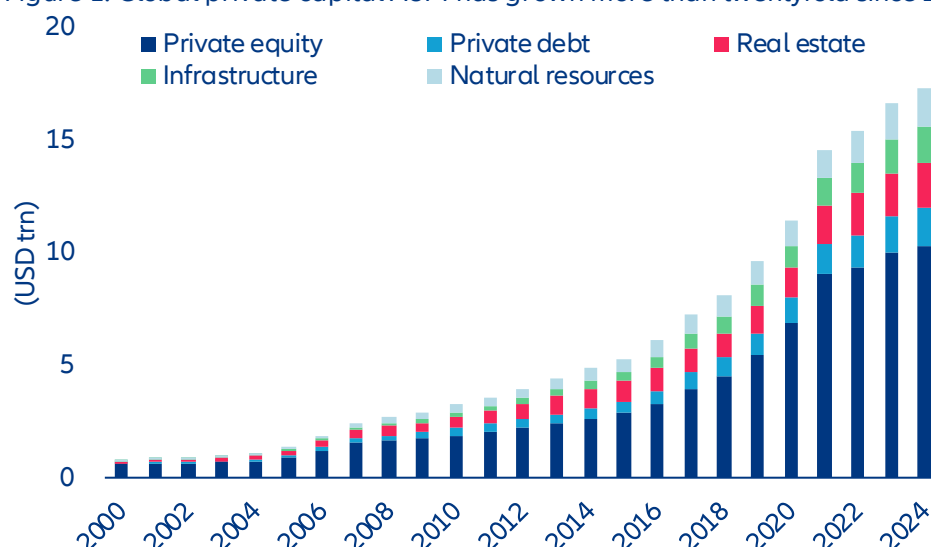
In Summary

- **Private markets are opening up to retail, and the product has to be rebuilt to fit.** Global private markets have grown more than twentyfold since 2000 to over USD17trn, propelled by institutional adoption of the Yale endowment model that tilted long-duration capital aggressively into illiquid assets. The next leg of growth runs through wealth channels and mass markets, where investors lack the governance, illiquidity tolerance and manager-selection capability. Semi-liquid evergreens have become the workhorse for wealth, while the mass market is reached primarily via DC plans, regulated retail funds, public-private hybrids, and insurance wrappers, structures in which a fiduciary, insurer, regulatory template or the product design itself makes the call rather than the end investor.
- **Three forces have converged to push democratization: savers need enhanced returns and diversification, governments need to mobilize private savings for public priorities and asset managers need new pools.** Diversified private-market exposure can improve portfolio efficiency, lifting expected returns from 6.2% to 7.9% while keeping stressed downside risk (CVaR) around -3% versus -5% for traditional liquid portfolios. Governments see a policy lever to address a USD400trn retirement gap by 2050, growing inequality as value creation stays private and infrastructure needs that public budgets cannot fund. For asset managers, the move is increasingly necessary: institutional fundraising is slowing as the traditional capital pool matures and large LPs reach their target allocations.
- **The pitch is real, but a closer look reveals a more nuanced picture.** First, the return story is compelling at face value: buyout funds have delivered 12-14% annually against 7-9% for public equities. But strip out leverage and most of the private-equity advantage disappears, leaving manager selection as the real driver of outperformance. Second, retail investors face a compounding triple disadvantage compared to institutional products: additional fee layers that erode the illiquidity premium, potentially weaker underlying assets, and no ability to select managers in an asset class where dispersion runs wide. Third, the semi-liquid promise is not illusory, but it comes with conditions; it works in normal market conditions but exposes investors to gates precisely when they want their capital back. Finally, retail money brings reactive behavior into an asset class increasingly tied to banks and life insurers, reintroducing the maturity transformation that closed-end structures had engineered away.
- **The early-2026 slowdown is a stress test, but structural drivers remain intact.** Expect product recalibration toward larger liquid buffers and tighter gating, consolidation toward mega-GPs with the scale and distribution to run retail wrappers and a new round of regulatory scrutiny on disclosure, leverage and liquidity buffers, adding cost but reinforcing product quality.
- **For investors, the dispersion between well- and poorly-built retail vehicles will widen.** Four markers separate institutional-grade retail product from repackaged versions: liquidity sized to the underlying asset, clean separation of retail and institutional pools, independent valuation governance and real manager-selection capability behind the wrapper. For institutional investors, the illiquidity premium will compress and deal-allocation conflicts will emerge as GPs spread a finite pipeline across vehicles with different liquidity profiles, making vigilance on cross-pool transfers essential.

A new investor base, a new product shelf

Private markets were built around institutions and family offices. The 1980s leveraged-buyout boom, fueled by the advent of junk bond markets, lit the fuse; institutional money turned the resulting industry into a colossus. Global private-capital assets under management (AUM) have grown from roughly USD800bn in the early 2000s to over USD17trn by 2024 (Figure 1), more than twentyfold in two decades.

Figure 1: Global private capital AUM has grown more than twentyfold since 2000

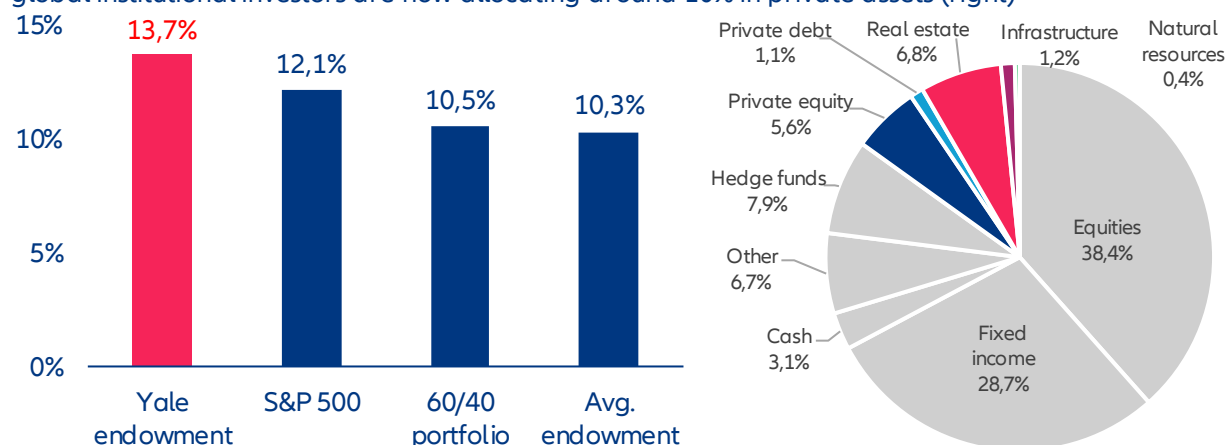


Sources: Preqin, Allianz Research

Note: Secondaries and funds of funds are excluded to avoid double-counting.

The pivotal driver behind the transformation of private markets into a mainstream institutional allocation is Yale's endowment model. When David Swensen took over the university's investment office in 1985, endowments ran the standard 60/40 mix. Over the next three decades he tilted the portfolio aggressively into private assets, arguing that long-duration capital should earn an illiquidity premium that liquid investors could not. By 2021, Yale held roughly 70% in alternatives, nearly ten times the pre-Swensen level, and the portfolio had returned 13.7% a year over Swensen's tenure against 10.5% for 60/40 and 10.3% for the average endowment (Figure 2). Yale's success rippled outward. By the 2010s, the biggest US university endowments held 40-60% in private markets on average. Sovereign wealth funds, public pension funds and insurers followed. Today, global institutional investors allocate around 10% to private assets (Figure 2), with endowments and public pensions allocating more than a third.

Figure 2: The Yale endowment has outperformed traditional benchmarks under Swensen's management (left) and global institutional investors are now allocating around 10% in private assets (right)

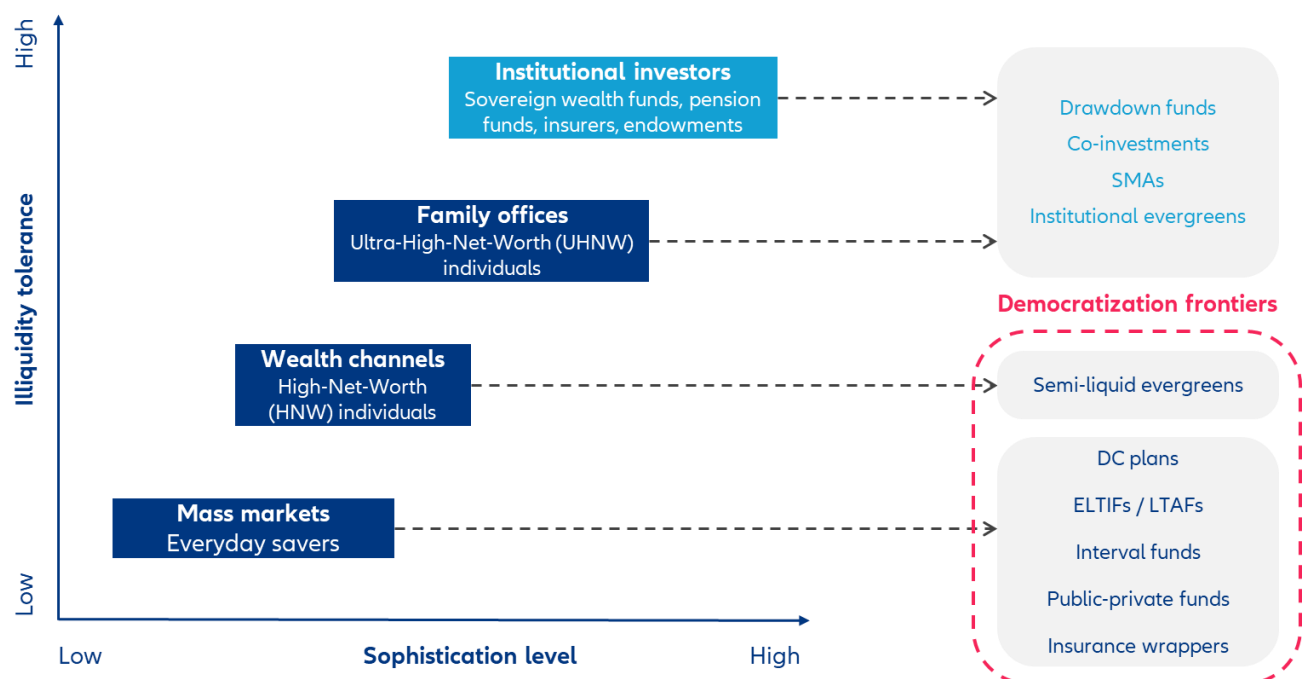


Sources: Bloomberg, The Economist, Preqin, Allianz Research

The fit is natural. Most institutional capital sits in drawdown funds run by general partners (GPs) who source, structure and manage the underlying deals. The 10-year-plus horizon suits pension funds and life insurers with liabilities measured in decades; in-house teams, investment committees and boards do the diligence that opaque assets demand and scale buys both fee leverage and access to the best managers. Beyond drawdown funds, institutions increasingly mix in co-investments (alongside the GP, often at low or zero fees), separately managed accounts (SMAs, typically USD100mn-plus, fully customized) and open-ended evergreens (operational simplicity and continuous liquidity, no J-curve). Together these tools let institutions calibrate exposure with a precision the conventional fund template cannot match. **Family offices, the private treasuries of the ultra-wealthy (a USD 30mn threshold, though in practice more is needed to access large GPs), follow the same model in miniature.** They vary in sophistication but use the same vehicles to reach the same managers, sometimes on slightly worse terms.

Now, the action has moved to the wealth and mass-market channels, where investors' needs are entirely different. "Democratization" means giving private-market access to those locked out of the institutional product suite and that requires more than lower minimums. It calls for fresh product design, with new trade-offs between liquidity, fees, transparency and asset quality (Figure 3).

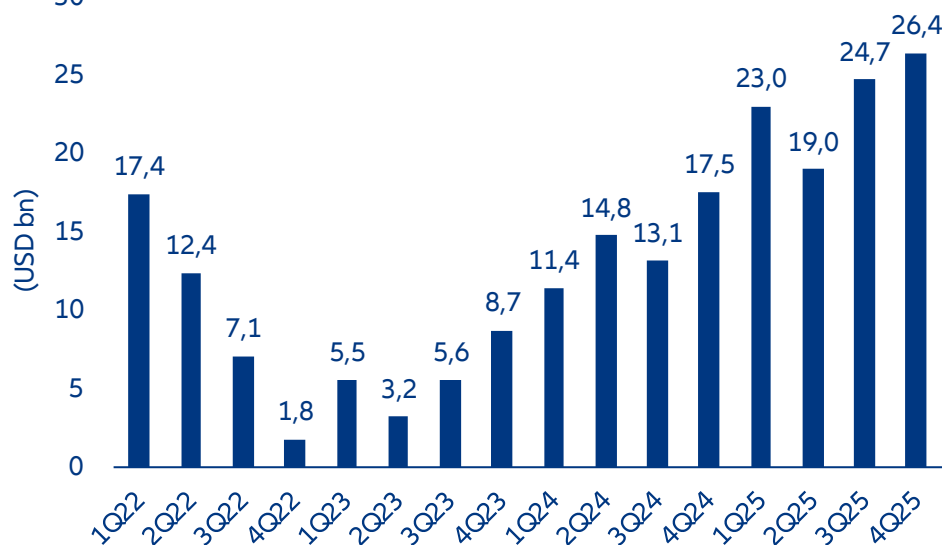
Figure 3: Investor segmentation and product structures across the wealth pyramid



Source: Allianz Research

The wealth channel covers the high- and very-high-net-worth segment, growing fast in both numbers and assets. Entry starts at USD1mn of investable assets; the more active VHNW tier kicks in at USD5mn. US households with USD5mn to USD20mn now control 40% of addressable wealth, up from 18% in 2013, turning them from once-niche cohort to a market force. They are too small to deal with GPs directly and reach private markets through private banks, wealth managers and financial advisers. This is the most commercially mature corner of democratization, and one of the fastest-growing. Inflows to the major listed alternative-asset managers have surged over the past three years, topping USD25bn in the fourth quarter of 2025 alone (Figure 4). US financial advisers now allocate USD1.9trn to less-than-fully-liquid strategies, but the average adviser still has only 2.3% of client portfolios in alternatives, and roughly half use none at all. The runway is long.

Figure 4: Net inflows to wealth products offered by major listed alternative managers have surged since early 2023



Sources: JP Morgan, Allianz Research

The mass market is a much bigger prize, and a wholly different machine. Its investors have less investment expertise than institutions, shorter effective horizons and no practical capacity to select managers themselves, which is why the channel is built around fiduciaries, insurers and regulated structures that do the selection for them. The US mass affluent and middle-market segments, 46.9m households in all, hold USD25trn of financial assets. The retirement system holds another USD49trn, of which USD14trn sits in employer-sponsored DC plans. In Europe, household savings are mostly parked in low-yielding bank deposits, around EUR10trn across the EU; the next-largest pool is unit-linked insurance, notably France's assurance-vie, Italy's PIR and Germany's equivalents.

Three structural differences set wealth-channel investors apart from institutions. They lack the governance infrastructure, they are less tolerant to illiquidity and they reach private markets through intermediaries whose operating models demand standardized features that fit into model portfolios. The result is a generation of products engineered for accessibility, with performance pursued within those constraints rather than at any cost. Semi-liquid evergreens have become the workhorse: quarterly subscription and redemption windows, no capital calls, simplified fees and standardized tax reporting. Each feature addresses a specific constraint. Well-engineered evergreens, with adequate liquidity buffers and disciplined valuation, can deliver a meaningful share of the institutional return profile to investors who previously had no access at all.

Reaching the mass market requires a different paradigm. The end investor no longer chooses the product directly; instead, the structure has to satisfy a fiduciary or regulator that it is suitable for an average participant. The result is a new range of vehicles, each tailored to a specific regulatory category or savings product rather than evaluated on its own merits. New structures are launching in every major jurisdiction; the main channels are defined contribution (DC) plans, ELTIFs and LTAFs, interval funds, public-private hybrids and insurance wrappers.

The deepest pools sit inside existing retirement and savings systems. US DC plans, accessed mainly through target-date funds with embedded private-market sleeves, are the largest mass-market pool in the world: USD14trn of employer-sponsored DC assets, including roughly USD10trn in 401(k)s. European insurance wrappers play the same role in the savings sector, with the insurer handling manager selection, valuation oversight and liquidity management for the policyholder. Both work precisely because they do not demand investment expertise of the end investor: the fiduciary or insurer takes on the discipline that retail investors lack and would struggle to acquire. The private exposure is bundled inside a familiar structure and the design ensures that no single policyholder's liquidity needs can destabilize the underlying assets. The retail experience is unchanged at the surface; private-market access, properly governed, sits one layer below. For long-horizon savers, this is one of the few realistic routes to the diversification and return premium of private markets without bearing institutional-style governance demands themselves.

Regulated retail vehicles take a more direct path, swapping regulators for fiduciaries. ELTIFs in the EU, LTAFs in the UK and interval funds in the US are standalone retail products an investor or adviser can pick actively, subject to rules on eligible assets, concentration, mandatory repurchases and disclosure. The regulatory framework replaces the institutional governance these products would otherwise lack, which is a large part of their appeal: the regulator has effectively vetted the structure. The cost is product flexibility and the channel has tilted strongly toward private debt and lower-volatility strategies that fit cleanly inside the templates.

Hybrid public-private funds are the newest design, aimed squarely at the liquidity problem. DC plans absorb redemption pressure through fiduciary scale; regulated retail vehicles use prescribed gates; hybrids embed a 30-60% public-asset buffer into the product itself. The public sleeve provides exit capacity; the private sleeve provides the alternative exposure that justifies the fund. The trade-off is return: a 40% private allocation inside a fund that is itself only a slice of a retail portfolio leaves effective private exposure of just a few percent at the investor level. Hybrids are a design solution to mass-market liquidity rather than a route to maximized private-market exposure. Their merit is extending some private-market characteristics to investors for whom pure private vehicles are not workable, accepting a return trade-off in exchange for genuinely daily-liquidity behavior.

The four channels share a single design principle, with different machinery. None assumes the end investor actively chooses the product. The decision rests with a fiduciary (DC plans), an insurer (insurance wrappers), a regulatory template (ELTIFs, LTAFs, interval funds) or the structural design itself (hybrids). This shift in who decides ultimately drives the outcome for investors, and means the routes to mass-market capital are likely to fragment by jurisdiction rather than converge on a single global model.

Why private markets are going mainstream now

For retail investors, the case rests on diversification and better risk-adjusted returns. The evidence is plain: private assets diversify; traditional ones increasingly do not. Since 2022, correlations between public equities and bonds have risen sharply, eroding the protective property of the standard 60/40 portfolio. Modern portfolio theory says that adding assets with low or negative correlations shifts the efficient frontier up and to the left; the correlation matrix in Table 1 backs this up. The numbers are forward-looking, not historical, and reflect today's regime. The arrows mark the largest moves, concentrated among traditional assets that have grown more closely correlated under stagflationary pressure.

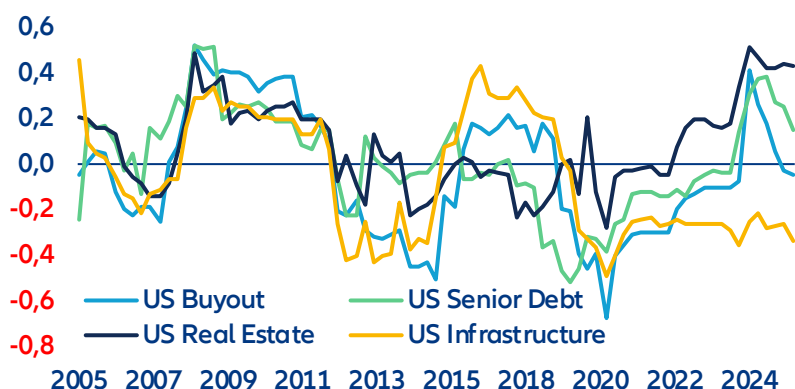
Table 1: Cross asset correlation matrix

	Euro Corporates	Euro High Yield	Euro Covered Bonds	USD Corporates	Equity BAA Aggregate	US Private Equity (Buyout)	US Private Debt (senior)	US Real Estate	US Infrastructure	Euro Government Bonds	Euro Agency bonds	Emerging Markets Bonds	Gold
Euro Corporates	1,00	0,82	0,86	0,77	0,63	0,20	0,31	0,24	0,17	0,80	0,83	0,87	0,47
Euro High Yield		1,00	0,53	0,54	0,82	0,24	0,33	0,20	0,30	0,42	0,48 ▲	0,83	0,43
Euro Covered Bonds			1,00	0,71	0,37	0,21	0,31	0,19	0,20	0,93	0,94	0,70	0,49
USD Corporates				1,00	0,43	0,17	0,23	0,30	0,12	0,72	0,73	0,76	0,53
Equity BAA Aggregate					1,00	0,14	0,26	0,09	0,24	0,28 ▲	0,33	0,74	0,46
US Private Equity (Buyout)						1,00	0,81	0,42 ▼	0,71	0,09	0,10	0,20	0,42
US Private Debt (senior)							1,00	0,43	0,62	0,22	0,25	0,31	0,37
US Real Estate								1,00	0,17 ▼	0,23	0,21	0,17	0,04
US Infrastructure									1,00	0,01	0,03	0,27	0,58 ▲
Euro Government Bonds										1,00	0,98	0,62	0,35
Euro Agency bonds											1,00	0,67	0,38
Emerging Markets Bonds												1,00	0,63
Gold													1,00

Source: LSEG Datastream, Allianz Research

Private assets are consistently uncorrelated with public ones, and barely correlated with each other. As public-market correlations climb, infrastructure stands out as the biggest beneficiary; pair correlations with euro government, agency and corporate bonds are all low. The more striking feature of the matrix is the heterogeneity within the private universe itself: real estate and infrastructure are not substitutes for private equity. The rolling four-year analysis in Figure 6 backs this up. From 2005 to 2026, all four American private asset classes, PE buyout, senior debt, real estate and infrastructure, have averaged correlations close to zero with a 60/40 public portfolio, none above 0.5. The dispersion has widened recently, with infrastructure the most decoupled, even as overall market correlations have climbed.

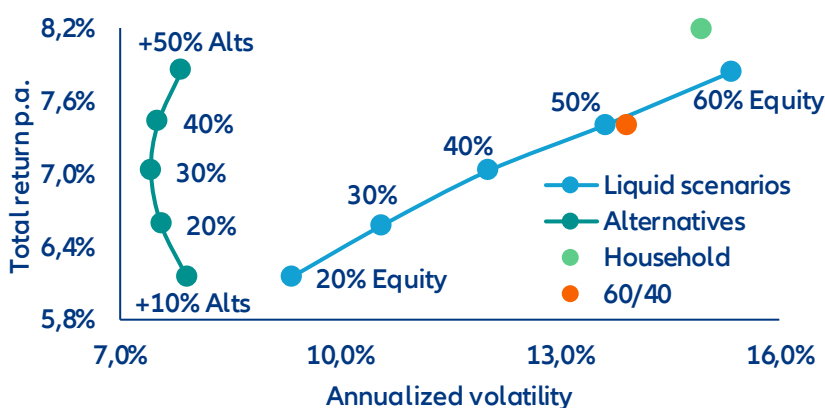
Figure 6: Cross-asset rolling correlations



Source: Pitchbook, Allianz Research

The most compelling quantitative argument for democratization sits in the efficient frontier itself. Figure 7 plots annualized volatility against total return for portfolios ranging from a pure traditional mix of 13 asset classes to ones with progressively larger alternative allocations (10-50%). As the alternatives weight rises, the frontier shifts unmistakably outward: the green alternatives frontier dominates the orange liquid frontier at every point, lower volatility and higher returns with no exceptions. Theory says low-correlation assets should improve the trade-off; the empirical gap is striking. Private assets earn an illiquidity premium without the concentration risk that comes from leaning on public equity, provided investors can stomach the illiquidity and keep enough liquid ballast for market stress. The classic 60/40 (orange dot) and the typical European household (green dot) sit on noticeably higher volatility, since bonds and equities have failed to diversify each other since 2022's stagflation. Owning a home does not fix this; a diversified slice of private markets does. The same pattern shows up in Conditional Value at Risk, which captures downside losses in bad markets and actually fits the non-normal distribution households better than volatility does.

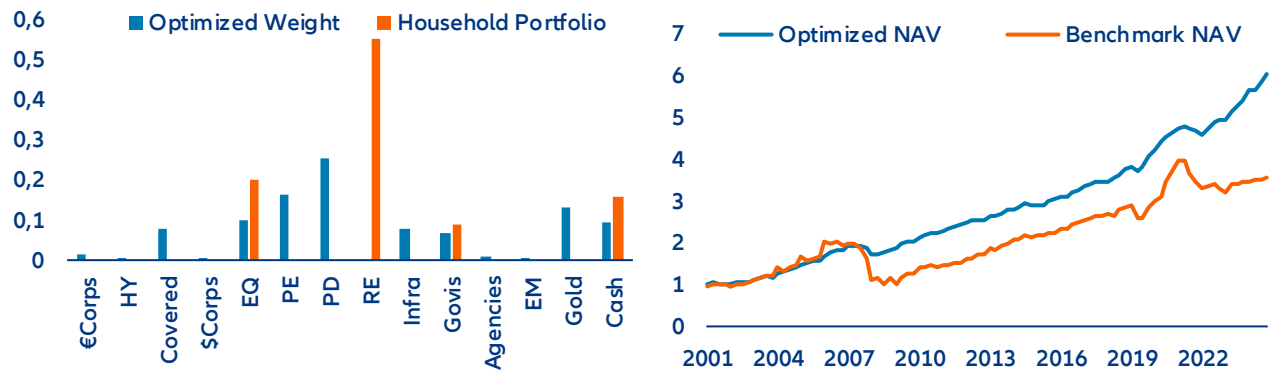
Figure 7: Efficient frontier liquid - illiquid assets



Source: Pitchbook, Allianz Research

An optimizer sharpens the picture. Figures 8 and 9 turn the theoretical frontier into a realized-performance comparison. The optimized portfolio (blue), allocating across alternatives and traditional assets, posts a Sharpe ratio of 2.85, against 0.64 for the typical household benchmark (orange). The optimizer solves one trade-off: maximize risk-adjusted return while penalizing both single-asset concentration and large tail losses (CVaR). It uses market-cap weights, forward-looking assumptions for returns, volatility and correlations, and alternative-history scenarios rather than historical averages, producing an allocation that holds up across regimes. The result: a balanced portfolio, half in private markets, complemented by low-risk bonds, gold and cash. Smaller private allocations still make sense, since the diversification benefit persists across weightings. The right share for any investor depends on their tolerance for risk.

Figure 8: Portfolio allocation in % (left) and portfolio performance (right)



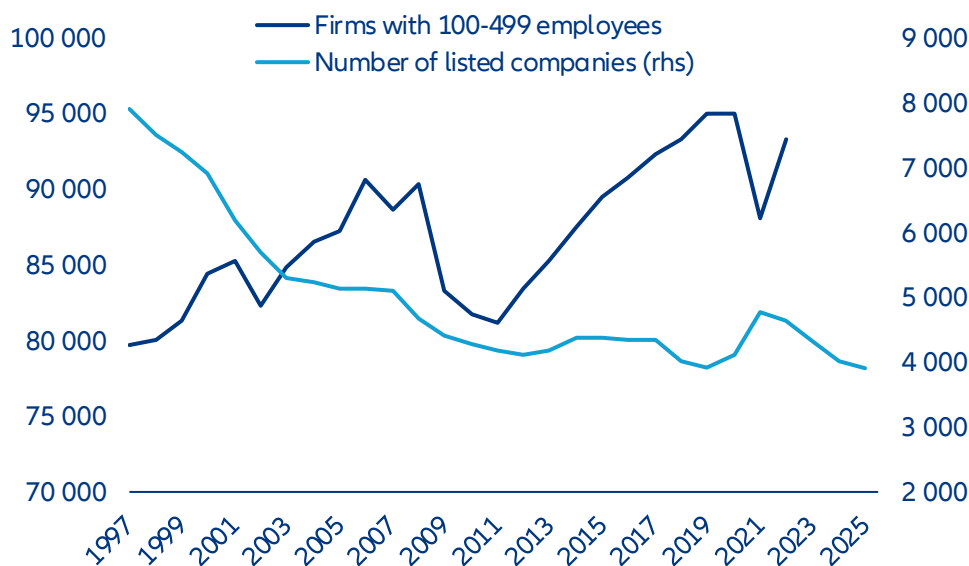
Source: Pitchbook, Allianz Research

For governments, democratization is more than an access question. It is a policy lever for retirement adequacy, distributional fairness and the funding of long-term economic priorities. With public finances constrained and traditional capital sources struggling to keep up, private markets are being repositioned as a way to mobilize household savings.

The most pressing motive is the retirement-adequacy crisis. The shift from defined-benefit to defined-contribution pensions has moved investment risk from employers and governments to individuals, leaving DC participants with portfolios dominated by public equities and bonds. DB plans have long held 15-30% in alternatives and benefited; DC participants have not, and the gap has compounded into a serious shortfall. The global retirement-savings gap is projected to reach USD400trn by 2050, up from USD70trn in 2023. With public-pension systems under strain, widening the DC investment universe is a way to narrow the gap without lifting public spending.

The second motive is inequality. As private markets stay walled off, a growing share of value creation slips out of reach of ordinary savers. The number of US-listed firms has roughly halved since the late 1990s, while the number with fewer than 500 employees has grown by nearly a fifth (Figure 10). Companies stay private for longer, so much of their growth, and the wealth that comes with it, is captured by sophisticated investors with broader access. Ordinary savers, stuck in public equities, watch from the sidelines.

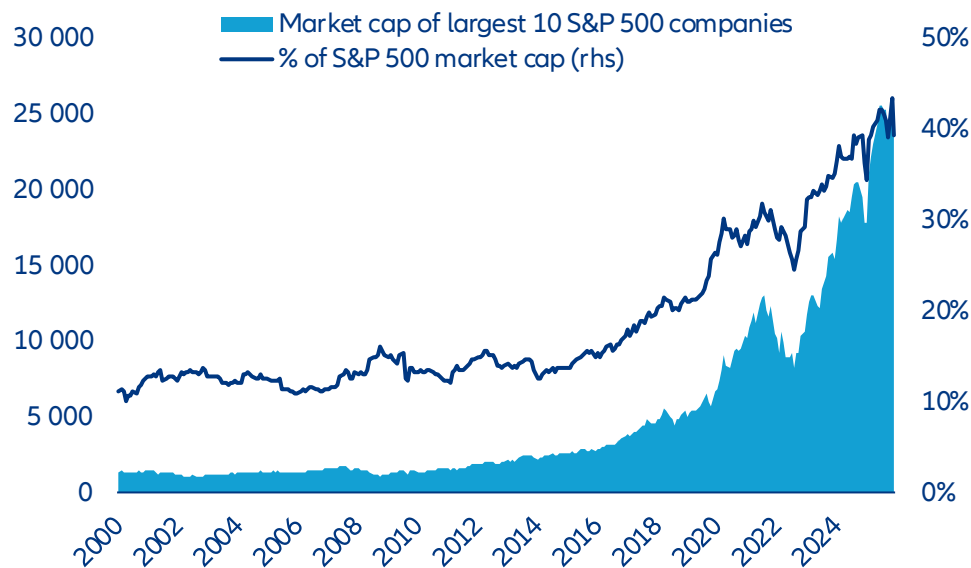
Figure 9: US listed companies have nearly halved while mid-market firms have grown by one-fifth over the past three decades



Sources: US Census Bureau, World Bank, Allianz Research

Public-market concentration is a third motive. The top 10 S&P 500 firms now make up 40% of the index, and US equity benchmarks are more concentrated than at any point since 2000 (Figure 11). Retail investors who hold passive index funds are heavily exposed to a narrow set of large-cap tech names; private assets reach sectors and stages that public indices underrepresent. The case for retail access is not just about returns. It is also about reducing the concentration risk that the passive revolution has, unintentionally, amplified.

Figure 10: S&P 500 concentration has reached its highest level since 2000



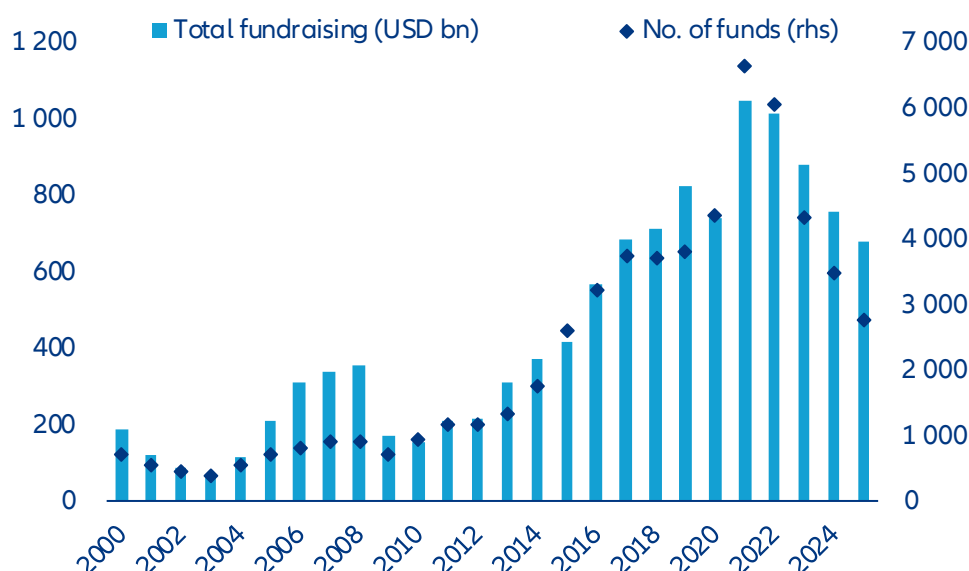
Sources: Macrobond, Allianz Research

Private capital is also being marshaled to fill gaps public budgets cannot. The EU's investment needs for infrastructure, the energy transition and digital transformation are projected at roughly 5% of GDP a year, well beyond what fiscally constrained governments can fund. Europe's EUR10trn or so of household bank deposits is largely unproductive capital. Channeling those savings into infrastructure, growth equity and energy projects through ELTIFs or LTAFs would do two things at once: lift household returns over the long term and deliver real-economy capital that the public purse cannot.

Jurisdictional competition is shaping both the design and the timing of these reforms. US, the EU, the UK, Australia and Singapore are all moving at once, none willing to see domestic savings drift offshore in search of alternatives. Britain's decision to make LTAFs ISA-eligible from 2026 was driven by competition with Luxembourg and Dublin as fund-domicile hubs; France's Green Industry Act was designed to keep retail savings inside the French insurance sector. The race is not just to channel retail capital into private markets, but to do so on one's own terms before a rival captures the flow.

For asset managers, the move into wealth and retail is no longer optional. After decades of rising inflows, institutional fundraising has hit a wall. In 2025, closed-end private-equity fundraising fell to a five-year low (Figure 12), as subdued exits left more than 16,000 portfolio companies held for over four years and LP distributions slowed to a trickle. The 2022 public-market drawdowns produced a denominator effect that left many institutions over their allocation targets, prompting a pause in fresh commitments. Beyond the cycle, the institutions that began private-markets programs in the 2000s have now reached, or nearly reached, their targets. With the traditional capital pool maturing, the industry has pivoted to investor groups that, until recently, had limited access.

Figure 11: PE fundraising volume and fund count have fallen sharply from their 2021 peaks



Sources: Preqin, Allianz Research

Box 1. Policy in motion: bringing private markets into 401(k)s

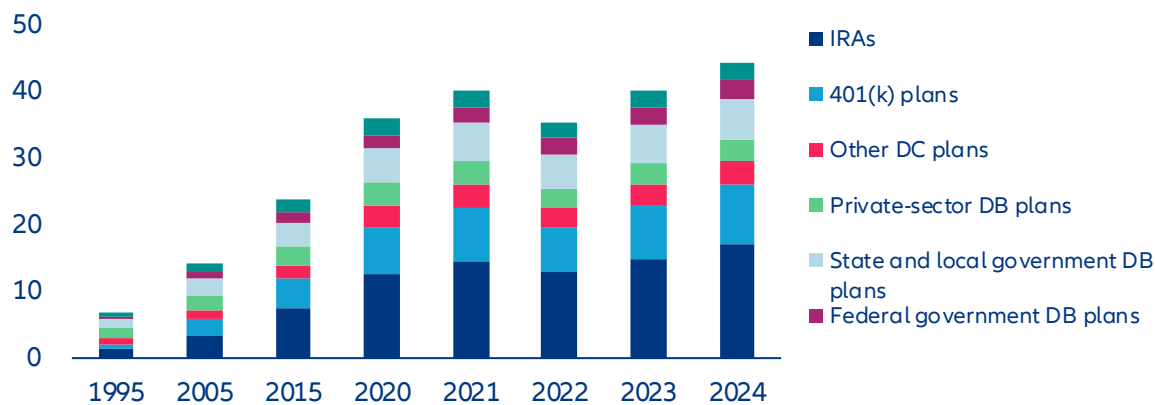
For most of their existence, American 401(k) and other defined-contribution (DC) plans have excluded private-market assets. ERISA fiduciary constraints, litigation risk and operational frictions (illiquidity, valuation complexity, fee opacity) made them poor fits. The SEC's accredited-investor rules further limited direct retail participation, reinforcing the structural divide between DC and defined-benefit (DB) plans. The first significant move came in June 2020, when the DOL issued an Information Letter allowing private equity inside diversified DC options such as target-date and balanced funds, provided it was managed prudently. A December 2021 Supplemental Statement then signaled a pause, noting that most plan sponsors lacked the expertise to evaluate such products properly.

By 2025 the policy had shifted. On 12 August 2025, the DOL formally rescinded its 2021 guidance discouraging 401(k) alternatives. The move aligns with the White House Executive Order on "Democratizing Access to Alternative Assets for 401(k) Investors" and clarifies that exposure to private-market assets may be permitted within diversified options under ERISA. The DOL is now signaling a neutral, principles-based approach: Fiduciaries must judge investment options on their merits and suitability, a clear departure from the earlier practice of singling out specific asset classes for scrutiny.

Congress has lent legislative support. The SECURE Act 2.0 of 2022 facilitated alternative access indirectly, by letting 403(b) plans use Collective Investment Trusts (CITs). CITs are often the preferred vehicle for private-market exposure in employer plans thanks to lower costs and more customization than mutual funds. Together these moves create a coherent policy environment in which private-market exposure can expand within DC plans under existing fiduciary and securities-law frameworks. An alternative investment option that sits inside a well-diversified fund managed by qualified professionals, and that the plan sponsor has documented thoroughly, can sit comfortably within ERISA's duty of prudence. Further sub-regulatory guidance from the DOL is expected, and the SEC may add guidance on valuation and disclosure as more products come to market.

The base effects are substantial. As of Q1 2025, the American DC system held roughly USD12.2trn in assets, compared with USD3.8trn in private and public DB plans. Alternatives make up around 30% of DB allocations but just 0.1% of DC portfolios, so even a modest reallocation would be material in absolute terms. A baseline scenario of 3-5% DC penetration by 2030 would deliver cumulative inflows of USD300bn-400bn; an upside of 10% penetration by 2035 would mean USD1.2trn-1.5trn in new allocations. Adoption is expected to start with large-plan default options that wrap diversified PE and private credit inside multi-asset structures, then diffuse gradually into mid-sized plans as operational frameworks standardize.

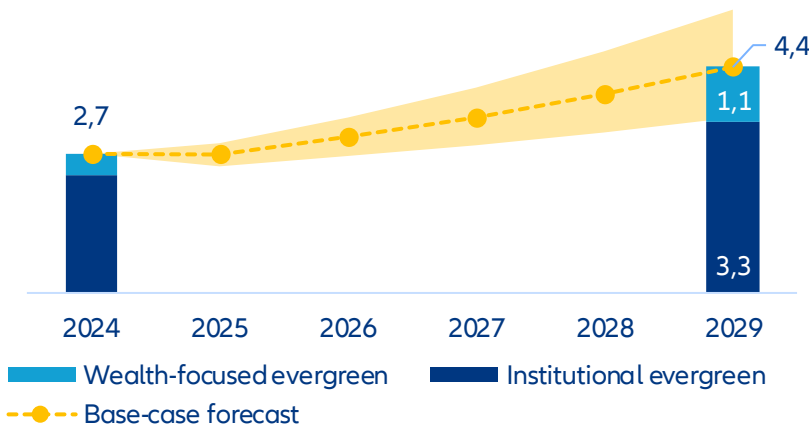
Figure 12: US retirement market assets (USD tn)



Sources: Investment Company Institute, The US Retirement Market, Fourth Quarter 2024, Allianz Research

Product evolution and opportunity set. Capital will flow along the path of least operational resistance. CITs are gaining ground because of their flexibility on liquidity and fees, smooth integration into plan recordkeeping and exemption from certain retail-disclosure requirements under the Investment Company Act. Semi-liquid '40-Act interval and tender-offer funds serve as secondary conduits for smaller, retail-aligned platforms, though their quarterly-liquidity features and higher cost structures may limit early scale. Across both channels, managers are re-engineering fund economics toward evergreen structures with quarterly or semi-annual rebalancing to accommodate DC flows. Pitchbook estimates that roughly \$2.7 tn is now managed in various indefinite-life formats globally, projecting growth to \$4.4 tn by the end of 2029 (Figure 13). Recordkeepers, custodians and administrators are investing in capabilities to support partial redemptions, performance-fee accruals and standardized daily valuation reporting.

Figure 13: Pitchbook estimate for Evergreen fund AUM forecast (USD tn)



Source(s): Pitchbook, Allianz Research. Geography: Global. Note: Note: Forecasts were generated on April 14, 2025. "Institutional evergreen" includes insurance AUM from Blackstone, KKR, Blue Owl Capital, The Carlyle Group, Ares Management, Apollo Global Management, and Brookfield.

Within diversified default options such as target-date funds, private equity is expected to dominate the growth allocation for younger cohorts, mainly through fund-of-funds or secondaries to achieve vintage diversification and reduce dispersion risk. Private credit is well placed to stabilize income for pre-retirement cohorts, focused on senior secured lending, infrastructure debt and core real-estate credit. Real assets (core real estate, infrastructure equity) will play a secondary role as inflation hedges and diversifiers, supported by established appraisal-based

valuation practices. Allocations will stay modest relative to liquid assets, typically capped at 10-15% of the default fund, to maintain daily-liquidity compliance.

The primary concerns at this stage are fiduciary risk, liquidity and valuation governance. Fiduciary exposure under ERISA remains real: sponsors must justify private-market inclusion as prudent on a net-of-fee basis, with documented due diligence and monitoring. The mismatch between daily participant flows and illiquid underlying holdings calls for hard caps, liquid sleeves and robust queue procedures, and under stress redemption gates can still be tested. A related vulnerability is the need for valuation fairness across transacting participants, which requires consistent third-party appraisal cycles, interim model-based pricing and audit transparency. Operational dependencies, recordkeeping, performance-fee calculation and participant communications introduce a layer of execution risk that must be addressed before a large-scale rollout.

For investors, the policy shift creates a durable, multi-year reallocation theme, not a short-term trade. Early strategies will favor diversified private equity and senior private credit that fit established liquidity and valuation criteria, with real assets in a supporting role. As operational systems and legal precedents mature, DC allocations to private markets could move from pilot programs to a structural component of retirement portfolios. Early participation should bring stable, long-duration capital to managers; investors should expect risk premia in yield-oriented segments to converge gradually as the DC channel scales.

Do returns justify the hype?

Whether private assets truly outperform is less straightforward than the headline numbers suggest. At first glance, the case is compelling. Over the past two decades, buyout funds have delivered net annualized returns of roughly 12-14%, against 7-9% for broad public benchmarks like the Russell 3000¹. But this obscures significant variation. Private debt has yielded steady mid- to high-single-digit returns; venture capital and real estate have been more cyclical, with periods of underperformance against public benchmarks.

The critical adjustment is leverage, and it transforms the comparison. Buyout funds typically run 50-70% loan-to-value at the initial stage to amplify equity returns; public benchmarks are unlevered. Once public markets are adjusted to comparable risk, the excess return collapses. Simulations from 2000-2024 show that a 20% buyout allocation generates only about 64bps of annual excess return over an unlevered benchmark, and roughly nothing once the benchmark is levered to match. Most of the apparent alpha turns out to be financial engineering, not investment skill².

Table 2: Private markets vs public markets (leveraged comparison, 2000–2024)

Asset Class	Portfolio Return	Benchmark (Unlevered)	Excess Return	Benchmark (Leveraged)	Excess Return (Leveraged)
Private Equity (Buyout)	7.54%	6.90%	0.64%	7.58%	-0.03%
Private Debt	7.37%	6.80%	0.57%	7.09%	0.29%
Venture Capital	5.71%	6.92%	-1.21%	,	Negative
Real Estate	6.89%	7.18%	-0.29%	,	Negative

Source: PitchBook simulations

The picture is one of conditional opportunities, not structural outperformance. Private markets can generate excess returns, but those returns are neither uniform nor guaranteed. Outcomes diverge sharply by asset class. Private debt emerges as the most consistent contributor, holding on to positive excess returns even on a leverage-adjusted basis. The buyout edge largely disappears; venture capital and real estate typically underperform. Results hinge on cycle, entry point and vintage, and the post-crisis years show notable regime sensitivity. The notion of a stable, persistent illiquidity premium is overstated; the return advantage is more fragile and context-dependent than the brochures suggest.

¹ Preqin, Burgiss, Cambridge Associates long-term private equity performance vs public benchmarks.

² PitchBook (2025), Are Private Markets Worth It?

Manager selection is the whole game. Private markets are defined less by an illiquidity premium than by a dispersion premium. Even modest selection skill, a moderately higher probability of landing in top-quartile funds, can lift total-portfolio return by 20-60bps a year, which is roughly 100-300bps of alpha inside the private allocation itself. The effect is largest in buyout and venture capital, where dispersion is widest, and smallest in private debt, where outcomes cluster. The risks of getting it wrong fall on direct and indirect investors alike, particularly those allocating at scale. Private markets reward access, discipline and underwriting capability far more than they reward passive exposure.

Table 3: Impact of manager selection skill (Average annualized alpha by strategy)

Asset Class	Underperformer Avoidance	Moderate Skill	High Skill
Private Equity (Buyout)	0.41%	0.35%	0.64%
Venture Capital	0.38%	0.32%	0.61%
Private Debt	0.25%	0.21%	0.38%
Real Estate	0.32%	0.28%	0.47%

Source: PitchBook simulations

Do retail investors get the B-product?

Three structural factors place retail vehicles on weaker terms than institutional ones. Higher fees eat into the return premium; retail investors risk being handed inferior underlying assets and in an asset class where the gap between best and worst is unusually wide, retail investors cannot meaningfully pick managers themselves. The three disadvantages compound, dragging down net returns.

Retail access comes with stacked fees that eat into the return premium. Institutional drawdown funds typically charge 1.5-2% on committed capital plus 20% carry over an 8% hurdle, and the largest LPs negotiate down within those bands, with within-fund management-fee dispersion of about 90bps. Wealth and retail vehicles carry additional cost layers, reflecting the work of getting private-market product to a broader audience: placement fees at the distribution platform, advisory fees from the adviser and a feeder-fund management fee on top of the underlying fund's economics (Table 4). Add it up and the gap between retail and institutional access often exceeds 200bps a year, enough, over a decade, to consume a meaningful share of the very illiquidity premium that justified the asset class to institutions in the first place.

Table 4: Fee layers in wealth channels

Fee type	Recipient	Purpose	Typical range	Notes
Servicing fees	Adviser (sometimes shared with brokerage)	Annual compensation for account management and client servicing	0.25% - 0.85% per year of client investment	Brokerage houses sometimes take a cut
Placement fees	Brokerage	One-time fee for offering the fund on the brokerage platform	Up to 0.5% of initial investment	Some funds cap combined servicing + placement at 0.85%
Brokerage commissions	Brokerage	Front-end commission charged at the point of sale	~2% on average (up to 3.5%)	Advisers have discretion to waive
Independent adviser fees	Independent adviser	Flat advisory fee on overall client assets	Variable (typically % of AUM)	Generally do not charge commissions or significant placement fees

Sources: FT, Allianz Research

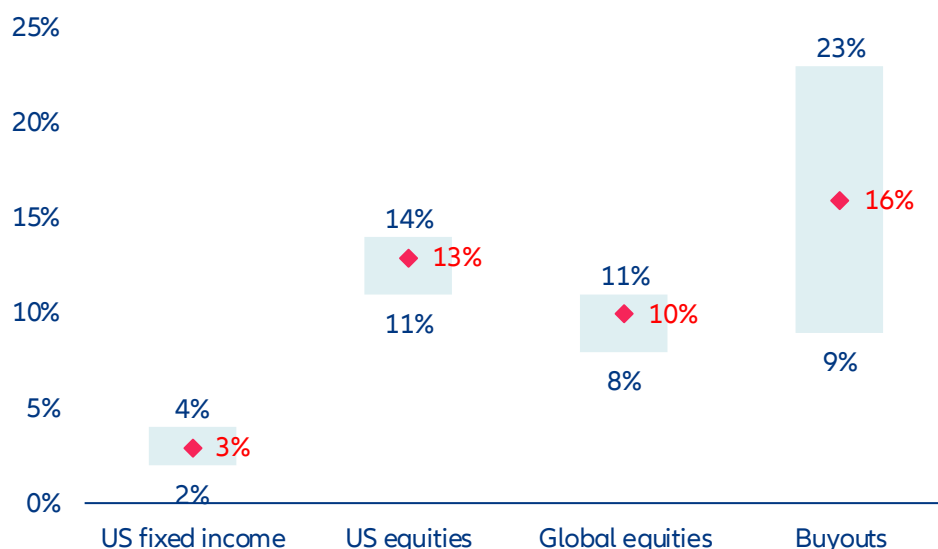
Notes: These fees sit on top of the underlying private capital fund's own management fee (typically 1.5%-2% of AUM) and performance fee (typically 20% of profits above an 8% hurdle)

Retail and institutional vehicles also tend to diverge in what they hold. Retail evergreens are naturally suited to perpetual-hold, lower-volatility assets that support NAV-based redemptions, while buy-to-exit strategies that need patient capital sit more comfortably in institutional drawdown funds. This is a structural feature of the architecture. The liquidity profile of the wrapper shapes what can sit inside it. The implication is that two vehicles carrying the same brand can deliver materially different return and risk profiles, and advisers and platforms will need to make that distinction legible. The rise of GP-led secondaries has created a channel for moving hard-to-exit assets between

funds. In these transactions, the GP is seller, buyer and valuation agent at once, an arrangement that puts the integrity of the price under strain.

The third disadvantage is the inability to choose. Manager selection is arguably the most important driver of private-market returns; the gap between top- and bottom-quartile private-equity managers is roughly 1,400bps, against around 300bps in public equity. For institutions, that dispersion is an opportunity. For retail investors, it is a risk. Mass-market participants in DC plans, insurance wrappers and platform-default products do not pick managers at all; intermediaries do, on the basis of their own incentives, including fee-sharing and distribution relationships, which can favor available managers over the most suitable ones (Figure 13).

Figure 14: Performance dispersion is especially wide for buyout funds



Sources: KKR, eVestment Alliance, Preqin, Allianz Research

Notes: Bars show median net IRRs; whiskers show first- and third-quartile thresholds. 15-year period through 31 December 2024. US equities include large- and small-cap indexes

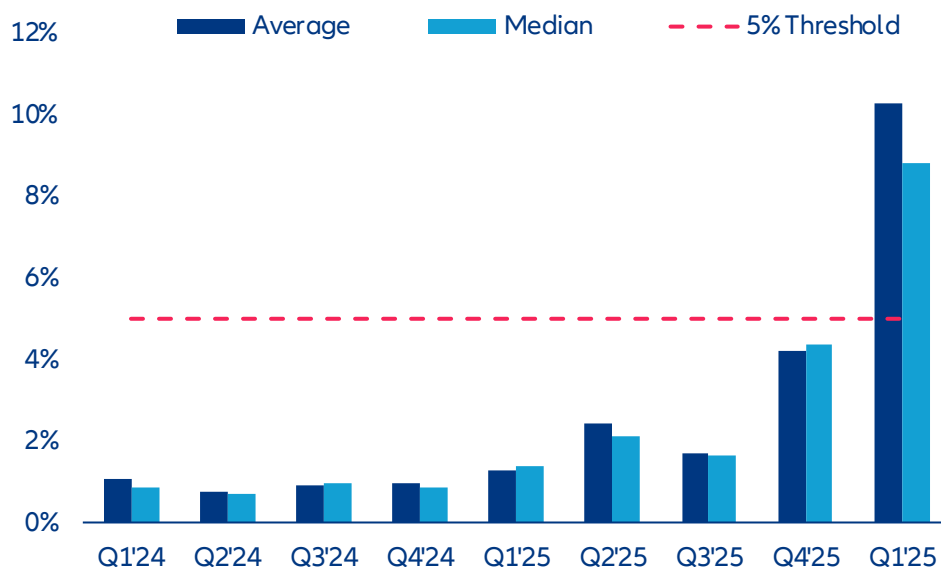
The three disadvantages mentioned above are interconnected and have a compounding effect. In order to achieve the same returns as institutions, retail investors must be willing to pay higher fees. Inferior underlying assets can negatively impact the gross return received. The inability to select top-quartile managers anchors the median retail outcome closer to the bottom of the dispersion than the top. When the effects compound, the median retail experience in an average private-market product can deliver a meaningfully lower net-of-fee return than an institutional investor in the same asset class, even though both are nominally invested in private markets. The gap is a function of product design, and well-built retail vehicles can close a portion of it.

The democratization narrative, therefore, comes with an important qualifier. Retail investors are gaining genuine access to assets that were previously out of reach, but the conditions of that access can compress the return premium that justifies the asset class in the first place. Whether the move from public to private markets pays off on a risk-adjusted basis depends on several factors: the specific product, the quality of the manager, the fee structure and the ability of the investor or adviser to distinguish well-engineered retail vehicles from those that travel less well outside the institutional setting they were designed for.

Is the “semi-liquid” promise an illusion?

An industry built on locking capital up is now selling products that promise to let it out. Retail investors face less predictable liquidity needs than institutions, and semi-liquid products try to bridge the gap with quarterly redemption windows. The bridge is narrow. Withdrawals are permitted only on set dates and after a holding period. If aggregate requests breach a preset threshold, typically 5% of NAV, the manager can invoke gates and honor only part of the request, leaving investors locked in for longer than they expected (Figure 15).

Figure 15: Redemption pressure is building for non-listed BDCs



Sources: SEC filings, Allianz Research

Notes: the data represent the mean and median redemption request rates for the eight largest non-listed BDCs

Disclosure has not always kept up with product complexity. Semi-liquid evergreens are typically marketed as a route into previously exclusive strategies, with the added convenience of periodic redemptions. Gates and conditional liquidity terms appear in the fund documentation, but get less prominence in marketing materials than the headline features. The gap between how the product behaves in calm markets and how it behaves under stress therefore surfaces only when redemption pressure rises, often to investors who had not fully absorbed the small print.

The underlying problem is that demand for liquidity peaks precisely when supply is lowest. Semi-liquid mechanisms work smoothly in normal markets. In periods of market stress, liquidity tightness or broad portfolio rebalancing, just when investors most want their money back, managers find it hardest to provide. Gates prevent the bank-run dynamic that would force fire sales and worse outcomes for everyone, but they trap investors at the worst possible time. Retail investors who expect access to their capital may discover that access depends on the same market conditions that drove them to seek it.

Once redemption pressure builds, the dynamics turn self-reinforcing. Initial requests draw headlines, which prompt more redemptions from anxious investors. Gating works as a brake, but the act of gating itself can be read as a distress signal, accelerating the next wave. If gates persist for several quarters, managers come under pressure to sell holdings or draw on subscription lines, producing losses for those who stayed and validating the concerns that drove the run. The non-traded BDC turmoil has played out exactly that way: redemptions rose, sponsors invoked caps, the caps generated more requests and gating spread across the category.

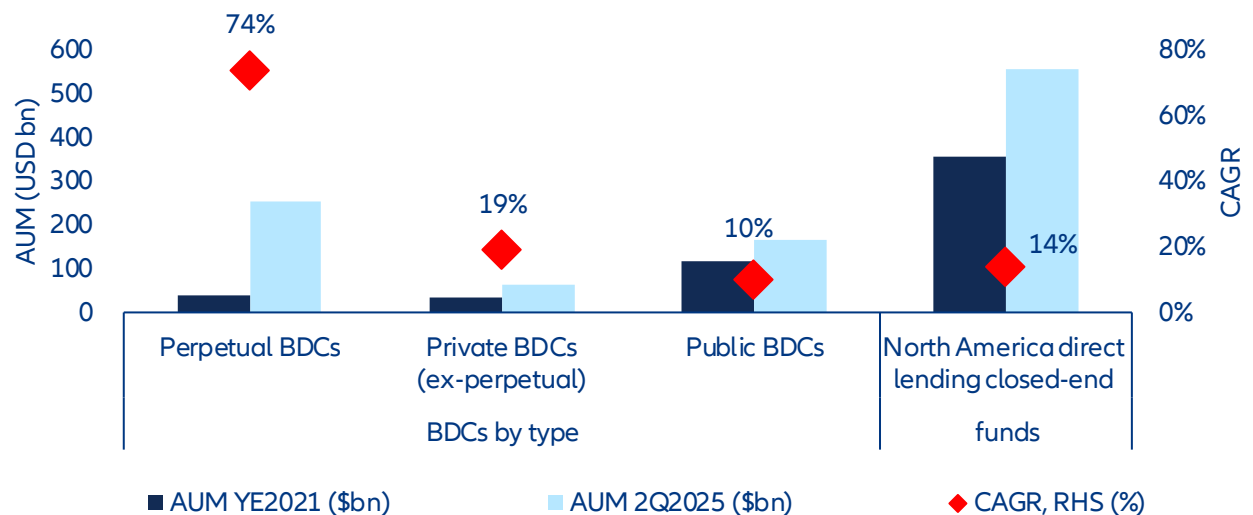
The semi-liquid promise is not an illusion; it is conditional. In normal markets, gates rarely trigger and redemptions clear smoothly. The mechanics work as intended. The case for democratization, however, depends on these structures holding up over time. The unfolding BDC episode is the first real-world stress test at scale, and a cautionary tale for the wider effort.

Box 2. The private BDC reckoning. A proof of concept for retail private credit or the proof of limits?

The Business Development Company (BDC) sector grew from USD190 bn in 2021 to over USD500bn by mid-2025, fueled by post-crisis bank retrenchment, floating-rate income and the push to bring private credit to retail and wealth-channel investors (Figure 16). The first quarter of 2026 provided the first real test of that expansion.

The result was not the broad credit crisis some had feared. It was something more telling. Wrappers designed for slow-moving institutional capital buckled when retail investors hit their limits, lagged marks and ran into limited liquidity in vehicles whose reported portfolios looked sound. This is not evidence of a failure in private credit; it is a reminder that product design, valuation architecture and investor base now matter alongside underlying credit quality. Any democratization vehicle built on periodic liquidity, manager-determined valuations and illiquid assets will carry some version of the same risk.

Figure 16. Perpetual BDCs have captured impressive AUM growth



Source Cliffwater Direct Lending Index, Preqin. As of Q2 2025

Perpetual non-traded BDCs pose a persistent design challenge. In Q1 2026, Blue Owl's technology vehicle received redemption requests totaling 40.7% of shares against a 5% quarterly cap, even though reported non-accruals were only 0.6%. The gap matters because quarterly liquidity sits on top of manager-marked, illiquid loans, giving investors an incentive to redeem before future marks absorb losses. The implication is durable: even if this queue clears, the wrapper stays vulnerable whenever confidence in reported NAV weakens.

Investors should know that reported income and reported credit quality can diverge for longer than commonly assumed. In Q4 2025, the share of private-credit loans classified as bad PIK reached 6.4%, against headline non-accrual rates of around 2%; Lincoln estimated a shadow default rate near 6%. The mechanism is straightforward: non-cash income preserves earnings optics while borrower stress is deferred, not resolved. As a result, dividend coverage, NAV stability and the perceived quality of the management team can look stronger than they really are until the recognition event arrives.

The issue is not just credit losses; it is delayed price discovery. BlackRock TCP Capital's roughly 19% NAV cut in late 2025 showed that loans previously carried near par could be repriced sharply once underlying weakness was acknowledged. This creates a feedback loop: stale marks dampen observed volatility but raise the incentive to exit before the revaluation lands. Future stress episodes are therefore likely to show up first as redemption pressure and only later as formal credit deterioration.

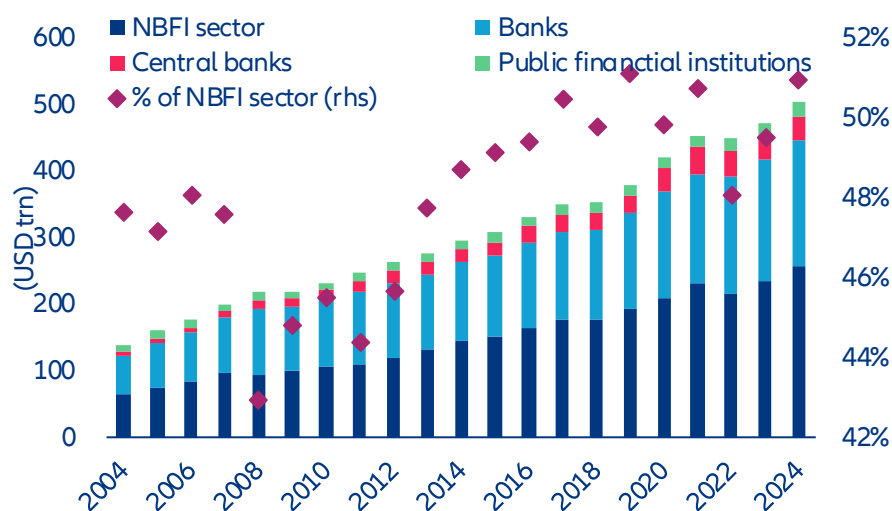
The 2026 episode points to structural vulnerability, not merely idiosyncratic manager error. According to Fitch, non-traded BDCs entered the period with lower leverage than their traded peers, yet sector-wide redemption requests for the largest vehicles averaged 12.1% in Q1 2026. The pattern suggests the pressure came from distribution and product design as much as from weak balance sheets. The implication for democratization is clear: expanding retail private credit will require a redesign of the framework, not just sharper marketing or stricter manager selection.

A new vector for systemic risk?

Even before retail money enters, private markets already inject several systemic risks into the financial system.

Start with banks. By end-2024, the non-bank financial intermediation (NBFI) sector held 51% of global financial assets (Figure 17), and the linkages have tightened: banks' credit exposure to private credit funds alone is estimated at USD270bn-500bn. In times of stress, private funds tend to draw on their committed credit lines just as banks are absorbing market volatility and rising loan losses. The drawdowns force previously off-balance-sheet exposures onto bank balance sheets, lifting capital requirements at the worst possible moment.

Figure 17: The NBFI sector has grown to 51% of global financial assets, tightening linkages with the banking system

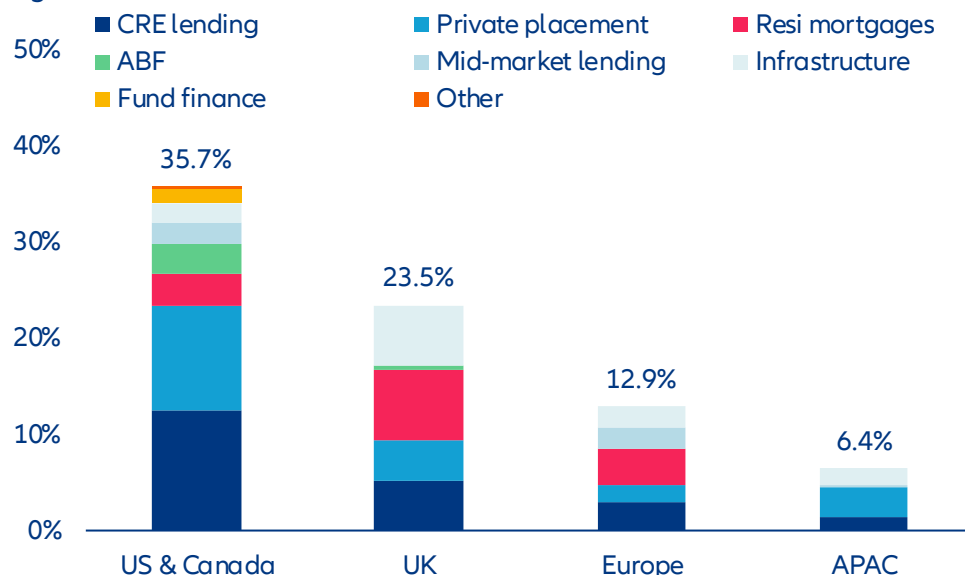


Sources: FSB, Allianz Research

Within the life sector, regulators have flagged a more specific concern: the rise of private-equity-backed insurers.

Private capital now controls roughly USD700bn of US life-insurance assets, and the market share of PE-backed insurers has climbed from 1% in 2012 to 13% by 2023. They have driven a broader shift toward private and structured credit, which now makes up 36% of North American life insurers' balance sheets (Figure 18). The IMF and the IAIS note that this subset of the industry tends to run thinner liquidity buffers and larger illiquid-credit positions than traditional, prudentially regulated life insurers. Stress at a PE-backed insurer can therefore feed pro-cyclical asset sales, as the 2023 collapse of Italy's Eurovita showed. The episode is a reminder that the case for retail private-market exposure rests on the quality of the wrapper as much as the quality of the asset, and that not all life-insurance balance sheets are built the same way.

Figure 18: Private credit accounts for 36% of North American life insurers' balance sheets



Sources: Moody's, Allianz Research

Democratization broadens these channels by tilting them toward a more reactive investor base. The institutional money that powered three decades of growth came with long horizons, contractual lock-ups and governance structures that resisted short-term liquidity pressure. Retail capital tends to chase peaks and redeem under stress, often on the same signals that drive broader volatility: news headlines, equity drawdowns, employment shocks. As retail's share of private-market AUM grows, each existing channel widens and the marginal investor becomes more skittish. The asset class may shift from absorbing market stress to amplifying it.

Democratization also introduces new risks. The mismatch between quarterly redemptions and decade-long assets revives run dynamics that the asset class had engineered away. Traditional institutional funds run on closed-end capital and multi-year commitments, which contain rapid redemption pressure: if a few funds suffer big losses, capital stays locked and sophisticated investors absorb the impact without a broader confidence shock. That mechanism weakens sharply once the investor base turns retail. Quarterly redemption rights on fundamentally illiquid assets reintroduce the maturity transformation that closed-end structures had removed. Gating can in principle prevent a bank-run dynamic, but the channels funds use to raise liquidity, credit lines and new loan issuance, are directly tied to banks and broader credit markets. If redemption pressure persists for several quarters, forced sales at distressed prices cannot be ruled out.

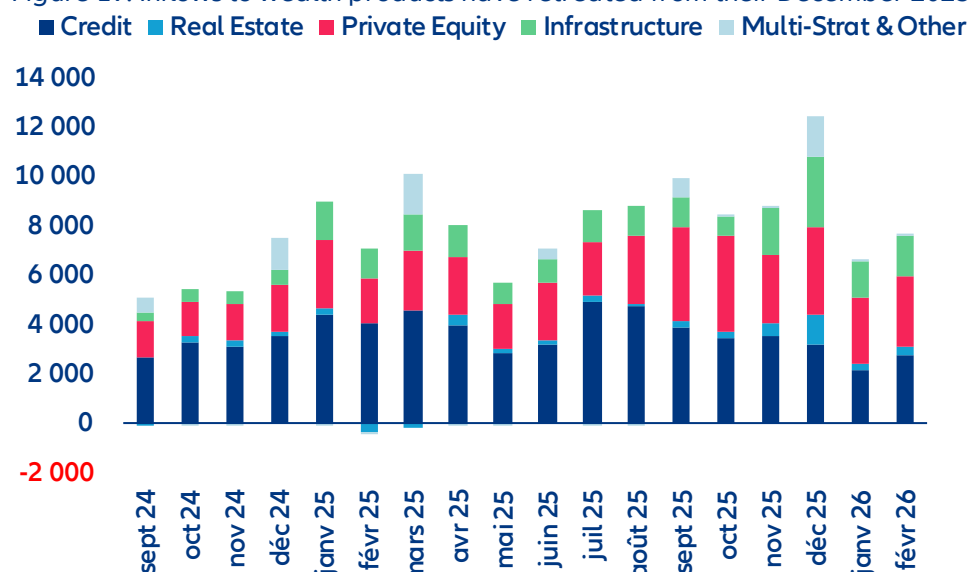
Valuation contagion is amplified when retail money is involved. Private assets are appraised by management, not by markets, and NAV marks can lag reality by several quarters. When a publicly traded vehicle holding similar private-credit loans trades at a persistent discount to its stated NAV, the market is signaling that it does not trust the valuation. That skepticism rarely stays local: it can prompt LPs to rethink commitments, lenders to tighten collateral terms and providers to pull subscription lines. Retail investors, who may not always tell well-marked vehicles from poorly-marked ones, tend to redeem indiscriminately when confidence breaks. Incentives in retail-focused funds can lean toward smoother, more optimistic NAV paths, since stable marks support both fee continuity and the credibility of the redemption mechanism. Where governance does not push back, this can build feedback loops of optimism that unwind sharply when tested.

Concentration among a handful of major managers can turn firm-level stress into a market-wide contagion. Six firms now control nearly half the retail-alternatives market, and the largest individual vehicles top USD100bn in assets. The GPs that have been most aggressive in raising retail capital also run multiple semi-liquid wrappers across product categories. Stress in one large manager's product can therefore spread quickly through correlated vehicles run by the same firm, and through the wider retail-alternatives universe, where so much AUM sits with so few entities.

What happens next?

Democratization has hit its first substantial obstacle. Inflows to wealth products at the major listed alternative managers have fallen sharply from their December 2025 peak (Figure 19). BDC sales were down roughly 40% year-on-year in early 2026, and redemption caps were triggered across major non-traded vehicles. The narrative has gone from "growth drivers" to "discipline and selectivity" in a single year.

Figure 19: Inflows to wealth products have retreated from their December 2025 peak



Sources: JP Morgan, Allianz Research

This looks like a stress test, not a fundamental shift. The BDC episode exposed real vulnerabilities in semi-liquid product design, and the fundraising slowdown reflects multi-year exit constraints and investor over-allocation. The policy scaffolding put in place over the past three years – the DOL safe-harbor rule, ELTIF 2.0 and LTAF ISA eligibility – is built and unlikely to be dismantled as the pressures that drove it – retirement-adequacy gaps, infrastructure funding needs and the hunt for broader investment opportunity – remain unresolved.

The next phase will be recalibration. Product designs are evolving in three visible directions. First, larger liquid-asset buffers inside semi-liquid wrappers, with some new launches reported to hold 25-40% in public securities to support redemptions. Second, tighter gating, longer notice periods, smaller per-quarter caps and clearer disclosure of restrictions at the point of sale. Third, a migration toward asset classes better suited to retail liquidity profiles.

The most likely structural outcome is consolidation. Mega-GPs are well placed to ride the coming wave of retail capital; smaller managers without distribution scale, brand recognition or the operational capacity to run semi-liquid wrappers will struggle to compete for retail flows. The projected parity of retail and institutional fundraising by 2030 is therefore likely to coincide with a sharp concentration of AUM among a few large managers, deepening the concentration risk noted above.

The regulatory environment has been benign so far, but the BDC turbulence will probably trigger a new round of scrutiny. Adding liquidity has resurfaced concerns that closed-end structures had previously sidestepped. Lock-ins removed deposit-like risk by stopping investors from pulling capital on demand, which limited the channels through which fund stress could spread. Open-ended structures bring back the duration mismatch and tilt the investor base toward less sophisticated participants more likely to react to market volatility. As private-markets funds push into territory once held by bank lending, regulators face a familiar dilemma: the same maturity transformation that justified the post-2008 reforms is being recreated outside the bank-regulatory perimeter, and now sold to retail investors.

Regulators themselves are divided. The SEC has explicitly rejected the idea that private credit is a systemic threat, prioritizing capital formation. The IMF, the Bank of England and the IAIS take a different view, flagging

interconnections, valuation opacity and less sophisticated investors. The disagreement is essentially about how much weight to place on tail risks that have not yet materialized.

Four developments will shape the next 12-24 months. The DOL's safe-harbor rule for 401(k) alternatives, in public comment through mid-2026, will decide whether plan sponsors get the legal certainty to offer alternatives broadly in DC plans. *Anderson v. Intel*, before the US Supreme Court in the autumn 2026 term, will decide whether ERISA fiduciary-breach claims require "meaningful benchmarks", with direct implications for plan-sponsor litigation risk. ELTIF 2.0 will get its first proper stress test as semi-liquid launches multiply and distribution widens across the EU. And the BDC gating events of early 2026 will set the tone for retail investor protection for years, depending on whether the SEC or FINRA pursues formal enforcement on point-of-sale disclosure.

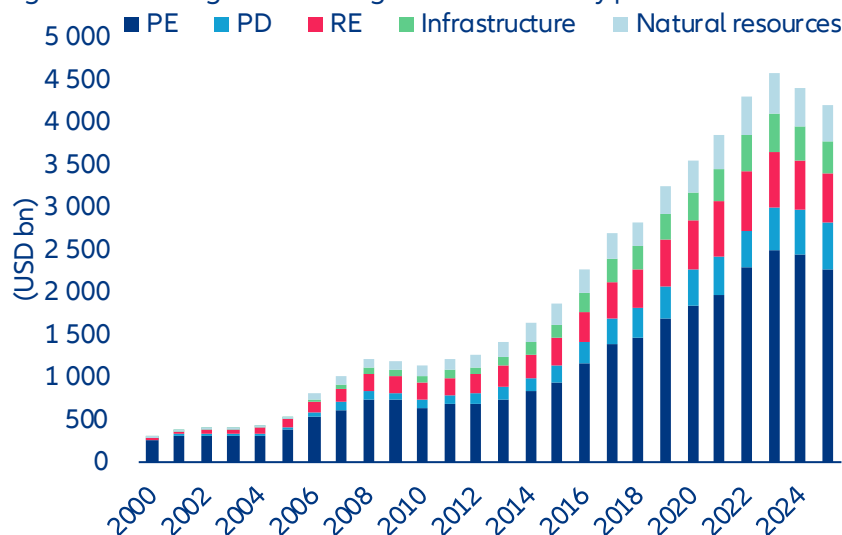
Whatever its form, the regulatory response will add cost and complexity. The BDC episode has already pushed the SEC to tighten oversight of how BDCs value illiquid holdings in falling markets. New disclosure rules, leverage limits, restrictions on retail-targeted marketing and mandatory liquidity buffers are all on the table in the major jurisdictions. Each addresses a real concern; each adds operational burden on GPs, costs that will be passed on to LPs and will erode the net-of-fee returns that justified the asset class in the first place. The deeper worry is that regulation will arrive after retail losses, eroding four decades of structural advantage.

What does this mean for investors?

Not all retail private-market vehicles are built the same way, and the distance between well-engineered and poorly-engineered product is likely to widen as the channel scales. Four markers separate the two. The first is liquidity architecture: adequate public-asset buffers, conservative gating thresholds and redemption windows calibrated to the underlying asset, rather than to the distribution channel. The second is clean separation between retail and institutional pools, so that allocation conflicts and cross-subsidies do not arise. The third is valuation governance: independent pricing, regular external review and transparency on NAV methodology. The fourth is manager selection capability, both at the GP level and, where the wrapper allows it, at the underlying-asset level. Investors and advisers will increasingly need to look past the brand and the headline strategy to these design features. For the industry, the opportunity is to build retail product that is genuinely institutional in its discipline, rather than institutional only in its marketing.

If retail capital arrives at scale, the illiquidity premium is likely to compress. Broader participation will erode the conditions that historically underpinned private-market outperformance, with more capital chasing an opportunity set that is already increasingly competitive. The pressure is acute now: managers are sitting on more than USD4trn of dry powder (Figure 20) to deploy in an environment of constrained exits and intense competition for quality deals. Institutions have historically captured the illiquidity premium through patient capital. That advantage is diluted as retail capital enters at scale.

Figure 20: Managers are sitting on USD4trn of dry powder



Sources: Preqin, Allianz Research

More fundamentally, the rise of retail products creates a structural conflict of interest in deal allocation. GPs must now spread a finite pipeline of deals across products with very different investor bases and liquidity profiles. That complicates life for institutional LPs, whose visibility on whether their interests are being protected, whether assets are being transferred between vehicles on unfavorable terms, and whether their original strategies are being reshaped to accommodate retail liquidity needs.

If GPs prioritize institutional vehicles, institutional returns are not directly compromised, but new risks emerge. Retail products receive systematically secondary deal flow while institutional LPs continue to see the most attractive transactions. A more troubling possibility is that retail vehicles end up serving as liquidity outlets for the institutional side, especially in tough exit environments, with hard-to-sell assets channeled into retail evergreens. In these trades the GP is seller, buyer and valuation agent at once, an arrangement whose integrity depends on a valuation discipline that is hard to sustain when the same firm is on both sides.

If GPs allocate deals on broadly equivalent terms across institutional and retail products, the critical question is whether retail and institutional capital is cleanly separated. In pooled vehicles, retail capital flows directly into vehicles with the same exposure as institutional LPs, which exposes institutional capital to the risk of more reactive retail behavior. In deployment, retail inflows tend to peak when valuations are high and competition for deals is fiercest, pushing GPs to deploy into transactions that would otherwise pace through. In the holding period, retail redemption pressure during market stress can force premature sales at distressed prices, with losses that institutional capital would have absorbed and recovered through patience. In distribution, the same pressure can disrupt the orderly exit timing institutional LPs rely on, with assets sold into weaker markets to meet retail liquidity needs rather than held until the optimal exit window. The institutional return premium earned from patient, locked-up capital is diluted in this scenario.

The corollary is that the firms best positioned in the next phase will be those that combine institutional underwriting with retail-aware product design: large balance sheets that can carry liquidity buffers without diluting returns, integrated insurance and asset-management capabilities that internalize valuation and governance discipline and distribution that reaches savers without adding unnecessary cost layers. For long-horizon retail investors, the structures that meet these tests are not a watered-down version of institutional product. They are a route to diversification and return outcomes that public markets, increasingly concentrated and increasingly correlated, can no longer deliver on their own.

These assessments are, as always, subject to the disclaimer provided below.

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Thinking fast, building slow: The energy cost of the US AI boom

12 May 2026

Content

Page 3-4

Executive Summary

Page 5-10

Can power markets handle the AI surge?

Page 11-12

Beyond the baseline: Agentic AI, adoption velocity and the rebound effect

Page 13-17

The hidden bill?: AI's impact on power prices

Page 18-19

Preparing AI for energy and energy for AI: policy recommendations

Executive Summary



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- **Artificial intelligence is about to impose the largest sustained demand shock on US electricity infrastructure in decades.** By 2030, data-center power consumption is expected to nearly double, lifting the sector's share of total US electricity demand from roughly 5% to around 9%. Although planned generation additions look sufficient on paper, data centers may absorb nearly half of projected new capacity, leaving thin margins if electric-vehicle adoption or industrial electrification accelerate faster than expected. Yet demand itself is evolving faster than forecasts can capture: generative AI reached 53% population-level adoption in just three years, agentic systems consume far more energy per interaction than conventional workloads and a 280-fold decline in inference costs since 2022 is driving a rebound effect that efficiency gains alone cannot offset.

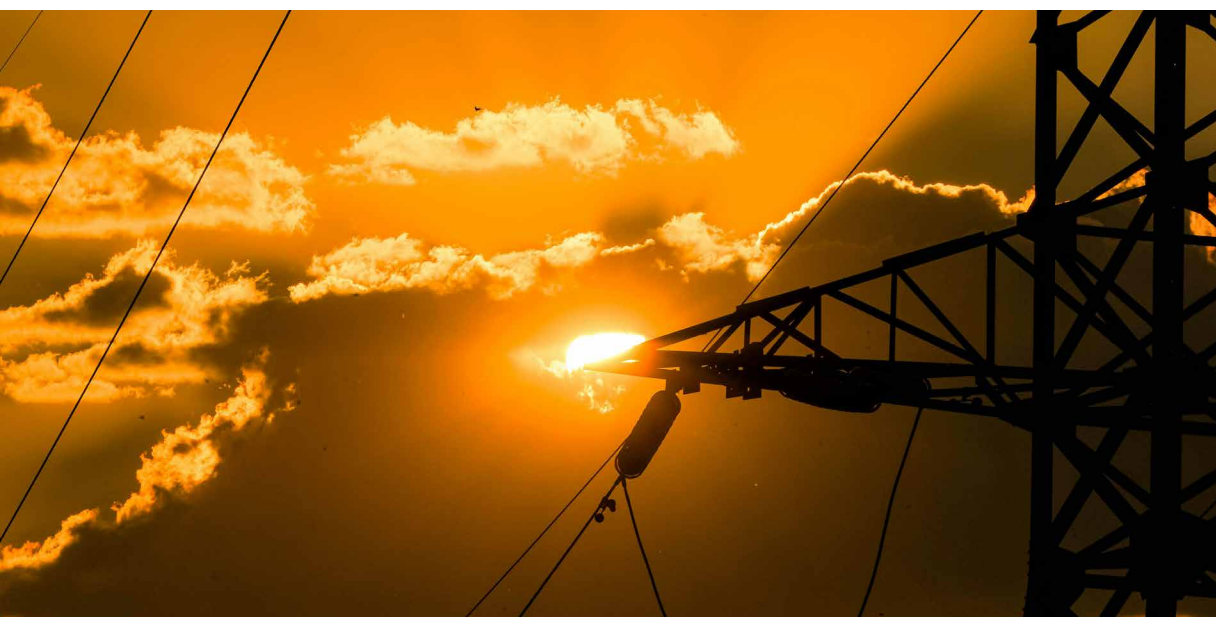
- **The critical bottleneck, however, is not generation capacity but the grid itself.** Data centers can be built in under two years, but grid connections can take up to seven years in congested markets such as Northern Virginia. Nationwide interconnection requests now total 1.84 terawatts, exceeding total installed US generating capacity. Supply-chain shortages have pushed lead times for critical grid equipment to several years, while projected demand would require building roughly 8,000 km of high-voltage transmission lines annually – around ten times the current pace. The strain is already showing: In early 2026, the Department of Energy invoked emergency powers to shift data centers onto backup generation during peak demand periods. Rising public opposition and legislative scrutiny add a further layer of uncertainty that conventional supply forecasts have yet to fully capture. The strain is already visible in project pipelines, with half of the 12 GW of US data-center capacity planned for 2026 being delayed or cancelled.

- **Aggregate electricity prices have not yet fully reflected these pressures, but a growing „data-center premium“ is emerging.** States hosting the highest concentration of data-center activity have so far seen price inflation below the national average, reflecting favorable grid conditions, economies of scale and the lagged structure of utility rate-setting. Beneath the surface, however, the impact has become increasingly visible since 2023. US residential customers are already paying USD1.4bn more per year on their electricity bills as a direct result of data-center demand, with just five utilities serving 4.4mn households in Northern Virginia, the Pacific Northwest and Arizona accounting for more than 40% of that total. The sales-weighted average price effect across all utilities sits at just 0.6%, but for the most exposed utilities roughly 7.8pps of a 24.5% cumulative price increase between 2020 and 2024 are directly attributable to data-center demand, adding 0.19pp to headline inflation over four years through the direct electricity channel alone. These markets historically benefited from below-average electricity costs, a

gap that has already narrowed from 5% to 3.7% since 2020 and is set to close further. Data-center investment grew 32% in 2025 and is set to rise a further 75% in 2026 alone, pointing to an additional electricity price effect of close to 14pps for the most exposed utilities over 2025–2026, nearly doubling the cumulative four-year effect in just two years. Meanwhile investor-owned utilities filed USD18bn in rate-increase requests in 2025, the highest since the mid-1980s, with costs largely falling on existing rate-payers rather than the facilities driving them.

- **Preparing energy infrastructure for AI and addressing community concerns is as urgent as building the AI infrastructure itself.** The immediate priority is interconnection reform: binding timelines, penalties for speculative queue filings and priority treatment for shovel-ready projects with firm power commitments would help relieve the most acute bottlenecks. Cost allocation is equally important. Unless data centers bear a proportionate share of the infrastructure costs they create, public opposition and permitting delays will intensify further. Mandatory energy-use disclosure, incentives to redirect investment away from saturated regions, stronger efficiency standards, demand-flexibility mechanisms and stricter additionality requirements for power-purchase agreements would fill the most critical gaps in a policy framework that remains largely inadequate to the scale of the challenge – and lay the foundation for AI ambitions that the grid can actually support.





Can power markets handle the AI surge?

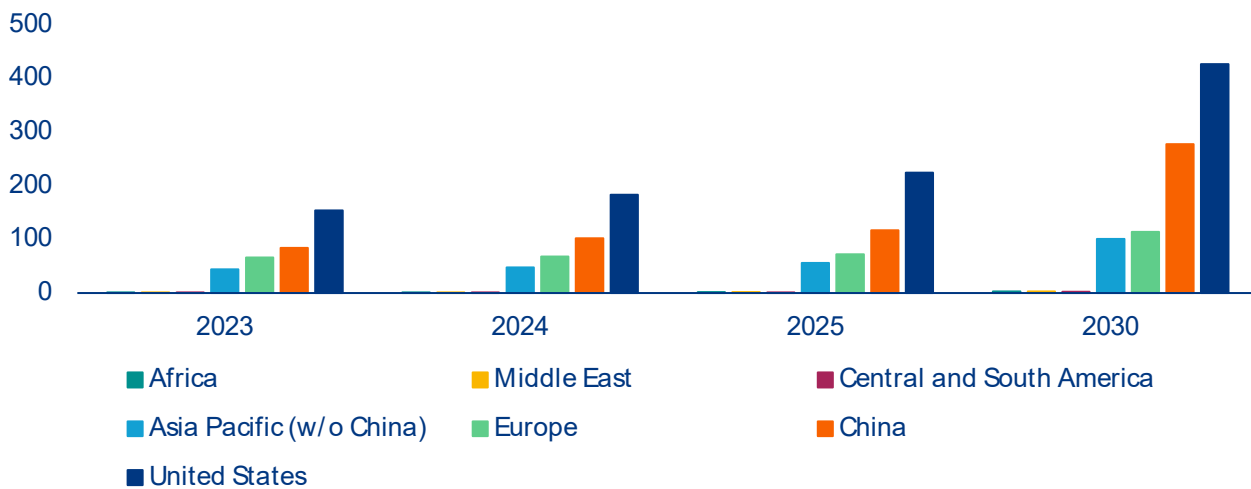
The AI revolution is reshaping global economies, starting with power markets. In the IEA's baseline scenario, AI-fueled growth in data-center power demand is expected to increase between 58% and 137% across major world regions between 2025 and 2030, supported by annual investment that now rivals the scale of the entire global oil and gas industry (Figure 1).¹ The shift is particularly pronounced in the US, where data-center power demand is projected to reach 426 TWh by 2030, nearly double the 2025 level. Based on long-term EIA projections, our base case estimates suggest this figure could double again by 2050, reaching roughly 810TWh. While the precise share attributable

to AI remains difficult to isolate, bottom-up modeling suggests AI inference already accounts for 12–14% of total consumption, with total AI data-center capacity reaching 29.6 GW by end-2025.² The implications for power systems are already visible in the US, where the data center share of total electricity consumption could jump from around 5% today to between 7% and 12% by 2028, with our estimate, based on current supply pipelines and baseline demand assumptions, pointing to a share of around 9%.³

¹ [Key Questions on Energy and AI \(IEA\)](#)

² [The 2026 AI Index Report | Stanford HAI](#)

³ [2024 United States Data Center Energy Usage Report | Energy Technologies Area](#)

Figure 1: Data-center power demand growth by region (in TWh)

Source: IEA baseline projection

This significant demand growth is fueling a massive expansion in generation capacity. According to the Energy Information Administration's (EIA) Short-Term Energy Outlook, power generation in the US is expected to grow by 441 TWh between 2023 and 2027, an increase of 10.5%.⁴ This would mean the five years since the breakthrough of generative AI see a larger increase than the preceding two decades, over which power generation grew by only 7.7%. 76% of the increase in power generation is expected to be driven by wind and solar (21.9% wind and 54.1% solar) with the remaining 23% supplied by gas (15.3%), nuclear (5.2%) and hydropower (3.0%). Coal remains on its post-2007 declining trend, though the pace has slowed markedly from -4% annual reductions before the AI era to just -0.5% today.

On paper, the capacity additions currently included in the EIA's pipeline of planned and under-construction projects appear broadly sufficient to accommodate rising data-center demand, with data centers expected to absorb around 47.1% of projected new supply.⁵ That apparent adequacy, however, rests on the fragile assumption that competing sources of electricity demand remain relatively contained. In reality, the grid is coming under pressure from

multiple directions simultaneously. Beyond data centers, electrified transport alone could materially tighten the balance. Under the IEA's baseline scenario, US electricity consumption from electric vehicles is projected to rise from 25.2TWh in 2024 to 91.4TWh by 2030. Near-term signals are mixed, but the risks remain tilted to the upside: Secondary-market EV sales continue to rise despite the expiry of the federal tax credit, while gasoline prices elevated by the escalation in the Middle East are reinforcing the economic case for adoption. Historically, sustained fuel-price pressure has been one of the strongest catalysts for accelerating EV uptake.⁶ Industrial electrification adds a further layer of uncertainty: Manufacturing reshoring and the electrification of industrial processes point towards structurally higher power demand that remains difficult to forecast with precision. Taken together, these competing demand vectors suggest that the supply cushion that seems comfortable could narrow far more quickly than current projections imply.

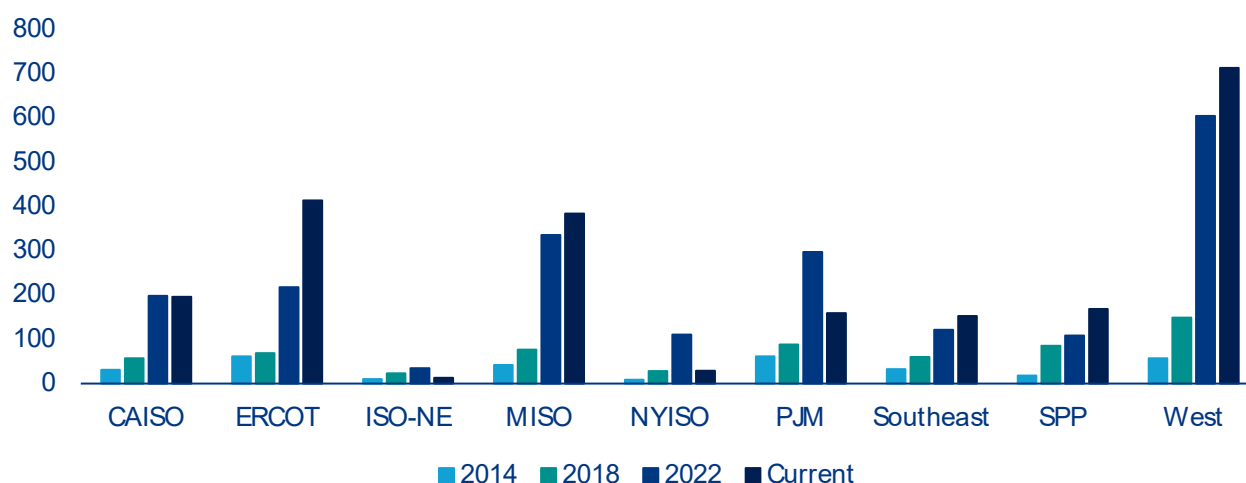
The bigger issue lies on the supply side and is rooted in US power grids, which struggle to keep pace with data-center interconnection demands. While physical construction typically takes under 24 months, grid-connection queues can more than double total

⁴ IEA STEO, April 2026 update⁵ Based on EIA-860M (February 2026 update)⁶ Global EV Outlook 2025 - IEA

project timelines, with hotspots like Northern Virginia reporting waiting times of up to seven years.⁷ This bottleneck reflects a deeper structural problem as both new generation capacity and new demand face the same overloaded interconnection queues (Figure 2). In Texas (ERCOT region), generation and storage projects queuing for grid connection nearly quadrupled between 2020 and 2024, while nationally active interconnection requests now stand at 1.84 TW, exceeding total US installed generating capacity. That figure overstates the genuine pipeline as only around 20% of requests ultimately reach commercial operation, with the remainder abandoned due to speculative filings, rising costs and mounting delays. Resolving these constraints requires grid expansion on a scale the US has not attempted in decades: In their 2024 National Transmission Planning study, the DOE estimates that in a lowest-cost scenario the total transmission system would increase by 2.1-2.6 times until 2050 and up to 3.3 times in a high demand case. This would imply building around 8000km of high voltage transmission lines per year, a pace that appears far beyond current construction rates, which have averaged well below 1,000 km annually in recent years.⁸

Getting there is complicated by constraints that span the entire supply chain. The grid equipment needed (transformers, switchgear and cables) faces lead times of up to four years alongside a 30% supply shortfall.⁹ Building new transmission lines compounds the problem further, taking four to eight years in advanced economies, a timeline fundamentally incompatible with the pace of data-center deployment. Beyond power infrastructure, the AI build-out is also running into semiconductor supply constraints: According to the IEA, HBM chip shortages have emerged as a binding bottleneck over the past six months, with insufficient supply of this specialized memory threatening to slow AI data-center deployment independently of grid availability.¹⁰ A construction workforce shortage of around 439,000 skilled workers and increasingly complex permitting processes compound the problem further. As a result, almost half of the approximately 12 GW of US data center capacity planned for 2026 is expected to be delayed or cancelled, with only around 5 GW currently under active construction.¹¹

Figure 2: US power capacity in queue for grid connection by electricity market region (in GW)



Source: LBNL, interconnection.fyi

⁷ [Epoch AI and Energy and AI](#)

⁸ [DOE and Grid Strategies](#)

⁹ [Wood Mackenzie](#)

¹⁰ [Key questions on energy and AI - IEA](#)

¹¹ [Slightline Climate Data and Bloomberg](#)

Meanwhile grid stress from data centers has moved from projection to reality, helping explain the prolonged integration assessments that slow their expansion. Unlike traditional commercial or residential load, data centers represent large, concentrated blocks of power demand that can disconnect or transfer to backup power in a coordinated manner during grid disturbances. Compounding this, AI-optimized data centers can exhibit sharper and more frequent load swings than conventional facilities: As workloads shift between high-intensity processing and lower-utilization periods, power draw can fluctuate rapidly, placing greater demands on frequency regulation and reserve markets. When a disturbance triggers a simultaneous disconnect, as occurred in Virginia in 2024, where 60 facilities representing 1.5 GW dropped off the grid at once, the sudden loss of load forces operators to curtail generation equally fast to avoid a frequency imbalance that can cascade into widespread blackouts.¹² The reverse risk is equally real: When data centers draw at full capacity during peak periods, they can overwhelm local transmission capacity, forcing power to reroute through congested lines and driving up congestion costs across the broader grid. This strain is increasingly forcing active intervention, as seen in January 2026 when the DOE invoked emergency powers to authorize PJM and ERCOT to shift data centers onto backup generators to prevent residential blackouts during a severe winter storm. Both failure modes reflect the same underlying mismatch: a grid designed for distributed, flexible demand is being asked to absorb loads that are geographically concentrated and operationally rigid.

These physical bottlenecks are now increasingly intersecting with political and social constraints.

Organized opposition to data-center development has emerged as an additional factor shaping the outlook. In 2025, an estimated USD156bn worth of data-center projects were blocked or delayed by local opposition, moratoriums and litigation in the US, with project cancellations more than quadrupling from six in 2024 to 25 in 2025.¹³ The opposition is largely rooted in concerns about electricity costs as the infrastructure required to accommodate data-center load growth is typically funded through broader rate-payer tariffs rather than charged directly to the facilities driving demand. In PJM alone, the grid's independent market monitor estimates that data-center load growth added USD16.6bn in capacity costs across the 2025/26 and 2026/27 delivery years, spread across 67mn consumers.¹⁴ Legislative responses are following, with moratorium bills introduced in at least 11 states in 2026, with some like Maine considering statewide construction pauses.¹⁵ While unlikely to halt the AI infrastructure buildout, rising opposition introduces permitting delays and regulatory uncertainty that compound existing supply-side constraints and underscores a broader question about who ultimately bears the cost of AI infrastructure.

¹² DCD

¹³ [Data Center Watch and Construction Dive](#)

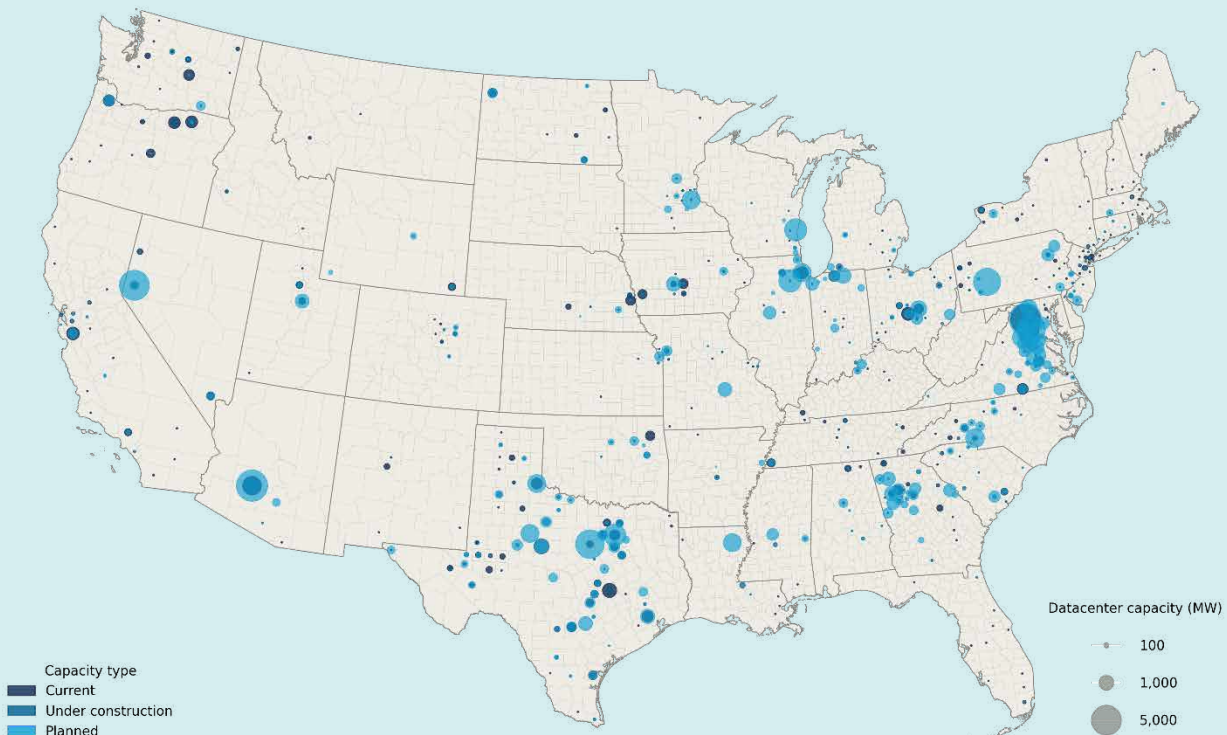
¹⁴ [Comment of the Independent Market Monitor for PJM](#)

¹⁵ [Axios](#)

Where are data centers built and what determines their location?

Data center location reflects a specific set of infrastructure, cost and regulatory requirements that have historically concentrated capacity in a small number of markets. Today, Texas and Virginia alone account for 42% of total US data center capacity, though the nature of that concentration differs markedly between the two. Virginia, and Northern Virginia's Loudoun County in particular, dominates current installed capacity and represents the largest data-center cluster in the world. Texas, by contrast, carries the largest pipeline of planned and under-construction capacity, reflecting its abundance of land, a deregulated electricity market, and aggressive state-level incentives (Figure 3).

Figure 3: US data center capacity by county (in MW)



Source: National Laboratory of the Rockies.

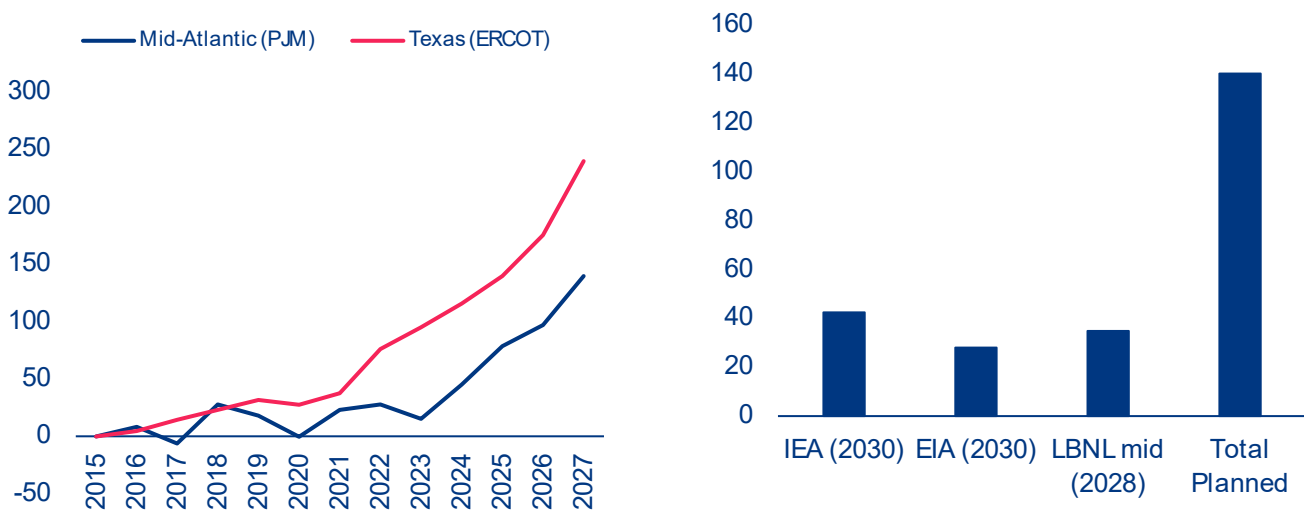
Four factors dominate siting decisions. Power availability and cost are the most fundamental as cheap and reliable electricity underpins the economics of large-scale compute. Fiber connectivity to major internet exchange points determines latency and operational viability. Land availability and permitting speed determine how quickly large campuses can be developed. Tax incentives, particularly data-center exemptions offered by Virginia, Texas and several other states, have actively directed investment toward specific markets.

The consequence of this concentration is both operational and systemic risk. When grid capacity tightens in saturated markets, the AI infrastructure buildout slows not just regionally but globally, given the role these clusters play in serving international demand. A regional grid failure in these clusters would ripple globally, translating into higher prices for AI and cloud services, slower model deployment and capacity-driven rate limiting for end users. Overclustered markets also face compounding vulnerabilities across power supply, water availability, cooling infrastructure and skilled labor, making geographic diversification of future capacity an increasingly urgent priority.

While aggregate power-supply growth appears sufficient on paper, the regional picture is more nuanced. In the two most exposed markets, ERCOT (Electric Reliability Council of Texas) and PJM (PJM Interconnection), power generation growth has accelerated sharply since 2022, jumping from 10 TWh p.a. to 22 TWh p.a. in Texas and from 4 TWh p.a. to 31 TWh p.a. in PJM (Figure 4 a). This momentum is expected to continue, with the EIA's Short-Term Energy Outlook projecting an additional 99 TWh of supply in Texas and around 60 TWh in PJM by end-2027. On the demand side, Texas and Virginia each carry approximately 30 GW of data center capacity in their pipelines, which, if fully implemented, would imply an increase in annual electricity demand of around 140 TWh per region. In practice, however, given typical build-out timelines and project attrition, a more realistic estimate puts incremental data center demand at 30–60 TWh per region by 2030 (Figure 4 b). At those levels, both markets can realistically absorb expected AI-driven load growth, provided their supply pipelines materialize.

Delivering on that supply potential is, however, not without obstacles. In ERCOT, the supply pipeline is theoretically comfortable, but interconnection queues have grown sharply in recent years, raising the risk that capacity additions lag behind the pace of data center buildout. In PJM, the challenge is more structural. Generation growth has been slower and surrounding regions offer limited import relief, given mostly flat supply growth expectations. Data centers are also not the only source of incremental demand, with electrification of transport and industry adding further pressure on available grid capacity. With 105 GW currently queued in PJM's interconnection queue, processing delays alone could push a significant share of planned capacity beyond the 2030 horizon. Should capacity additions in either market fail to keep pace with demand, persistently tight grid conditions would translate into additional upward pressure on regional electricity prices.

Figure 4: Estimated power generation growth by grid region vs 2015 (lhs, TWh) and estimated data center power demand in Texas and Virginia (rhs, TWh)



Source: Allianz research based on EIA (STEO), IEA, LBNL and National Laboratory of the Rockies; Note: Planned power demand was estimated based on state level NLR capacity data for facilities planned or under construction. IEA, EIA and LBNL country level estimates were downscaled to regions based on planned capacity shares.



Beyond the baseline: Agentic AI, adoption velocity and the rebound effect

The future trajectory of AI's impact on power markets is shaped by two opposing forces. Rapid efficiency improvements are materially reducing the energy cost per token, while surging adoption, growing usage complexity and the emergence of agentic workflows are pushing demand in the opposite direction. Which force dominates will determine whether current supply projections prove adequate or significantly understated.

The energy efficiency of AI systems has improved dramatically across all layers of the technology stack... Compute performance per watt has increased by around 34% per year since 2008, with successive GPU generations delivering compounding gains.¹⁶ At the model level, architectural innovations such as Mixture-of-Experts designs, quantization, speculative decoding and model distillation have each contributed further reductions in energy per task, in some cases cutting compute

requirements by several multiples without meaningful loss in output quality. On the infrastructure side, hyperscale and AI-specialized data centers have become significantly more efficient at converting electricity into useful computation, with overhead losses from cooling and power distribution falling from around 40–50% of total facility energy in older facilities to just 10–20% in modern ones.¹⁷ According to the IEA, widespread adoption of existing AI applications across industry, buildings and transport could deliver aggregate energy savings of over 13 EJ by 2035, equivalent to around 3% of global final energy consumption.¹⁸ Realizing this potential, however, is far from guaranteed, as widespread adoption remains contingent on overcoming persistent barriers including fragmented data, limited digital skills and inadequate regulatory incentives.

¹⁶ [Epoch AI](#)

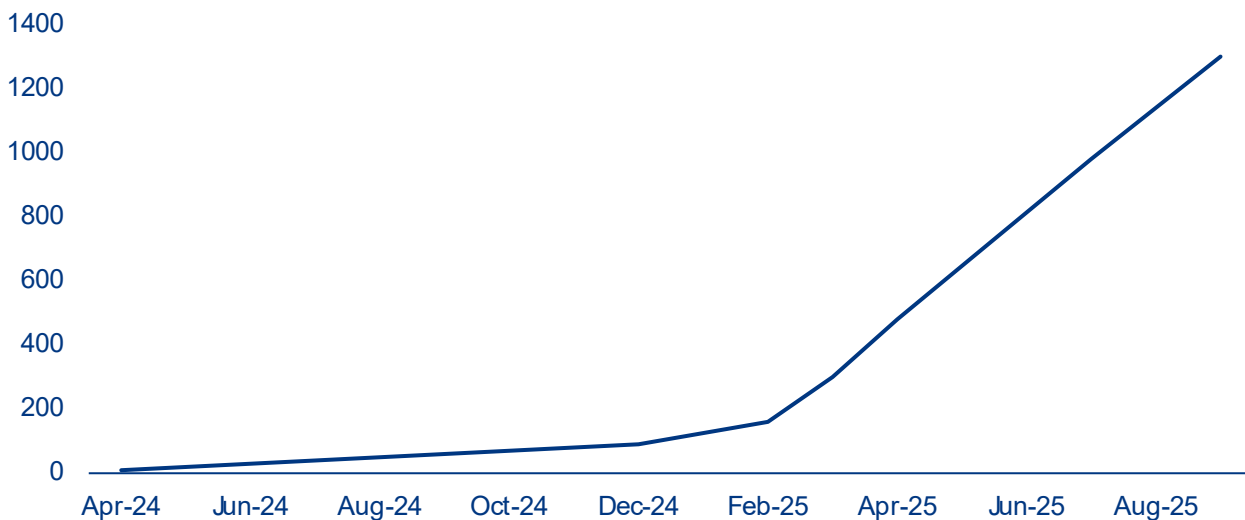
¹⁷ [LBNL](#)

¹⁸ [Key Questions on Energy and AI - IEA](#)

...yet demand is expanding at least as fast, driven by rapid adoption, growing model capability and a structural shift in how AI is used. Today, generative AI has reached 53% population-level adoption within three years of its mass-market introduction, faster than the personal computer or the internet, with 88% of companies reporting AI use in 2025 according to Stanford's AI Index.¹⁹ The models powering this growth have scaled at an equally striking pace, with the compute required to train frontier AI models growing at roughly 4–5x per year since 2020 and GPT-4 estimated at around 1.8trn parameters, roughly 10 times larger than GPT-3 released just three years prior.²⁰ Usage

intensity has followed the same trajectory. According to OpenRouter's platform data, average prompt length has grown roughly fourfold since early 2024 as users shifted from simple queries to context-heavy, multi-step workflows, with reasoning models growing from a negligible share to over 50% of token usage by late 2025.²¹ This growth is reflected in token consumption figures across major AI companies. OpenAI's API alone grew from 300mn tokens per minute in 2023 to over 6bn by late 2025, a 20-fold increase in under two years, while Google's Gemini token consumption increased approximately sevenfold between February and September 2025 (Figure 5).²¹

Figure 5: Google Gemini monthly token consumption (trillion tokens)



Sources: Google DeepMind; Alphabet Q2 2025 earnings.

The primary driver behind this acceleration is a fundamental shift in how AI is being used. Unlike conventional chat interactions, agentic systems plan and execute multi-step tasks autonomously, dramatically increasing token consumption per interaction. The energy implications are substantial, with an agentic task with reasoning consuming around 50Wh compared with roughly 0.3Wh for a standard text query, roughly 150 times more energy per interaction. Another reinforcing factor is the sharp decline in inference

costs, which fell more than 280-fold between late 2022 and late 2024.²³ As lower prices stimulate broader and more intensive use, efficiency gains at the model and hardware level risk being offset by a classic rebound effect, whereby falling costs per unit of consumption drive total consumption higher. Taken together, these trends suggest that AI-driven electricity demand is likely to continue growing faster than efficiency improvements can offset in the near term, reinforcing the supply-side pressures on regional power markets.

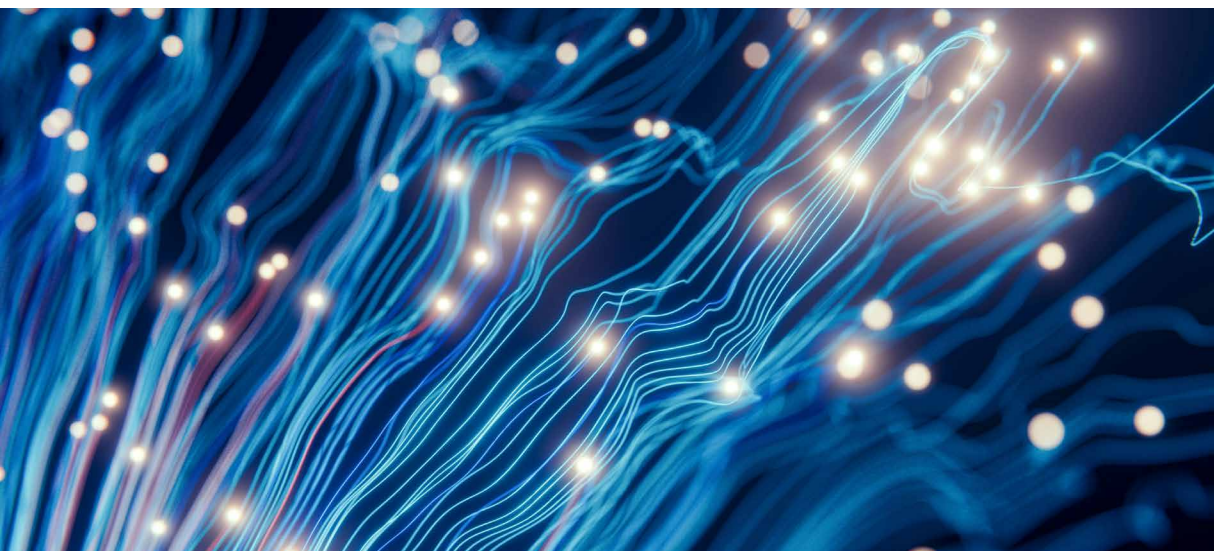
¹⁹ [The 2026 AI Index Report - Stanford HAI](#)

²⁰ [Epoch AI](#)

²¹ [OpenRouter](#)

²² Gemini's token growth partly reflects its integration into Google Search, which significantly expanded query volumes beyond standalone AI product usage.

²³ [The 2025 AI Index Report - Stanford HAI](#)

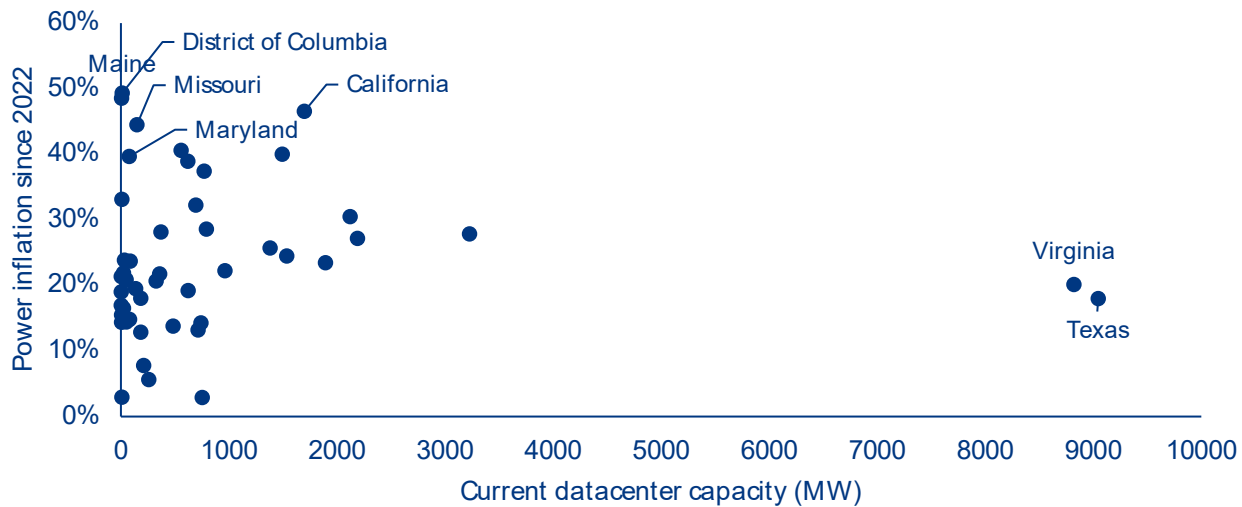


The hidden bill?: AI's impact on power prices

Data-center demand affects electricity prices through several compounding channels. Large concentrated loads drive up capacity market auction prices as grid operators must procure additional firm generation to meet anticipated peak demand, with costs socialized across all ratepayers. Grid-connection infrastructure represents a second channel as transformers, switchgear and dedicated transmission lines are typically recovered through broader utility tariffs rather than charged to the facilities driving demand. These pressures land on grids already strained by decades of underinvestment, amplifying the cost impact of each new interconnection. A third channel operates through generation mix: Data centers require firm, around-the-clock power that renewables alone cannot provide, pushing utilities toward more expensive gas and nuclear capacity. Finally, the renewable energy that data centers rely on for sustainability commitments is often located far from load centers, creating congestion costs that ripple across the broader grid. Network charges have so far remained relatively stable, but are structurally bound to rise in the regions where AI infrastructure is most concentrated. Because each channel operates on a different timescale and affects different consumer groups, the overall price impact is neither immediate nor evenly distributed across consumers and regions.

So far, aggregate state-level data tell a surprisingly benign story. Despite hosting more than 40% of total US data-center capacity, Texas and Virginia have seen power price inflation of only 18% and 20% respectively since 2022, below the 24% US average (Figure 6). This is consistent with research from Lawrence Berkeley National Laboratory (2026), which finds that state-level load growth from 2019 to 2025 was generally associated with lower all-sector average retail electricity prices as rising fixed infrastructure costs are spread across a larger base of consumers and kilowatt-hours sold.²⁴ A further moderating factor is that data centers have historically clustered in regions with favorable grid conditions and low base energy costs, insulating those markets from the price pressures that concentrated demand might otherwise create. Regulatory lag compounds this dynamic: Utility rate cases can take well over a year from filing to approval, and capital investments are recovered over asset lifetimes rather than immediately, meaning that retail prices today largely reflect costs incurred years earlier. That lag is, however, beginning to unwind: Investor-owned utilities filed USD18bn in rate increase requests in 2025, the highest level since the mid-1980s, with regulators approving 64% of requests.

²⁴ [Energy Markets & Planning](#)

Figure 6: Data-center capacity and retail electricity price inflation since 2022, by state

Source: Allianz Research based on EIA and NLR.

State-level data mask considerable heterogeneity

across individual utilities. A utility directly serving a major data-center cluster faces fundamentally different demand dynamics than a rural cooperative in the same state, yet both are folded into the same state-level price average. To move beyond the aggregate picture, we estimate the effect of data-center demand growth on residential electricity prices at the utility level, exploiting the fact that utilities entered the AI investment boom with very different pre-existing exposure to data center capacity.²⁵ The intuition is straightforward. Utilities that happened to serve counties with significant data center infrastructure were more directly in the path of the subsequent investment wave than those that did not. By interacting each utility's data center footprint with the national surge in construction investment – a design known as a Bartik shift-share instrument – we can estimate the price effect of data center demand while accounting for time-invariant utility characteristics and aggregate macroeconomic trends. The analysis covers approximately 1,200 US utilities over 2020–2024, drawing on EIA utility data matched to county-level data center capacity data from the National Laboratory of the Rockies (NLR) and national construction investment figures from the Census Bureau.

The results suggest that price pressures are more visible at the utility level than aggregate data imply, with three findings standing out. First, the price effect is concentrated entirely in the post-2023 period, precisely when AI-driven investment accelerated sharply, while the pre-2023 period shows no significant relationship between data center exposure and price changes (Table 1, column 2). This timing is difficult to explain by anything other than the AI demand shock itself, and provides confidence that we are capturing a genuine effect rather than a coincidental correlation with pre-existing trends. Second, the demand-side results validate the approach: utilities with higher data center exposure show markedly stronger commercial electricity consumption growth post-2023, consistent with hyperscale facilities connecting to utility grids and driving up load (Table 1, column 3). The effect is broad-based across utilities rather than driven by any single outlier. Third, in terms of magnitude, a utility with 4% of national data center capacity in its service territory, broadly representative of the most exposed markets outside the largest hubs, is associated with a cumulative residential price increase of approximately 3% over 2020–2024, based on the USD22bn growth in national data-center construction investment observed over that period. Modest in isolation, this effect is best understood

²⁵ Data center capacity shares reflect 2025 installed capacity, the large majority of which was planned and permitted before the generative AI boom took hold in 2023, making them a reasonable proxy for pre-existing exposure rather than a response to the shock we are trying to measure.

as an early signal rather than a final estimate – given the regulatory lag discussed above, much of the cost pressure already absorbed by utilities is still working its way through to consumers.

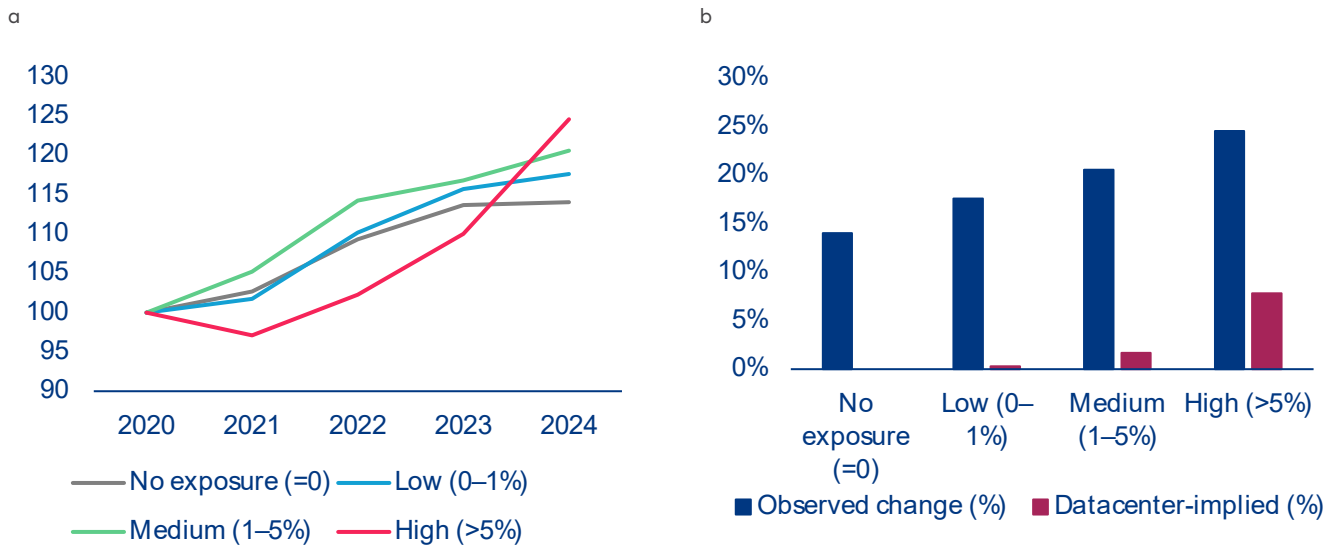
Table 1: Estimated effect of data-center demand on electricity prices

	Dep. var: log(Residential Price)		Dep. var: log(Commercial Sales/Customer)
	(1) Baseline	(2) Pre/Post Split	(3) Post-2023
Bartik IV	0.0342* (0.018)	—	—
Bartik IV × Post-2023	—	0.0321** (0.0156)	0.2416*** (0.0556)
Bartik IV × Pre-2023	—	0.0145 (0.021)	—
Observations	6,586	6,586	6,388
Within R²	0.0098	0.0115	0.033

Source: Allianz Research; Note: All specifications include utility fixed effects and year fixed effects. Standard errors are clustered at the utility level and reported in parentheses. The Bartik instrument is $B_{ut} = z_u \times G_t$, where z_u is utility u 's predetermined share of national data-center capacity and G_t is annual US data center construction investment (US Census Bureau). Specifications (1) and (2) use log residential prices as the dependent variable; (1) uses the full-period IV while (2) splits it at 2023 to test for pre-trends. Specification (3) uses log commercial sales per customer as the dependent variable with the post-2023 IV, providing demand-side validation of the instrument. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Data-center demand is leaving a measurable imprint on electricity prices, with effects concentrated in the most exposed markets. Figure 7a) shows that while all utility groups experienced meaningful price increases over 2020–2024, the pattern is far from uniform. The no-exposure group saw the smallest increase at 14%, while the high-exposure group ended the period 24.5% above its 2020 level, a gap that widened noticeably after 2023 as AI-driven investment accelerated. Of that 24.5% rise, approximately 7.8pps are directly attributable to

data-center demand based on our regression estimates (Figure 7b)), with the remainder reflecting the broader energy cost pressures that affected all utilities over the period, including fuel-price volatility, rising infrastructure investment costs and general inflation. The difference between the two groups of around 10pps therefore reflects both the data-center premium and other structural differences between high and low-exposure utility markets.

Figure 7: a) Price index by exposure group (2020=100) and b) Observed vs data center-implied price change 2020-2024 (in %)

Source: Allianz Research based on EIA and NLR; Note: Exposure groups defined by utility share of national data center capacity: no exposure (n=718), low 0-1% (n=442), medium 1-5% (n=73), high >5% (n=5). Data center-implied effect = $\beta \times \text{avg. group capacity share} \times \text{DC investment growth (USD21.91bn)}$

Across the US, residential customers are already paying around USD1.4bn more per year on their electricity bills as a direct result of data-center demand. Just five utilities, serving 4.4mn households – roughly 4% of residential customers – in Northern Virginia, the Pacific Northwest and Arizona, account for more than 40% of the total, with their customers facing an average extra annual bill of USD139, attributable to data-center growth. A wider group of 73 medium-exposure utilities pays around USD28 per customer per year, while the vast majority of US utilities are not affected. Strikingly, the most affected markets are not pricier than the national average to begin with. Their historical 5% cost advantage has already narrowed to 3.7% by end-2024, and the gap is set to close further as investment accelerates, eroding a buffer that has so far masked the true scale of the data-center premium.



Table 2: Estimated financial impact of data center demand on residential electricity prices by utility exposure group in 2024

Metric	No exposure	Low (0–1%)	Medium (1–5%)	High (>5%)
Utility characteristics				
Number of utilities	718	442	73	5
Avg. exposure share (%)	0	0.17	2.3	10.47
Market size				
Avg. residential price (USD/kWh)	0.143	0.161	0.159	0.147
Total sales (TWh)	203.2	724.5	247.4	52.5
Total customers (mn)	17.3	71.42	24.08	4.37
Estimated price impact				
Estimated price effect (%)	0	0.13	1.72	7.84
Extra bill per customer (USD/yr)	0	2.1	28.05	138.65
Extra bill total (USD mn)	0	150	675	605
Inflation impact				
Direct CPI contribution (pp)	0	0.0032	0.0427	0.1945

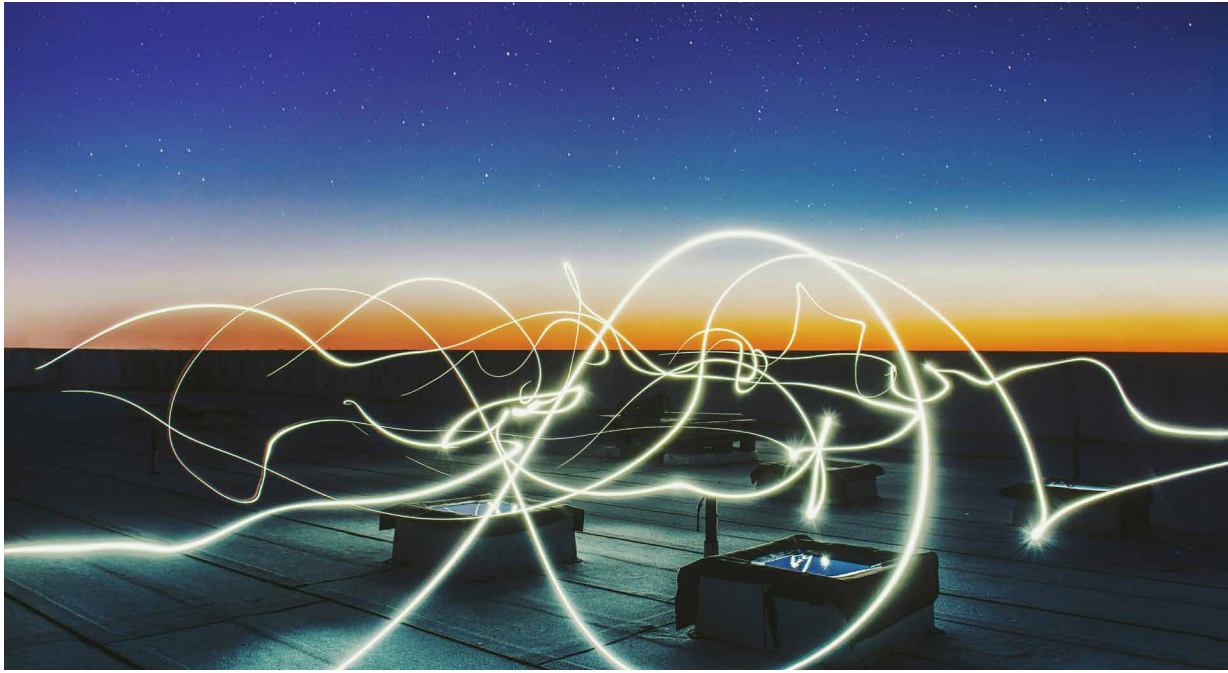
Source: Allianz Research based on EIA and NLR

Translated into aggregate consumer prices, our findings confirm an overall modest but highly skewed data-center price premium. Across the full sample, the sales-weighted average residential price increase attributable to data centers amounts to around 0.6% over 2020–2024, translating into a direct headline CPI contribution of just 0.015pp. For the most exposed utilities, however, the same channel adds roughly 0.19pp to headline CPI, around thirteen times the full-sample figure.²⁶ Because these estimates capture only residential electricity and exclude commercial pass-through, they should be read as a lower bound on the true inflation impact, with the full effect on aggregate consumer prices likely materially larger once higher input costs for electricity-intensive sectors are reflected in final goods prices.

This premium is likely to grow as the investment wave continues. Data-center investment already grew by 32% in 2025 and is set to increase by a further 75% in 2026 alone.²⁷ As pending rate cases translate accumulated cost pressures into retail tariffs, our estimates point to an additional price effect of close to 14pps for the most exposed utilities over 2025–2026, equivalent to a further CPI contribution of around 0.34pp, nearly doubling the cumulative four-year effect in just two years. With infrastructure costs still largely socialized across all rate-payers rather than borne by the facilities driving demand, the question of who ultimately pays for the AI buildout is set to become increasingly central to the policy debate.

²⁶ Direct CPI contribution estimates assume an electricity share of 2.48% in the consumer price basket, based on [BLS CPI](#) relative importance weights (December 2024)

²⁷ [Key Questions on Energy and AI - IEA](#)



Preparing AI for energy and energy for AI: policy recommendations

The scale and pace of AI-driven power demand growth points to a set of policy priorities that are urgent, tractable and largely absent from current frameworks, which would help lay the foundation for AI ambitions that the grid can actually support.

1) Expand and accelerate

- **Reform interconnection queues.** Interconnection queues are the single most binding near-term constraint on both generation capacity and data-center deployment. Streamlining queue processes, introducing binding timelines for interconnection studies, strengthening financial penalties for speculative filings that inflate backlogs and delay genuine projects and prioritizing shovel-ready projects with firm power commitments would significantly accelerate the pace at which new capacity reaches the grid.

2) Govern & allocate

- **Improve transparency and demand tracking.** Mandatory disclosure of data-center energy consumption, water use and interconnection requests would materially improve capacity planning. Incorporating large-load interconnection requests into publicly accessible data systems would make the pace and geography of data-center expansion visible to regulators, utilities and the public in real time.
- **Address overclustering risks in saturated markets.** The concentration of approximately 42% of US data-center capacity in Texas and Virginia creates systemic risks that extend beyond local grid stress. Overclustered markets face compounding vulnerabilities across power supply, water availability, cooling infrastructure, and skilled labor, while their position as critical nodes in

global AI value chains means that a regional grid failure or prolonged supply disruption could have disproportionate consequences for AI deployment globally. Policymakers and grid operators should assess concentration thresholds beyond which new approvals in saturated markets require enhanced scrutiny, consider incentive structures that actively redirect investment toward underutilized regions with available grid capacity, and prioritize locations with proximity to renewable energy resources to reduce long-run dependence on firm fossil fuel generation.

- **Require data centers to internalize their infrastructure costs.** The socialization of capacity costs onto existing ratepayers is both economically inefficient and politically unsustainable. Policymakers should establish frameworks requiring large new loads to bear a proportionate share of the transmission and capacity infrastructure their demand necessitates. Time-of-use and peak-pricing tariffs that reflect system stress would more accurately align data-center costs with the grid impacts they create, reducing the cross-subsidization of large loads by residential ratepayers. Where market-based pricing mechanisms are insufficient, direct compensation requirements, obliging data centers to offset the costs their demand imposes on host communities, provide an alternative route to addressing the equity dimension and reducing the political friction that is increasingly delaying project approvals.

3) Optimize & decarbonize

- **Implement mandatory energy-efficiency standards.** No binding energy performance standards currently apply to data centers in most jurisdictions despite their growing share of national electricity consumption. Minimum PUE thresholds, water-usage effectiveness requirements and mandatory energy intensity reporting would establish a baseline accountability framework, with stricter requirements for new hyperscale builds where best-in-class efficiency is already technically and economically achievable.²⁸
- **Develop demand-flexibility frameworks.** Full curtailment of data center operations is economically impractical given facility capital intensity, but partial demand flexibility is more tractable and largely untapped. Batch workloads such as model training and non-time-sensitive inference can in principle be shifted to off-peak periods, reducing peak demand pressure and congestion costs without compromising operational continuity. Regulators and grid operators should develop demand-response frameworks that reward this flexibility through capacity market mechanisms or dynamic pricing signals.
- **Ensure AI powers the energy transition rather than slowing it.** The risk that AI-driven data-center growth locks in years of additional fossil-fuel generation is real and underappreciated. Behind-the-meter generation, power purchase agreements tied to additionality requirements, and prioritization of nuclear and long-duration storage for firm power needs would reduce this risk. AI infrastructure and decarbonization are not inherently in tension. The former should be designed to accelerate the latter.

²⁸ [Data Centers and Their Energy Consumption - Congressional Research Service](#)

A photograph showing a group of diverse hands of various skin tones stacked on top of each other, resting on a thick, textured tree branch. The background is a lush green forest with sunlight filtering through the leaves. The text "Our team" is overlaid in the center, with "Our" in white and "team" in orange.

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